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CYBERSECURITY AND ARTIFICIAL INTELLIGENCE"

Evaluating the Robustness of AI-Based Forensic Tools under Adversarial and Anti-Forensic Attacks

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*If you can keep your head when all about you
Are losing theirs and blaming it on you, ...*

*If you can dream—and not make dreams your master;
...*

*If you can force your heart and nerve and sinew
To serve your turn long after they are gone, ...*

Yours is the Earth and everything that's in it. ...

— Rudyard Kipling, *If*— (1910)

*A Pietro,
mia Bussola, mia Luce, mia Àncora.
Mia Vita.*

*A Francesca,
mio grande Amore, mia casa.*

*A Silvia,
preziosa consigliera lungo il viaggio.*

*A Davide,
guida in un percorso di cui intravedo appena l'inizio.*

*A Mario,
esempio e ispirazione per il cambiamento.*

*Alla mia famiglia d'origine
e alla mia famiglia acquisita,
per il sostegno e l'aiuto che non mi hanno mai fatto
mancare.*

*A tutte le persone che ho incontrato lungo il cammino,
a chi mi ha aiutato a diventare ciò che sono.*

Abstract

The increasing integration of Artificial Intelligence (AI)-based systems into Digital Forensics (DF) workflows is changing how large collections of digital images are classified, retrieved, and prioritized for human examination. Although these systems may reduce analyst workload and accelerate the identification of potentially relevant content, their operational value depends on whether their behavior remains stable when inputs are degraded, distributionally different, or deliberately manipulated.

This thesis evaluates the operational robustness of transparent proxy models and selected black-box software configurations used for automated image classification and semantic retrieval in forensic contexts. Rather than proposing a new classifier or ranking commercial products, the study examines how heterogeneous systems behave under clean, Out-of-Distribution (OOD), adversarial, and anti-forensic conditions. The evaluation is organized through Forensic AI Robustness Lab (FAIRLab), a task-specific methodological protocol connecting human-validated dataset construction, deterministic partitioning, transparent proxy-model evaluation, perturbation generation, blind-bundle preparation, black-box output normalization, quantitative analysis, qualitative explainability, and integrity controls. Errors are interpreted in terms of missed-target risk, human-review workload, directional transitions, configuration sensitivity, and source-to-output traceability.

The frozen source population contains 1 500 images: 500 **weapon** images, 500 **non_weapon** images, and a separate 500-image OOD branch. The OOD population is not treated as a third supervised class. It is used to characterize the assignments produced when closed-set systems process images that do not conform to the nominal binary task.

The 1 000-image binary subset provides the common basis for generating 5 000 adversarial or adversarial-style artifacts and 5 000 anti-forensic artifacts. Together with the 1 000 clean binary inputs and 500 clean OOD inputs, these derived populations form an 11 500-item forensic evaluation bundle. The 10 000 manipulated artifacts remain source-condition variants of the same 1 000 binary images and do not represent independent underlying cases.

The transparent level comprises EfficientNet-B0, ResNet18, and a Contrastive Language-Image Pre-training (CLIP)-based proxy. The black-box level comprises four software systems evaluated through six frozen configurations: Magnet AXIOM/Magnet.AI, Magnet Griffeye/T3K CORE, Excire Foto 2025 at D20, D50, and D80, and Celebrite Inseyets. Their tags, bookmarks, exported labels, and semantic-retrieval results are interpreted through documented normalization rules without assuming architectural or semantic equivalence.

Four model-dependent attacks—Fast Gradient Sign Method (FGSM), One Pixel Attack, Sigma Zero Attack, and SuperDeepFool—were generated against fold-specific EfficientNet-B0 checkpoints. Their evaluation on EfficientNet-B0 constitutes direct-target assessment, whereas their application to the other proxies and black-box configurations measures empirical transfer. The model-agnostic Color Shift condition and five anti-forensic transformations are interpreted separately. The analysis combines conventional performance metrics with directional error transitions, OOD assignment behavior, architecture-specific maximum predicted-class probabilities, and five qualitative Integrated Gradients (IG) case studies. Probability and explainability outputs are used only as model-specific diagnostic evidence.

The results show that strong clean performance does not imply uniform operational robustness. Clean proxy-model accuracies are 0.978 for EfficientNet-B0, 0.964 for ResNet18, and 0.927 for the CLIP-based proxy. Under direct-target evaluation, EfficientNet-B0 accuracy decreases to 0.575 under FGSM and to 0.015 under Sigma Zero Attack. The same artifacts produce more limited and heterogeneous effects on the non-target systems, demonstrating that empirical transfer must be measured rather than assumed. The anti-forensic conditions generally produce smaller aggregate variations than the strongest direct-target attacks but still introduce operationally relevant sample-level errors.

The separate OOD analysis further shows that out-of-task inputs may still be assigned to the firearm-oriented operational state. At the proxy level, the fold-aggregated weapon assignment rates are 0.3696 for EfficientNet-B0, 0.4376 for ResNet18, and 0.8356 for the CLIP-based proxy. These values represent closed-set assignment behavior rather than supervised correctness.

The black-box evaluation produces 69 000 normalized decisions and reveals distinct configuration-dependent operating profiles. No evaluated configuration simultaneously minimizes missed detections, false-positive review workload, and OOD weapon assignments. In particular, changes to the Excire Foto 2025 **Distance limit** materially alter recall, selectivity, and OOD behavior. The results therefore characterize the frozen software versions, enabled functions, configurations, observable output semantics, and normalization mappings.

The thesis concludes that AI-based image-analysis systems may provide useful support for forensic triage but must not be treated as autonomous mechanisms for evidentiary decision-making. Their defensible use requires task-specific validation, explicit threat modeling, documented versions and configurations, separate analysis of OOD behavior, source-to-output traceability, and continued expert verification. FAIRLab contributes a unified and auditable procedure for making these requirements experimentally observable across transparent proxy models and selected black-box software configurations evaluated on the same frozen experimental basis.

Keywords: Digital Forensics; Artificial Intelligence; AI-based forensic tools; image classification; operational robustness; adversarial attacks; anti-forensic techniques; out-of-distribution analysis; explainability.

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Acronyms

Adv-ML Adversarial Machine Learning

AI Artificial Intelligence

AUROC Area Under the Receiver Operating Characteristic Curve

BLIP Bootstrapping Language-Image Pre-training

CGI Computer-Generated Imagery

CLIP Contrastive Language-Image Pre-training

CNN Convolutional Neural Network

DF Digital Forensics

DL Deep Learning

FAIRLab Forensic AI Robustness Lab

FGSM Fast Gradient Sign Method

FN False Negative

FNR False Negative Rate

FP False Positive

FPR False Positive Rate

GDPR General Data Protection Regulation

GPU Graphics Processing Unit

IG Integrated Gradients

JPEG Joint Photographic Experts Group

MD5 Message Digest Algorithm 5

ML Machine Learning

OOD Out-of-Distribution

OSINT Open Source Intelligence

PGD Projected Gradient Descent

PRNU Photo Response Non-Uniformity

SHA-256 Secure Hash Algorithm 256-bit

SVM Support Vector Machine

TN True Negative

TP True Positive

UFED Universal Forensic Extraction Device

ViT Vision Transformer

xAI Explainable Artificial Intelligence

Chapter 1

Introduction

The integration of Artificial Intelligence (AI)-based systems into Digital Forensics (DF) workflows addresses the need to examine growing volumes of digital data and identify content that may be relevant to an investigation. When automated support is used to classify, organize, or retrieve images, however, its operational value depends on the stability and interpretability of the outputs produced under realistic conditions. Degraded, distributionally different, or intentionally manipulated inputs may prevent relevant content from receiving appropriate review priority or may increase the amount of irrelevant material presented to the analyst.

Against this background, the present thesis evaluates the operational robustness of transparent proxy models and selected black-box software configurations used for automated image classification or semantic retrieval of images depicting firearms and related visual content. The evaluation compares behavior across clean binary inputs, a separate Out-of-Distribution (OOD) branch, adversarial conditions, and anti-forensic transformations. Its purpose is not to validate automation as an autonomous evidentiary decision-making mechanism, but to examine the conditions under which it may provide controlled and traceable support for forensic triage. Human oversight, source-to-output traceability, explicit configuration management, and cautious interpretation of automated results remain central principles throughout the study [1, 2, 3].

1.1 Digitalization, Threats, and the Central Role of Digital Forensics

The progressive digitalization of contemporary society has transformed how information is produced, stored, transmitted, and analyzed. Cloud infrastructures, distributed services, mobile devices, social platforms, and heterogeneous information systems generate increasing volumes of data that may be relevant to personal, institutional, investigative, and judicial activities. The same technological evolution has expanded the points at which digital information may be altered, concealed, destroyed, or otherwise compromised [4, 5].

Within this context, DF provides the scientific and technical methods required to identify, acquire, preserve, examine, analyze, and present digital evidence in a traceable and technically defensible manner. The value of an analytical result depends not only on the information obtained, but also on the possibility of reconstructing which data were examined, which operations were performed, and which limitations affected the analysis [6, 7, 8, 9].

These requirements also apply to the tools used during examination and analysis. When automated functions influence the selection, organization, or prioritization of evidence, their reliability and documentation may directly affect the completeness and verifiability of the forensic workflow. The present study is positioned within this problem: it examines automated

image-analysis outputs as operational support whose value depends on controlled validation, traceability, and continued expert review.

1.2 Automation of Forensic Analysis and the Role of AI

The quantitative and qualitative expansion of the data subject to investigation has made exclusively manual forensic analysis increasingly difficult to sustain. Traditional methods remain essential for technical control, verification, and final interpretation, but they may be difficult to scale when analysts must examine large collections of images, videos, logs, conversations, system files, and heterogeneous artifacts. Under these conditions, automation can support the organization, prioritization, and preliminary examination of evidence collections [1, 4].

Beyond scalability, repeated exposure to disturbing or sensitive multimedia content may also affect the well-being of practitioners involved in forensic review. A survey of investigators working with child sexual abuse material highlights the value of filtering and safer-presentation technologies not only for improving examination efficiency, but also for reducing unnecessary and prolonged exposure to potentially harmful content [10].

The integration of AI, Machine Learning (ML), and Deep Learning (DL) techniques into forensic tools has expanded this support through semantic classification, visual similarity analysis, grouping of recurring patterns, anomaly detection, and retrieval of potentially relevant material. These functions do not merely accelerate existing procedures. By influencing which items receive early attention and which remain outside the initial review set, they may alter the allocation of analyst time and the organization of human examination [3, 11].

The benefits of automation therefore cannot be assessed exclusively in terms of speed or workload reduction. A false negative may prevent relevant content from being prioritized, whereas a false positive may increase analytical noise and the amount of material requiring manual verification. The operational consequences depend on both the frequency and the direction of the errors. Evaluation must consequently extend beyond nominal accuracy and consider unexpected or manipulated inputs, system versions and configurations, the semantics of observable outputs, the rules used to normalize them, and the traceability of each automated result.

Automated classifications, tags, bookmarks, and semantic retrieval results must remain subordinate to expert verification. They may support triage and evidence organization, but they do not independently establish the relevance, authenticity, or evidentiary weight of an image. This distinction is especially important for proprietary or otherwise non-transparent systems, whose internal models, thresholds, and preprocessing procedures may not be available to the analyst.

1.3 Robustness of AI-Based Systems and Adversarial and Anti-Forensic Threats

Strong performance under standard conditions does not necessarily imply stable behavior in the presence of altered, degraded, distributionally different, or hostile inputs. This limitation is particularly relevant to DF, where automated image-analysis systems may process content affected by compression, resizing, color alteration, platform-specific processing, degradation, or intentional manipulation.

Anti-forensic techniques broadly include strategies intended to obstruct the identification, acquisition, preservation, examination, or interpretation of potentially relevant data. They may involve concealment, obfuscation, metadata manipulation, or image-level transformations independently of whether an AI-based classifier is present. Steganography represents one example of the broader domain, although the experimental anti-forensic branch of this thesis

is restricted to image-level transformations directly relevant to the robustness of automated visual analysis [11, 12].

Adversarial attacks are instead defined in relation to their effect on a model or decision process. The literature on Adversarial Machine Learning (Adv-ML) has shown that deliberately optimized inputs may induce erroneous predictions even when the applied alteration is constrained or visually limited. These findings challenge the assumption that high clean-data accuracy is sufficient evidence of reliability under realistic or hostile conditions [13, 14, 15, 16].

The two domains remain conceptually distinct, but they intersect when a manipulation reduces the ability of an automated system to identify, retrieve, or prioritize content of investigative interest. A manipulation directed at a classifier may therefore also produce an anti-forensic consequence by affecting triage and subsequent human review [2, 3].

In operational scenarios, adversarial or anti-forensic manipulations may also be introduced deliberately by criminals or cybercriminals to reduce the likelihood that relevant content is detected or prioritized; the increasing availability of general-purpose generative AI tools may further lower the technical barrier to producing or adapting code and digital content for concealment or evasion purposes [11].

The threat model is essential to interpreting these effects. Direct-target evaluation must be distinguished from empirical transfer to a different architecture or software system. White-box and score-based access represent different attacker capabilities, whereas deterministic model-agnostic transformations follow a separate experimental rationale. The resulting effects must not be treated as equivalent forms of attack success.

Operational robustness is defined in this thesis as the observable stability of a specific system, version, configuration, task, and output mapping across the frozen input conditions included in the protocol. It is not an absolute or version-independent property. A system may perform strongly on clean inputs while increasing missed-target risk under a specific manipulation, generating additional false-positive review workload, producing closed-set assignments on out-of-task content, or exhibiting sensitivity to thresholds, queries, and normalization rules.

1.4 Research Problem and Relevance in Realistic Forensic Scenarios

Despite the extensive literature on adversarial attacks and model robustness, the evaluation of these issues in operationally oriented forensic scenarios remains comparatively limited. Many studies rely on standardized benchmarks, general-purpose *computer vision* datasets, or experimental conditions that do not reproduce the organization and interpretive constraints of a forensic workflow.

Studies more closely aligned with forensic practice indicate that automated image analysis must address large evidence collections, heterogeneous sources, task-specific semantic categories, ambiguous content, and outputs intended to support analyst review. Silva et al. propose an AI-based system for classifying images contained in criminal evidence, including firearms and other investigatively relevant categories. More recent work by Sanna et al. shows that automated functions integrated into forensic software may exhibit operationally relevant limitations when processing manipulated or unexpected content [2, 3, 17].

The research problem is therefore not limited to demonstrating that a classifier can be induced to fail. It is necessary to determine how an error affects the support provided to the analyst: whether relevant content is no longer prioritized, whether irrelevant material is routed toward review, whether recall and selectivity change, and whether out-of-task inputs receive closed-set assignments. It is also necessary to distinguish effects produced by attacks optimized against a source model from empirical transfer to other architectures or proprietary software, and to determine how configuration and output-normalization choices influence the observed behavior.

Addressing these questions requires evaluation populations aligned with the defined operational task. Such populations should include realistic positives, realistic negatives, manipulated variants, borderline cases, and content that does not conform to the nominal classification problem. This principle is consistent with research on scenario-oriented evaluation of automated visual systems, including the PHASER framework, which emphasizes alignment between evaluation data and the intended deployment context [18].

The present thesis applies this principle to a controlled task centered on realistic individual firearms and clearly recognizable firearm-related visual content. The supervised component distinguishes **weapon** and **non_weapon**, while a separate OOD branch contains borderline, anomalous, degraded, synthetic, semantically adjacent, and otherwise out-of-task content. The OOD branch is not treated as a third supervised class. It is used to characterize the assignments produced when closed-set systems process inputs that do not conform to the nominal binary task [17, 19].

Adversarial and anti-forensic artifacts are generated exclusively from the binary subset, preserving the distinction among supervised correctness, clean-to-perturbed error transitions, and behavior on the separate OOD branch. The evaluation combines transparent proxy models with selected black-box software configurations on the same frozen source population and derived perturbation suite. The proxy models provide an inspectable experimental reference, whereas the proprietary systems are assessed through outputs observable to the analyst. No assumption is made that the proxy models reproduce or explain their inaccessible internal components.

Within this framework, a **weapon-to-non_weapon** transition may prevent potentially relevant content from being prioritized, while the opposite transition may increase manual review workload. An OOD assignment to **weapon** is not an ordinary supervised false positive, but it may still route out-of-task material toward a firearm-oriented review category.

The research problem is therefore located at the intersection of algorithmic robustness, software configuration, output interpretation, and operational reliability. The scientific contribution lies in defining and applying an integrated, task-specific evaluation protocol, while the operational relevance lies in identifying limitations, error profiles, configuration dependencies, and traceability requirements that must remain visible when automated tools are used to support evidence analysis [2, 3, 17, 18].

1.5 Research Objectives

This thesis aims to evaluate systematically the operational robustness of transparent proxy models and selected black-box software configurations used for the automated classification or semantic retrieval of images of potential investigative interest in DF. The general objective is to determine whether behavior observed under clean reference conditions remains stable when the evaluated systems process distributionally different, degraded, transformed, or intentionally perturbed images.

The experimental scope distinguishes the supervised **weapon/non_weapon** task from a separate OOD branch and treats adversarial and anti-forensic conditions as distinct evaluation branches. The adversarial branch comprises model-dependent attacks and the deterministic, model-agnostic Color Shift condition, which is interpreted separately from direct-target and transferred attack success. The anti-forensic branch comprises plausible image-level transformations that may arise through ordinary media processing or may be used intentionally to reduce the effectiveness of automated analysis [20, 21].

Within this scope, the work is organized around four main objectives.

The first objective is to define a rigorous, documented, and reproducible methodological protocol for evaluating operational robustness under controlled but operationally relevant forensic image-analysis conditions. The protocol connects human-validated dataset construction, deterministic partitioning, proxy-model evaluation, perturbation generation, blind-bundle

preparation, black-box output normalization, quantitative and qualitative analysis, and traceability controls.

The second objective is to evaluate heterogeneous systems on the same frozen source population and derived perturbation suite. The transparent level comprises EfficientNet-B0, ResNet18, and a Contrastive Language-Image Pre-training (CLIP)-based proxy consisting of a frozen visual encoder and a supervised binary classification head. The black-box level comprises the selected software configurations evaluated exclusively through observable outputs and explicitly documented normalization rules.

The third objective is to characterize robustness through more than aggregate accuracy. The analysis considers false negatives, false positives, clean-to-perturbed error transitions, OOD weapon assignment rates, architecture-specific maximum predicted-class probabilities, and configuration-dependent changes in recall and selectivity. Direct attacks against fold-specific EfficientNet-B0 source models are distinguished from empirical transfer to ResNet18, the CLIP-based proxy, and the selected black-box configurations. A complementary qualitative Explainable Artificial Intelligence (xAI) analysis of selected EfficientNet-B0 cases supports case-level diagnostic inspection but is not treated as quantitative robustness evidence or as an explanation of the black-box systems.

The fourth objective is to relate the experimental findings to the operational use of automation in DF. Automated classifications, tags, bookmarks, and semantic retrieval results are treated as triage indicators rather than as autonomous evidentiary determinations. The analysis therefore identifies the limitations, error profiles, configuration dependencies, traceability requirements, and forms of human verification necessary for compatibility with a *human-in-the-loop* forensic workflow [1, 2, 22].

1.5.1 Research Questions

The experimental design addresses the following research questions:

- RQ1.** To what extent do the transparent proxy models maintain their clean-baseline performance under the separate OOD, adversarial, and anti-forensic conditions?
- RQ2.** How do direct source-model attacks, cross-architecture transfer, and the model-agnostic Color Shift condition differ in their effects on the evaluated proxy models?
- RQ3.** How do the selected black-box software systems respond to the same frozen experimental corpus, and how do observable output semantics and configuration choices affect the resulting operational profiles?
- RQ4.** Which error transitions, robustness degradations, OOD assignment behaviors, and unknown output states are most relevant to forensic triage and continued human review?

1.6 Contribution of the Thesis

The main contribution of this thesis lies in the definition and application of a methodological protocol for evaluating the operational robustness of transparent proxy models and selected black-box software configurations used for automated image classification or semantic retrieval in DF. Unlike evaluations focused exclusively on standard *computer vision* benchmarks or aggregate clean-condition performance, the study adopts a forensic and operational perspective across clean, OOD, adversarial, and anti-forensic conditions [2, 3, 21, 23].

The contribution is articulated across four interconnected dimensions.

The first contribution is the formalization of the **Forensic AI Robustness Lab (FAIRLab)** protocol. FAIRLab coordinates human-validated dataset construction, deterministic partitioning, transparent proxy-model evaluation, controlled perturbation generation, blind-bundle preparation, black-box output normalization, error-oriented metrics, qualitative explainability, and traceability controls. It is not proposed as a new classifier, defensive mechanism, standalone

forensic application, or general theoretical architecture, but as a documented and auditable procedure for evaluating heterogeneous systems under controlled but operationally relevant conditions.

The second contribution is the construction of a frozen experimental corpus aligned with the selected forensic image-analysis task. The source population contains 1 500 images: 500 **weapon** images, 500 **non_weapon** images, and 500 images reserved for the separate OOD branch. The 1 000-image binary subset is used to generate five adversarial or adversarial-style populations and five anti-forensic populations. Together with the clean binary and OOD inputs, these derivatives form an 11 500-item forensic evaluation bundle comprising 1 000 clean binary items, 500 clean OOD items, 5 000 adversarial artifacts, and 5 000 anti-forensic artifacts. The 10 000 manipulated artifacts remain derived source-condition combinations rather than independent underlying cases.

Sample selection follows a documented *human-in-the-loop* procedure. Manual validation is treated as an explicit methodological component, and the adopted criteria, reviewer decisions, and associated limitations remain traceable [24].

The third contribution is the parallel evaluation of transparent proxy models and selected black-box software configurations on the same frozen experimental basis. The transparent level comprises EfficientNet-B0, ResNet18, and a CLIP-based proxy. The black-box level comprises four software systems evaluated through six frozen configurations: Magnet AXIOM/Magnet.AI, Magnet Griffeye/T3K CORE, Excire Foto 2025 at D20, D50, and D80, and Cellebrite Insecrets. Their tags, bookmarks, exported classifications, and semantic-query results are normalized through explicit operational rules without assuming equivalent internal taxonomies or decision mechanisms.

The proxy models provide an inspectable reference and the controlled generation layer for model-dependent adversarial artifacts. The protocol distinguishes direct-target attacks generated against fold-specific EfficientNet-B0 checkpoints from empirical transfer to ResNet18, the CLIP-based proxy, and the selected black-box configurations. The deterministic and model-agnostic Color Shift condition remains separate, preventing transfer results from being interpreted as direct robustness evaluations of systems against which no matched attack was optimized.

The fourth contribution is an auditable and error-oriented interpretation of operational robustness. The evaluation considers false negatives affecting firearm-oriented content, false positives increasing analyst workload, directional clean-to-perturbed transitions, closed-set assignments on the separate OOD branch, configuration-dependent recall and selectivity, and residual embedded-metadata sensitivity. Stable identifiers, manifests, integrity records, and normalized outputs preserve source-to-result traceability throughout the experimental workflow and support procedural reproducibility and artifact-level auditability.

A complementary qualitative xAI analysis applies Integrated Gradients (IG) to five purposively selected EfficientNet-B0 cases. The attribution maps support case-level diagnostic inspection but are not treated as quantitative robustness evidence, causal explanations, semantic localization ground truth, or explanations of the selected black-box systems.

Overall, the thesis does not propose a universally robust firearm classifier or rank the evaluated software products. Its contribution consists in defining and applying a traceable procedure for revealing how heterogeneous automated image-analysis systems behave when the assumptions of clean, closed-set evaluation no longer hold. Automated outputs remain triage indicators whose value depends on task-specific validation, explicit configuration and mapping rules, source-to-result traceability, and continued expert review [1, 2, 3].

1.7 Technical-Forensic Relevance and Operational Reliability

In forensic contexts, the performance of an automated tool cannot be assessed independently of the task for which it is used and of the consequences that its outputs may have on evidence analysis. A false negative on firearm-oriented content may prevent a potentially relevant image from receiving appropriate review priority, whereas a false positive may increase the amount of irrelevant material requiring examination. An OOD image assigned to **weapon** is not an ordinary supervised false positive, but it may still trigger a flag or increase the material routed toward review [11, 22].

The relevance of an automated output therefore depends on more than aggregate accuracy. It also depends on the system version, active configuration, observable output semantics, adopted normalization rule, input population, and threat model. This is particularly important for black-box systems, whose tags, bookmarks, exported classifications, percentages, or semantic-query results do not necessarily represent equivalent binary decisions or comparable uncertainty measures.

Operational reliability is closely connected to traceability, verifiability, and critical oversight. A defensible result should remain linked to the source item and its integrity record, the model checkpoint or software version, the active configuration, the observable output, the normalization rule, and the condition under which processing occurred. Automated outputs must remain reviewable components of a *human-in-the-loop* workflow rather than autonomous sources of technical truth.

The study remains deliberately limited from a legal and evidentiary perspective. It does not determine the admissibility, probative value, or evidentiary weight of an automated classification, nor does it certify any evaluated software product. Such determinations require case-specific legal analysis, independent validation, chain-of-custody assessment, disclosure of the applicable technical procedures, and consideration of the relevant procedural framework. The technical-forensic relevance of the work instead lies in distinguishing the controlled instrumental use of AI from uncritical reliance on automated outputs [25, 26, 27].

1.8 Structure of the Thesis

The thesis is organized into seven chapters, progressing from the theoretical and research context to the methodological protocol, its implementation, the experimental results, and their operational interpretation. A separate appendix documents the principal repository and reproducibility artifacts.

Chapter 1 introduces the research problem, defines the objectives and Research Questions, presents the main contributions, and outlines the technical-forensic relevance of the study.

Chapter 2 presents the theoretical foundations of digital evidence, automated forensic analysis, AI-based image classification, Adv-ML, anti-forensic techniques, OOD behavior, and xAI.

Chapter 3 reviews the relevant literature and identifies the gap concerning integrated and operationally oriented robustness evaluation of transparent models and black-box software systems.

Chapter 4 defines the high-level FAIRLab protocol, including the research design, evaluation branches, threat model, metrics, explainability approach, and principles governing traceability, integrity, and reproducibility.

Chapter 5 documents the concrete experimental setup, data and model configurations, perturbation-generation procedures, forensic evaluation bundle, software versions, output-normalization rules, and technical validation controls.

Chapter 6 presents the clean, OOD, adversarial, anti-forensic, black-box, sensitivity, comparative, and qualitative explainability results and interprets their operational implications.

Chapter 7 answers the Research Questions, summarizes the findings and contributions, discusses limitations and forensic implications, and identifies directions for future work.

The separate **Appendix A** documents the repository structure, canonical artifacts, official pipeline sequence, validation assets, and boundaries of controlled reproducibility.

Chapter 2

Technical Background Concepts

This chapter introduces the theoretical concepts necessary to understand the evaluation of the operational robustness of proxy models and AI-based software employed in DF for the automated classification or semantic retrieval of images. The theoretical framework is built upon the foundations of DF and digital evidence. The chapter then examines the role of AI in forensic workflows, the relevance of images as digital evidence, the principles of ML and DL applied to computer vision, Adv-ML, anti-forensic techniques applied to visual content, and the role of xAI in assessing model reliability. In this chapter, these topics are framed at a conceptual level; the rationale for their selection, the adopted configurations, and the procedures used to apply them within the experimental protocol are discussed in the methodology chapter.

2.1 Digital Forensics and Digital Evidence

DF is the technical and scientific discipline concerned with the identification, acquisition, preservation, examination, analysis, and presentation of digital data that may be relevant in an investigative or judicial context. It arises from the need to handle information stored or transmitted through electronic devices according to procedures suitable for preserving its evidentiary value. From this perspective, digital data do not automatically acquire the status of evidence: they must be collected, documented, and interpreted through procedures that are controllable, repeatable, and compatible with the requirements of integrity, authenticity, and traceability demanded by forensic investigation [4, 5, 8, 28].

Digital evidence presents distinctive characteristics compared with traditional physical evidence. It is intangible, easily replicable, often distributed across devices, cloud services, removable media, and communication systems, and may be altered even by apparently ordinary operations, such as powering on a device, opening a file, or executing an application. This fragility makes it necessary to distinguish between the mere technical availability of data and their forensic usability. An image, a log, a message, or a multimedia file can contribute to the reconstruction of an event only if its investigative life cycle is documented in a way that makes it possible to verify the operations performed and any transformations it may have undergone.

The requirements of integrity, authenticity, and traceability therefore constitute the methodological foundation of DF. Integrity concerns the ability to demonstrate that the analyzed data have not been modified in an uncontrolled manner with respect to the acquired source. Authenticity refers to the possibility of linking the evidence to a specific device, subject, system, or operational context. Traceability, finally, requires every phase of forensic activity to be documented through records, logs, technical notes, hash values, and chain-of-custody procedures. These elements do not serve only an administrative function; rather, they make it possible to verify the path that leads from the original digital item to the information presented in an investigative or judicial setting [9, 27, 29, 30].

From an operational perspective, the literature and technical guidelines describe the digital forensic process as a sequence of logically distinct phases: collection, examination, analysis, and reporting [31]. The collection phase concerns the identification and acquisition of relevant digital sources, with particular attention to preserving the state of the evidence. The examination phase includes the extraction and organization of potentially relevant data, for example through file system analysis, artifact recovery, indexing, and filtering. The analysis phase consists of the technical interpretation of the extracted data, the correlation among different sources, and the reconstruction of relevant events or behaviors [32]. Finally, the reporting phase requires clear, verifiable, and understandable documentation of the activities performed, the tools used, the results obtained, and the limitations encountered [5, 7, 8, 33].

Figure 2.1 summarizes this general workflow and places the present thesis primarily within the *examination* and *analysis* phases.

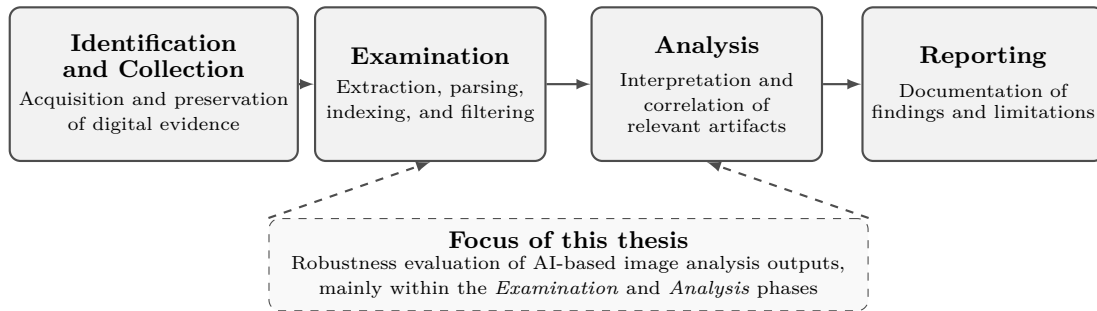


Figure 2.1: Simplified digital forensic workflow

2.1.1 Acquisition and Preservation of Digital Evidence

The acquisition phase aims to make copies or extractions of digital evidence available for examination while preserving, as far as possible, their state, integrity, and traceability. The operational procedures depend on the nature of the source, its state at the time of intervention, and the type of information being sought. In general terms, a distinction can be made between the acquisition of persistent data from storage media and the acquisition of volatile data from active systems. In both cases, the operations performed must be documented in a way that enables the subsequent reconstruction of the acquisition process [5, 8, 9].

In the case of storage media accessible under controlled conditions, a central technique is physical acquisition, or *bit-stream imaging*, through which a sector-by-sector copy of the original medium is produced. Unlike a simple logical copy of visible files, a bit-stream image may also preserve unallocated space, *slack space* areas, file system structures, and remnants of deleted data that may be relevant to the investigation. When technically applicable, the use of *write blocking* devices makes it possible to prevent accidental write operations on the source medium during acquisition. The integrity of the copy is then verified through cryptographic hash functions, such as Secure Hash Algorithm 256-bit (SHA-256), with the values obtained, tools used, operators, date, and conditions of the activity documented within the chain of custody [9, 27, 33].

Logical acquisition, by contrast, consists of extracting the data made available by the operating system, the application, or the device under analysis, without necessarily including the entire physical structure of the medium. This mode may be relevant, for example, for acquiring application content, multimedia archives, or data accessible from mobile devices and digital services. It may allow information of interest to be obtained rapidly, but it has limitations with respect to the completeness of the recovered data and the availability of unallocated areas or deleted artifacts.

When the system is powered on or contains information that would be lost upon shutdown, a *live* acquisition may be necessary. This may include volatile memory, running processes,

active network connections, open sessions, mounted encrypted volumes, or other transient elements. Live acquisition inevitably involves interaction with the system under analysis and must therefore be justified and documented with particular caution, specifying which changes were introduced by the operator’s activity or by the tools employed [5, 8].

For the purposes of this thesis, these principles are relevant because the images subjected to automated classification must be considered digital artifacts whose provenance, integrity, and processing history affect the interpretation of the results. Robustness evaluation therefore does not replace the procedures for acquiring and preserving evidence, but is situated downstream from them, examining the reliability of automated support applied to content that has already been acquired and organized under documented conditions.

2.1.2 Examination, Analysis, and Triage of Digital Content

Once evidence has been acquired and preserved, the *examination* phase aims to identify, extract, and organize potentially relevant artifacts. Commonly used techniques include file system analysis, recovery of deleted files, *file carving*, metadata examination, keyword searching, content indexing, hash-based comparison, and the extraction of artifacts generated by operating systems or applications. These activities make it possible to reduce the volume of raw data and isolate sets of content to be subjected to subsequent technical and investigative assessment [5, 8, 33].

The *analysis* phase instead concerns the interpretation of the extracted artifacts and their correlation with the investigative hypotheses. It may include the temporal reconstruction of events through *timeline analysis*, comparison across different sources, interpretation of metadata, the identification of relationships among files, users, or devices, and the assessment of the investigative significance of multimedia content. In this phase, the mere presence of a file is not sufficient: its relevance, context, reliability, and relationship with the other available elements must be established.

In the case of large collections of images and videos, examination and analysis activities may be preceded or supported by triage procedures. Triage aims to prioritize potentially relevant content without replacing the analyst’s verification. In this context, semantic search, visual grouping, and automated classification functionalities may help highlight images compatible with investigative categories of interest, such as the presence of a weapon. However, the operational value of these functionalities depends on their reliability: a false negative may reduce the likelihood that relevant content will be examined promptly, whereas a false positive may unnecessarily increase the review burden.

The present thesis is positioned precisely at this point in the forensic workflow. It does not evaluate acquisition techniques or propose a method for authenticating digital evidence, but analyzes the operational robustness of automated classification and semantic image retrieval as forms of support for triage and the examination of previously acquired visual content. This distinction is methodologically essential: proper acquisition ensures preservation of the artifact, whereas the evaluation conducted in this work concerns the reliability of subsequent automated selection when the image is clean, out-of-distribution, degraded, or intentionally manipulated.

2.2 Artificial Intelligence in Digital Forensics

The integration of AI into DF addresses the increasing quantitative and qualitative complexity of digital investigations. The growing volume of data acquired from personal devices, cloud infrastructures, mobile systems, messaging applications, and multimedia archives makes an exclusively manual analysis progressively less sustainable. In this scenario, automation techniques, intelligent indexing, and assisted classification can support the analyst in the preliminary selection of potentially relevant information, reducing the operational workload and accelerating the examination and analysis of digital evidence [1, 4].

In forensic contexts, AI can be employed for heterogeneous tasks, including pattern recognition,

the grouping of similar content, semantic search, the automated classification of images and documents, anomaly detection, and the prioritization of artifacts to be submitted to human review. These functionalities do not replace the operator’s evaluative activity, but they can support the navigation of large datasets, especially when the evidence contains large amounts of multimedia content or poorly structured information. The relevance of AI in DF should therefore be understood in terms of investigative assistance, rather than as an automated delegation of evidentiary decision-making [1, 22].

The use of ML and DL models, however, introduces a discontinuity with respect to traditional forensic tools. Many classical digital analysis procedures rely on deterministic rules, known signatures, hashes, metadata, keyword searches, or file system structures. Machine learning models, by contrast, produce probabilistic outputs based on representations learned from data. This characteristic makes it possible to process complex content, such as images, videos, and natural language text, but it also requires closer attention to the quality of the training data, the distribution of inputs, model generalization, and the stability of predictions under non-ideal conditions.

In forensic settings, this distinction is particularly relevant because the output of an AI-based system can influence how the analyst explores the evidence. An incorrect classification does not necessarily determine an autonomous evidentiary conclusion, but it may affect the priority assigned to specific content, the visibility of relevant artifacts, or the overall efficiency of the investigative workflow. Consequently, the reliability of an automated module cannot be assessed solely through aggregate accuracy metrics, but must also be considered with respect to false positives, false negatives, distributional bias, and scenarios involving intentional manipulation.

Recent literature has shown that the use of AI-based functionalities in software employed for forensic analysis raises a specific validation problem: it is not sufficient to verify that the system works correctly on ordinary data; it is also necessary to understand its behavior in the presence of degraded, manipulated, or out-of-distribution data [2, 3]. This aspect is especially important when the analysis concerns visual content, since images and videos can be altered through compression, resizing, blurring, contrast modification, or adversarial perturbations. Under such conditions, a system that appears accurate on clean data may exhibit a significant reduction in its classification capability.

For these reasons, the adoption of AI in DF must be accompanied by a paradigm based on human supervision, technical documentation, and experimental evaluation of robustness. The analyst must be able to understand the role played by the automated system, the limitations of the classification produced, and the conditions under which the model may generate errors. From this perspective, AI does not represent a substitute for forensic expertise, but rather a support tool whose usefulness depends on the possibility of verifying its performance, stability, and reliability within the operational context in which it is used.

The case of automated image classification constitutes a particularly significant domain for analyzing these issues. Images may contain information directly relevant to an investigation, but they are also subject to technical transformations and manipulations that can modify their appearance, metadata, or statistical characteristics. For this reason, before introducing the principles of ML and DL applied to computer vision, it is necessary to clarify the role of images as digital evidence and the meaning of their automated classification in the context of *image forensics*.

2.3 Image Forensics and Automated Classification

Digital images represent a particularly relevant category of evidence in contemporary forensic investigations. They may document places, objects, persons, behaviors, tools, or circumstances that are potentially significant for the reconstruction of an event. Unlike other digital artifacts, however, an image combines a visual dimension that is immediately interpretable by the human observer with a complex technical structure, consisting of pixel values, metadata, compression parameters, information about the acquisition device, and possible traces of

subsequent processing. For this reason, forensic image analysis requires consideration of both the depicted content and the digital properties of the file that represents it [34, 35].

Image forensics focuses on the study of digital images with the aim of assessing their origin, integrity, authenticity, and possible manipulations. Classical techniques in this domain include, among others, metadata analysis, the study of compression artifacts, the search for statistical inconsistencies, the identification of resampling traces, and the examination of signals associated with the image acquisition process. These signals include traces produced by the photographic sensor of the acquisition device, such as the CMOS or CCD sensor integrated into a camera or smartphone. In particular, Photo Response Non-Uniformity (PRNU) represents a characteristic noise component attributable to non-uniformities in sensor response and may be used, in specific forensic analyses, to support the attribution of an image to an acquisition device [36]. Such analyses do not aim merely to establish whether an image is visually plausible, but to verify whether the file contains technical elements compatible with a specific origin or a specific processing history [34, 37].

In the context of the present thesis, *image forensics* is considered primarily as a conceptual framework for understanding the complexity of images as digital evidence. The focus is not the direct detection of forgeries or manipulations through specialized forgery detection techniques [38], but the evaluation of automated classification and semantic retrieval of visual content in a forensic context. This distinction is important: determining whether an image has been manipulated and classifying what it represents are two different tasks, although they may interact. A manipulation or degradation of the image may not be the primary object of the analysis, but it may nevertheless affect the ability of an automated tool to correctly recognize the depicted content.

Automated image classification consists of assigning an image to one or more semantic categories on the basis of its visual content. In forensic contexts, such categories may correspond to objects, scenes, persons, documents, sensitive content, or elements considered relevant to an investigation. This form of automation can support the analyst in managing large multimedia collections, making it possible to filter, sort, or prioritize potentially relevant images. However, automated classification must not be interpreted as an autonomous evidentiary determination: it produces a technical indication that requires contextualization, verification, and human supervision.

The reliability of automated classification also depends on the technical characteristics of the image. Resolution, visual quality, illumination, occlusions, perspective, Joint Photographic Experts Group (JPEG) compression, resampling, blurring, and color modifications can alter the digital representation of the content and reduce the stability of predictions. Some transformations are common in normal image production and sharing flows, for example when content is uploaded to social platforms, forwarded through messaging applications, or saved multiple times in compressed formats. Other transformations may instead be introduced deliberately to degrade the image, mask details, or make automated analysis more difficult.

This sensitivity is particularly critical in forensic contexts, because a classification error may affect the preliminary evidence selection process. A false negative may prevent a relevant image from being brought to the analyst’s attention, whereas a false positive may increase the review burden and generate irrelevant investigative paths. In both cases, the problem concerns not only the technical performance of the classifier, but also the relationship among automation, investigative priority, and the reliability of the forensic workflow.

Therefore, automated image classification in forensic contexts must be evaluated by considering not only the correctness of the prediction on clean data, but also its stability with respect to realistic conditions of acquisition, compression, degradation, and manipulation. This aspect directly connects *image forensics* to the evaluation of the robustness of AI-based models. Before discussing adversarial attacks and anti-forensic techniques, it is necessary to introduce the principles of ML and DL that enable computer vision systems to learn visual representations and produce automated classifications.

2.4 Machine Learning and Deep Learning for Computer Vision

Computer vision encompasses the set of computational methods aimed at the automated analysis of images and videos, with the objective of extracting structured information from visual content. In this domain, ML and DL techniques have progressively replaced or complemented approaches based on manually designed features, enabling models to learn visual representations directly from data. In the case of image classification, the problem consists of associating one or more semantic labels with an image on the basis of the observable characteristics of its visual content [39, 40].

Within the computer vision domain, it is useful to distinguish among related but non-equivalent tasks. *Image classification* assigns one or more semantic labels to the entire image; *object recognition* concerns the recognition of the presence or category of an object represented in the visual content; *object detection* adds to this recognition the spatial localization of the object, typically through regions or *bounding boxes*; finally, *semantic search* enables the retrieval of images consistent with a textual description or concept. In the present thesis, the main experimental task is formulated as binary semantic image classification, based on the presence or absence of firearm-related visual content of investigative relevance, such as clearly recognizable realistic individual firearms. Accordingly, the **weapon** class does not require the geometric localization of the object within the image, but rather the correct identification of the content as relevant to the forensic triage task. This setting is consistent with studies that have applied AI-based systems to the classification of images extracted from criminal evidence, including operational categories such as firearms and ammunition [17].

In the supervised learning paradigm, a model is trained on a set of annotated examples, each consisting of an input and a corresponding class label. During training, the model learns a decision function capable of associating new inputs with one of the classes defined by the task. The objective is not to memorize the observed examples, but to generalize correctly to unseen data. This distinction is central in forensic contexts, because a classifier used on real evidence may encounter images that differ from those present in the training data in terms of quality, provenance, resolution, compression, visual context, or semantic distribution.

The quality of generalization depends on several factors, including the representativeness of the dataset, the consistency of the annotations, class separability, data balance, and the model's ability to learn features that are genuinely relevant to the task. In the absence of these conditions, a system may achieve good performance on a controlled test set while being less reliable in realistic scenarios. This problem is particularly relevant in forensic tasks, where images may originate from heterogeneous sources, be degraded by compression or sharing processes, or contain ambiguous and out-of-distribution cases.

Traditional ML approaches applied to image classification are often based on a separation between feature extraction and classification. Visual features are computed through descriptors designed or selected according to the problem, while a classifier learns a decision rule in the feature space. Models such as Support Vector Machine (SVM) have long represented an effective solution for supervised classification problems, owing to their ability to construct robust decision surfaces in high-dimensional spaces [39, 41]. General-purpose libraries such as scikit-learn provide standardized implementations of classical supervised-learning algorithms and evaluation utilities, supporting reproducible experimentation across conventional ML workflows [42]. However, the quality of the result depends strongly on the choice of features and on their ability to represent image content in a discriminative manner.

DL has profoundly changed this paradigm by introducing models capable of automatically learning hierarchical representations of data. In Convolutional Neural Network (CNN) architectures, the first layers tend to capture simple visual structures, such as edges, textures, and local contrasts, whereas deeper layers combine this information into more abstract and semantically meaningful representations. Architectures such as ResNet have shown the effectiveness of residual connections for training deep networks, mitigating the degradation

problem encountered during optimization and improving the ability to learn complex visual representations [43]. Later models, such as EfficientNet, introduced more balanced scaling strategies across depth, width, and image resolution, with the objective of improving the relationship between performance and computational cost [44].

Alongside convolutional architectures, vision-language models capable of learning joint representations of images and text have become increasingly prominent in recent years. In these systems, an image is not analyzed only with respect to predefined classes during training, but can also be associated with textual descriptions or more general semantic concepts. Models such as CLIP have shown the possibility of learning transferable visual representations from large-scale language supervision, opening the way to zero-shot or text-guided classification [45]. This approach is particularly interesting in contexts in which operational categories do not perfectly correspond to the classes available in traditional datasets, but it also introduces new challenges related to prompt formulation, semantic coverage of the classes, and prediction stability.

Alongside CLIP, generative vision-language models such as Bootstrapping Language-Image Pre-training (BLIP) also make it possible to associate images with textual descriptions, introducing a complementary perspective with respect to closed-set classification based on predefined labels [46]. In forensic contexts, such models may support forms of automated description of visual content and contribute to the preliminary prioritization of multimedia artifacts. However, their use requires particular caution, because the generated description may be sensitive to image quality, data distribution, prompt formulation, and the presence of ambiguous or out-of-distribution content.

In the experimental protocol of the present thesis, BLIP is not used as a main proxy model. Its introduction in the background serves a contextual function, as it helps frame the evolution of vision-language models and their possible use in scenarios involving automated image analysis. The experimental evaluation instead focuses on proxy models that are more directly comparable in the binary `weapon/non_weapon` task and on AI-based software selected for operational evaluation in a forensic context. For the latter, analyzed in a black-box setting, the thesis does not assume any specific internal architecture for recognition, retrieval, tagging, or object localization, but evaluates the observable behavior of the outputs available to the analyst, such as classifications, tags, semantic bookmarks, or visual and semantic retrieval results, with respect to the experimental conditions considered.

From a forensic perspective, model choice cannot be assessed exclusively as a function of average accuracy. An automated classification system must also be considered with respect to its behavior on heterogeneous, degraded, ambiguous, or out-of-distribution data.

Such inputs may produce incorrect closed-set assignments or predictions associated with high maximum predicted-class probabilities. Evaluating OOD behavior therefore makes it possible to observe how frequently content outside the nominal task is assigned to the known operational classes and how the corresponding probability outputs are distributed [20, 47].

These probability values remain architecture-specific and may not be calibrated. They must therefore be interpreted as intra-model diagnostic outputs rather than as direct measures of correctness, certainty, or forensic reliability.

This setting motivates the introduction of a separate experimental OOD branch, distinct from the two supervised operational classes. In the present thesis, this branch represents borderline, anomalous, degraded, synthetic, or otherwise out-of-task content with respect to the `weapon/non_weapon` task.

The branch is evaluated separately and is not used to train a three-class classifier. Its inclusion also does not presuppose that the black-box software systems implement a dedicated OOD-detection mechanism or expose an explicit unknown state.

A further aspect concerns the probabilistic interpretation of outputs. Many classification models produce scores or probabilities associated with classes, but these values do not necessarily

correspond to a direct measure of evidentiary reliability. High confidence may result from spurious correlations, dataset bias, or overfitting to the training distribution. For this reason, the use of ML and DL models in a forensic workflow requires a cautious assessment in which automated predictions are considered analytical support rather than autonomous determinations.

Forensic computer vision must therefore confront a methodological tension: on the one hand, DL models make it possible to analyze large quantities of images with performance that is often superior to traditional approaches; on the other hand, their reliability may decrease in the presence of visual variations, anti-forensic transformations, adversarial perturbations, or out-of-distribution inputs. This tension justifies the need to study the robustness of classifiers not only under clean and controlled conditions, but also in realistic scenarios in which visual content may be degraded, manipulated, or deliberately altered.

2.5 Adversarial Machine Learning

Adv-ML studies the behavior of machine learning systems in the presence of inputs constructed or modified with the aim of altering their operation. Unlike errors due to random noise, poor data quality, or generalization limits, adversarial attacks presuppose the presence of an actor who deliberately modifies the input in order to induce the model to produce an incorrect prediction. In the computer vision domain, this problem is particularly relevant because small variations in digital content may be sufficient to change the classification produced by a model, even when the image remains apparently similar to a human observer [13, 48].

An adversarial example can be described, in general terms, as an input modified with respect to an original sample, but constructed so as to induce the classifier to produce an incorrect label. Let x denote the original image, y its correct class, and $f(\cdot)$ the classifier. The attacker's objective is to generate a perturbed version x' , such that x' remains sufficiently close to x according to a distance or perceptibility constraint, but is classified differently by the model. In simplified form, the problem can be expressed as the search for a perturbation δ such that:

$$x' = x + \delta \quad \text{and} \quad f(x') \neq y.$$

The perturbation δ is often constrained by distance metrics or ℓ_p -norm constraints, expressed as $\|\delta\|_p \leq \epsilon$, where ϵ represents the maximum allowed perturbation budget and, in the most common cases, $p \in \{0, 2, \infty\}$. These constraints make it possible to formalize sparse attacks, limited-energy perturbations, or modifications that are uniformly bounded across pixels, respectively, although they do not exhaust the problem of visual perceptibility in real-world scenarios [14, 49].

This formulation highlights the dual nature of the attack: on the one hand, the perturbation must be limited; on the other hand, it must be sufficiently effective to change the system's decision.

Figure 2.2 conceptually represents this mechanism in the context of an evasion attack.

Adversarial attacks can be classified according to several criteria. A first distinction concerns the phase of the model life cycle in which the attack is conducted. Poisoning attacks occur during the training phase, by altering the data or labels with the aim of compromising the resulting model. Evasion attacks, by contrast, occur at inference time and consist of manipulating the inputs submitted to an already trained model [15, 50]. In the context of automated image classification in forensic settings, the main focus is on evasion attacks, because the most relevant scenario is one in which images already present on a device, in an archive, or within digital evidence are altered to reduce the likelihood of being correctly identified by an automated tool.

A second distinction concerns the level of knowledge available to the attacker. In white-box scenarios, the attacker has complete or extensive information about the model, including its architecture, parameters, loss function, and gradients. In these cases, the perturbation can be

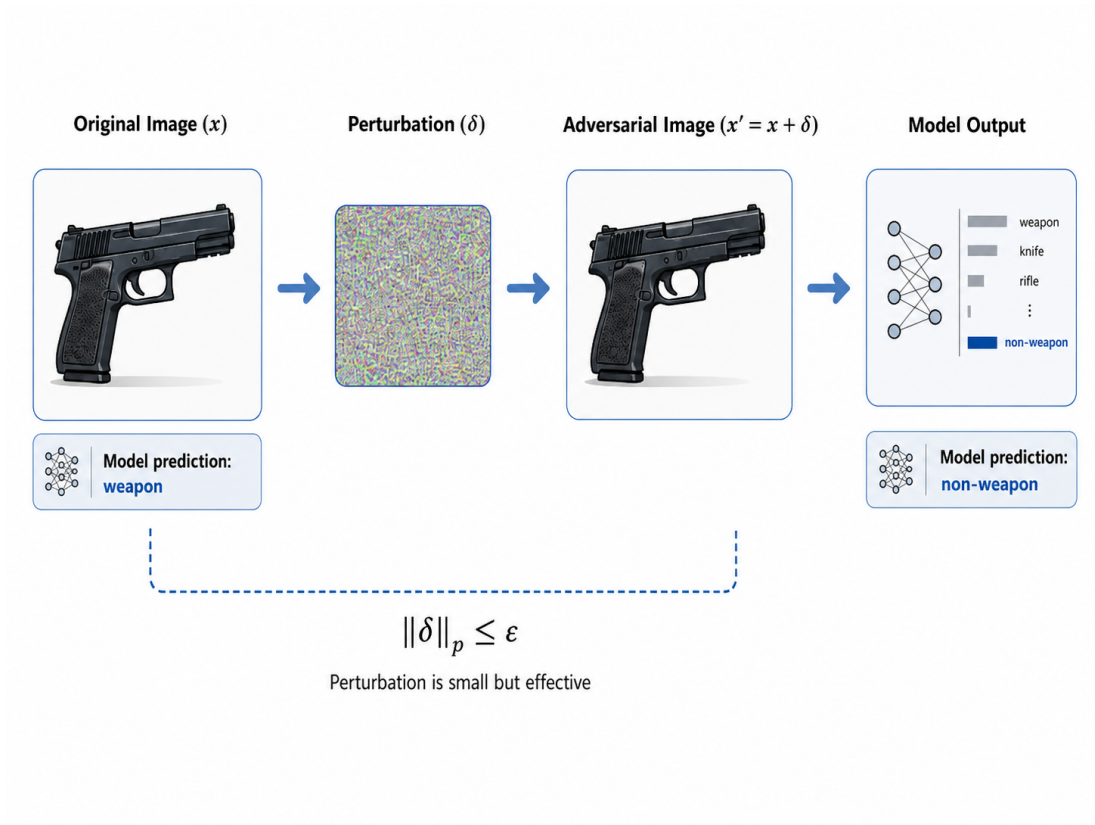


Figure 2.2: Conceptual representation of an adversarial evasion attack. A bounded perturbation δ is added to the original input x to obtain an adversarial image x' , which induces a wrong classifier prediction while preserving visual coherence for a human analyst

optimized directly with respect to the internal behavior of the classifier. In black-box scenarios, by contrast, the attacker does not know the details of the model and may rely only on observable outputs, repeated queries, surrogate models, or the transferability of adversarial examples across different classifiers [14, 50, 51]. This distinction is particularly important in the present thesis, because the commercial software selected for operational evaluation is observed in a black-box setting and does not allow assumptions about access to the parameters or internal logic of any AI-based components.

A further distinction concerns the objective of the attack. In untargeted attacks, the intent is simply to induce the model to produce a class different from the correct one. In targeted attacks, by contrast, the objective is to push the classifier toward a specific class chosen by the attacker. In the case of forensic images, an untargeted attack may be sufficient to reduce the ability of the tool to retrieve relevant content, whereas a targeted attack could aim to make sensitive content appear as if it belonged to an innocuous category. However, in an evidentiary context, even less controlled forms of error may have significant operational consequences, because they affect the selection, prioritization, and review of digital evidence.

The literature has proposed numerous techniques for generating adversarial examples, differentiated by perturbation constraint, computational cost, required knowledge, and transferability. Gradient-based methods, such as Fast Gradient Sign Method (FGSM), exploit the information provided by the loss function to identify an effective perturbation direction [13]. Other approaches aim to produce more optimized, sparser, or more localized perturbations, reducing the number of modified pixels or approximating minimal perturbations capable of crossing the classifier’s decision boundary [14, 52, 53, 54]. In this chapter, these techniques are framed at a conceptual level; the rationale for their selection, the adopted configurations, and the procedures used to apply them within the experimental protocol are discussed in the methodology chapter.

In forensic contexts, the adversarial problem does not concern only the abstract vulnerability of models, but also their operational reliability within an investigative process. If an AI-based system is used to filter or classify large quantities of images, a perturbation capable of altering its prediction may reduce the likelihood that relevant content will be promptly brought to the analyst’s attention. The risk does not necessarily consist in the automated production of an incorrect judicial decision, but rather in the possibility that automation may modify the path of discovery, prioritization, and interpretation of evidence. For this reason, adversarial robustness must be considered a component of the broader technical-forensic reliability of the system.

2.5.1 Threat Model and Attacker Assumptions

The threat model defines the assumptions concerning the attacker’s capabilities, objectives, and constraints. In Adv-ML, it is necessary to rigorously delimit the attack scenario and to avoid robustness evaluations that lack an operational context. Without an explicit threat model, the result of an adversarial test risks being difficult to interpret, because it is not clear which type of adversary the system is able to tolerate and which conditions are instead excluded from the analysis [14, 51].

In the context of automated image classification in forensic settings, the attacker can be modeled as an actor interested in reducing the likelihood that specific content will be correctly identified by an automated tool. The main objective is not necessarily to compromise the entire forensic workflow, but to alter the classifier’s response on specific visual inputs. In operational terms, this may translate into the attempt to transform a relevant image into one classified as non-relevant, ambiguous, or belonging to a category different from the expected one. This scenario is consistent with an evasion attack, because the manipulation affects the analyzed input rather than the model training phase.

The attacker’s capabilities may vary depending on the level of access to the target system. In a white-box scenario, the attacker has complete or extensive information about the model, including its architecture, parameters, loss function, and gradients, and can therefore construct perturbations directly optimized with respect to the classifier. In a black-box scenario, by contrast, the attacker does not have access to the internal logic of the target system and may

rely on observable outputs, surrogate models, or the transferability of adversarial examples. In the protocol of the present thesis, white-box access is relevant for the controlled evaluation of proxy models when required by the specific attack technique, whereas the selected commercial software is observed in a black-box setting, since its internal classification logic is not available. This distinction affects the type of attack that can be applied, the required computational cost, and the operational plausibility of the threat.

The threat model must also specify the constraints imposed on the perturbation. In the image domain, the modification may be limited in terms of numerical magnitude, perceptual distance, number of altered pixels, affected image area, or admissible transformations. In forensic contexts, these constraints acquire a specific meaning: an excessively visible manipulation may be detected by the human analyst or by integrity-control procedures, whereas a subtler modification may preserve the visual plausibility of the image while still altering the model prediction. The notion of imperceptibility must therefore be treated with caution, because human perception, file quality, and the investigative relevance of visual details do not necessarily coincide with a simple mathematical distance metric.

A further aspect concerns the attacker’s knowledge of the forensic domain. In realistic scenarios, the attacker may not know the specific tool used by the analyst, but may be aware that images and multimedia content are often subjected to automated classification. This partial knowledge may be sufficient to motivate transformations aimed at reducing the automatic recognizability of the content, especially when such transformations do not clearly compromise the general readability of the image. Therefore, the threat model must consider not only a highly informed attacker, but also scenarios in which the adversary operates with limited knowledge of the model and exploits generic or transferable transformations. The operational relevance of this assumption is consistent with the literature on adversarial examples in physical or otherwise non-purely digital scenarios, which shows that some perturbations may retain effectiveness even outside strictly controlled experimental conditions [55].

The threat model outlined in this subsection specifically concerns adversarial attacks, namely perturbations constructed with the objective of altering the classifier’s response. The anti-forensic transformations discussed in the following section instead constitute a conceptually distinct category: they may be applied even in the absence of knowledge about the model or about any automated classification system, and they are more generally aimed at concealing, degrading, or making digital evidence less informative. In the experimental protocol of the present thesis, the two categories are therefore evaluated separately: on the one hand, adversarial perturbations oriented toward inducing errors in the model decision; on the other hand, realistic anti-forensic transformations not optimized with respect to the classifier, applied to observe the stability of automated support in the presence of plausible image manipulations. Their relationship is considered only at the level of operational effects, since both may reduce the likelihood that content of investigative relevance will be correctly flagged or prioritized for subsequent analyst review.

Table 2.1 summarizes the main dimensions of the adversarial threat model adopted in the experimental evaluation, distinguishing between the theoretical formulation of the attacks and the operational-forensic focus of the present thesis.

2.5.2 Adversarial and Adversarial-Style Perturbation Families

Within the general framework of Adv-ML, this thesis considers a set of perturbations included in the adversarial evaluation, comprising both attacks constructed with respect to model behavior and a model-agnostic *adversarial-style* transformation. The choice is not intended to cover the entire landscape of existing techniques exhaustively, but rather to include perturbations that are heterogeneous in terms of computational cost, level of knowledge required about the model architecture, alteration constraint, and operational plausibility. In particular, gradient-based, sparse, iterative, and model-agnostic perturbations are considered, in order to observe the behavior of AI-based systems with respect to manipulations characterized by different properties.

Dimension	Theoretical scenarios	Thesis operational focus
Attack phase	Poisoning vs evasion	Evasion , since the manipulation affects already acquired or analyzed images at inference time.
Adversary knowledge	White-box vs black-box	White-box for controlled proxy-model evaluation, where required by the attack; black-box for commercial software , whose internal AI-based image-analysis logic is not available.
Attack objective	Targeted vs untargeted	Untargeted , with emphasis on performance degradation, missed detection, and reduced classification reliability.

Table 2.1: Adversarial threat model dimensions considered in the experimental robustness evaluation

The **Fast Gradient Sign Method (FGSM)** represents a simple and efficient adversarial baseline. Given a classifier f , a loss function $\mathcal{L}$, an image x , and the corresponding correct label y , the attack generates a perturbed image by moving in the direction of the sign of the gradient of the loss with respect to the input:

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f(x), y)).$$

The parameter ϵ controls the maximum magnitude of the perturbation. In the experimental context of the thesis, FGSM is relevant because it makes it possible to rapidly evaluate the classifier’s sensitivity to a single-step white-box perturbation constructed directly with respect to the model’s loss function [13].

The **One Pixel Attack** instead represents a family of sparse and black-box attacks, in which the perturbation is constrained to modify a very limited number of pixels. In general form, the objective is to find a version x' of the image such that:

$$f(x') \neq f(x) \quad \text{with} \quad \|x - x'\|_0 \leq k,$$

where k denotes the maximum number of pixels that can be modified. In the extreme case of the One Pixel Attack, $k = 1$. The attack is relevant because it shows how highly localized variations can alter the decision of a neural network, even without direct access to the model gradients [52].

Sigma Zero Attack belongs to the family of attacks oriented toward sparsity and the L_0 norm. Unlike attacks that limit the global magnitude of the perturbation through L_2 or L_∞ norms, L_0 attacks aim to reduce the number of modified input components. The objective can be expressed as the search for an adversarial image x^{adv} such that:

$$f(x^{adv}) \neq f(x) \quad \text{while minimizing} \quad \|x - x^{adv}\|_0.$$

This setting is particularly significant in forensic contexts because sparse perturbations may be difficult to identify through ordinary visual inspection and may not visibly degrade the perceptual quality of the image [56].

SuperDeepFool is regarded as an iterative attack inspired by the DeepFool family. The general idea behind DeepFool is to iteratively estimate a perturbation capable of crossing the classifier’s decision boundary, seeking a modification of limited magnitude that is sufficient to change the model’s prediction. In abstract form, the objective is to find a perturbation δ such that:

$$x' = x + \delta \quad \text{and} \quad f(x') \neq f(x),$$

with δ constructed through successive updates guided by the local geometry of the decision function. In the context of the thesis, this family of attacks is relevant because it represents iterative perturbations that are more structured than a single-step attack such as FGSM, without attributing to the technique any general ability to overcome every possible defense or form of pre-processing [53, 54]. This caution is consistent with the literature showing that some defenses may produce only apparent robustness, for example through gradient masking or obfuscated gradients, without ensuring genuine resistance to adaptive attacks [57].

A further attack frequently used in the evaluation of adversarial robustness is Projected Gradient Descent (PGD), which extends the idea of gradient-based attacks through iterative updates constrained within a perturbation neighborhood and has been widely used as a reference for formulating adversarial robustness as a model security problem [58]. In the present thesis, however, PGD is not included in the main experimental protocol. This choice stems from the need to keep the workflow computationally sustainable and focused on operational robustness in forensic contexts, using SuperDeepFool and Sigma Zero Attack as representatives of iterative and sparse perturbations, respectively, in line with the objectives of the experimental protocol.

Alongside attacks constructed with respect to model behavior, the thesis also considers a model-agnostic perturbation based on shifting color channels. **Color Shift** globally modifies the chromatic representation of the image, for example through offsets or multiplicative factors applied to the R , G , and B channels. It represents an intermediate case with respect to the classical distinction between model-dependent adversarial attacks and anti-forensic transformations. In this thesis, it is treated as a model-agnostic *adversarial-style* perturbation: it does not use gradient information and does not directly optimize a model loss function, but it is applied with the experimental purpose of evaluating the effect of semantically conservative alterations on automated classification. For this reason, although it shares some visual characteristics with realistic image transformations, it is kept within the experimental group of adversarial perturbations, explicitly qualifying it as a model-agnostic *adversarial-style* transformation and not as an attack optimized with respect to the model.

Table 2.2 summarizes the adversarial and *adversarial-style* perturbations considered in the experimental protocol.

Technique	Family	Methodological relevance
FGSM	Gradient-based, single-step	Efficient white-box baseline for evaluating the model's sensitivity to the gradient of the loss function.
One Pixel Attack	Sparse, black-box	Highly localized perturbation, useful for evaluating vulnerability to minimal modifications in the pixel domain.
Sigma Zero Attack	Sparse, L_0 -oriented	Attack oriented toward minimizing the number of modified pixels, relevant to scenarios in which the perturbation must remain localized.
SuperDeepFool	Iterative boundary-based	Iterative attack inspired by DeepFool, useful for exploring decision stability with respect to crossing the decision boundary.
Color Shift	Adversarial-style, model-agnostic	Semantically conservative global chromatic alteration, not optimized with respect to the model, included among adversarial perturbations to evaluate the effect of controlled color variations on automated outputs.

Table 2.2: Adversarial and adversarial-style perturbations considered in the experimental protocol

2.6 Anti-Forensic Techniques Applied to Digital Images

Anti-forensic techniques comprise the set of strategies aimed at hindering, degrading, or making the forensic analysis of digital content less reliable. In the image domain, such techniques may affect both the visual properties of the content and the statistical and structural characteristics of the file, with the objective of reducing the possibility of reconstructing the image history, detecting manipulations, identifying acquisition traces, or correctly classifying the depicted content. In general terms, anti-forensics does not necessarily coincide with the explicit forgery of an image, but also includes transformations capable of altering or masking information relevant to technical analysis [12, 59].

In the context of *image forensics*, many analyses rely on the presence of statistical and structural traces associated with the formation or subsequent processing of the image file. Such traces may include compression artifacts, patterns introduced by resampling operations, inconsistencies among image regions, and signals attributable to the acquisition device, such as the PRNU of the photographic sensor. Anti-forensic transformations may aim to reduce the visibility or reliability of these indicators, making it more difficult to distinguish between original, degraded, or manipulated content. For example, JPEG recompression may modify the artifacts produced by previous saving operations; resampling may alter the spatial correlations introduced by resizing operations; histogram or contrast modifications may change the distribution of image intensity values [34, 36, 37].

In the present thesis, these techniques are discussed to frame the domain of *image forensics* and the possible manipulations of visual evidence; the experimental evaluation does not concern source attribution through PRNU, but rather the stability of semantic image classification with respect to selected transformations applied in a controlled manner.

A first family of transformations concerns compression and recompression. The JPEG format, widely used in digital devices and sharing platforms, applies lossy compression that modifies the frequency components of the image and introduces artifacts depending on the selected quality level. From a forensic perspective, such artifacts may be informative, but they may also interfere with subsequent analyses. Intentional recompression may attenuate or overwrite previous traces, reduce perceptual quality, or modify statistical characteristics exploited by automated tools. For this reason, compression is not only a normal technical step in image management, but may also become a relevant transformation from an anti-forensic standpoint [35, 59].

A second category includes geometric and resampling operations, such as resizing, cropping, rotation, and resampling. These transformations may be common in ordinary image production and sharing workflows, but they may also be used to modify the scale of objects, remove relevant regions, alter scene composition, or introduce statistical patterns different from the original ones. Resampling, in particular, may leave traces detectable through forensic analysis, but it may also be combined with additional transformations to make its identification more difficult [34, 37].

Additional anti-forensic transformations concern controlled visual degradation. Blurring, noise, resolution reduction, sharpness alteration, and color modifications may decrease the readability of relevant details without necessarily making the image unusable for a human observer. In an AI-assisted forensic workflow, these transformations are particularly important because they may modify the visual features used by classifiers. An object that remains recognizable to the analyst may become less discriminative for the model, especially if the transformation alters edges, textures, local contrasts, or spatial relationships among the components of the scene.

Alongside visible or statistically detectable transformations, a further category of techniques attributable to the anti-forensic domain concerns steganography, namely the concealment of information within digital content. In the case of images, additional data may be embedded by altering poorly perceptible portions of the representation, such as the least significant bits of pixel values. Although steganography has different purposes from visual degradation or recompression, it is relevant within the framework of *image forensics* because it may introduce hidden content, modify the statistical properties of the file, and require specialized analyses to

distinguish between an apparently ordinary image and an image used as a concealment vector [12, 35].

In the present work, steganography is mentioned as a broader component of the anti-forensic domain, whereas the experimental evaluation focuses on visual and statistical transformations directly relevant to the robustness of automated classification.

It is important to distinguish anti-forensic techniques from the adversarial attacks discussed in the previous section. Adversarial attacks are generally designed with respect to the behavior of a model and aim to induce an incorrect prediction through optimized perturbations. Anti-forensic transformations, by contrast, may be model-independent and more generally oriented toward degrading, masking, or altering digital evidence. This distinction does not imply an absolute separation: an anti-forensic transformation may produce adversarial effects on a classifier, just as an adversarial perturbation may have anti-forensic consequences when it reduces the investigative usefulness of an image. However, from a methodological standpoint, the two categories must be kept distinct because they presuppose different attacker capabilities, objectives, and assumptions.

In the case of automated image classification, anti-forensic techniques are relevant because they may affect the operational robustness of AI-based systems even in the absence of an attack optimized against the model. A recompressed, resized, blurred, or contrast-altered image may represent a realistic input, compatible with ordinary acquisition, storage, or sharing processes. However, the same transformations may be intentionally exploited to reduce the likelihood that content will be correctly classified. The difficulty therefore lies in distinguishing between ordinary degradations within the digital image life cycle and transformations applied with the purpose of concealment or reduction of analytical effectiveness.

From a forensic perspective, this ambiguity requires cautious evaluation. Not every compressed, resized, or degraded image can be considered the result of anti-forensic conduct; on the contrary, many transformations are ordinary consequences of the use of mobile devices, social networks, messaging applications, and storage systems. However, when the objective is to evaluate the reliability of automated tools, it is necessary to include such transformations within the space of realistic scenarios. The robustness of a forensic classifier depends not only on its accuracy on clean images, but also on its ability to maintain stable behavior in the presence of plausible, intentional, or accidental degradations.

From this perspective, anti-forensic techniques applied to digital images constitute a bridge between traditional *image forensics* and the evaluation of the robustness of computer vision models. They highlight that an image is not a static and neutral input, but a digital artifact subject to technical transformations that may affect both its human interpretability and its automated processing. For this reason, the evaluation of AI-based models and software employed in the context of DF must consider not only formally defined adversarial attacks, but also manipulations and degradations compatible with realistic operational scenarios.

2.6.1 Image-Level Anti-Forensic Transformation Families

In the context of the present thesis, anti-forensic transformations are considered as realistic manipulations of the image artifact, not necessarily optimized with respect to a specific classifier. They act on the visual, statistical, or geometric properties of the file and may derive both from ordinary acquisition, saving, and sharing processes and from intentional applications aimed at reducing the effectiveness of automated analysis. For this reason, such transformations constitute a bridge between traditional *image forensics* and the evaluation of the operational robustness of AI-based systems.

The **JPEG recompression** transformation consists of saving an image again using lossy compression. Since the JPEG format operates on the frequency components of the image and introduces artifacts depending on the selected quality level, recompression may modify or overwrite previous traces, alter local statistics, and degrade visual details relevant to classifiers. From a forensic perspective, this transformation is relevant because it may interfere both with

analyses based on compression artifacts and with AI-based models that are sensitive to textures, edges, and local patterns [35, 59].

The **Resample + Resize** transformation modifies the resolution and spatial grid of the image. Let x denote the original image. A resizing transformation can be modeled as:

$$x' = \mathcal{R}(x; s, \mathcal{I}),$$

where s is the scaling factor and $\mathcal{I}$ represents the interpolation method. When an image is reduced and subsequently restored to its original size, part of the spatial information is lost and replaced by interpolated values. This process may alter pixel correlations, geometric traces, and local patterns used by both forensic tools and classification models [34, 37].

The **Gaussian Blur** transformation introduces controlled blurring through convolution with a Gaussian kernel. In simplified form, the operation can be described as:

$$x' = x * G_\sigma,$$

where G_σ is a two-dimensional Gaussian function and $*$ denotes convolution. This transformation attenuates high frequencies, reduces edge sharpness, and may decrease the discriminative value of local details. In the context of automated classification, blur is relevant because many CNN models learn features in the first convolutional layers that are strongly related to edges, textures, and local intensity variations.

The **Histogram Modification** transformation acts on the distribution of image intensity values. In the case of *histogram equalization*, the objective is to remap intensity levels using the cumulative distribution function, according to a transformation of the following form:

$$x'_i = \lfloor (L - 1) \cdot \text{CDF}(x_i) \rfloor,$$

where L denotes the number of intensity levels and CDF is the normalized cumulative distribution. This transformation may improve perceived contrast, but it may also modify global and local statistics used by image forensics tools and automated classifiers [60].

The **Contrast Stretching** transformation applies a linear normalization of the dynamic range. Let $I_{\min}$ and $I_{\max}$ denote two intensity bounds, possibly computed through percentiles to reduce the influence of outliers. The transformation can be expressed as:

$$x'_i = \begin{cases} 0, & \text{if } x_i \leq I_{\min}, \\ 255, & \text{if } x_i \geq I_{\max}, \\ \frac{x_i - I_{\min}}{I_{\max} - I_{\min}} \cdot 255, & \text{otherwise.} \end{cases}$$

This operation modifies the global contrast of the image and may alter the response of models with respect to dark regions, overexposed areas, or regions with low tonal separation.

Table 2.3 summarizes the anti-forensic transformations considered in the experimental protocol.

Table 2.4 summarizes the conceptual distinction between *adversarial attacks* and *anti-forensic transformations* in the context of automated image classification.

2.7 Explainability, Robustness, and Model Reliability

The evaluation of an AI-based model cannot be reduced exclusively to the measurement of its average accuracy. In a forensic context, the usefulness of an automated system also depends on the stability of its predictions, its ability to handle degraded or out-of-distribution inputs, the transparency of the information provided to the analyst, and the possibility of documenting, in a verifiable manner, the role played by the model within the investigative workflow. Accuracy, robustness, interpretability, and reliability are therefore complementary dimensions that must

Transformation	Methodological relevance
JPEG recompression	Modifies compression artifacts, frequency components, and local image details.
Resample + Resize	Alters the spatial grid, introduces interpolation, and may degrade geometric correlations and local patterns.
Gaussian Blur	Attenuates high frequencies, edges, and textures, reducing the discriminative value of potentially relevant details.
Histogram Modification	Remaps the distribution of intensity levels, modifying global and local image statistics.
Contrast Stretching	Normalizes the dynamic range and alters global contrast, with possible effects on the features learned by classifiers.

Table 2.3: Anti-forensic transformations considered in the experimental protocol

Dimension	Adversarial attacks	Anti-forensic transformations
Primary target	Decision function of the AI-based model.	Visual, statistical, or structural properties of the digital image file.
Nature	Perturbations designed to alter the model prediction.	Manipulations or degradations such as JPEG recompression, resizing, blur, or contrast modification.
Model dependency	Usually direct or surrogate-based, especially in white-box or transfer-based settings.	Usually low or indirect, since the transformation can often be applied without knowing the classifier.
Visibility	Often designed to be imperceptible, sparse, or localized.	Often compatible with visible degradations or ordinary image processing operations.
Forensic relevance	May affect automated evidence selection by inducing false positives or false negatives.	May alter the evidentiary quality of the image and reduce the reliability of automated analysis.

Table 2.4: Comparison between adversarial attacks and anti-forensic transformations in the context of automated image classification

be considered jointly when an automated classification system is used to support the analysis of digital evidence [61, 62].

Robustness can be defined, in general terms, as the ability of a model to maintain stable and coherent behavior in the presence of input variations. Such variations may be accidental, such as noise, compression, low resolution, or non-ideal acquisition conditions, or intentional, such as adversarial perturbations and anti-forensic transformations. In the computer vision domain, robustness does not concern only resistance to small numerical perturbations, but also the classifier’s ability to handle realistic changes in the visual properties of the image, the data distribution, and the quality of the analyzed content [21, 63].

Reliability is a broader concept than robustness. A model may be robust with respect to a specific family of perturbations, but not necessarily reliable in a forensic sense. Reliability requires the system to produce results that are sufficiently stable, verifiably documented, and interpretable with respect to the context of use. In particular, a classifier employed in a DF workflow must be evaluated with respect to its ability to support the analyst without replacing human judgment, to signal uncertainty or ambiguous cases, to avoid overestimating its own confidence, and to maintain coherent performance even on data different from those observed during training. This distinction is essential because an output that is formally correct on average may still be unreliable if it is not stable, verifiable, or adequately contextualized.

The role of xAI is situated within this framework. Explainability techniques aim to provide tools for making model behavior more understandable, especially when models operate as complex or partially opaque systems. In computer vision, these techniques may include saliency maps, attribution methods, visualizations of the image regions that most influence the prediction, and local explanations of classifier behavior. Well-known examples include model-agnostic approaches, such as LIME and SHAP, and attribution or visual localization techniques, such as IG and class activation mapping [64, 65, 66, 67].

These approaches are not methodologically equivalent. Gradient-based techniques, such as IG, require access to the model and to the gradient of the prediction with respect to the input, and are therefore applicable to the controllable proxy models considered in the present thesis. Model-agnostic approaches, such as LIME or some formulations of SHAP, can instead operate through input-output queries, but their application to commercial software evaluated in a *black-box* setting depends on the availability of adequate outputs, queryable interfaces, and reproducible experimental conditions. For this reason, the operational choice of the explainability technique used in the protocol is described in Chapter 4.

In the case of commercial black-box software, outputs such as tags, classifications, retrieval results, or semantic bookmarks must therefore be interpreted as observable operational signals rather than as local explanations of the internal decision-making process. They can be normalized and evaluated against the experimental ground truth, but they do not make it possible to directly infer which features, image regions, or internal representations determined the system’s decision.

The use of xAI in forensic contexts must nevertheless be interpreted with caution. A visual explanation does not constitute autonomous proof of the correctness of the model, nor does it guarantee that the classifier has learned concepts that are semantically coherent with human reasoning. Saliency maps, for example, may indicate image regions relevant to the prediction, but they do not necessarily demonstrate that the model recognized the object of interest in the expected way. Moreover, explanations generated by different methods may diverge from one another, and their stability may be influenced by input perturbations or model variations [61, 68].

Despite these limitations, xAI can provide useful support in the evaluation of operational robustness. Comparing explanations produced on clean and manipulated images may help identify variations in model behavior, shifts in attribution toward irrelevant regions, or dependence on visual artifacts that are not relevant to the task. In this sense, explainability should not be understood as a conclusive validation tool, but as an auxiliary component of the

analysis, useful for formulating hypotheses about classifier behavior and documenting possible anomalies in the system's response.

The relationship between robustness and explainability is particularly important in cases in which a model preserves the same label but changes the apparent rationale of the prediction, or produces an incorrect classification while focusing on irrelevant visual regions. Such situations show that accuracy alone is not sufficient to understand system behavior. In a forensic context, the possibility of observing how the model reacts to degradations, anti-forensic transformations, or adversarial perturbations can contribute to a more complete assessment of its reliability, especially when the automated output is used to filter or prioritize large quantities of images.

A further element concerns confidence calibration. DL models may produce predictions with high scores even in the presence of ambiguous, degraded, or out-of-distribution inputs. This phenomenon is problematic in AI-assisted workflows, because apparently high confidence may lead the user to overestimate the reliability of the output. Operational robustness therefore requires not only that the model maintain good performance on manipulated data, but also that uncertainty be handled in a coherent and non-misleading manner, especially in cases in which the image does not clearly belong to the classes expected by the system [20, 47].

Within the framework of the present thesis, explainability, robustness, and reliability are considered complementary dimensions in the evaluation of proxy models and in the interpretation, to the extent observable, of the outputs produced by AI-based software employed in the forensic context. Robustness makes it possible to measure the stability of predictions with respect to adversarial perturbations and anti-forensic transformations; xAI, applied to proxy models, makes it possible to observe, with due caution, local attributions associated with predictions; reliability integrates these elements within the operational context, considering the role of the analyst, the documentability of the process, and the possible consequences of automated errors in evidence selection. This setting makes it possible to move beyond a purely performance-based evaluation and to situate model behavior within the verifiability requirements that characterize DF.

2.8 Legal and Regulatory Profiles of AI in Forensic Contexts

The use of AI in forensic contexts raises not only technical issues, but also affects the assumptions of reliability, verifiability, and control that characterize digital evidence. In investigative and judicial proceedings, an automated system may contribute to the selection, classification, or prioritization of relevant content, but it cannot replace the assessment of the forensic operator or that of the judicial authority. The legal dimension of AI in DF must therefore be traced back to the principle that any technical result used for evidentiary purposes must be documented, contextualized, and, as far as possible, verified through transparent and repeatable procedures [5, 9, 69].

The framework of digital evidence is also situated within a supranational regulatory context in which cybercrime and attacks against information systems are governed by instruments such as the Budapest Convention on Cybercrime and Directive 2013/40/EU, which reinforce the centrality of digital evidence in investigative activities and judicial cooperation [70, 71].

Digital evidence is characterized by particular vulnerability to modification, loss of context, and attribution difficulties. For this reason, international standards and guidelines emphasize the need to preserve integrity, authenticity, traceability, and the chain of custody throughout the entire evidence management life cycle [9, 29]. When analysis is supported by AI-based modules, these requirements do not disappear; rather, they also extend to the functioning of the automated tool. It therefore becomes necessary to document not only the origin and handling of the data, but also the role played by the model, the conditions of use, the available metrics, the known limitations, and any human intervention in the assessment of the result.

From this perspective, case law from the Court of Cassation has progressively emphasized

the need for seizure and acquisition activities involving computer data to be supported by adequate reasoning, by criteria of relevance with respect to the object of the investigation, and by safeguards suitable for avoiding acquisitions or retentions that are disproportionate to evidentiary purposes [72, 73].

In the context of criminal investigations conducted by competent authorities, the specific European framework on personal data protection is represented by Directive (EU) 2016/680, transposed into the Italian legal system by Legislative Decree No. 51 of May 18, 2018. This legal framework concerns processing carried out for the purposes of the prevention, investigation, detection, or prosecution of criminal offenses or the execution of criminal penalties, including automated processing of personal data. In this context, the use of AI-based systems to support forensic analysis must be assessed with respect to the principles of lawfulness, necessity, proportionality, and security of processing, as well as the requirements of documentation and control of the operations performed. The General Data Protection Regulation (GDPR) remains relevant for processing operations that do not fall within the specific scope of *law enforcement* activities, for example where similar systems are used by private entities or in different organizational and application contexts not attributable to the purposes of prevention, investigation, detection, or prosecution of criminal offenses. Therefore, references to automated decision-making and impact assessments must be applied according to the specific legal basis and purpose of the processing [74, 75, 76].

A further reference is the AI Act, which introduces a risk-based regulatory model and pays particular attention to systems used in sensitive contexts, including *law enforcement* activities and the administration of justice. In these areas, the use of AI may affect fundamental rights, procedural guarantees, and the ability of the parties to understand or contest the outcome of an assisted decision-making process. For this reason, systems classified as high-risk are subject to requirements concerning data governance, technical documentation, logging, transparency, human oversight, accuracy, robustness, and cybersecurity [26]. These requirements are particularly consistent with the needs of DF, in which every relevant step must be reconstructable and justifiable.

However, the concrete regulatory classification of the software considered in the present thesis depends on its intended purpose, the role actually played within the investigative workflow, and the specific conditions of use; this work does not automatically assume that the evaluated systems fall within a specific risk category under the AI Act.

In the specific context of automated image classification, the legal issue does not necessarily consist in the model producing a final decision on the evidentiary value of the image. More realistically, the risk concerns the indirect influence of automation on the investigative path: a false negative may reduce the likelihood that a relevant image will be examined promptly, whereas a false positive may direct the analyst's attention toward irrelevant content. In both cases, the output of the automated system may affect evidence selection and prioritization, even though it does not formally constitute a judicial decision. This makes it necessary to adopt a cautious approach, in which automated classification is regarded as a support tool rather than an autonomous source of assessment.

Human oversight therefore represents a central requirement. It should not be understood as a merely formal control, but as the actual possibility for the operator to understand the role of the system, assess its output, recognize conditions of uncertainty, and intervene when the automated result appears incoherent or insufficiently reliable. In forensic contexts, oversight also has a documentary function: the analyst must be able to explain whether and how the model output contributed to evidence selection, which checks were performed, and which limitations were considered in the overall assessment.

The opacity of DL models introduces further challenges. If a system produces a classification without making the technical reasons for the prediction understandable, the ability to challenge, replicate, or critically evaluate the result may be reduced. xAI techniques can partially mitigate this problem by providing indications about the image regions or features that most influenced the prediction. However, as discussed in the previous section, such explanations must not be

confused with proof of the model's correctness. They constitute interpretive support tools, useful for documentation and review, but they do not replace technical validation or human judgment [61, 62].

In light of these considerations, the robustness of AI-based systems also has legal relevance, not only technical relevance. A model vulnerable to adversarial perturbations, realistic degradations, or anti-forensic transformations may compromise the quality of the support provided to the analyst and reduce the reliability of the investigative workflow. Experimental robustness evaluation therefore helps clarify the limits within which an automated system may be used responsibly, the conditions that may generate errors, and the safeguards that should accompany its use in forensic scenarios.

In conclusion, the use of AI in DF requires a balance between operational efficiency and procedural safeguards. Automation may improve the ability to analyze large quantities of data, but it must be integrated into a controlled, documented, and supervised process. In the case of automated image classification, this balance requires the model to be evaluated not only with respect to its accuracy on clean data, but also with respect to its robustness, explainability, traceability, and ability to operate under realistic conditions. These elements constitute the theoretical basis for the experimental methodology developed in the following chapters.

Chapter 3

State of the Art

This chapter critically examines the scientific literature relevant to the main thematic areas of the thesis, following the organizational distinction introduced in Chapter 2. For each area, it discusses the principal contributions, the evidence they provide, and the limitations that remain unresolved, without repeating the concepts already presented in the technical background.

The objective is to identify the specific research gap that motivates the experimental approach adopted in this thesis: the limited availability of integrated and operationally oriented protocols for evaluating transparent proxy models and selected black-box software configurations on a common, task-specific experimental basis under clean, OOD, adversarial, and anti-forensic conditions.

3.1 AI and Automation in Digital Forensics

The volume of digital data constitutes a central point of departure for the literature on automation in DF. Garfinkel [4], in a work that significantly shaped the research agenda of the following decade, documented how the rapid growth of digital data volumes was rendering manual analysis increasingly unsustainable, identifying automation as one of the priority directions for investment. The work did not propose specific solutions, but rather a mapping of unresolved problems: among them, the lack of methods for the automated triage of large multimedia corpora, the absence of shared benchmarks, and the difficulty of evaluating tools under realistic operational conditions. This agenda influenced subsequent research, which responded by progressively integrating ML and DL techniques into forensic workflows.

A first systematic attempt to map this integration was made by Marturana et al. [77], who surveyed the state of the art in artificial intelligence applied to DF at a time when the field was still predominantly experimental. Their work showed that existing applications were fragmented across domains—file system analysis, content classification, and pattern matching—and that a unified approach capable of covering the entire investigative workflow was lacking. The models proposed in the literature achieved good performance in controlled settings, but had not been systematically evaluated on data drawn from real-world scenarios or tested for reliability in the presence of anomalous inputs.

This research direction matured over the following decade. Costantini et al. [1] proposed a framework integrating automated reasoning and ML techniques to support the forensic analyst in managing complex evidence, with applications to event correlation and multimedia content classification. The work demonstrates how AI can reduce the operational workload in the examination and analysis phases, but leaves the question of robustness unaddressed: the framework assumes well-formed data and does not address scenarios in which inputs have been manipulated or degraded. This limitation is relevant because real forensic data are often of variable quality and may have been intentionally altered before acquisition.

A more recent and operationally oriented perspective is offered by Faqir [22], who provides an overview of AI-based applications in contemporary digital investigations, including automated triage, classification of illicit content, facial recognition, and communication analysis. The work highlights that the adoption of these tools in investigative practices has grown significantly, but also draws attention to a persistent tension: most systems are evaluated in terms of aggregate accuracy on controlled datasets, without adequately considering the stability of their predictions under manipulated or out-of-distribution input conditions. This gap between performance measured under controlled conditions and operational reliability is among the most pertinent issues motivating the positioning of the present thesis.

A further recent perspective broadens the operational framework by jointly considering the opportunities offered by AI in digital investigations and the risks arising from its possible use for evasive or anti-forensic purposes. Along these lines, Sanna et al. [11] discuss the possible support provided by AI to cybercrime detection and DF activities, while also highlighting how general-purpose generative tools may facilitate the production of images or code intended for the steganographic concealment of information. The empirical evaluation of the robustness of AI-based tools with respect to manipulated visual content is instead addressed more directly by the studies on *Magnet.AI* and *Excire Photo AI*, discussed in the following section [2, 3].

Overall, this line of research has produced three main findings. First, AI-based automation in DF is motivated by a genuine scaling challenge that traditional methods can no longer address sustainably [1, 4]. Second, AI serves as operational support for the analyst rather than a substitute, and evidentiary conclusions remain a human responsibility [11, 22]. Third, the validation of AI-based systems in DF remains partial: the literature tends to test models on clean and controlled data, without systematically examining their behavior when inputs are degraded, manipulated, or out of distribution [2, 3].

It is precisely this third finding that motivates the specific problem addressed in the following section: if validation of AI-based models in DF remains incomplete even in the academic literature, the situation becomes even more critical for commercial software used or evaluated in the context of forensic analysis, where AI-based functionalities may be proprietary, opaque, and difficult to verify independently.

3.2 AI-based Software for Forensic Image Analysis and Operational Validation

The integration of AI-based functionalities into software used for the analysis of digital evidence represents one of the main operational developments in DF. In the context of visual content, such functionalities can support triage, semantic search, automated categorization, and the prioritization of images for analyst review. However, the availability of an automated function does not, in itself, imply that its behavior is adequately robust in the presence of degraded, ambiguous, or intentionally manipulated inputs. This problem is particularly relevant when AI components are proprietary and the software can be observed only through the outputs it produces, without access to the underlying models, training datasets, or decision thresholds [1, 2, 3].

The literature specifically devoted to the operational robustness of AI-based software applicable to forensic contexts remains limited. Sanna et al. conducted an initial evaluation of *Magnet.AI*, integrated into *Magnet AXIOM*, and *Excire Photo AI*, examining the behavior of their automated functionalities with respect to anomalous or manipulated content [2]. A subsequent study further investigated the response of the same tools to facial deepfakes and morphed images, identifying critical limitations in the robustness of the AI-based functionalities examined, and emphasizing the particular forensic relevance of false negatives, since potentially relevant content may fail to reach the analyst’s attention [3]. These works constitute the most directly relevant prior contributions for the present thesis, while differing from it in the specific task considered: the present work concerns the operational `weapon/non_weapon` classification task, focused on

the identification of images containing firearms or related content, and includes a separate analysis of system behavior in the presence of ood inputs.

3.2.1 Magnet AXIOM and Magnet.AI

Magnet AXIOM is commercial software designed for the processing and analysis of evidence from heterogeneous digital sources. Among the functionalities relevant to automated image analysis, the *Magnet.AI* component enables the categorization of multimedia content that may be relevant to an investigation, including categories related to weapons, drugs, nudity, and other sensitive content [2, 78]. This functionality is directly relevant to the task addressed in the present thesis, as it allows one to observe whether images containing firearms or related content are flagged within a forensic workflow.

Previous studies have shown that *Magnet.AI* can be subjected to operational evaluation in a black-box setting, by observing the categories returned by the software without assuming knowledge of the underlying model. In particular, Sanna et al. examined the behavior of the software with respect to manipulated content and, subsequently, with respect to facial images generated through deepfake and morphing techniques, identifying limitations in the robustness of certain automated functionalities [2, 3]. The primary methodological limitation remains the opacity of the proprietary component: the researcher can measure the behavior of the outputs available to the analyst, but cannot reconstruct with certainty the architecture, training data, preprocessing, or internal decision logic.

3.2.2 Magnet Griffeye and T3K CORE

Magnet Griffeye is a commercial platform used for the triage, review, and organization of large volumes of multimedia content in investigative and forensic contexts. In the context of the present thesis, the tool is relevant not as a controllable academic classifier, but as commercial black-box software capable of producing observable semantic outputs on images through automated analysis modules.

The *T3K CORE* component enables the generation of automated semantic bookmarks associated with the analyzed multimedia content. These bookmarks can be used by the analyst to filter, prioritize, or review potentially relevant images. From a methodological perspective, this type of output differs from the categories returned by *Magnet.AI*, from the semantic retrieval results produced by *Excire Foto 2025*, and from the classifications exported in the Cellebrite workflow. Nevertheless, it belongs to the same family of observable operational signals that can be normalized against an experimental ground truth.

The relevance of *Magnet Griffeye* to this thesis therefore lies in the possibility of extending the black-box evaluation to an additional output paradigm: not a class probability or an explicit supervised decision, but a set of automatically generated semantic bookmarks. This characteristic reinforces the general methodological problem addressed in the thesis: different commercial tools may expose heterogeneous outputs, yet all must be mapped to a common representation through explicit, documented normalization rules that are consistent with the experimental task.

3.2.3 Excire Foto 2025

Prior literature has analyzed *Excire Photo AI* as AI-based software oriented toward the search and recognition of photographic content, including in controlled forensic analysis contexts [2, 3]. In these studies, Excire was evaluated primarily with respect to identity search functionalities and its behavior on manipulated facial images. The reported results show that deepfakes and morphing can substantially degrade the reliability of automated image retrieval associated with specific identities [3].

The scope of the present thesis is different. The software considered is *Excire Foto 2025*, used as a standalone general-purpose application equipped with AI-based functionalities for organizing

and searching visual content, and not as a native component of a tool developed exclusively for forensic purposes [79]. Its inclusion in the study reflects the methodological interest in verifying whether a visual semantic search system can be employed, in a controlled forensic analysis context, to retrieve images consistent with a concept of investigative relevance such as the presence of a weapon.

This distinction is essential for interpreting the results: *Excire Foto 2025* should not be described as a native forensic **weapon/non_weapon** classifier, but rather as AI-based software whose observable behavior can be mapped to an operational decision of retrieval or non-retrieval of an image with respect to the queries and configurations defined in the experimental protocol. Here too, the evaluation remains black-box: the thesis measures the outputs available to the analyst, without assuming knowledge of the model, the training data, or the internal similarity criteria.

3.2.4 Cellebrite Inseyets and Automated Media Classification

The Cellebrite ecosystem is widely associated with mobile forensics and digital investigation activities. The vendor documentation describes *Cellebrite Inseyets* as an integrated suite for accessing, extracting, processing, analyzing, and reporting on digital evidence originating primarily from mobile devices [80]. From this perspective, Inseyets does not represent a single isolated module, but an operational environment that combines extraction, analysis, triage, and investigative workflow management functionalities.

With respect to visual content, Cellebrite documentation also describes media classification functionalities oriented toward the automated categorization of images and videos through AI-based techniques [81, 82]. These functionalities reflect a broader trend in contemporary DF, in which commercial tools integrate automated modules to support the triage, selection, and prioritization of large volumes of multimedia content.

In the context of the present thesis, *Cellebrite Inseyets* is therefore relevant as an example of commercial black-box software in which AI-based functionalities are integrated into professional mobile forensic workflows. Compared with desktop or general-purpose tools primarily oriented toward photographic search and categorization, Cellebrite extends the evaluation of operational robustness to a mobile forensic context, where the analyst may need to examine large volumes of images and videos extracted from devices or digital storage media.

As with the other commercial tools considered in this section, the integration of automated functionalities does not eliminate the need to assess their reliability, limitations, and behavior in the presence of degraded, manipulated, or not fully representative inputs relative to ideal operating conditions. For this reason, *Cellebrite Inseyets* is discussed as a further significant case of commercial black-box software with respect to the general problem of the operational robustness of AI-based systems in forensic contexts.

3.2.5 Common Limitations of Black-Box Evaluation

The tools considered exhibit distinct functional characteristics and should not be treated as equivalent systems. *Magnet.AI* exposes operational categories applicable to image content; *Excire Foto 2025* is considered as a system for semantic search and retrieval of visual content; *Cellebrite Inseyets* is evaluated through the specific media-analysis workflow and exportable outputs used in the experiment; *Magnet Griffeye*, through *T3K CORE*, produces automated semantic bookmarks. Accordingly, any operational comparison across these tools cannot assume equivalence of task, internal models, decision thresholds, or output modalities.

A common limitation concerns the proprietary or otherwise non-observable nature of the AI components employed. In a black-box evaluation, it is not possible to directly control the model architecture, training dataset, update procedures, preprocessing, confidence calibration, or decision criteria. Validation must therefore rely on external and verifiable elements: an independent ground truth, a documented dataset, common experimental conditions,

reproducible manipulated inputs, exportable outputs, and matching procedures based on identifiers or hashes.

In light of these limitations, the contribution of the present thesis does not consist of internally characterizing the proprietary algorithms of the analyzed software, nor of proposing a new classification tool. Rather, it consists of evaluating, through a reproducible experimental protocol, the operational robustness of observable outputs in the presence of clean images, out-of-distribution content, adversarial perturbations, and anti-forensic transformations. This approach extends the prior literature to a different visual task of investigative interest and to a comparison between controllable proxy models and AI-based software evaluated in a black-box setting.

Software	Relevant function	Relevance to the present thesis	Methodological limitation
Magnet AXIOM / Magnet.AI	Automated categorization of visual content, including images associated with weapons	Enables observation of the operational flagging of weapon images in software designed for forensic analysis	Proprietary AI component evaluable only through observable outputs
Magnet Griffeye / T3K CORE	Automated generation of semantic bookmarks on multimedia content	Enables evaluation of an additional black-box output paradigm, based on semantic bookmarks that can be normalized with respect to the experimental task	Commercial black-box tool; comparability depends on the operational mapping between exported bookmarks and the experimental weapon class
Excire Foto 2025	AI-based semantic search and retrieval of images	Enables evaluation of the retrieval of images consistent with the firearm-related concept through standalone software assessed in a controlled forensic context	Not a native forensic classifier; behavior depends on operational queries and configurations
Cellebrite Inseyets	AI-based categorization of images and videos within mobile analysis	Enables verification of the applicability of the experimental protocol to a mobile analysis workflow, based on the outputs concretely exportable from the software	Commercial black-box tool; comparability depends on the format of exportable outputs and on the operational recoding of classifications

Table 3.1: AI-based software relevant to the operational evaluation and main limitations of black-box analysis

3.3 Automated Image Classification in Forensic Scenarios

Automated image classification constitutes one of the main points of contact between computer vision and DF. In the forensic literature, this problem is not treated as a simple general visual recognition task, but as an activity supporting the selection and prioritization of potentially relevant evidence within large multimedia collections. In this context, the objective is not only to assign a correct label to an image, but also to reduce the analyst’s workload, increase the likelihood of identifying content of investigative relevance, and limit the risk that relevant elements remain undetected within very large datasets [1, 2].

Early contributions in this direction applied ML and DL models to semantic classification, object recognition, and multimedia triage tasks, with particular attention to categories of investigative interest such as faces, objects, scenes, illicit content, or elements potentially associated with criminal conduct. Costantini et al. [1] situate these functionalities within a broader evidence analysis workflow, in which automated classification does not replace forensic interpretation, but contributes to organizing and filtering the acquired material. This framing is relevant because it clarifies that visual classification, in forensic contexts, serves a primarily operational function: directing the analyst’s attention toward portions of the evidence that warrant more detailed review.

In the specific domain of weapon recognition, Hao et al. [19] proposed DeepFirearm, an approach based on deep representations for fine-grained firearm recognition. Their work demonstrates that the classification of images containing firearms or related content cannot be reduced to a trivial visual problem: the variability of poses, perspectives, occlusions, image quality, and morphological similarity across different weapon models requires discriminative representations capable of capturing relevant visual detail. However, works of this type are primarily oriented toward recognition or retrieval under controlled experimental conditions, and do not systematically address the behavior of models when images are degraded, manipulated, or drawn from the heterogeneous sources typical of a digital investigation.

The availability of public datasets for weapon detection, such as the *Weapon Detection Dataset* [83], has supported the development of models capable of recognizing firearms and related content in static images. These resources have significant experimental value, but present limitations when transferred to realistic forensic scenarios. Public datasets are often designed for visual recognition or surveillance applications, rather than for the operational validation of AI-based models and software applicable to forensic contexts. As a result, they may not adequately represent borderline cases, degraded images, out-of-distribution content, screenshots, synthetic images, or material originating from social platforms and messaging applications. This gap between benchmark datasets and real investigative material constitutes one of the most relevant issues in the evaluation of AI-based systems applied to DF.

From an architectural perspective, the computer vision literature has demonstrated the effectiveness of deep CNN architectures for image classification. Architectures such as ResNet [43] and EfficientNet [44] have provided solid foundations for many visual classification systems, owing to their ability to learn hierarchical representations of image content. More recently, vision-language models such as CLIP [45] and BLIP [46], together with Vision Transformer (ViT) architectures [84], have broadened the scope of automated classification by enabling text-guided, zero-shot, or semantic-description-based classification. In forensic contexts, this development is potentially relevant because investigative categories may not align precisely with the classes of traditional datasets; however, the literature has not yet adequately clarified how stable these models are in the presence of degraded, ambiguous, or manipulated images.

A recurring issue in the literature concerns the differing operational severity of classification errors. In forensic tasks, a false positive may increase the analyst’s review burden, but a false negative may prevent relevant content from reaching human review. This asymmetry is particularly significant in triage systems, in which automated output does not directly produce an evidentiary conclusion, but influences the order in which evidence is examined. Consequently, the evaluation of a forensic classifier cannot be limited to average accuracy: it must also consider recall on critical investigative classes, the distribution of errors, the stability of predictions, and the ability to avoid assigning a known class with excessive confidence to out-of-distribution inputs.

The most recent literature on AI-based software applicable to forensic contexts confirms this concern. Sanna et al. [2] demonstrate that automated classification functionalities integrated into operational software may produce unstable results when visual content deviates from expected conditions. Although these studies do not exhaust the problem of image classification in forensic contexts, they indicate clearly that the transition from models that are accurate

on controlled datasets to tools that are reliable in investigative scenarios requires more rigorous evaluation protocols. In particular, the combination of visual classification, content of investigative relevance, out-of-distribution images, and intentional manipulations remains insufficiently explored.

Overall, the literature on automated image classification offers useful models and datasets, but leaves open a specific problem for DF: the limited availability of integrated evaluations that simultaneously consider the investigative relevance of the classes, the heterogeneity of sources, the presence of borderline or OOD cases, and robustness with respect to degradations, anti-forensic transformations, or adversarial perturbations. This gap motivates the transition to the following section, devoted to benchmarks and results from the literature on the robustness of computer vision models.

3.4 Robustness Evaluation: Benchmarks and Findings from the Literature

Recent literature in computer vision has systematically demonstrated that accuracy obtained on clean and distributionally homogeneous data is not sufficient to estimate the operational reliability of a model. This finding is particularly relevant for DF, because images analyzed in an investigative context rarely correspond to the controlled conditions typical of academic benchmarks: they may originate from heterogeneous devices, have been compressed by social platforms or messaging applications, exhibit low resolution, contain occlusions, or have been intentionally altered before acquisition. Within this framework, robustness is not an incidental property of the classifier, but a necessary condition for automation to serve as reliable support for evidence triage and prioritization.

A central contribution in this direction is the work of Hendrycks and Dietterich [21], who introduced a systematic benchmark for evaluating the behavior of classification models under common corruptions and perturbations. Their work showed that models with high performance on clean data may suffer significant degradation when exposed to noise, blur, compression, chromatic variations, or other realistic alterations. The point relevant to the present thesis is not the specific nature of the benchmark, but the empirical result it makes evident: nominal performance does not necessarily predict model stability under non-ideal operational conditions. This finding undermines the use of accuracy on clean data as the sole evaluation criterion for systems intended for real-world environments.

Similar findings have emerged in the object detection domain. Michaelis et al. [85] showed that models trained and evaluated under favorable conditions may lose effectiveness when exposed to environmental variations, visual degradations, or scenarios distant from the training distribution. Although their application domain is not forensic, the contribution is methodologically relevant because it confirms that robustness must be evaluated with respect to plausible perturbations of visual data, and not solely with respect to standardized datasets. In forensic contexts, this observation becomes even more critical: an apparently ordinary degradation may affect the system's ability to recognize content of investigative relevance and, consequently, the priority assigned to the evidence.

A second line of literature concerns the detection of out-of-distribution inputs. Hendrycks and Gimpel [20] showed that neural classifiers may produce high-confidence predictions even on inputs that do not belong to the classes expected by the task. This behavior is problematic in operational contexts, because an automated system may forcibly assign an anomalous image to a known class rather than signaling uncertainty. Fort et al. [47] further highlighted the limitations of OOD detection approaches, showing that the ability to recognize anomalous inputs depends strongly on the type of distribution considered and on the degree of distance from the training data. For DF, this implies that borderline, synthetic, heavily degraded, or task-irrelevant images should not be treated as simple classification errors, but as a structural component of any reliability evaluation.

The literature on robustness benchmarks has also shown that model instability may emerge both in the presence of realistic degradations and in the presence of intentionally constructed perturbations. Ford et al. [63] examined the relationship between error on noisy data and adversarial vulnerability, suggesting that model fragility with respect to intentional attacks is not entirely separable from sensitivity to natural or realistic input variations. This relationship is important for the present thesis because it justifies an integrated evaluation encompassing both anti-forensic transformations and adversarial perturbations.

A further relevant contribution is RobustBench [86], which provided a standardized reference for comparing models and defenses with respect to adversarial attacks. Although this benchmark belongs to the general computer vision domain and is not designed for forensic scenarios, it demonstrates the importance of shared protocols, comparable metrics, and controlled experimental conditions. The absence of an equally consolidated reference specifically oriented toward DF constitutes a meaningful gap: models and tools used in investigative contexts are often evaluated according to non-uniform criteria, on datasets that are difficult to compare, and without systematic verification of their robustness with respect to realistic or adversarial manipulations.

AutoAttack belongs to the same research direction, proposing an automated and more standardized evaluation procedure for estimating the adversarial robustness of classifiers, and reducing the risk of overestimating defenses through attack configurations that are insufficiently strong or not comparable [87]. Here too, the contribution does not directly concern DF, but reinforces the methodological need for explicit, reproducible, and sufficiently stringent evaluation protocols.

The need to extend robustness evaluation beyond general benchmarks also emerges from studies applied to automated systems used in security contexts. Nowroozi et al. [88], although working in a domain different from forensic image analysis, show that adversarial perturbations can substantially degrade the performance of systems that appear reliable under ordinary conditions. More directly in the context of forensic analysis of visual content, Sanna et al. [2, 3] show that AI-based software applicable to forensic workflows may exhibit fragility when exposed to manipulated inputs. These findings confirm that the robustness problem affects both experimentally evaluated models and the operational outputs of software observed in a black-box setting.

Overall, the literature on robustness benchmarks yields three key indications. First, accuracy on clean data is not sufficient to evaluate the reliability of a classifier intended for real-world scenarios. Second, the presence of degraded, manipulated, or out-of-distribution inputs must be treated as an integral component of the evaluation protocol, not as a marginal condition. Third, the forensic domain requires specific benchmarks and experimental protocols capable of connecting algorithmic robustness to the operational impact on triage and evidence selection. This need naturally leads to the literature on Adv-ML applied to images, in which degradation of model behavior is no longer caused by realistic or accidental variations, but by perturbations intentionally constructed to induce classification errors.

3.5 Adversarial Machine Learning Applied to Image Classifiers

The literature on Adv-ML has shown that image classifiers based on ML and DL may be vulnerable to inputs intentionally constructed to alter their predictions. Chapter 2 introduced the fundamental concepts of adversarial examples, threat models, and input perturbations; in this section, the focus shifts to the main findings of the literature and to their relevance for visual classifiers that may be employed in forensic scenarios.

The seminal works by Szegedy et al. [48] and Goodfellow et al. [13] established that high-accuracy deep classifiers can produce incorrect decisions in the presence of intentionally constructed perturbations, even when such alterations are perceptually limited. For DF, this

finding is relevant because it shows that an automated system used in visual triage can be induced to underestimate or overlook content of investigative relevance if the input has been manipulated before acquisition or analysis.

A second line of literature has investigated the construction of attacks that are more precise, iterative, or optimized with respect to the minimum perturbation required to alter the model's decision. DeepFool [53] showed that it is possible to estimate small-magnitude perturbations capable of moving an image across the classifier's decision boundary, thereby providing a more systematic means of measuring the distance between a clean input and an adversarial one. SuperDeepFool [54] belongs to the same research direction, proposing an approach oriented toward generating effective perturbations against deep models through a more targeted optimization process. These works were not developed specifically for DF, but they are relevant because they provide a methodological basis for measuring how unstable a visual classifier may be with respect to non-random image variations.

The literature has also shown that adversarial vulnerability is not limited to perturbations distributed across many pixels. Su et al. [52] introduced the One Pixel Attack, demonstrating that, in some cases, modifying an extremely small number of pixels may be sufficient to induce an incorrect classification. More recently, approaches based on sparse constraints, such as Sigma Zero Attack [56], have further explored the possibility of constructing attacks with localized alterations and limited perceptual impact. This line of research is particularly significant for forensic contexts, because it shows that effective image manipulation does not necessarily require extensive or visually evident transformations: even localized modifications can interfere with the behavior of automated models.

A further line of research has shown that adversarial perturbations may also take localized and visually more explicit forms, as in the case of adversarial patches. Brown et al. [89] demonstrated that an artificial region applied to an image can induce classification errors without modifying the entire visual content. For the present thesis, this result is relevant primarily at the conceptual level: it shows that classifier vulnerability is not limited to distributed and nearly imperceptible perturbations, but extends to localized alterations that can interfere with the model's decision-making process.

The literature also includes generative attacks, in which generative networks are trained to produce transferable adversarial perturbations. A representative example is AdvGAN, which employs a generator to construct adversarial examples efficiently and with transferability across models [90]. This family of attacks is not included in the experimental protocol of the present thesis, as the work prioritizes perturbations that are more directly interpretable in the operational context of DF.

A crucial aspect for operational scenarios is the distinction among white-box, black-box, and transfer-based attacks. Papernot et al. [50] showed that model vulnerability can transfer across different classifiers, while subsequent literature on black-box attacks has systematized scenarios in which the attacker has only limited access to model outputs or uses surrogate models to generate transferable perturbations [91]. Carlini and Wagner [14] contributed to strengthening the security evaluation of classifiers by demonstrating the limitations of several defenses proposed against adversarial attacks. The surveys by Akhtar and Mian [49] and Yuan et al. [51] confirm that the transferability of adversarial examples is one of the most significant problems for the use of AI-based models in environments that are not fully controlled. In the case of commercial software evaluated in forensic contexts, this observation is particularly important: even when the model integrated into the software is proprietary and non-inspectable, an attacker could generate perturbations on accessible models and verify their effectiveness through transfer.

Biggio and Roli [15] situated these vulnerabilities within the paradigm of *adversary-aware security*, arguing that the evaluation of a machine-learning-based system must explicitly account for the presence of a strategic adversary. This principle is central to the present thesis: a forensic classification system does not necessarily operate in a neutral environment, but may be exposed to data prepared or modified by actors interested in hindering the analysis. Adversarial

robustness is therefore not only a technical property of the model, but a component of its operational reliability.

In the domain closer to *image forensics*, Barni et al. [60] showed that neural-network-based models for detecting digital manipulations may be vulnerable to adversarial attacks. Nowroozi [23] further investigated the problem of applying ML techniques to *image forensics* in adversarial scenarios, highlighting how models designed for forensic analysis can be degraded by targeted perturbations. These contributions are particularly important because they shift the problem from generic computer vision examples to systems used for tasks closer to the forensic domain. The emerging finding is that adversarial vulnerability does not affect only generic classifiers trained on standard datasets, but also models developed to recognize manipulations or relevant characteristics in digital images.

The connection with AI-based software applicable to forensic contexts has been reinforced by more recent work on operational robustness. Nowroozi et al. [88] showed, in a different security context, that robustness verification with respect to adversarial perturbations must be an integral part of the evaluation of automated systems. Sanna et al. [2, 3] brought this perspective more directly into the forensic analysis domain, showing that AI-based functionalities integrated into operational software may exhibit fragility when exposed to manipulated content. Although these studies do not cover the full spectrum of adversarial attacks on images, they indicate clearly that the validation of AI-based software applicable to forensic contexts cannot be limited to its effectiveness on ordinary inputs.

A further line of literature has analyzed the use of input transformations as a possible defense against adversarial examples. Techniques such as recompression, resizing, or image transformations may reduce the effectiveness of some perturbations, but do not eliminate the need for systematic robustness evaluation [92]. This aspect is relevant to the present thesis because some transformations considered in the literature as defense mechanisms may take on a different meaning in a forensic context: they are not employed to protect the model, but to simulate realistic anti-forensic stress conditions and evaluate the operational stability of classifiers and commercial tools.

Overall, the literature on Adv-ML applied to images yields three indications relevant to this thesis. First, visual classifiers may be vulnerable to intentional perturbations even when they achieve high performance on clean data. Second, this vulnerability may manifest across scenarios that differ in attacker knowledge, perturbation magnitude, and transferability across models. Third, works specifically addressing *image forensics* show that even systems designed for forensic tasks are not immune to these fragilities. However, an important gap remains: most studies consider academic models or controlled benchmarks, whereas integrated evaluations combining visual classifiers, AI-based software observed in a black-box setting, realistic datasets, and adversarial perturbations within the same experimental protocol remain limited.

This gap motivates the transition to the following section. Adversarial perturbations represent manipulations designed with respect to model behavior; in forensic contexts, however, images may also be altered through more ordinary or anti-forensic transformations, such as recompression, resizing, blurring, and tonal modifications. Such transformations do not necessarily require knowledge of the classifier, but may nonetheless affect the stability of automated analysis.

3.6 Anti-Forensics and Realistic Transformations: Impact on Automated Classification

Building on the conceptual distinction introduced in Chapter 2 between adversarial perturbations oriented toward the model’s decision and anti-forensic transformations applied to the digital artifact, this section examines contributions that have studied the impact of realistic image manipulations on forensic analysis. The point relevant to the present thesis is that transformations not optimized with respect to the classifier may nonetheless affect input quality

and the stability of the automated outputs used in visual triage.

Yaacoub et al. [12] and the taxonomic literature on anti-forensics [93] frame anti-digital forensics as the set of techniques aimed at making the identification, acquisition, preservation, or interpretation of digital evidence more difficult. In the image domain, this perspective encompasses both manipulations aimed at deleting or masking forensic traces and apparently ordinary transformations that may alter the characteristics of the file. The contribution of this literature is significant because it shifts attention from the authenticity of the image alone to its reliability as an input for analysis: an image may remain semantically interpretable to a human observer while becoming more problematic for an automated classification system.

A particularly relevant line of work concerns JPEG recompression. Stamm and Liu [59] showed that anti-forensic operations applied to compression can alter or mask traces useful for image analysis, reducing the ability to identify prior manipulations. More generally, the literature on *image forensics* has shown that compression modifies the statistical and frequency structure of the image, affecting details, textures, and artifacts that can be exploited by both forensic analysts and automated models [34, 35]. This observation is directly relevant to automated classification: if a model learns features that are sensitive to compression quality, subsequent recompression may affect prediction stability even without visibly altering the semantic content of the image.

A second group of works concerns resizing, resampling, and geometric transformations. Popescu and Farid [37] showed that resampling leaves detectable statistical traces, establishing a central research direction in the analysis of digital manipulations. From the perspective of automated classification, however, the problem concerns not only the possibility of detecting resampling, but also its effect on the representations learned by models. Successive resizing operations, interpolations, cropping, or changes in aspect ratio may modify proportions, edges, and visual details, with particularly significant consequences in tasks where the object of interest occupies only a limited portion of the image. In forensic scenarios, this effect may influence the system’s ability to recognize objects of investigative relevance, especially when they are partially occluded, small, or embedded in visually complex contexts.

The literature on realistic image transformations has also considered alterations such as blur, noise, chromatic variations, histogram modifications, and contrast stretching. These transformations are not necessarily anti-forensic in the strict sense: they may result from low-quality acquisition, forwarding through messaging platforms, uploading to social networks, repeated saving, or ordinary editing processes. However, their relevance to classifier robustness has been highlighted by benchmarks on common corruptions, in which degradations of this type produce significant performance reductions even in models with high accuracy on clean data [21]. In forensic contexts, this evidence carries operational significance: realistic transformations do not represent marginal cases, but frequent conditions in the digital material actually submitted to analysis.

Information hiding techniques, such as steganography, remain relevant in the forensic domain, but are treated in this thesis as part of the general theoretical framework rather than as the primary object of the experimental evaluation. The empirical focus is instead placed on transformations that directly affect automated visual classification, such as recompression, resampling, blurring, and tonal modifications.

Recent studies on AI-based software applicable to forensic contexts confirm that such transformations must also be considered in operational validation. Sanna et al. [2, 3] show that manipulated content, or content that does not conform to ordinary acquisition conditions, may reduce the reliability of the automated outputs produced by the analyzed software. A key observation is that anti-forensic and realistic transformations do not necessarily require a sophisticated attack against the model: they can be applied with common tools, introduced by sharing platforms, or result from ordinary image processing steps. This makes them particularly plausible in real investigative scenarios.

Overall, the anti-forensic literature and the literature on robustness to realistic transformations

converge on a single point: the digital image should not be regarded as a stable and neutral input. It may undergo alterations that modify forensic traces, perceptual quality, and the features exploited by automated classifiers. However, most existing works analyze manipulation detection, model robustness, or tool validation separately, without integrating these aspects into a unified protocol oriented toward DF. Accordingly, the need remains to evaluate how realistic and anti-forensic transformations affect the automated classification of content of investigative relevance and, more generally, the quality of the support provided by AI-based systems to the analyst.

This need motivates the transition to the following section, devoted to explainability. If adversarial perturbations and anti-forensic transformations can modify model behavior, tools that allow at least partial analysis of which image regions or visual characteristics contribute to the automated decision become necessary. xAI techniques do not resolve the robustness problem, but they can provide critical support for interpreting and diagnosing the fragilities of AI-based systems.

3.7 Explainability Applied to the Reliability Evaluation of AI-based Systems

The growing adoption of AI-based systems in high-stakes domains has made the explainability of algorithmic decisions a central concern. In the context of DF, this need assumes specific relevance: the output of an automated model may guide triage, influence the priority assigned to specific content, and contribute to the selection of items to be submitted to human review. For this reason, the literature on xAI is relevant not because it renders models automatically reliable, but because it provides tools for critically analyzing their behavior, particularly when predictions change in the presence of degraded, manipulated, or out-of-distribution inputs.

The surveys by Adadi and Berrada [61], Guidotti et al. [94], Barredo Arrieta et al. [95], and Doshi-Velez and Kim [62] have helped clarify that model interpretability is not a unitary concept, but depends on the intended recipient of the explanation, the type of model, the required level of granularity, and the purpose of the evaluation. This distinction is particularly important in forensic contexts. An explanation intended for a researcher evaluating a classifier is not equivalent to one intended for an operational analyst or for a party required to verify the technical correctness of a procedure. It follows that, in the forensic context, xAI should be understood as support for verification and diagnosis, rather than as a substitute for experimental validation [11, 61, 62].

In the domain of image classification, foundational works on local explanations and visual attributions have provided methodologically distinct tools: LIME operates through local model-agnostic approximations, Integrated Gradients uses the gradient of the model with respect to the input, and Class Activation Mapping techniques highlight regions associated with the prediction in specific visual architectures [64, 66, 67]. The limitation, with respect to the problem addressed in the present thesis, is that these contributions were not designed to evaluate the operational robustness of proprietary AI-based software in forensic contexts, nor to establish whether an explanation remains diagnostically useful in the presence of manipulated inputs. For this reason, the explainability analysis included in the experimental protocol is applied to controllable proxy models and used as qualitative support for interpreting failure cases, rather than as a primary metric or as an equivalent explanation of the software evaluated in a black-box setting.

In the case of forensic image classification, this type of analysis is useful primarily for verifying whether, after a manipulation, local attributions remain associated with the object of investigative relevance or instead shift toward secondary regions, textures, backgrounds, or artifacts introduced by the transformation. This diagnostic use does not assign autonomous evidentiary value to the visual explanation, but allows the quantitative degradation of performance to be connected to observable changes in classifier behavior.

The literature has nonetheless highlighted important limitations of visual explanations. Carvalho et al. [68] emphasize that model interpretability depends strongly on the method used, and that different explanation methods may produce results that do not fully overlap. Moreover, many attribution techniques are sensitive to parameter choices, baselines, local noise, or preprocessing decisions. This implies that an apparently coherent saliency map does not constitute autonomous proof of the model’s correctness. In forensic contexts, this caution is essential: a visual explanation may support the analyst in understanding system behavior, but it cannot replace manual verification of the evidence or quantitative validation of the classifier.

A further limitation concerns the applicability of xAI to commercial software evaluated in forensic contexts. Many attribution methods require access to the model, to gradients, or at least to the possibility of querying the classifier repeatedly. In proprietary software, these conditions are not always satisfied. Consequently, while xAI can be applied in a relatively controlled manner to academic or open-source models, its use on commercial AI-based software is more complex and sometimes limited to indirect forms of analysis. This aspect reinforces the issue already discussed in Section 3.2: the opacity of proprietary modules hinders not only the reproducibility of classifications, but also their technical explainability.

In the specific context of DF, the literature that systematically applies xAI techniques to the robustness evaluation of forensic classifiers remains limited. Works on xAI provide general tools for interpreting DL models, while studies on adversarial robustness and anti-forensics focus primarily on performance metrics. A fully developed convergence among forensic validation, robustness evaluation, and explainability is therefore still lacking. This gap is significant because, in an investigative scenario, it is not sufficient to know that a model fails after a manipulation: it is also useful to understand whether the error is associated with shifts in local attributions toward irrelevant regions, with artifacts introduced by the transformation, or with a structural fragility of the learned representation.

In summary, xAI contributes to the reliability evaluation of AI-based systems in three ways. First, it enables qualitative analysis of which image regions influence the prediction. Second, it allows the behavior of the model on clean and manipulated images to be compared, highlighting possible shifts in the regions considered relevant. Third, it supports the diagnosis of model fragilities, while providing no autonomous guarantees of correctness. For these reasons, explainability constitutes a complement to the quantitative evaluation of robustness, not a substitute for it.

The following section concludes the state-of-the-art review by highlighting the overall gap that emerges from the research directions analyzed. The literature has produced significant contributions on forensic automation, AI-based software applicable to forensic contexts, visual classification, robustness, adversarial attacks, anti-forensic techniques, and explainability; however, these areas have been addressed predominantly in isolation. The present thesis is positioned precisely within the space left open by this fragmentation.

3.8 Literature Gap and Thesis Positioning

The review conducted in the previous sections shows that the literature has produced significant contributions on each of the axes relevant to this thesis, but has addressed them largely in isolation. Works on automation in DF have highlighted the need for tools capable of managing large volumes of digital data and supporting the analyst in the triage, examination, and analysis phases [1, 4, 22]. Studies on AI-based software applicable to forensic contexts have begun to document concrete problems of opacity, independent validation, and fragility in the presence of manipulated content [2, 3]. The literature on automated image classification has shown the effectiveness of computer vision models for recognizing objects and relevant content, but often in controlled scenarios and not specifically designed for forensic validation [19, 43, 44, 45, 46]. In parallel, works on robustness have demonstrated that accuracy on clean data is not sufficient to assess the operational reliability of a classifier [20, 21, 47, 86].

A second body of work addresses the threats that may alter the behavior of automated systems.

The literature on Adv-ML has shown that intentionally constructed perturbations can induce errors even in high-performance models [13, 14, 15, 48, 50]. Works more closely related to *image forensics* have further highlighted that even models designed for forensic tasks may be vulnerable in adversarial scenarios [23, 60]. In parallel, the anti-forensic literature has studied transformations applied directly to the digital artifact, such as recompression, resampling, statistical alterations, and changes in visual quality, showing that these operations may affect both forensic traces and the digital representation of the image [34, 35, 37, 59]. Finally, the literature on xAI has provided useful tools for qualitatively analyzing model behavior, but these techniques have rarely been integrated into complete forensic protocols for robustness evaluation [61, 62, 64, 66, 67].

Table 3.2 summarizes the works most relevant to the positioning of the present thesis, highlighting their main contribution and the primary limitation with respect to the experimental design adopted in this thesis.

The main limitation emerging from this framework is not the absence of research on the individual topics, but the lack of an integrated evaluation. Figure 3.1 schematically illustrates this fragmentation: the seven reviewed research directions converge on a methodological space that the literature does not yet address in a unified manner.

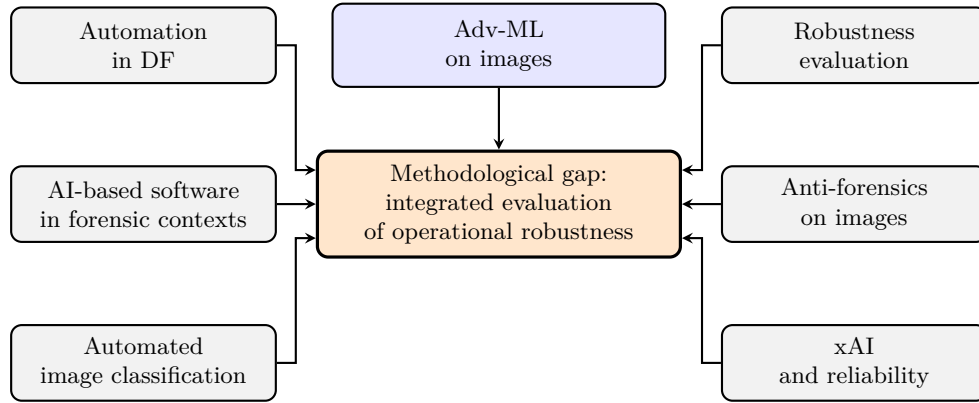


Figure 3.1: Structuring of the main research directions in the literature with respect to the methodological gap addressed by the present thesis

The specific gap identified by the present thesis therefore concerns the limited availability of integrated experimental protocols that combine, within the same evaluation design, automated image classification or semantic image retrieval, *human-in-the-loop* dataset validation, heterogeneous proxy models, AI-based software selected for operational evaluation in a forensic context, adversarial perturbations, anti-forensic transformations, quantitative metrics, and explainability-based analysis of model behavior. The relevance of this integration derives from the fact that, in a forensic context, the reliability of an automated system cannot be evaluated only as an isolated property of the classifier. It also depends on the quality and representativeness of the data, the presence of borderline or out-of-distribution cases, the nature of the manipulations applied, the transparency of the software used, and the effects that potential errors may have on triage and evidence selection.

The present thesis is positioned precisely within this gap. Its contribution consists of conducting an experimental evaluation of the operational robustness of proxy models and AI-based software employed, in the context of DF, for the automated classification or semantic retrieval of images of potential investigative interest, jointly considering the algorithmic and operational-investigative dimensions. The objective is not only to measure performance degradation under adverse conditions, but also to understand how such degradation may affect the system's ability to support the analyst in the selection of relevant content. From this perspective, robustness is treated not as an abstract attribute of the model, but as a functional requirement for the controlled use of AI in DF workflows.

Table 3.2: Representative works and positioning of the present thesis with respect to the literature

Representative works	Main contribution	Limitation with respect to the present thesis
Sanna et al. [2, 3]	Evaluate the robustness of AI-based software applicable to forensic contexts, with attention to <i>black-box</i> evaluations and manipulated multimedia content.	They constitute the closest reference to the present thesis, but focus on preliminary scenarios or different visual domains. They do not incorporate within a single protocol a frozen source population with weapon and non_weapon classes, a separate OOD evaluation branch, adversarial perturbations, anti-forensic transformations, and a unified bundle for forensic evaluation. The comparative normalization of heterogeneous black-box outputs, such as categories, classifications, retrieval results, and semantic bookmarks, also remains underexplored.
Nowroozi and Barni et al. [23, 60]	Analyze <i>image forensics</i> in adversarial scenarios and show that forensic models based on ML and DL may be vulnerable to targeted perturbations.	They provide the theoretical basis for adversarial image forensics, but focus primarily on academic models or controlled tasks, rather than on the operational evaluation of AI-based software observed in a <i>black-box</i> setting.
Biggio and Roli [15]	Formalize the paradigm of adversarial machine learning and the need to evaluate classifiers in the presence of a strategic adversary.	This contribution is foundational for the threat model, but remains general with respect to DF and does not directly address forensic workflows, operational software, or the classification of digital evidence.
Costantini et al. and Faqir [1, 22]	Show the role of AI as support for analysis, evidence organization, and the formulation of investigative hypotheses.	They contextualize automation in DF, but do not systematically evaluate the robustness of AI-based models or software with respect to degraded, manipulated, or out-of-distribution inputs.
Yaacoub et al., Stamm, and Farid [12, 34, 59]	Describe anti-forensic techniques and image transformations capable of altering traces, metadata, visual quality, and the statistical structure of the file.	They are essential for motivating the anti-forensic dimension of this thesis, but do not systematically connect these transformations to automated classification or semantic image retrieval through AI-based software applicable to forensic contexts.
Hendrycks et al., Fort et al., AutoAttack, and Robust-Bench [20, 21, 47, 86, 87]	Demonstrate that accuracy on clean data, robustness to corruptions, OOD behavior, and adversarial robustness must be evaluated separately.	They provide general benchmarks and methodological principles, but are not designed for the data, constraints, and workflows specific to DF.
Ribeiro et al., Sundararajan et al., and Zhou et al. [64, 66, 67]	Propose xAI and visual attribution techniques useful for interpreting the behavior of image classifiers.	They support the qualitative analysis of failure cases, but do not replace quantitative validation or solve the opacity problem of commercial software evaluated in a <i>black-box</i> setting.

The positioning of the thesis is therefore twofold. On the one hand, it connects to the literature on Adv-ML, robustness evaluation, and xAI, on the premise that classification systems must also be evaluated in the presence of hostile, degraded, or anomalous inputs. On the other hand, it is situated within the domain of DF, where the quality of automation must be assessed with respect to its operational usefulness, the verifiability of the results, and the capacity to maintain human control over the analysis process. This framing makes it possible to move beyond a purely performance-based evaluation and to address robustness as a methodological, technical, and forensic issue. This research space motivates the FAIRLab experimental protocol adopted in the present thesis and described in the following chapter.

Chapter 4

Methodology

This chapter defines the methodology adopted to evaluate the operational robustness of AI-based tools for automated image classification in forensic contexts. It addresses the gap identified in Chapter 3 by formalizing FAIRLab as a reproducible and auditable experimental protocol that combines controlled experimental conditions with operationally relevant forensic scenarios.

The chapter presents the overall methodology and research assumptions, followed by the procedures for dataset construction, human-in-the-loop validation, dataset freezing and partitioning, proxy-model and black-box software evaluation, perturbation testing, metric computation, explainability, and traceability. Implementation-specific data structures, scripts, architectures, parameters, software versions, and configurations are documented in Chapter 5; quantitative outcomes and their interpretation are reported in Chapter 6.

4.1 Overview of the FAIRLab Methodology

The FAIRLab protocol defines a structured methodological procedure for evaluating automated image-classification systems under clean, OOD, adversarial, and anti-forensic conditions. Its scope extends beyond clean classification accuracy because automated outputs may affect evidence prioritization and the allocation of human analytical resources. Operational robustness is therefore examined in relation to missed relevant content, additional review workload, and unstable behavior on inputs that differ from the expected operational distribution.

The protocol integrates two complementary evaluation levels. The first uses fully inspectable proxy classification models as controlled experimental references; these models are not treated as replicas or functionally equivalent substitutes for proprietary systems. The second evaluates selected AI-based software systems under black-box or near-black-box access conditions, using only their observable inputs and outputs and making no assumptions about inaccessible architectures, training data, calibration procedures, or decision logic [2, 3].

Both levels rely on the same frozen experimental corpus. Its binary in-domain component supports clean and perturbed evaluation, whereas a separate clean OOD branch is evaluated outside model training and perturbation generation, as detailed in Section 4.6. The clean binary samples, clean OOD inputs, adversarial artifacts, and anti-forensic artifacts are combined into the forensic evaluation bundle, enabling comparison under aligned experimental conditions.

Quantitative metrics characterize classification performance, robustness degradation, error direction, and OOD behavior. Explainability provides qualitative diagnostic information for selected cases involving an inspectable proxy model and is not extended to the selected black-box systems. Traceability controls preserve the relationships among source samples, derived artifacts, experimental conditions, observable outputs, and reported results.

The protocol specifies the methodological requirements and information that must be preserved without prescribing particular model architectures, software products, file formats, directory structures, databases, hashing algorithms, or explainability techniques. These choices belong to the concrete implementation documented in Chapter 5. Figure 4.1 summarizes the resulting workflow.

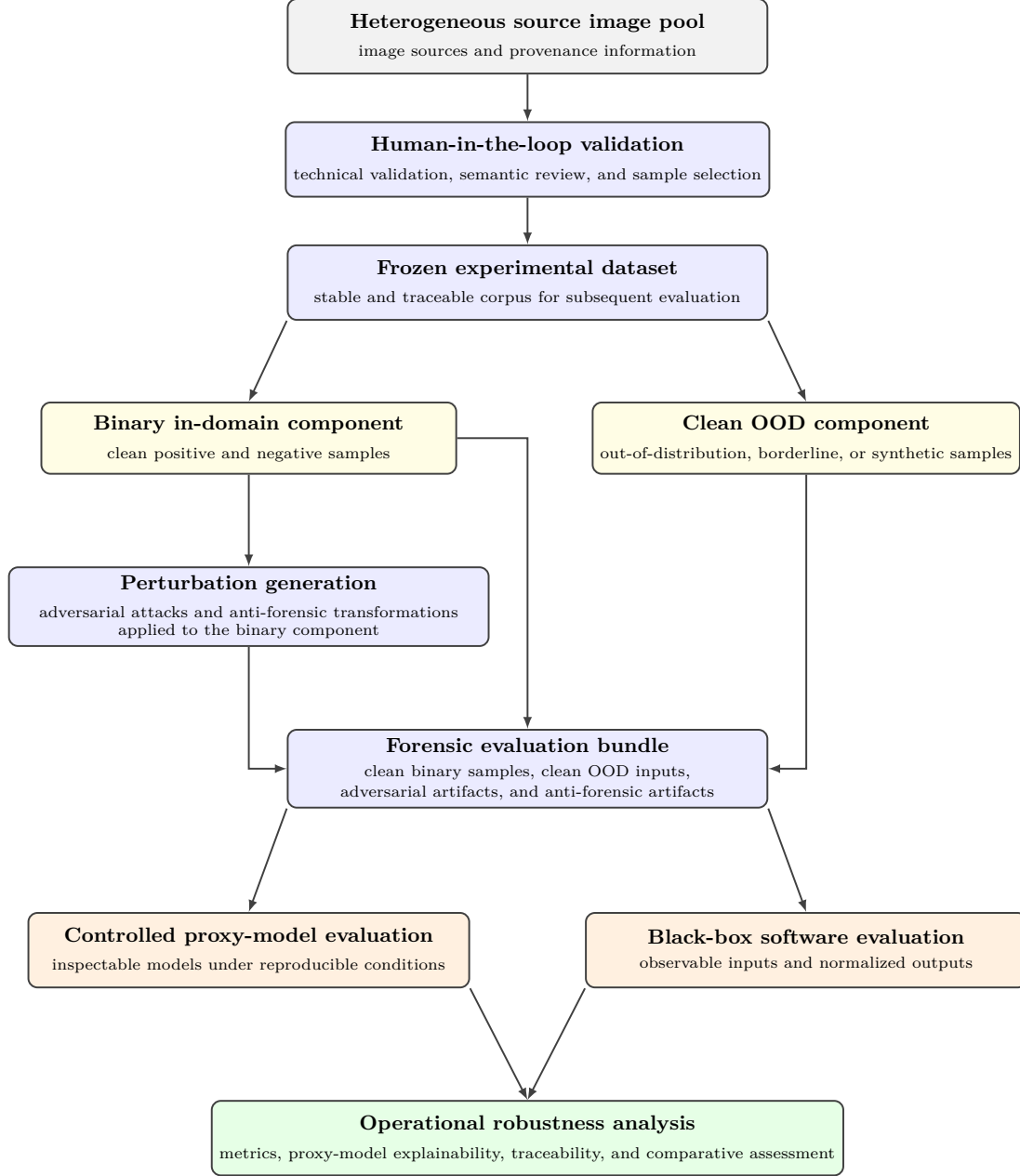


Figure 4.1: FAIRLab experimental protocol. The frozen corpus is divided into a binary in-domain component and a separate clean OOD branch. Perturbations are applied only to the binary component, and the resulting common bundle is evaluated through complementary proxy-model and black-box software analyses

4.2 Research Design and Experimental Assumptions

The experimental design adopts forensic visual triage as its reference scenario, in which automated classification supports the preliminary identification and prioritization of potentially

relevant images. The case study defines a controlled firearm-related classification task rather than attempting to cover the entire domain of weapon detection. This restricted scope provides a clear target category while retaining sufficient visual variability for robustness assessment. The adopted designations are operational categories established for the experiment and must not be interpreted as legal qualifications or universal semantic classes.

The binary task distinguishes between **weapon**, defined as the positive class, and **non_weapon**, defined as the in-domain negative class. A separate OOD branch is reserved for distributional evaluation and is not modeled as a third supervised class. The assignment criteria were formalized before dataset freezing and govern human-in-the-loop validation, sample selection, and the documentation of inclusion and exclusion decisions. Table 4.1 summarizes these operational definitions.

Table 4.1: Operational definitions adopted in the experimental protocol

Designation	Role in the protocol	Operational assignment criterion
weapon	Positive class of the binary task	Images containing clearly recognizable and realistic firearm-related visual content. Representative examples include pistols, rifles, submachine guns, and shotguns. The target object must be sufficiently visible and reliably attributable to the experimental category. Marginal, strongly occluded, excessively ambiguous, synthetic, toy-like, replica-like, or airsoft-like objects are excluded from this class.
non_weapon	In-domain negative class of the binary task	Realistic, usable, and visually interpretable images that do not contain an object belonging to the experimental target category. Examples may include people, environments, vehicles, ordinary objects, and indoor or outdoor scenes. Borderline, synthetic, or anomalous content is excluded. Images containing knives, swords, replicas, airsoft weapons, toy weapons, videogame weapons, or rendered weapon-like objects are also excluded because they may introduce semantic ambiguity.
OOD	Separate branch for distributional evaluation	Borderline, anomalous, or non-representative images that do not constitute clean positive or negative samples. Examples include knives, swords, replicas, toys, airsoft weapons, videogame weapons, CGI content, synthetic images, generated images, tanks, missiles, explosions, and war scenes. The branch may also include source images that are strongly degraded, unusual, or semantically ambiguous. These inputs are retained without experimental manipulation and are not equivalent to the adversarial or anti-forensic artifacts generated in later stages. Such content may be contextually related to weapons or violence without constituting a valid sample for the binary task.
exclude	Review-only exclusion designation	Images that are corrupted, unreadable, unusable, non-interpretable, or semantically irrelevant to the experimental protocol. This designation is used during human review to document the removal of unsuitable samples. Excluded images do not contribute to the frozen dataset or to any subsequent evaluation branch.

Figure 4.2 presents representative examples of the two binary classes and the separate OOD branch. The examples are illustrative and do not exhaust the internal variability of the corresponding groups.

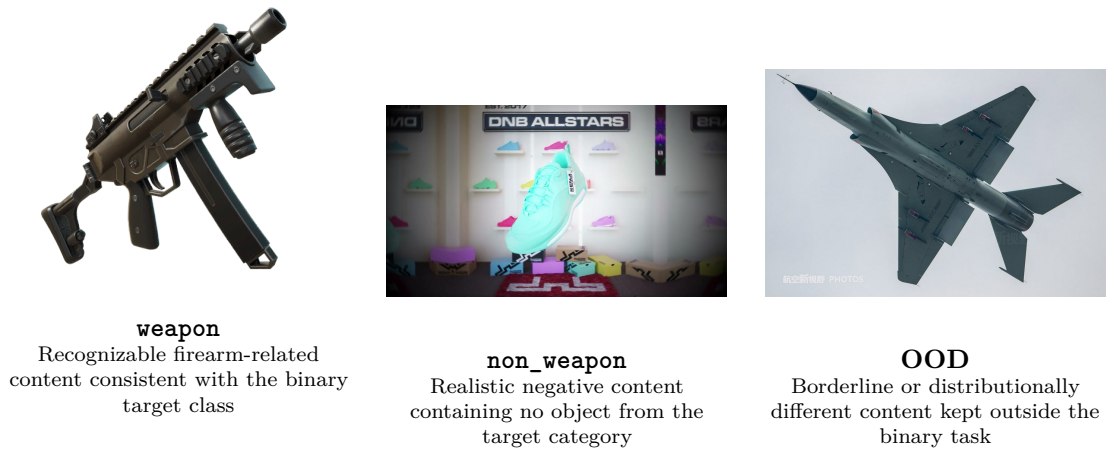


Figure 4.2: Representative examples of the binary operational classes and the separate OOD branch adopted during human-in-the-loop review

The **exclude** designation is used exclusively as a quality-control disposition during human review and does not constitute an additional analytical branch. Likewise, the OOD branch is not a homogeneous semantic class or a subclass of **non_weapon**. It contains borderline, anomalous, degraded, synthetic, or otherwise distributionally different inputs that do not constitute clean instances of either binary class. Since classifiers may produce predictions with high maximum predicted-class probabilities outside their training distribution [20, 47], these inputs must be assessed separately from ordinary binary errors. Their exclusion from model training and perturbation generation is formalized in Section 4.6.

These operational definitions provide the semantic basis for the dataset construction, review-documentation, and freezing procedures described in Sections 4.3, 4.4, and 4.5. They ensure that final assignments are derived from explicit and traceable review criteria rather than inherited directly from source labels, filenames, or directory structures.

4.3 Dataset Construction and Source Consolidation

Dataset construction separates source collection, technical validation, semantic review, and final dataset freezing. This distinction is methodologically necessary because the technical validity of an image file does not establish its semantic suitability for the forensic classification task.

The process begins with the consolidation of heterogeneous image sources into a candidate pool. The objective is not to create a visually homogeneous collection, but to capture part of the variability encountered in contemporary digital investigations. Depending on the investigated domain and data availability, the candidate pool may include public datasets, specialized repositories, open-web content, Open Source Intelligence (OSINT)-oriented collections, and material obtained from less structured or non-indexed online environments.

The composition and relative contribution of these sources may vary, but the provenance of each candidate sample must remain identifiable throughout the preparation and review process. Table 4.2 summarizes the methodological roles of the source categories considered in the protocol.

Table 4.2: Methodological roles of the image-source categories

Source category	Methodological role
Public datasets	Provide structured and reproducible collections that can support the initial construction of the candidate pool. Their original labels are retained as source information and are not automatically inherited as experimentally validated reference assignments.
Specialized repositories	Provide domain-specific images that can improve the representation of candidate content related to the target visual category. Samples obtained from these sources remain subject to the same validation and assignment criteria applied to the rest of the candidate pool.
Open-web collections	Introduce greater variability in image quality, composition, context, and acquisition conditions. They reduce dependence on curated benchmark distributions.
OSINT-oriented sources	Represent less structured content obtained from social, multimedia, or other openly accessible investigative sources. They contribute operationally relevant variability.
Non-indexed or unconventional online sources	Provide heterogeneous material that may differ from conventional public datasets in content, quality, context, or distribution. Their inclusion supports the evaluation of less standardized acquisition contexts.

After source consolidation, candidate images undergo technical validation before any semantic assignment is made. The validation process verifies that each file can be processed reliably and may include readability, structural validity, dimensional suitability, duplicate detection, and stable sample identification. The resulting records must preserve sufficient information to associate each retained sample with its provenance, technical-validation outcome, and integrity information.

Source location, directory structure, filename, and pre-existing annotations may be retained as contextual or audit information, but they must not be treated as definitive reference assignments. Final semantic membership must instead be determined through the operational criteria defined in Section 4.2. This separation limits the propagation of source-specific assumptions and labeling errors into the frozen experimental dataset.

The relative size of the original source collections must also be considered during sample selection. A large or highly asymmetric source must not automatically dominate the final corpus. Composition decisions should instead reflect the experimental objectives, the required class structure, and the need to preserve meaningful source diversity.

Following technical validation, the candidate pool is submitted to the documented semantic-review process described in Sections 4.4 and 4.5. Figure 4.3 summarizes the transition from heterogeneous sources to the frozen experimental dataset.

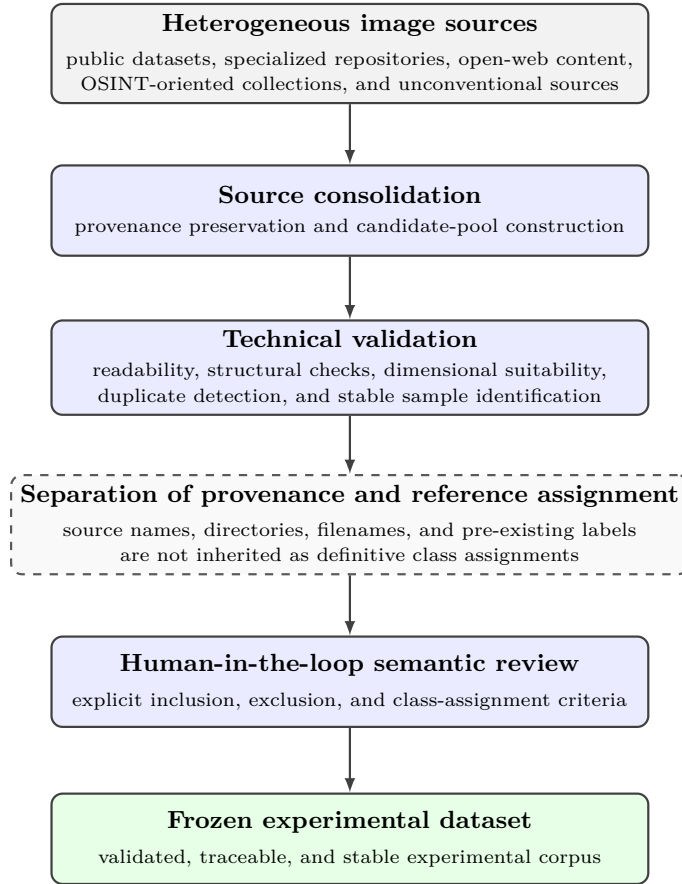


Figure 4.3: Dataset source consolidation. Heterogeneous sources are combined into a traceable candidate pool, subjected to technical validation, and reviewed according to explicit semantic criteria. Final class assignments are not inherited from source directories, filenames, or pre-existing annotations

4.4 Review Documentation and Semantic Decision Records

Following technical validation, each candidate sample is associated with a structured review record that connects its technical preparation to the subsequent human-in-the-loop semantic assessment. The record must preserve two distinct categories of information: sample identity, provenance, technical validity, and integrity information; and the semantic decision produced during manual review, including review status, operational assignment, exclusion disposition, explanatory notes, and any information required to document ambiguous or OOD cases.

Regardless of its technical representation, the review record must maintain a stable association among the candidate sample, its provenance, the documented semantic decision, and any experimental artifact subsequently derived from it. This association enables the decision process and the transition from the candidate pool to the frozen experimental dataset to be reconstructed.

Labels inherited from source material or generated through automated preprocessing may be retained as contextual information, but they must not replace the explicit semantic judgment produced during human-in-the-loop review. Final operational assignments are determined exclusively according to the criteria defined in Section 4.2.

The separation between technical-validation information and semantic decision records

preserves the distinction between file suitability and relevance to the experimental task. The resulting documentation supports the consistent application of the assignment criteria, preserves the rationale for inclusion, exclusion, and class-assignment decisions, and provides the traceability required for the review and dataset-freezing procedure described in Section 4.5.

4.5 Human-in-the-Loop Review and Final Dataset Freezing

The technically validated candidate pool is subjected to a documented *human-in-the-loop* review to determine the semantic suitability of each sample and assign the corresponding operational designation. The criteria defined in Section 4.2 must be applied consistently and independently of source directories, filenames, inherited annotations, or automated pre-labels.

Although the review itself does not rely on automated pre-labels, retaining explicit human control over semantic assignments is consistent with trustworthy-AI guidance, which emphasizes human agency and oversight in the use of AI systems [96].

The review follows a conservative decision rule. Samples are included in the binary task only when they can be assigned to either **weapon** or **non_weapon** with sufficient semantic clarity. Borderline, anomalous, synthetic, replica-like, degraded, or otherwise distributionally different content is retained within the separate OOD branch rather than being forced into either binary class. Unusable or semantically irrelevant samples are assigned to the review-only **exclude** disposition.

These assignments are operational designations defined for the experimental protocol and must not be interpreted as definitive legal, material, or evidentiary determinations regarding the depicted content. In particular, assignment to the OOD branch represents a methodological decision for cases that do not constitute clear instances of either binary class.

Each review decision must remain reconstructable through the semantic decision records defined in Section 4.4. These records preserve the association among the reviewed sample, its provenance, the assigned disposition, any explanatory notes, and the experimental artifacts subsequently derived from it.

After all candidate samples have been reviewed, the selected corpus is frozen. Dataset freezing establishes a stable reference corpus whose composition remains unchanged throughout the subsequent evaluation phases. Any correction or modification introduced after freezing must be explicitly documented and treated as a new version of the experimental dataset.

The frozen dataset provides the common source corpus for the controlled proxy-model evaluation and the black-box evaluation of the selected AI-based software systems. Its subsequent separation into the binary evaluation subset and the clean OOD branch is defined in Section 4.6.

The number of reviewers and the review workflow may vary across implementations. When the protocol is conducted by a single reviewer, the absence of inter-annotator agreement must be acknowledged as a limitation. Explicit assignment criteria, structured decision records, and dataset freezing support procedural auditability and reproducibility, but do not eliminate the subjectivity inherent in semantic assessment. Dataset freezing therefore acts as a methodological control that distinguishes changes in systems, perturbations, and metrics from undocumented changes in the evaluated corpus.

4.6 Data Partitioning and OOD Evaluation Strategy

After dataset freezing, the experimental corpus is divided into two distinct branches: an in-domain binary subset and a separate clean OOD evaluation set. The binary subset supports proxy-model training, clean evaluation, and robustness testing, whereas the OOD branch examines system behavior on borderline, semantically atypical, or distributionally different inputs.

The binary subset is organized into mutually exclusive experimental partitions, hereafter referred to as folds. Their number and size may vary according to the available dataset, class distribution, and experimental requirements. Regardless of the adopted configuration, each sample must belong to only one fold, class distributions should remain sufficiently consistent whenever stratification is feasible, sample identity and provenance must remain traceable, and the allocation procedure must be deterministic or documented in sufficient detail to permit regeneration.

Within this protocol, folds are not introduced primarily as a conventional cross-validation mechanism for hyperparameter selection or general performance estimation. Instead, they define controlled experimental holdout units. For each target fold, the corresponding proxy-model instance is trained on the remaining folds and evaluated on the held-out samples, thereby preventing overlap between model fitting and clean or robustness evaluation.

The fold-aware organization also preserves the relationship among each held-out sample, the corresponding proxy-model checkpoint, and any model-dependent adversarial artifact generated from that sample. Such an artifact must be produced using a checkpoint that was not trained on its source sample. Model-agnostic adversarial-style conditions and anti-forensic transformations remain associated with the originating sample and its fold but do not depend on a specific checkpoint.

The clean OOD set remains outside the binary folds, is not used for proxy-model training, and is not treated as a third supervised class. No experimental adversarial perturbation or anti-forensic transformation is applied to these samples. This separation allows the study to examine how automated systems assign inputs that do not constitute ordinary **weapon** or **non_weapon** instances without conflating distributional mismatch with deliberate or incidental input manipulation.

The partitioning records must associate each sample with its evaluation branch and, where applicable, its fold and corresponding proxy-model checkpoint. They must also preserve the relationship between each original image and any derived experimental artifact, consistently with the traceability requirements defined in Section 4.15.

Figure 4.4 summarizes the resulting partitioning strategy.

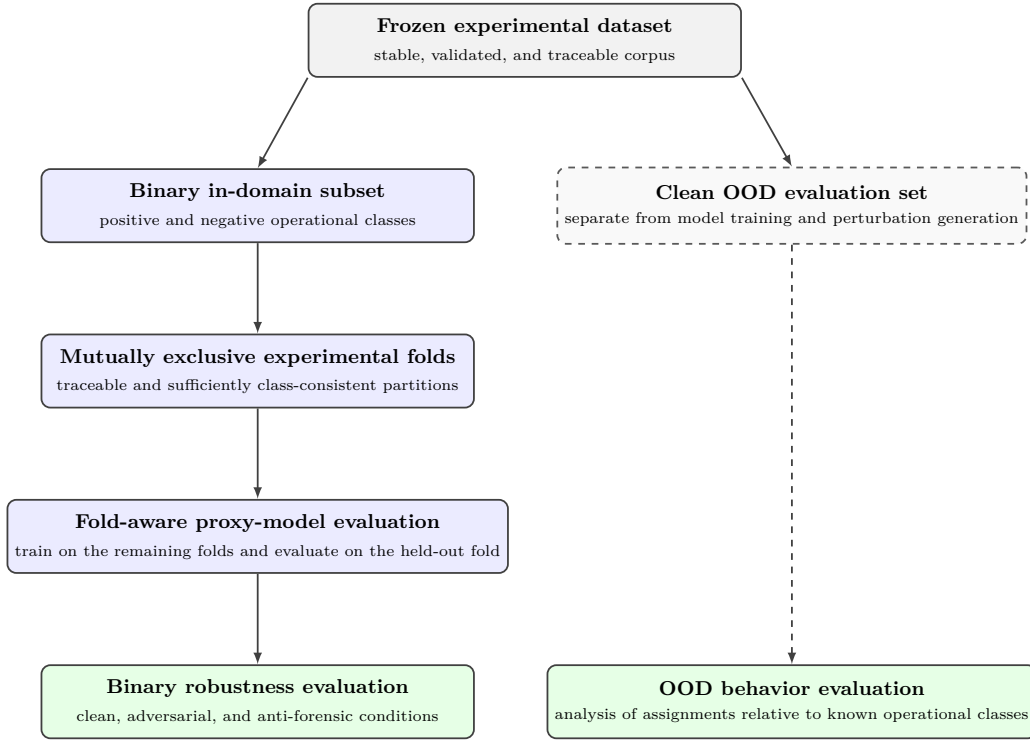


Figure 4.4: Data partitioning and OOD evaluation strategy. The binary subset is organized into mutually exclusive folds for fold-aware training and held-out evaluation, while the clean OOD branch remains outside model training and perturbation generation

4.7 Proxy-Model Evaluation Methodology

The controlled component of the methodology relies on fully inspectable classification models, hereafter referred to as *proxy classification models*. These models provide experimental references for analyzing classification behavior, robustness degradation, maximum predicted-class probability patterns, error direction, and transferability under reproducible conditions. They are not replicas, functionally equivalent substitutes, reverse-engineered reconstructions, or architectural reproductions of the selected black-box software systems.

The proxy-model evaluation methodology serves three methodological objectives. First, it provides a transparent reference for measuring classification performance and robustness degradation. Second, it enables the generation of model-dependent adversarial examples under explicit white-box or score-based access assumptions. Third, it supports the evaluation of whether perturbations generated against one controlled model transfer to other proxy architectures or to the selected black-box software systems.

The proxy models implement the binary classification task defined in Section 4.2. The mapping between `weapon`, `non_weapon`, and their numerical encodings must remain explicit and consistent throughout training, evaluation, and metric computation. The clean OOD branch is excluded from training and is not treated as a third supervised class, as established in Section 4.6.

The methodology does not prescribe specific model architectures. Instead, the controlled evaluation should include models with complementary algorithmic characteristics, such as conventional convolutional architectures, efficiency-oriented architectures, or models based on large-scale pretrained visual representations. The architecture, initialization strategy, preprocessing procedure, class mapping, optimization method, training configuration, and resulting checkpoints must be documented. The concrete architectures and configurations adopted in this thesis are reported in Chapter 5.

Proxy-model training follows the fold-aware strategy defined in Section 4.6. For each target fold, the corresponding model instance is trained exclusively on the remaining folds and evaluated on the held-out samples. Training conditions must remain consistent across folds for a given model configuration, and any variation in architecture, preprocessing, initialization, or optimization must be documented.

The same separation applies to model-dependent perturbation generation. An adversarial artifact associated with a held-out sample must be generated using a fold-specific checkpoint that was not trained on that sample. This requirement prevents information leakage between model fitting, adversarial generation, and robustness evaluation.

For model-dependent adversarial attacks, one controlled model is designated as the *primary proxy target*. This model provides the source system against which adversarial examples are generated under the access assumptions required by each attack. White-box attacks may use gradients, decision-boundary information, or other internal properties, whereas score-based attacks rely on observable model scores without requiring direct gradient access. Selection of the primary proxy target is an implementation decision and does not imply that the chosen architecture is universally more accurate, representative, or robust than the alternatives.

Using a single documented primary target limits unnecessary multiplication of experimental variants and produces a coherent adversarial test set. The resulting model-dependent artifacts are evaluated under three conditions:

1. *direct source-model evaluation*, in which adversarial artifacts are assessed against the fold-specific proxy-model checkpoint used to generate them under the corresponding access assumptions;
2. *cross-model transfer evaluation*, in which the same artifacts are assessed against different proxy architectures;
3. *black-box transfer evaluation*, in which the same artifacts are submitted to the selected black-box software systems.

Model-agnostic adversarial-style conditions and anti-forensic transformations are generated independently of a designated source model. Their evaluation across proxy architectures and black-box systems therefore measures response stability under shared manipulated conditions rather than transferability from a source model. The corresponding access assumptions and operational rationale are detailed in Section 4.8.

The proxy-model and black-box evaluation levels must remain analytically distinct. Proxy models are inspectable and configurable components whose training conditions, checkpoints, and internal behavior are available for controlled analysis. The selected software systems are evaluated only through their observable inputs and outputs. Exposing both levels to the same clean samples, clean OOD inputs, adversarial artifacts, and anti-forensic artifacts enables comparative assessment without assuming architectural or functional equivalence.

4.8 Perturbation Selection Rationale and Operational Relevance

The perturbation methodology evaluates classification behavior under controlled input modifications selected according to their access assumptions, computational cost, perceptual impact, reproducibility, and operational plausibility. These criteria are particularly relevant to forensic visual triage, where automated systems may process large image collections under operational time constraints. Computationally intensive perturbations remain useful for controlled vulnerability analysis, but may be less representative of manipulations that can be applied rapidly and at scale.

The methodology distinguishes among three categories of input modification:

1. *model-dependent adversarial attacks*, which are generated against a designated source model under explicit white-box or score-based access assumptions;
2. *model-agnostic adversarial-style conditions*, which modify an image according to predefined rules without being optimized against a specific classifier;
3. *anti-forensic transformations*, which emulate plausible modifications of the image artifact or its visual representation.

Model-dependent attacks provide controlled tests of classifier vulnerability and may support direct and transferability analyses. Although they do not necessarily represent the most probable manipulations in ordinary forensic scenarios, they demonstrate how classification behavior may change when an actor exploits information obtained from, or interaction with, a designated source model.

Model-agnostic adversarial-style conditions require neither access to internal model properties nor interaction with observable scores. Their application across proxy architectures and black-box software systems therefore supports comparison under a shared manipulated condition rather than transferability analysis from a designated source model.

Anti-forensic transformations emulate image-processing operations that may occur deliberately or incidentally before forensic acquisition or analysis. They generally require no access to the evaluated classifier and may support efficient batch processing, making them relevant to both deliberate manipulation and ordinary image-handling workflows.

The perturbation set should provide complementary experimental coverage rather than multiple near-duplicate variants of the same mechanism. Its selection should account for diversity in access assumptions and algorithmic mechanisms, while also considering computational feasibility, scalability, and reproducibility across experimental runs.

The selected conditions should preserve sufficient task-relevant semantic content and retain operational plausibility. They should also support cross-system comparison and, where applicable, transferability analysis, while remaining compatible with the evaluated image formats and software systems.

Table 4.3 summarizes the general relationship between perturbation category, computational cost, and operational relevance.

Table 4.3: Perturbation categories, computational cost, and operational relevance

Perturbation category	Typical cost	Methodological and operational role
Single-step white-box attacks	Low to medium	Provide efficient baselines for evaluating sensitivity to gradient-informed input modifications.
Iterative or optimization-based white-box attacks	Medium to high	Support controlled analysis of decision-boundary vulnerability and minimal or targeted input modifications.
Score-based attacks	Medium to high	Evaluate vulnerability through repeated interaction with observable model scores without requiring direct access to gradients or internal parameters.
Model-agnostic adversarial-style conditions	Low to medium	Enable source-model-independent testing through predefined changes to image appearance or statistical characteristics.
Lossy re-encoding and resampling	Low	Represent common modifications caused by editing, forwarding, format conversion, or platform-mediated processing.
Filtering and degradation	Low	Assess whether the reduction or alteration of discriminative visual details affects classification behavior.
Tonal and chromatic transformations	Low	Evaluate sensitivity to changes in color, contrast, luminance, or statistical image properties.

The cost levels in Table 4.3 constitute a qualitative methodological assessment rather than measured execution times. Actual cost depends on the algorithm, hardware, image dimensions, parallelization, software libraries, and implementation choices. Direct transformations and single-step attacks generally require less computation, whereas iterative, optimization-based, and score-based attacks may require repeated inference, queries, or gradient computation. The concrete attacks, transformations, parameters, and execution configurations adopted in this thesis are documented in Chapter 5.

Perturbations are generated exclusively from the binary in-domain subset. The clean OOD branch remains outside this process because it addresses distributional mismatch rather than robustness to input manipulation, as established in Section 4.6.

Each generated artifact must remain associated with its source image, perturbation category, configuration, and evaluation condition. For model-dependent attacks, this association must also preserve the source-model checkpoint and the corresponding access assumption. These records support reproducibility and allow the same artifacts to be evaluated across proxy models and selected black-box software systems.

Figure 4.5 summarizes the resulting workflow.

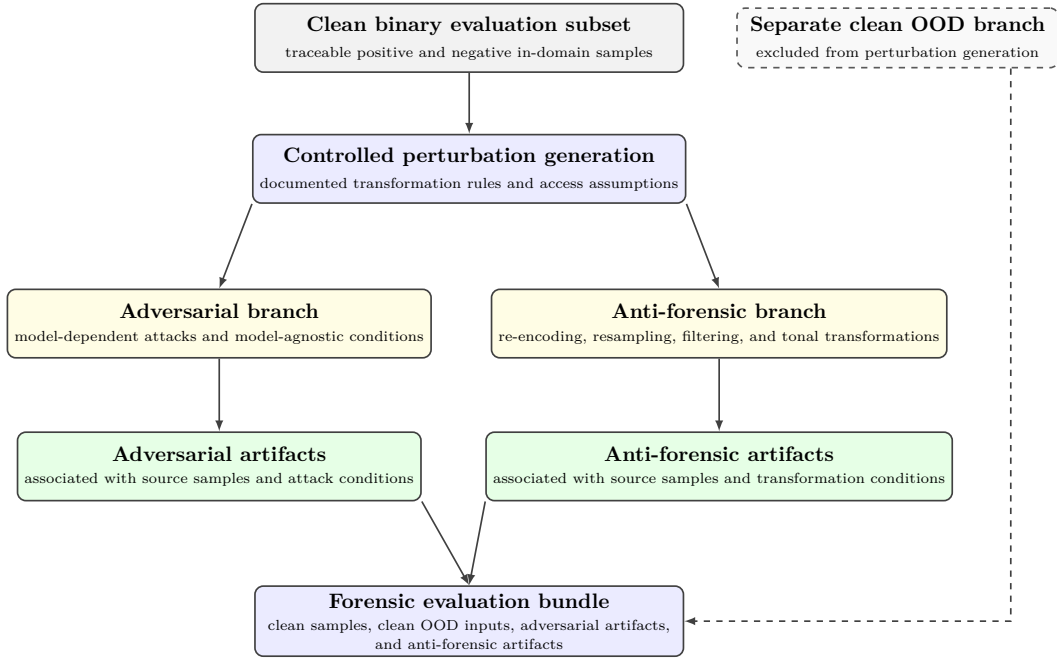


Figure 4.5: Perturbation generation workflow. Adversarial and anti-forensic artifacts are derived exclusively from the binary subset and are integrated into the forensic evaluation bundle together with the clean binary samples and the separate clean OOD inputs

4.9 Adversarial Evaluation Methodology

The adversarial component of the FAIRLab protocol evaluates whether controlled input modifications can alter the behavior of automated image classifiers while preserving the relationship between each generated artifact and its corresponding clean sample. Adversarial artifacts are generated exclusively from the binary in-domain subset; the clean OOD branch remains excluded because it addresses distributional mismatch rather than robustness to deliberate input manipulation, as established in Section 4.6.

Consistently with Section 4.8, the methodology distinguishes model-dependent attacks from model-agnostic adversarial-style conditions. Model-dependent attacks are generated against a designated source model under explicit access assumptions that may include gradients, internal model properties, decision-boundary information, or observable output scores. Model-agnostic conditions instead apply predefined modifications without querying or optimizing against a specific classifier.

The selected model-dependent attacks should provide complementary coverage of different adversarial mechanisms, including single-step gradient perturbations, sparse or localized score-based modifications, iterative optimization procedures, and decision-boundary-based attacks [13, 52, 53, 56]. The protocol does not aim to provide an exhaustive adversarial-robustness benchmark or a comprehensive comparison of all available attack algorithms, which would require extensive parameter sweeps and standardized attack suites [58, 87]. Instead, the attack set is selected to be computationally feasible, reproducible, sufficiently heterogeneous, and suitable for direct vulnerability and transferability analyses.

Model-dependent adversarial examples are generated against the primary proxy target defined in Section 4.7. For each held-out sample, generation must use the corresponding fold-specific checkpoint trained without that sample. This requirement preserves the separation between model fitting, attack generation, and evaluation and prevents information leakage.

Each generated artifact must remain explicitly associated with the original clean sample, its reference operational class, and the corresponding experimental fold. The retained information

must also identify the attack family or model-agnostic condition, as well as the source model, checkpoint, and access assumption when applicable. In addition, the generation parameters and resulting output artifact must be recorded together with all information necessary to reconstruct the generation condition and to verify the identity, integrity, and provenance of the artifact.

The output representation should minimize changes unrelated to the selected adversarial mechanism. Any additional compression, encoding, or representation effect introduced during artifact generation must be documented and considered when interpreting the resulting classification behavior.

Attack generation and robustness evaluation are treated as separate phases. Generation metadata may record whether an attack changed the prediction of the source model, but such an outcome does not by itself constitute a final robustness result. The generated artifact must subsequently be evaluated through the common metric and comparison framework.

Model-dependent artifacts are assessed through the direct source-model, cross-model transfer, and black-box transfer conditions defined in Section 4.7. The black-box condition measures only observable transfer behavior and does not imply access to the internal properties of the selected software systems.

Model-agnostic adversarial-style artifacts are evaluated across the proxy models and selected black-box software systems under the same manipulated condition. Because they are generated independently of a designated source model, differences in system behavior are interpreted as response variation under a shared condition rather than as adversarial transferability.

The specific attack algorithms, model configurations, parameter values, output representations, and traceability artifacts adopted in this thesis are documented in Chapter 5.

4.10 Anti-Forensic Evaluation Methodology

The anti-forensic component of the FAIRLab protocol evaluates the stability of automated image classification under controlled and operationally plausible modifications of the image artifact. Unlike the model-dependent attacks described in Section 4.9, these transformations are not optimized against the gradients, loss function, observable scores, or decision boundary of a specific classifier. They are applied through predefined image-processing rules and can therefore be evaluated consistently across proxy models and selected black-box software systems.

The term *anti-forensic* is used here in an experimental and operational sense and does not necessarily imply malicious intent or deliberate evidence concealment. It identifies transformations that may alter or destabilize automated analysis by modifying the visual, statistical, spatial, or encoding-related properties of an image. Such modifications may be introduced deliberately or arise from ordinary processes such as repeated saving, format conversion, platform-mediated recompression, resizing, forwarding, filtering, enhancement, or normalization.

The methodology considers transformation families that affect complementary properties of the image artifact: lossy re-encoding, spatial resampling and resizing, blurring and local-detail degradation, tonal-distribution modification, and contrast adjustment. The specific algorithms and parameter values adopted in this thesis are documented in Chapter 5. Table 4.4 summarizes the methodological role of each family.

Table 4.4: Methodological roles of the anti-forensic transformation families

Transformation family	Methodological role
Lossy re-encoding	Evaluates classification stability when the image representation is altered by compression artifacts or repeated encoding.
Spatial resampling and resizing	Examines sensitivity to resolution changes, interpolation, and repeated downsampling or restoration operations.
Blurring and local-detail degradation	Assesses the dependence of automated classification on edges, textures, and other high-frequency visual information.
Tonal-distribution modification	Evaluates sensitivity to changes in the statistical distribution of image intensities.
Contrast adjustment	Examines whether normalization or enhancement of the intensity range alters the classification outcome.

The transformations are applied exclusively to the binary in-domain subset. The clean OOD branch remains outside this process because it addresses distributional mismatch rather than robustness to image manipulation, as established in Section 4.6.

Within each experimental condition, transformation rules and parameter values must remain fixed and be applied consistently across the selected samples, unless parameter-sensitivity analysis is explicitly included in the research design. This requirement supports comparison across models, folds, operational classes, and software systems.

The selected settings should produce controlled and interpretable modifications while preserving sufficient task-relevant semantic content for meaningful evaluation. The objective is not to make the images unrecognizable, but to determine whether operationally plausible changes can alter automated classification behavior. The transformation set need not provide an exhaustive catalog of anti-forensic techniques, but should offer complementary coverage of encoding, spatial, filtering, tonal, and contrast effects while remaining reproducible, computationally feasible, scalable, and compatible with the evaluated systems.

Each generated artifact must remain associated with its clean source image, reference operational class, experimental fold, transformation family, applied configuration, and resulting output. The corresponding records must also preserve the information required to reconstruct the transformation and verify artifact identity, integrity, and provenance.

Artifact generation and robustness evaluation are treated as separate phases. Successful generation and completion of technical integrity checks establish that the intended artifacts were produced correctly, but do not demonstrate robustness degradation. Their impact must be determined through subsequent evaluation using the common metric and comparison framework.

The generated artifacts are evaluated on both the proxy classification models and the selected black-box software systems. This common evaluation supports a structured comparison between controlled model behavior and the observable operational behavior of the selected software systems under the same anti-forensic conditions.

4.11 Forensic Evaluation Bundle and Blind-Input Design

The operational evaluation of black-box software systems requires experimental inputs to be transferred without exposing descriptors that reveal their reference class, provenance, perturbation condition, or expected operational assignment. For this purpose, the FAIRLab protocol introduces a *forensic evaluation bundle*: a controlled and traceable collection of clean binary samples, clean OOD inputs, adversarial artifacts, and anti-forensic artifacts prepared for black-box evaluation.

Bundle construction does not alter the clean samples, generated artifacts, or reference operational assignments. Its methodological purpose is to separate the files submitted to the evaluated software systems from the information required for subsequent verification, normalization, and analysis.

The bundle therefore provides two complementary views of the same experimental corpus:

1. a *blind input view*, containing only the files submitted to the evaluated software systems;
2. a *control and audit view*, preserving the association between each submitted file and its source sample, reference operational assignment, experimental condition, provenance, and integrity information.

Files in the blind input view must use neutral identifiers and must not expose the reference operational class, fold, attack family, transformation type, proxy model, or other experimental descriptors through filenames, directory names, or paths. The corresponding mapping remains available only within the separate control records.

This design reduces the risk that either the software system or the analyst infers the experimental role of a sample from its organization rather than from the submitted image. The blindness requirement concerns descriptor-level information controlled by the experimental pipeline; it does not conceal the visual content of the image or the observable effects of an applied perturbation, since these constitute the actual inputs under evaluation.

The control and audit view must preserve a stable relationship among the neutral bundle identifier, source-image identifier, reference operational assignment, experimental family, perturbation condition, fold or evaluation partition, provenance information, integrity information, and the identifier used in the blind input view.

Table 4.5 summarizes the principal components of the forensic evaluation bundle.

Table 4.5: Methodological components of the forensic evaluation bundle

Component	Methodological function
Blind input view	Contains only neutrally identified files submitted to the evaluated software systems. It must not expose reference classes, perturbation families, folds, model names, or other experimental descriptors through filenames, paths, or directory organization.
Control metadata	Preserves the mapping between each blind identifier and the corresponding source sample, reference operational assignment, experimental condition, provenance, and integrity information.
Structured audit view	Provides an internal organization of experimental inputs and derived artifacts for technical verification and inspection. It is not submitted to the evaluated software systems.
Integrity and consistency records	Document file identity, uniqueness, completeness, and correspondence with the source records produced during the preceding experimental phases.
Embedded-information audit	Examines whether image files contain metadata or other machine-readable fields that could reveal provenance, processing history, or experimental information not concealed through neutral filenames and paths.

Only the blind input view is submitted to the evaluated software systems. Control metadata, structured audit records, and reference assignments remain outside the import and processing environment and are used only after the observable outputs have been exported.

Blindness primarily concerns filenames, paths, directory organization, and accompanying control records. Image files may nevertheless contain embedded information concerning acquisition devices, processing software, provenance, or content descriptors [97, 98, 99]. Bundle

construction should therefore include an audit of metadata and other machine-readable fields that could reveal experimental information.

The audit should identify potentially informative fields without modifying the evaluated files unless metadata removal or normalization is explicitly included in the research design. Residual embedded metadata must be documented as a limitation of the blind-input design rather than interpreted as evidence of classification performance or operational robustness.

Completeness, uniqueness, and correspondence checks must verify that all required clean and perturbed inputs are present, that no unintended duplicate has been introduced, and that every blind identifier is correctly associated with its control record. These checks establish the technical validity of the evaluation bundle but do not constitute a robustness result.

The forensic evaluation bundle thus provides the controlled interface between the reproducible experimental pipeline and the black-box software environments. Its blind input view is processed according to the protocol described in Section 4.12, while the separate control and audit view supports subsequent output matching, normalization, traceability, and metric computation.

4.12 Black-Box Software Evaluation Protocol

The operational component of the FAIRLab methodology evaluates software systems that provide automated or AI-assisted functionality for image classification, semantic retrieval, multimedia triage, or forensic review. Unlike the proxy classification models described in Section 4.7, these systems are evaluated under black-box or near-black-box access conditions. No access is assumed to their internal architectures, training data, model weights, decision thresholds, calibration procedures, or proprietary inference logic. The evaluation is based exclusively on submitted inputs and observable or exportable outputs.

The purpose of this phase is not to validate or certify the selected software systems or to rank heterogeneous tools in absolute terms. It examines their observable operational behavior under controlled and reproducible conditions using the same clean binary samples, clean OOD inputs, adversarial artifacts, and anti-forensic artifacts considered in the controlled experimental pipeline.

Processing begins with the blind input view defined in Section 4.11. Only neutrally identified image files are imported or indexed, while reference assignments, fold allocations, perturbation descriptors, and control records remain outside the software environment until output matching and normalization.

The evaluated systems may expose heterogeneous outputs, including assigned categories, semantic tags, bookmarks, textual or visual retrieval results, system-specific scores or rankings, detected-item lists, exported reports, and missing, ambiguous, unsupported, or non-interpretable states. These outputs do not necessarily correspond directly to the operational classes defined by the experimental protocol and must therefore be normalized before quantitative comparison.

Normalization comprises three stages:

1. matching each exported result to the corresponding blind bundle input through a stable identifier;
2. interpreting the observable output through a documented software-specific mapping rule;
3. recoding the interpreted result into common operational prediction states suitable for metric computation.

The matching procedure must preserve the relationship between each exported result and its control record without exposing the reference assignment during software processing. The software-specific mapping rule must be established and documented before final quantitative

analysis and must identify which categories, tags, bookmarks, retrieval outcomes, or other observable outputs constitute evidence that the target class was flagged.

The absence of a target-related output indicates only that the software did not produce the relevant observable result under the tested configuration. It must not be interpreted as independent semantic evidence that the target object is absent. Likewise, outputs that cannot be mapped reliably must not be forced into positive or negative predictions.

The protocol therefore preserves an explicit **unknown** state for missing, ambiguous, unsupported, unmapped, or non-interpretable outputs. Unknown outcomes and explicit negative predictions represent different observable states and must remain separately reportable. Under a documented risk-oriented analysis, an unknown outcome associated with a **weapon** sample may contribute to an operational missed-detection count, but its original unknown status must remain preserved. Conversely, broadly related semantic outputs must not be automatically mapped to positive detections, since doing so may inflate false-positive counts.

Table 4.6 summarizes the general normalization principles applied to heterogeneous black-box outputs.

Table 4.6: General normalization principles for black-box software outputs

Observable output condition	Operational normalization principle
Target-related category, tag, or bookmark	Recoded as a positive detection only when the exported output is explicitly associated with the target category according to the documented mapping rule.
Semantic retrieval result	Recoded according to a predefined retrieval rule that specifies the accepted queries, configurations, and aggregation procedure.
No target-related output	Recorded as no target detection under the tested configuration. This state does not constitute independent semantic evidence that the target object is absent.
Ambiguous or broader semantic output	Preserved as unknown unless the mapping rule establishes a sufficiently precise relationship with the target class.
Missing, unsupported, unmapped, or non-interpretable output	Preserved as unknown in the normalized records. For weapon samples, this state may contribute to an operational missed-detection count under a documented risk-oriented rule while remaining separately reportable.

Software configuration is implementation-specific. For each evaluated system, the category, query, module, plugin, or processing function that most closely corresponds to the experimental target must be selected and documented. When multiple reasonable configurations are available, sensitivity analysis may be used to examine their effects on detection coverage and false-positive behavior. The concrete software systems, versions, mappings, prompts, thresholds, and configurations adopted in this thesis are reported in Chapter 5.

For each evaluated system, the protocol follows the same general workflow. Only the blind evaluation inputs are imported or indexed, after which the relevant automated analysis or retrieval functionality is enabled. The complete blind input view is then processed under a documented configuration, and all observable outputs required for matching and interpretation are exported. These outputs are subsequently aligned with the experimental control metadata and normalized into common operational prediction states. Comparable metrics are then computed while preserving any outcomes classified as **unknown**.

Raw exports, normalization records, and aggregated metrics must remain separate. Raw exports preserve the software outputs as originally produced; normalization records document input matching, interpretation, and recoding; and metric records contain only the quantitative results derived after normalization. This separation preserves the traceability chain from the submitted bundle input to the observable output, mapping rule, normalized prediction state, and final metric, while allowing ambiguous mappings or normalization errors to be reviewed without repeating software execution.

Categories, tags, bookmarks, classifications, and retrieval results produced by black-box systems must not be interpreted as local explanations of their internal decision processes. They are treated solely as observable outputs, and the methodology does not infer which internal features, models, or proprietary mechanisms produced them.

The evaluation is therefore anchored to externally verifiable behavior. It measures whether the selected systems produce operationally relevant outputs for the submitted inputs without attributing internal properties that cannot be independently verified [2, 3].

Figure 4.6 summarizes the resulting black-box evaluation workflow.

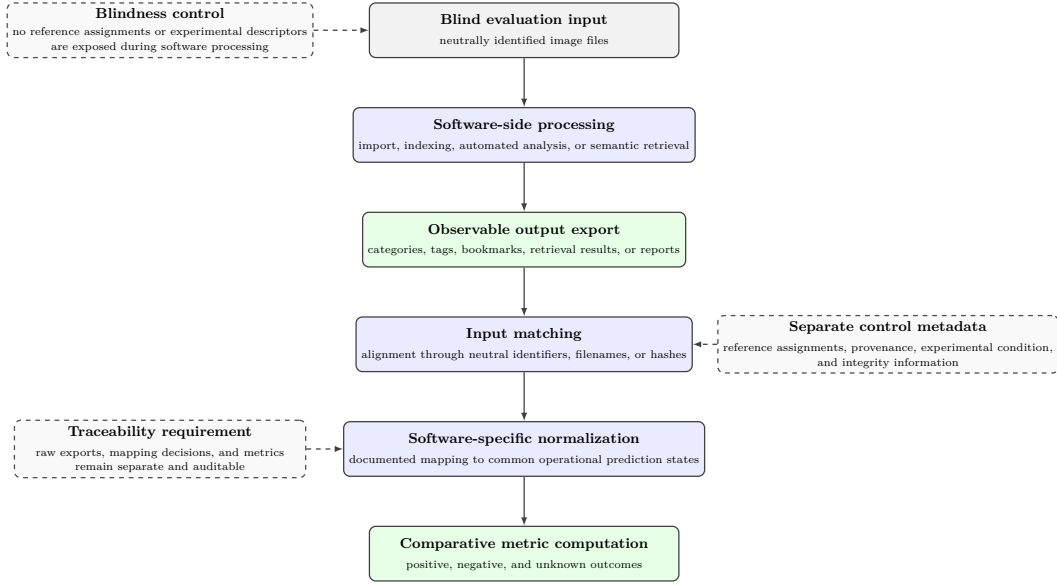


Figure 4.6: Black-box software evaluation workflow. Neutrally identified inputs are processed without access to reference operational assignments or experimental descriptors. Observable outputs are subsequently matched to the control metadata, normalized through documented software-specific rules, and converted into common operational prediction states

4.13 Evaluation Metrics

The FAIRLab evaluation metrics cover three complementary dimensions of operational robustness: binary classification performance under clean conditions, robustness under adversarial and anti-forensic conditions, and system behavior on clean OOD inputs. It also supports the evaluation of black-box software systems after their heterogeneous outputs have been normalized into common operational prediction states.

Aggregate accuracy alone is insufficient for operational interpretation. Binary metrics are therefore computed separately for clean, adversarial, and anti-forensic conditions and interpreted according to class and error direction. The OOD branch is evaluated through separate behavioral indicators and is not incorporated into the binary confusion matrix.

The protocol does not adopt Area Under the Receiver Operating Characteristic Curve (AUROC) as a central cross-system metric. Although score-based measures are applicable to proxy models, the selected black-box systems may expose only categories, semantic tags, bookmarks, retrieval results, or system-specific scores whose meaning and calibration cannot be independently verified. The protocol therefore prioritizes measures that can be derived consistently from both inspectable models and normalized black-box outputs, without implying a theoretical limitation of AUROC or other score-based metrics.

For the binary task, the principal metrics are accuracy, balanced accuracy, precision, recall, F1-score, false-negative rate, false-positive rate, and the confusion matrix. Let TP , TN , FP ,

and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. Accuracy is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.1)$$

Recall for the positive **weapon** class is defined as:

$$\text{Recall}_{\text{weapon}} = \frac{TP}{TP + FN} \quad (4.2)$$

The corresponding false-negative rate is:

$$\text{FNR}_{\text{weapon}} = \frac{FN}{TP + FN} \quad (4.3)$$

The false-positive rate is:

$$\text{FPR} = \frac{FP}{FP + TN} \quad (4.4)$$

Balanced accuracy accounts for performance on both operational classes:

$$\text{Balanced Accuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (4.5)$$

Precision and F1-score are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.6)$$

$$\text{F1} = 2 \frac{\text{Precision} \cdot \text{Recall}_{\text{weapon}}}{\text{Precision} + \text{Recall}_{\text{weapon}}} \quad (4.7)$$

Robustness under manipulated conditions is evaluated through a paired comparison between each perturbed artifact, its corresponding clean sample, and the reference operational class. This design distinguishes errors already present under clean conditions from those induced by a perturbation.

For a perturbation condition p , the accuracy drop is defined as:

$$\Delta \text{Accuracy}_p = \text{Accuracy}_{\text{clean}} - \text{Accuracy}_p \quad (4.8)$$

Positive values indicate degradation relative to the clean baseline. Zero or negative values indicate no measured degradation under the evaluated condition and must not be interpreted as evidence of general robustness.

Let $\mathcal{C}_{\text{clean}}$ denote the set of samples classified correctly under the clean condition. The induced error rate for condition p is defined as:

$$\text{InducedErrorRate}_p = \frac{\left| \left\{ i \in \mathcal{C}_{\text{clean}} : \hat{y}_i^{(p)} \neq y_i \right\} \right|}{|\mathcal{C}_{\text{clean}}|} \quad (4.9)$$

The numerator represents the induced error count. For model-dependent adversarial conditions, this rate may also be reported as the attack success rate because it measures the proportion

of clean-correct samples rendered incorrect by the attack. Samples already misclassified under clean conditions do not contribute to this quantity.

Particular attention is given to the transition:

$$\text{weapon} \rightarrow \text{non_weapon}$$

because it represents the loss of a positive detection for target-category content. The reverse transition:

$$\text{non_weapon} \rightarrow \text{weapon}$$

is also measured because it may increase the volume of non-target content prioritized for human review.

When continuous proxy-model outputs are available, the analysis also considers the mean maximum predicted-class probability and its variation relative to the corresponding clean condition. These quantities are diagnostic intra-model indicators and must not be interpreted as calibrated probabilities or compared directly across proxy architectures.

For each proxy model and manipulated condition, the maximum-probability shift is defined as:

$$\Delta c = c_{\text{perturbed}} - c_{\text{clean}} \quad (4.10)$$

where c denotes the mean maximum predicted-class probability under the indicated condition. Because the predicted class may change, this quantity does not necessarily compare the probabilities assigned to the same semantic class.

The OOD evaluation remains conceptually separate from the binary task. Since OOD samples are not treated as a third supervised class, binary or multiclass accuracy is not computed for this branch. Instead, the protocol characterizes how frequently these inputs are assigned to the known operational classes through:

- the proportion of OOD inputs assigned to **weapon**;
- the proportion of OOD inputs assigned to **non_weapon**;
- the mean maximum predicted-class probability for proxy models;
- the proportion of proxy-model predictions whose maximum probability meets or exceeds a predefined high-Max-P threshold;
- the proportion of unknown outputs exposed after normalization of black-box software outputs.

The OOD weapon flag rate is defined as:

$$\text{WeaponFlagRate}_{\text{OOD}} = \frac{N_{\text{OOD} \rightarrow \text{weapon}}}{N_{\text{OOD}}} \quad (4.11)$$

This measure describes the operational tendency of a system to map distributionally different inputs to the target class. It is not a conventional false-positive rate because OOD inputs do not belong to the supervised negative class, nor does it establish that an individual assignment is semantically correct, incorrect, relevant, or irrelevant.

Maximum-probability-based OOD indicators are available only for proxy models. They remain diagnostic intra-model quantities because the selected black-box systems do not expose homogeneous numerical outputs suitable for an equivalent cross-system probability analysis.

Black-box evaluation is based on the normalized states defined in Section 4.12: positive detection, no target detection, and **unknown**. Only matched records with interpretable positive

or negative decisions contribute to the conventional binary confusion matrix; unknown outputs must not be forced into either class.

For a **weapon** sample, an interpretable no-target decision constitutes a false negative. An unknown output may also contribute to a separately reported operational miss count under a documented risk-oriented rule, but it must not be recoded as a conventional false negative. For a **non_weapon** sample, an interpretable target-related output constitutes a false positive, whereas an unknown output is not automatically treated as one because it does not represent an explicit target detection.

For the selected black-box software systems, the protocol evaluates the matching rate between exported outputs and blind-bundle inputs and computes the true-positive, false-negative, true-negative, and false-positive counts on interpretable records. These counts support the calculation of weapon recall, the weapon false-negative rate, the false-positive rate, and, where permitted by the normalized outputs, accuracy and balanced accuracy. The evaluation also records the number and rate of **unknown** outcomes, together with any system-specific not-detected or uncategorized rates exposed by the software and any supplementary operational miss counts that are explicitly defined.

The matching rate is a technical coverage measure rather than a classification metric. It represents the proportion of submitted bundle inputs for which a corresponding software output can be identified:

$$\text{MatchingRate} = \frac{N_{\text{matched}}}{N_{\text{submitted}}} \quad (4.12)$$

The unknown output rate is defined as:

$$\text{UnknownRate} = \frac{N_{\text{unknown}}}{N_{\text{matched}}} \quad (4.13)$$

Unmatched inputs are represented through the matching rate and are not included again in the numerator of the unknown rate. These coverage measures are reported because incomplete matching and non-interpretable outputs may affect apparent black-box performance, but they must not be conflated with ordinary classification errors.

Table 4.7 summarizes the metric categories adopted in the protocol.

Table 4.7: Metric categories adopted in the evaluation protocol

Category	Metrics and operational meaning
Binary classification	Accuracy, balanced accuracy, precision, recall, F1-score, and confusion matrix for the weapon and non_weapon classes.
Robustness under manipulation	Clean and perturbed accuracy, accuracy drop, induced error count and rate, prediction transitions, and intra-model maximum-probability variation when available.
Critical operational errors	False negatives on weapon samples and false positives on non_weapon samples, interpreted according to their different effects on forensic prioritization and review workload.
OOD behavior	Weapon flag rate, in-domain assignment distribution, and proxy-model maximum-probability indicators. The branch remains excluded from binary classification metrics.
Black-box software coverage	Matching rate, unknown output count and rate, and system-specific missing, not-detected, or uncategorized states.
Black-box classification behavior	True positives, false negatives, false positives, true negatives, weapon recall, weapon false-negative rate, false-positive rate, accuracy, and balanced accuracy computed on interpretable normalized outputs where applicable.
Supplementary operational-risk analysis	Unknown outcomes on weapon samples may be included in a separately reported operational miss count without being recoded as conventional false negatives.

Figure 4.7 summarizes the metric structure across the proxy-model and black-box evaluation levels.

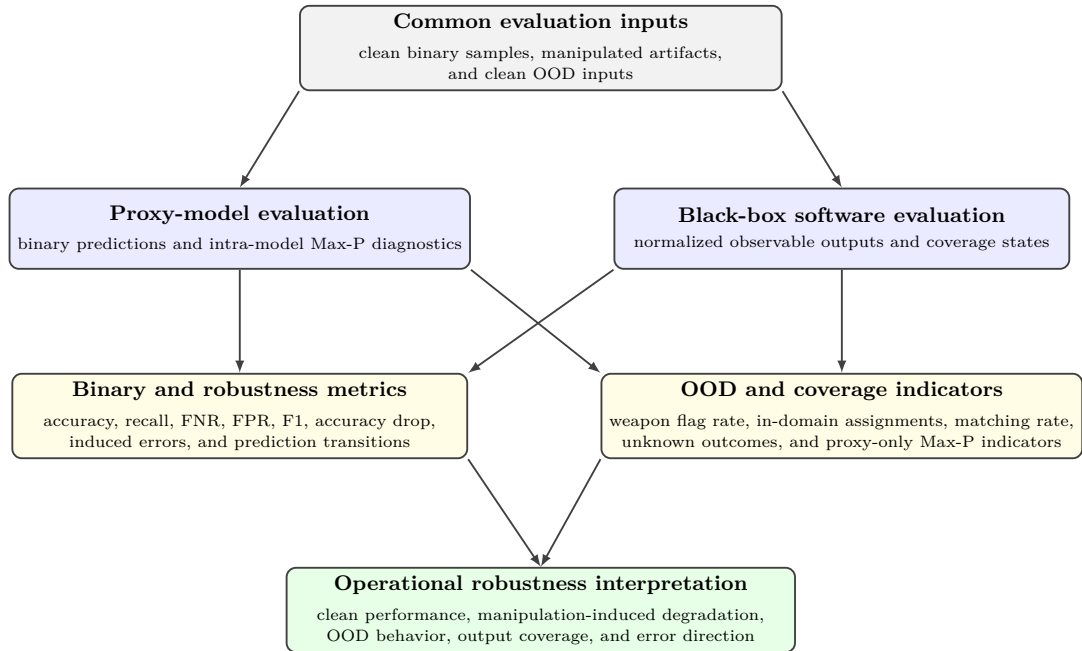


Figure 4.7: Metric structure. Condition-specific binary, robustness, OOD, and coverage indicators are computed across the proxy-model and black-box evaluation levels and integrated only during the operational interpretation

These metrics do not replace human review or establish reliability across all forensic scenarios. Their purpose is to identify conditions in which automated support may fail to surface

target-category content, increase review workload, or produce unstable outputs on manipulated or distributionally different inputs.

4.14 Explainability Approach

The xAI component of the FAIRLab methodology serves an exclusively diagnostic and qualitative function. It complements the quantitative robustness metrics by supporting the examination of selected cases involving an inspectable proxy classification model. Explainability outputs are not treated as stand-alone evidence, definitive explanations of algorithmic decisions, or demonstrations of model reliability. They are used to characterize attribution patterns and formulate hypotheses regarding changes in model behavior across clean, adversarial, anti-forensic, and OOD conditions.

The explainability analysis is restricted to one selected fully inspectable proxy model. The adopted technique must be compatible with its architecture, weights, preprocessing procedure, and inference pipeline and may require access to gradients, intermediate activations, or other internal components. The same internal attribution analysis cannot be applied to the selected black-box or near-black-box software systems because their architectures, representations, decision thresholds, and scoring procedures are not accessible. Their observable outputs must not be interpreted as explanations of the internal processes that produced them.

The methodology adopts a case-study approach rather than an aggregate explainability evaluation. Representative samples are selected according to predefined diagnostic criteria to examine behaviors that cannot be fully characterized through corpus-level metrics. The selected cases are not used to estimate general model performance.

High-Max-P criteria are applied only within the selected proxy model and serve exclusively as case-selection indicators. They must not be interpreted as calibrated probabilities or compared directly with numerical outputs from other architectures or software systems.

Table 4.8 summarizes the five case categories and their diagnostic functions.

Table 4.8: Case-selection criteria for explainability analysis

Case category	Diagnostic function
Correctly classified clean reference	Provides a qualitative reference for examining whether the attribution pattern is associated with visually relevant regions under an unmanipulated condition.
Clean classification error	Examines an error already present under the clean condition and whether the attribution pattern appears localized, diffuse, partial, or associated with contextual image regions.
High-Max-P OOD assignment	Examines the attribution pattern associated with an in-domain assignment produced for a distributionally different input. The assignment is treated as an operational behavior rather than as an ordinary supervised error.
Anti-forensic induced error	Examines whether an operationally plausible image transformation is associated with a change in the attribution pattern and the resulting predicted class.
Adversarially induced error	Examines the attribution pattern associated with an incorrect prediction produced after a model-dependent adversarial perturbation, including cases in which the model retains a high maximum predicted-class probability.

Case selection and the resulting explainability artifacts must remain traceable to the corresponding source sample, prediction record, evaluation condition, model configuration, and

checkpoint. Each record must also preserve the predicted class, reference operational class, and rationale for case selection, consistently with the controls defined in Section 4.15.

The methodology does not prescribe a single explainability technique for all implementations. Gradient-based attribution, activation-based visualization, local surrogate methods, or other feature-attribution approaches may be used when their assumptions and limitations are compatible with the selected model and research questions. The specific technique and configuration adopted in this thesis are documented in Chapter 5.

The adopted procedure should be compatible with the selected inspectable proxy model and should explicitly document the attribution target and reference baseline whenever required by the chosen method. The attribution process must be reproducible, applied consistently across the selected clean and manipulated cases, and traceable to the corresponding model checkpoint, input sample, and prediction record. The resulting outputs must also be presented as diagnostic attribution evidence rather than as definitive causal explanations.

Attribution maps and related visualizations characterize relationships between input features and a selected model output under the assumptions of the adopted method. They do not establish that highlighted regions reproduce the model’s internal reasoning or demonstrate that a prediction is correct, reliable, or forensically valid.

When the selected method provides completeness, approximation, or convergence diagnostics, these quantities must be recorded and considered during interpretation. They act as validation aids and do not automatically establish that an attribution is complete or reliable. A strict convergence claim may be made only when supported by the recorded diagnostic values and an explicitly defined acceptance criterion.

Explainability remains distinct from the quantitative robustness assessment. Classification metrics characterize the frequency, magnitude, and direction of errors across the experimental corpus, whereas xAI supports the qualitative examination of selected decisions and failure cases. Within the FAIRLab protocol, it therefore acts as a supporting diagnostic layer while preserving the distinction among technical attribution, experimental robustness evaluation, and evidentiary assessment by a human forensic analyst.

4.15 Traceability and Reproducibility Controls

Traceability is a cross-cutting requirement of the entire FAIRLab protocol. It applies to technical preparation, human-in-the-loop review, dataset freezing, partition generation, proxy-model training, perturbation generation, bundle construction, black-box software evaluation, metric computation, and xAI analysis.

Consistent with established DF principles, every processing step that creates or modifies an experimental file must be documented so that the relationships among the original sample, derived artifacts, model or software outputs, and final experimental results can be reconstructed [5, 8, 9]. Each sample must remain identifiable throughout its lifecycle even when its filename, encoding, directory location, or visual representation changes.

The traceability structure distinguishes four complementary elements. The first is *sample identity*, which establishes a stable reference to the original image. The second is *artifact lineage*, which records the relationship between an original sample and any derived adversarial, anti-forensic, blind-bundle, or explainability artifact.

The third element is *experimental context*, which associates each input or derived artifact with its reference operational assignment, partition, model configuration, perturbation condition, and evaluation branch. The fourth is *output provenance*, which connects model predictions, software exports, normalization decisions, metrics, and qualitative case studies to the corresponding evaluation input.

Integrity verification should rely on cryptographic identifiers or other mechanisms capable of detecting unintended changes to experimental files. The selected mechanism must be

documented and applied consistently. Additional identifiers may be retained for interoperability with black-box software systems, but they must not replace the primary integrity mechanism unless they provide appropriate security and collision-resistance properties.

Traceability records should be progressive: each stage preserves the identifiers inherited from the preceding stage and adds only the information required to describe the newly created artifact, configuration, or experimental condition. This approach allows the lifecycle of a sample to be reconstructed without relying exclusively on filenames or directory structures.

Table 4.9 summarizes the principal categories of traceability records required by the methodology.

Table 4.9: Principal traceability records across the FAIRLab protocol

Record category	Role in traceability
Technical preparation records	Document sample identity, provenance, technical validity, and integrity information for the candidate image pool.
Human-review records	Associate each candidate sample with its review status, operational assignment, exclusion disposition, and supporting notes.
Frozen-dataset records	Define the stable experimental corpus and preserve the reference operational assignments used in subsequent phases.
Partition records	Associate each sample with its binary or OOD evaluation branch and, where applicable, its experimental fold.
Model records	Document model architecture, training configuration, class mapping, checkpoint identity, and the relationship between each checkpoint and its training partitions.
Derived-artifact records	Link adversarial and anti-forensic artifacts to the corresponding clean sample, generation condition, parameters, and model configuration when applicable.
Evaluation-bundle records	Map neutral blind identifiers to the corresponding source sample, reference operational assignment, experimental condition, and integrity information.
Prediction and normalization records	Connect proxy-model predictions and observable software outputs to the evaluated input and document any matching, recoding, or mapping decision.
Metric and explainability records	Associate quantitative results and qualitative case studies with the underlying samples, model checkpoints, software configurations, and experimental conditions. Explainability records refer only to the selected inspectable proxy model.

The minimum traceability chain must support navigation in both directions: from an original sample to its evaluation inputs, derived artifacts, and experimental outputs, and from a reported result back to the exact input, configuration, and processing conditions that produced it.

Figure 4.8 summarizes this bidirectional traceability chain.

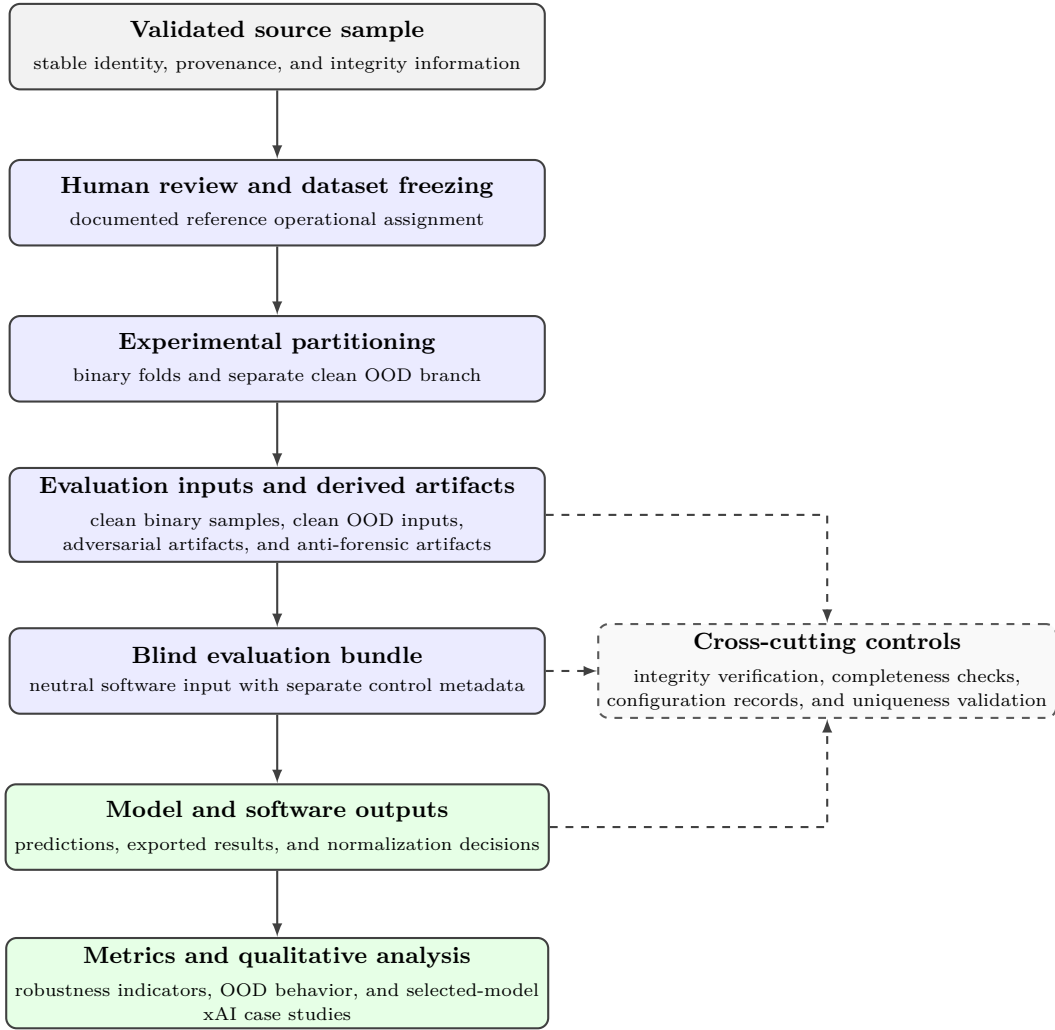


Figure 4.8: Traceability chain across the FAIRLab protocol. Stable identifiers, provenance information, integrity controls, and structured records preserve the relationships among source samples, evaluation inputs, derived artifacts, observable outputs, and final analyses

Reproducibility requires more than source-code availability. The experimental configuration must be documented in sufficient detail to recreate the relevant processing conditions, including the dataset version, partitioning strategy, randomization controls, model configuration, preprocessing rules, perturbation parameters, software version and enabled functionality, normalization rules, and metric definitions. When randomness affects partitioning, model initialization, optimization, perturbation generation, or case selection, the corresponding random state or deterministic procedure must be recorded.

Model and artifact records must preserve the associations required by the fold-aware and perturbation-generation procedures. In particular, each checkpoint must remain linked to its training partitions, and a checkpoint used to evaluate or perturb a held-out sample must not have been trained on that sample. Model-dependent adversarial artifacts must identify their source model and checkpoint, whereas model-agnostic adversarial-style and anti-forensic artifacts must preserve the shared generation condition and applied configuration. These records support the distinction among direct source-model evaluation, transfer evaluation, and comparison under shared model-agnostic conditions.

Black-box evaluation introduces an additional separation requirement. Reference assignments, perturbation descriptors, folds, model identifiers, and other controlled experimental metadata

must remain outside the software input during processing while remaining available after export for matching, normalization, and metric computation. The evaluation records must make this separation verifiable.

Automated consistency controls should verify, where applicable, the correspondence between expected and generated artifacts, the uniqueness of stable and derived identifiers, and the integrity of the generated files. They should also detect missing or unresolved paths and confirm consistency among source records, derived artifacts, and evaluation outputs. In addition, the controls should assess the completeness of software exports and normalized predictions while preserving a clear separation between blind inputs and experimental metadata.

These controls establish the technical coherence of the experimental workflow; they do not demonstrate classification correctness or operational robustness. Their purpose is to ensure that reported results derive from the intended inputs, artifacts, and configurations rather than from undocumented substitutions, incomplete runs, or incorrect associations.

For black-box systems, reproducibility concerns the documented external configuration, submitted inputs, observable outputs, and normalization procedures rather than inaccessible internal models, thresholds, or inference logic. The protocol therefore combines *procedural reproducibility*, which documents how the experiment can be repeated, with *artifact-level traceability*, which identifies the exact inputs, models, software configurations, and derived files contributing to each reported result.

The concrete implementation of these controls is documented in Chapter 5, whereas the corresponding experimental outcomes are reported in Chapter 6.

Chapter 5

Implementation and Experimental Setup

This chapter documents the concrete implementation of the FAIRLab methodology introduced in Chapter 4. It describes the preparation and freezing of the image corpus, deterministic data partitioning, fold-aware proxy-model training, adversarial and anti-forensic artifact generation, construction of the forensic evaluation bundle, normalization of black-box software outputs, explainability analysis, and the controls used to preserve artifact identity, provenance, integrity, and reproducibility.

The chapter reports how the protocol was instantiated and which technical artifacts were produced. Quantitative outcomes and their operational interpretation are reported separately in Chapter 6, preventing implementation choices, software configurations, and data-management procedures from being conflated with measured robustness.

Both evaluation levels operate on the same frozen source population. The 1 000 clean binary images and 500 clean OOD images constitute the source corpus, while 10 000 adversarial and anti-forensic artifacts are derived exclusively from the binary subset. The proxy models process manifest-linked inputs and derived artifacts, whereas the selected black-box software systems receive neutrally identified, byte-identical copies through the forensic evaluation bundle. Equivalence between these views is verified through SHA-256 digests.

5.1 Experimental Workflow and Implementation Environment

The experimental workflow was implemented as a staged pipeline in which each phase produces explicit records or artifacts used by subsequent phases. The core numbered execution sequence comprises scripts 00–21, covering controlled data restoration, acquisition, dataset preparation, human review, partitioning, model training, perturbation generation, evaluation, bundle construction, explainability, black-box normalization, and reporting. Scripts 22–24 provide public-artifact support, frozen-result validation, and reporting-asset auditing without modifying the canonical experimental inputs, derived artifacts, or result records.

Table 5.1 summarizes the main implementation phases, their entry points, and their technical outputs.

Table 5.1: Main implementation phases and corresponding pipeline entry points

Implementation phase	Primary entry points	Main outputs
Controlled restoration and source acquisition	<code>datasets/scripts/acquisition/00_download_raw_datasets_bundle.py</code> and scripts 01–07	Restored or acquired source corpora, acquisition metadata, and locally preserved source structures.
Technical preparation and human review	<code>datasets/scripts/prepared/08_build_prepared_dataset.py</code> , <code>datasets/scripts/prepared/09_generate_review_manifest_full.py</code> , and <code>datasets/scripts/final/10_manual_selection_protocol_reviewer.py</code>	Validated candidate pool, duplicate and invalid-file reports, review manifests, manual decisions, and the frozen-dataset definition.
Dataset partitioning	<code>datasets/scripts/splits/11_generate_clean_and_ood_splits.py</code>	Deterministic clean folds, the separate OOD evaluation manifest, and the split-generation summary.
Proxy-model training	<code>models/scripts/12_train_proxy_models.py</code>	Fold-aware checkpoints, training logs, training summaries, and the model registry.
Perturbation generation	<code>datasets/scripts/attacks/13_generate_anti_forensic_attacks.py</code> and <code>datasets/scripts/attacks/14_generate_adversarial_attacks.py</code>	Anti-forensic and adversarial artifacts, generation manifests, perturbation parameters, source-image references, and integrity information.
Proxy-model evaluation and bundle construction	<code>evaluation/scripts/15_evaluate_proxy_models.py</code> and <code>datasets/scripts/bundle/16_build_forensic_evaluation_bundle.py</code>	Proxy-model predictions, evaluation metrics, and the blind and auditable 11 500-file forensic evaluation bundle.
Explainability and black-box normalization	<code>explainability/scripts/17_generate_integrated_gradients_case_studies.py</code> , <code>explainability/scripts/18_xai_interactive_launcher.py</code> , and <code>evaluation/scripts/19_normalize_forensic_ai_tool_predictions.py</code>	Attribution maps, human-reviewed case selections, normalized black-box predictions, matching reports, and software-version records.
Experimental reporting and sensitivity analysis	<code>results/scripts/20_generate_experimental_reporting_assets.py</code> and <code>results/scripts/21_generate_embedded_metadata_sensitivity_check.py</code>	Thesis-ready tables and figures, consolidated reporting assets, and embedded-metadata sensitivity outputs.
Validation and public-artifact audit	Scripts 22–24 under <code>results/scripts/</code>	Optional privacy-reduced sensitivity outputs, frozen-result validation reports, and reporting-asset usage audits.

At the repository level, the implementation is divided into functional components. Dataset preparation, split generation, and bundle construction are maintained under `datasets/`; perturbation manifests and reports under `attacks/`; proxy architectures, checkpoints, and training records under `models/`; proxy predictions and normalized black-box outputs under `evaluation/`; commercial-tool documentation and sanitized extracts under `forensic_tools/`; explainability scripts and artifacts under `explainability/`; and consolidated metrics, reporting assets, and validators under `results/`. The authoritative thesis source is maintained under `docs/LatexThesis/`.

The implementation was managed through Python virtual environments and executed across Windows PowerShell and Linux/Kali environments according to the requirements of each stage. Black-box processing additionally required licensed installations of the selected software systems. The repository-level `requirements.txt` file defines the Python dependency set, including the libraries used for data processing, image manipulation, proxy-model training and inference, adversarial generation, source acquisition, visualization, and explainability. PyTorch and Torchvision provide the deep-learning framework, `open_clip_torch` supports the CLIP-based proxy model, and Captum supports the IG workflow.

The code supports CPU and CUDA-enabled execution. Manifest inspection, output normalization, metric validation, and reporting audits can be performed without dedicated acceleration, whereas proxy-model training, iterative adversarial generation, and explainability regeneration constitute the most computationally demanding stages.

The repository is distributed as a research artifact rather than as a self-contained software package or a byte-identical operating-system image. Procedural reproducibility and artifact-level auditability are supported through numbered scripts, frozen manifests, fold-aware checkpoints, prediction tables, explicit parameters, integrity digests, validation reports, and controlled experimental artifacts. The public repository provides the material required for

structural inspection, metric recomputation, reporting validation, and audit, while the complete image corpora, exact frozen black-box input bundle, and complete proprietary software exports remain subject to controlled access.

5.2 Dataset Preparation and Organization

5.2.1 Technical Preparation and Source Consolidation

The initial candidate pool was assembled from five source collections maintained under `datasets/raw/`: a public Kaggle-based collection, the specialized DeepFirearm corpus, open-web images, content acquired from Telegram and YouTube, and material collected from non-indexed web sources. The collections remained organized according to acquisition provenance, while source names, directory structures, and inherited labels were retained only as provenance information and were not used as final operational assignments.

Technical preparation was performed by `datasets/scripts/prepared/08_build_prepare_d_dataset.py`. The script processes the five configured source directories in a fixed order, recursively enumerates supported image files, and constructs the unified candidate pool under `datasets/prepared/final_pool/`. It does not perform semantic annotation or final sample selection, which are handled by the human review procedure described in Section 5.2.2.

Candidate discovery was restricted to files with the extensions `.jpg`, `.jpeg`, `.png`, `.bmp`, `.webp`, `.tif`, and `.tiff`. Each file was opened and structurally verified using Pillow. Images that could not be decoded or verified were recorded as `unreadable_image`, whereas images whose width or height was below 300 pixels were excluded under the `below_min_size` condition. Unexpected failures were recorded with the corresponding exception type.

For every technically valid image, the script computed a SHA-256 digest over the original file bytes. Exact-duplicate detection was performed globally across all five source collections. Because the source directories and their contents were processed deterministically, the first occurrence of each digest was retained, while every subsequent occurrence was recorded as a discarded duplicate and linked to the retained sample.

Accepted images were assigned stable identifiers following the format `img_00000001` and copied to `datasets/prepared/final_pool/images/`. The original file extension was preserved, and no resizing, recompression, or format conversion was performed. After each copy operation, the destination file was hashed again and compared with the source digest. A mismatch caused execution to terminate, preventing modified, corrupted, or incomplete copies from entering the prepared pool.

Table 5.2 reports the statistics produced by the final execution of the preparation script.

Table 5.2: Technical preparation statistics by source collection

Source collection	Candidates	Valid unique	Invalid	Duplicates
01_kaggle_weapon	719	462	69	188
02_deepfirearm	15 755	12 158	2 565	1 032
03_google_scraped	903	846	52	5
04_telegram_youtube	973	898	19	56
05_deepweb	1 419	957	419	43
Total	19 769	15 321	3 124	1 324

Note. *Valid unique* indicates technically valid images remaining after global exact-duplicate removal. Duplicate counts are attributed to the source collection in which the discarded occurrence was encountered.

The resulting prepared pool contained 15 321 technically valid and byte-distinct images from 19 769 candidate files. Of the remaining files, 3 124 were excluded because they were unreadable, structurally invalid, or below the minimum dimensional threshold, and 1 324 were discarded as exact duplicates.

The preparation stage generated four principal audit records:

- `datasets/prepared/final_pool/metadata.csv`, containing one record for each retained image;
- `datasets/prepared/final_pool/reports/invalid_images.csv`, documenting technically excluded files and their exclusion reasons;
- `datasets/prepared/final_pool/reports/duplicates_discarded.csv`, linking each discarded duplicate to the retained sample and its SHA-256 digest;
- `datasets/prepared/final_pool/reports/prepared_build_summary.json`, containing global and source-specific preparation statistics.

For each retained image, `metadata.csv` records the stable image identifier, prepared filename and relative path, source group, source dataset, original relative path and filename, SHA-256 digest, dimensions, file size, extension, and technical-validity state. These fields preserve the relationship between the prepared sample and its acquisition source.

The output of this stage is a technically validated and globally deduplicated candidate pool, not the final experimental dataset. Reduction of the 15 321-image prepared pool to the frozen experimental corpus was performed only through the human review and final-selection procedure described in the following subsection.

5.2.2 Review Documentation and Human-in-the-Loop Validation

The transition from technical preparation to semantic assessment was implemented through the full review manifest generated by `datasets/scripts/prepared/09_generate_review_manifest_full.py` and stored at `datasets/prepared/manifests/review_manifest_full.csv`. The script creates one review record for each of the 15 321 images retained in the prepared pool.

Identity, provenance, and technical metadata are propagated from `datasets/prepared/final_pool/metadata.csv`, including the stable image identifier, SHA-256 digest, prepared path, source dataset, original source path, dimensions, file size, extension, and technical-validity state. The digest is not recomputed because it was already established and verified during technical preparation.

The manifest extends these fields with separate columns for semantic assessment and review-state management. It also reserves fields for optional automatic prelabeling. These fields were initialized without model outputs, and `prelabel_status` was set to `pending`. No automatic score, model prediction, or class suggestion was used to determine the final human assignments.

Table 5.3 summarizes the principal groups of fields maintained in the full review manifest.

Table 5.3: Principal field groups in the full review manifest

Field group	Representative fields	Implementation role
Identity and provenance	<code>image_id</code> , <code>sha256</code> , <code>relative_path</code> , <code>source_dataset</code> , <code>source_relative_path</code>	Maintains the association among the review decision, the technically prepared file, and its acquisition source.
Technical metadata	<code>width</code> , <code>height</code> , <code>size_bytes</code> , <code>extension</code> , <code>is_valid_image</code>	Preserves the technical characteristics established during preparation without mixing them with semantic decisions.
Optional prelabel fields	<code>auto_label</code> , <code>auto_confidence</code> , <code>weapon_score</code> , <code>non_weapon_score</code> , <code>prelabel_status</code>	Provides schema support for optional auxiliary prelabeling. These fields were not used to determine the frozen reference assignments.
Manual review fields	<code>final_label</code> , <code>review_state</code> , <code>review_notes</code> , <code>reviewer_id</code> , <code>review_timestamp</code>	Records the operational assignment or exclusion disposition and the state of the manual assessment.
OOD and exclusion documentation	<code>ood_flag</code> , <code>ood_notes</code> , <code>exclusion_reason</code>	Supports explicit documentation of borderline, distributionally different, or otherwise unsuitable samples.

Manual assessment was performed through the interactive application implemented by `datasets/scripts/final/10_manual_selection_protocol_reviewer.py`. At its first execution, the application initializes the persistent working database `datasets/final/manifests/manual_selection_protocol_db.csv` from the full review manifest. Subsequent executions resume the existing database and the previously saved session state.

The interface presents images in batches of ten arranged in a five-column grid and supports mouse- and keyboard-based assignment to **weapon**, **non_weapon**, and **ood**, together with the review-only **exclude** disposition. Additional controls support zooming, page navigation, undo operations, explicit assignment removal, saving, and resumption.

Two operating modes were implemented:

- in **review_pending** mode, the reviewer inspects pending images from a selected source and assigns an operational class or exclusion disposition;
- in **review_selection** mode, the reviewer re-examines images already assigned to one selected experimental group, optionally filters them by source, and may remove an unsuitable assignment by returning the sample to the pending pool.

The second mode provides an explicit quality-control mechanism for reviewing the selected corpus without treating an initial assignment as immutable.

Figure 5.1 documents the interactive environment used during the human-in-the-loop selection process.

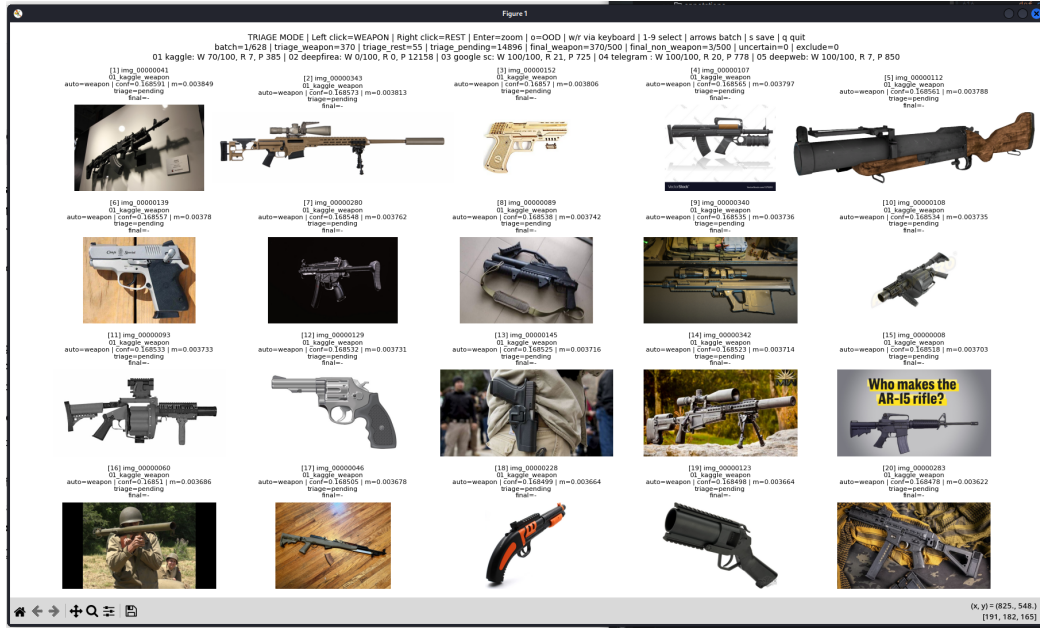


Figure 5.1: Human-in-the-loop review interface used during final dataset selection. The interface supported manual inspection and assignment to **weapon**, **non_weapon**, or **ood**, together with the review-only **exclude** disposition

The interface displays the operational criteria established in Chapter 4. Clear and realistic firearm-related content is assigned to **weapon**; realistic and interpretable negative content is assigned to **non_weapon**; borderline, synthetic, replica-like, strongly degraded, anomalous, or otherwise out-of-task content is assigned to **ood**; and unusable or irrelevant content is assigned to **exclude**. Borderline samples are not forced into either binary class.

The application enforces global targets of 500 images for each of the **weapon**, **non_weapon**, and **ood** groups. Once a target is reached, further assignments to that group are blocked until an

existing selection is removed. Soft source-level targets of 100 samples per group and source are displayed as advisory balancing alerts and do not constrain reviewer decisions.

The implemented procedure is therefore target-driven rather than an exhaustive semantic annotation of the prepared pool. The global targets define the final sample counts, while the source-level alerts support composition control without requiring equal representation from every source.

Each assignment, clearing, removal, or undo operation records the timestamp, reviewer identifier, session mode, source filter, image identifier, source dataset, preceding assignment, new assignment, and review state in `datasets/final/reports/manual_selection_log.csv`. The current position, operating mode, active filters, last action, and recent undo history are stored in `datasets/final/reports/manual_selection_state.json`.

The application automatically saves the working database, updates the session state, and regenerates the principal output records after review actions. Existing CSV files are backed up with timestamped names before replacement, and temporary files followed by atomic replacement are used to reduce the risk of partial or corrupted writes.

Table 5.4 summarizes the records produced by the application.

Table 5.4: Records produced by the manual review application

Record	Role
<code>datasets/final/manifests/manual_selection_protocol_db.csv</code>	Persistent working database containing reviewed and pending candidate records.
<code>datasets/final/manifests/manual_selection_final_1500.csv</code>	Manifest containing the samples assigned to the three final experimental groups.
<code>datasets/final/manifests/manual_selection_adversarial_subset.csv</code>	Binary subset containing the selected weapon and non_weapon samples used for partitioning, training, clean evaluation, and perturbation generation.
<code>datasets/final/manifests/manual_selection_reviewed.csv</code>	Separate record of samples assigned to the review-only exclude disposition.
<code>datasets/final/reports/manual_selection_log.csv</code>	Chronological record of reviewer actions and assignment transitions.
<code>datasets/final/reports/manual_selection_state.json</code>	Saved interactive-session state used to resume the review.
<code>datasets/final/reports/manual_selection_summary.json</code>	Summary of group counts, pending and reviewed records, target completion, output locations, and consistency checks.

The final working database contained 1 500 reviewed records and 13 821 records that remained pending. The prepared pool was therefore not reviewed exhaustively. Human assessment continued until the predefined targets of 500 **weapon**, 500 **non_weapon**, and 500 **ood** images had been reached. The resulting binary subset contained 1 000 samples. The final summary confirmed exact completion of both the 1 500-image corpus and the 1 000-image binary subset.

The implemented process separates technical file validity, semantic suitability for the experimental task, and final inclusion in the frozen corpus. The final dataset composition and the consistency controls applied to its canonical manifests are described in Section 5.2.3.

5.2.3 Final Dataset Composition and Freezing

The manual selection process concluded when the three enforced global targets had been reached. The final reviewed corpus contained exactly 500 **weapon** images, 500 **non_weapon** images, and 500 **ood** images, for a total of 1 500 samples. Candidates not assigned to one of these groups remained pending in the working review database and were not implicitly included in the experimental corpus.

The canonical definition of the frozen dataset is stored in `datasets/final/manifests/manual_selection_final_1500.csv`. Each row preserves the stable image identifier, repository-relative prepared path, source dataset and source group, SHA-256 digest, final human assignment,

review state, reviewer identifier, review timestamp, and source priority. The records are sorted deterministically by final assignment, source dataset, and image identifier.

Table 5.5 reports the final composition and downstream role of each experimental group.

Table 5.5: Composition of the final frozen dataset

Experimental group	Images	Role in the experimental pipeline
weapon	500	Positive class of the binary image-classification task and source of the target-category samples used for clean and perturbed evaluation.
non_weapon	500	Negative class of the binary task, comprising realistic and interpretable content without objects belonging to the firearm-related target category.
ood	500	Separate OOD evaluation branch used to characterize system behavior on borderline, anomalous, synthetic, source-degraded, or semantically out-of-task inputs.
Total	1 500	Frozen experimental corpus supporting the complete evaluation protocol.

Table 5.6 reports the source-level composition of the frozen corpus. The observed distribution results from the target-driven semantic review process and was not generated through rigid source quotas.

Table 5.6: Distribution of the final frozen dataset by source and experimental group

Source	weapon	non_weapon	ood	Total
01_kaggle_weapon	100	0	100	200
02_deepfirearm	100	0	0	100
03_google_scraped	100	30	71	201
04_telegram_youtube	100	57	225	382
05_deepweb	100	413	104	617
Total	500	500	500	1 500

The source distribution is non-uniform. In particular, 413 of the 500 **non_weapon** samples originate from **05_deepweb**. This composition is preserved explicitly in the frozen records and is considered when interpreting the experimental results in Chapter 6.

The **ood** group remains a separate evaluation branch and is not used as a third supervised class. The review application therefore also exports `datasets/final/manifests/manual_selection_adversarial_subset.csv`, which contains only the 500 **weapon** and 500 **non_weapon** samples.

This balanced 1000-image subset provides the source population for clean-fold generation, proxy-model training, clean binary evaluation, and adversarial and anti-forensic artifact generation. The 500 OOD samples are excluded from training and perturbation generation and remain available as clean evaluation inputs for both the proxy architectures and the selected black-box software systems.

Table 5.7 summarizes the canonical manifests produced by the freezing stage.

Table 5.7: Canonical manifests produced by the dataset-freezing stage

Manifest	Rows	Downstream role
<code>datasets/final/manifests/manual_selection_final_1500.csv</code>	1 500	Canonical definition of the complete frozen dataset, including the <code>weapon</code> , <code>non_weapon</code> , and <code>ood</code> groups.
<code>datasets/final/manifests/manual_selection_adversarial_subset.csv</code>	1 000	Balanced binary subset used for fold generation, proxy-model training, clean evaluation, and generation of adversarial and anti-forensic artifacts.
<code>datasets/final/manifests/manual_selection_removed.csv</code>	0	Record reserved for candidates assigned to the review-only <code>exclude</code> disposition. No sample received this disposition in the frozen execution.

The application generates these manifests directly from the persistent selection database. Records assigned to `weapon`, `non_weapon`, or `ood` are exported to the complete manifest, while the binary manifest is obtained by retaining only the two binary-class assignments. No additional relabeling or automated semantic transformation is performed between the two exports. The binary manifest is therefore a strict subset of the complete manifest, and its reference assignments remain identical to the corresponding human-reviewed decisions.

The final state is documented in `datasets/final/reports/manual_selection_summary.json`. The report records the output locations, group counts, reviewed and pending records, remaining targets, source-level advisory targets, and final consistency checks. In the frozen execution, `final_1500_exact` and `adversarial_subset_1000_exact` were both set to `true`, confirming the expected sizes of the complete corpus and binary subset.

Freezing was implemented at the manifest level. After the target counts and consistency checks had been satisfied, the exported manifests became the canonical inputs for all downstream stages. Subsequent scripts consume their image identifiers, paths, assignments, source information, and SHA-256 digests without modifying the human-reviewed decisions.

Any replacement, removal, or reassignment of a frozen sample therefore requires regeneration of all dependent partitions, derived artifacts, bundle records, predictions, and metrics. An isolated downstream modification would no longer correspond to the canonical dataset definition.

The frozen manifests define which images belong to the experiment and their reference operational assignments. The deterministic generation of the binary folds and the separate OOD evaluation manifest is described in Section 5.2.4.

5.2.4 Dataset Partitioning and Fold Generation

The frozen dataset was converted into operational evaluation partitions by `datasets/scripts/splits/11_generate_clean_and_ood_splits.py`. The script consumes the complete frozen manifest `datasets/final/manifests/manual_selection_final_1500.csv` and the balanced binary manifest `datasets/final/manifests/manual_selection_adversarial_subset.csv`.

Before generating any output, the script verifies the presence of the required fields, accepted reference assignments, expected sample counts, uniqueness of the image identifiers and SHA-256 digests within each input manifest, and availability of the corresponding prepared image files. The complete manifest must contain exactly 500 `weapon`, 500 `non_weapon`, and 500 `ood` records, while the binary manifest must contain exactly 500 samples from each supervised class. Execution terminates when any validation condition is not satisfied.

The 1 000-image binary subset was partitioned into five deterministic folds using `seed = 42`. Each fold contains exactly 200 images, comprising 100 `weapon` and 100 `non_weapon` samples. Class balance is enforced as a hard constraint, whereas source balance is optimized as a secondary objective because the source distributions of the two classes are not uniform.

Fold assignment is performed independently within each class-source stratum. The records in each stratum are shuffled using a deterministic pseudo-random generator initialized from the global seed, class assignment, and source identifier. Each sample is then assigned to a fold that has not reached its class quota. Among eligible folds, the implementation selects the fold with

the lowest existing count for the same class and source, followed by the lowest current class occupancy and the fold identifier as deterministic tie-breaking criteria.

This procedure enforces the class distribution reported in Table 5.8 while approximating source balance without relaxing the mandatory class quotas.

Table 5.8: Composition of the deterministic clean folds

Fold	weapon	non_weapon	Total
fold_1	100	100	200
fold_2	100	100	200
fold_3	100	100	200
fold_4	100	100	200
fold_5	100	100	200
Total	500	500	1 000

The corresponding files were copied to class-aware directories following the pattern `datasets/splits/clean/fold_<n>/<class>/`, where `<n>` ranges from 1 to 5 and `<class>` is either `weapon` or `non_weapon`.

During each copy operation, the script computes both SHA-256 and Message Digest Algorithm 5 (MD5) digests. The copied SHA-256 value is compared with the digest stored in the frozen manifest, and any mismatch terminates execution. SHA-256 remains the primary integrity control, whereas MD5 is retained as an auxiliary interoperability identifier because it may be exposed by forensic software exports and hash-based workflows.

The clean partition is documented in `datasets/splits/manifests/clean_folds_manifest.csv`. For each sample, the manifest records its image identifier, assigned fold, reference operational assignment, source dataset and source group, original source information, prepared path, split path, SHA-256 and MD5 values, file size, extension, and experimental-condition descriptors. Clean records are marked with `sample_type = clean`, `attack_family = none`, and `attack_name = clean`.

The 500 OOD samples are not assigned to the binary folds. They are copied to `datasets/splits/ood/ood_eval_set/ood/` and documented in `datasets/splits/manifests/ood_eval_manifest.csv`. These records preserve the same identity, provenance, path, integrity, and file-level information used for the clean partitions. They are assigned to the logical partition `ood_eval_set` and marked with `sample_type = ood`, `attack_family = none`, and `attack_name = ood`.

Table 5.9 summarizes the outputs produced by the partition-generation stage.

Table 5.9: Outputs produced by the partition-generation stage

Output	Role
<code>datasets/splits/clean/fold_1/</code> through <code>datasets/splits/clean/fold_5/</code>	Class-aware directories containing the 1 000 clean binary images assigned to the five deterministic folds.
<code>datasets/splits/ood/ood_eval_set/ood/</code>	Separate directory containing the 500 clean OOD inputs.
<code>datasets/splits/manifests/clean_folds_manifest.csv</code>	Canonical mapping among image identifiers, folds, binary reference assignments, provenance, copied paths, and integrity values.
<code>datasets/splits/manifests/ood_eval_manifest.csv</code>	Canonical mapping for the separate OOD evaluation set.
<code>datasets/splits/manifests/split_generation_summary.json</code>	Machine-readable summary of the seed, number of folds, input and output paths, class and source counts, and final validation checks.

The final summary records 1 000 clean binary samples and 500 OOD samples. It confirms that every fold contains exactly 100 samples from each binary class and that the image identifiers

and SHA-256 digests are unique within the clean and OOD output branches.

Existing split directories and manifests are not overwritten by default. A complete rebuild requires the explicit `-force` option, which removes the previous outputs before regeneration. This safeguard prevents artifacts generated from different versions of the frozen dataset from being combined unintentionally.

The resulting folds define the common partitioning structure used by the fold-aware proxy-model implementation described in Section 5.3. The separate OOD branch remains outside binary model training and perturbation generation and is retained for its dedicated behavioral evaluation.

5.3 Proxy-Model Implementation

5.3.1 Architectures and Preprocessing

The proxy-model layer was implemented through three fully inspectable binary classifiers: ResNet18, EfficientNet-B0, and a CLIP-based classifier built on a ViT-B/32 visual encoder. The two convolutional architectures were adapted through supervised fine-tuning, whereas the CLIP proxy uses frozen pretrained visual representations and a trainable binary head. This implementation provides three controlled architectural references without assuming equivalence with the selected black-box software systems.

All three models use the same class-to-index mapping:

$$\text{non_weapon} \mapsto 0, \quad \text{weapon} \mapsto 1.$$

The OOD branch is not represented as a third output and is excluded from proxy-model training. Its images are processed only during the dedicated OOD evaluation stage.

The model suite is implemented in `models/scripts/12_train_proxy_models.py`, while its frozen architecture definitions, checkpoint locations, and checkpoint digests are recorded in `models/model_registry.json`.

Table 5.10 summarizes the implemented architectures and their adaptation to the binary task.

Table 5.10: Proxy-model architectures implemented for the binary classification task

Proxy model	Implementation	Binary adaptation
ResNet18	<code>torchvision.resnet18</code> initialized with <code>IMAGENET1K_V1</code> weights	The original fully connected layer is replaced by <code>Linear(in_features, 2)</code> . The residual convolutional architecture provides the conventional supervised baseline [43].
EfficientNet-B0	<code>torchvision.efficientnet_b0</code> initialized with <code>IMAGENET1K_V1</code> weights	The final classifier element is replaced by <code>Linear(in_features, 2)</code> . This architecture serves as the primary proxy target for model-dependent adversarial generation [44].
CLIP	<code>open_clip</code> ViT-B/32 visual encoder initialized with the pretrained <code>openai</code> weights	The visual encoder is frozen and connected to a trainable linear head with two outputs. The resulting proxy classifies pretrained visual representations rather than using the original image-text similarity interface [45].

For ResNet18 and EfficientNet-B0, the implementation explicitly requires the corresponding `IMAGENET1K_V1` weight definitions exposed by Torchvision. Training terminates if these definitions are unavailable rather than silently falling back to random initialization.

The CLIP-based proxy is constructed from the `open_clip` ViT-B/32 visual encoder with the `openai` pretrained weights. All encoder parameters are frozen. For each image, the encoder produces a visual feature vector that is normalized to unit length and passed to a trainable linear layer with two outputs.

The fold-specific CLIP checkpoints store the parameters of the binary head rather than a complete copy of the pretrained visual encoder. Reconstructing the full proxy therefore requires the external ViT-B/32 model and pretrained weights identified in the model registry.

A common image-loading procedure is used for all three architectures. Each file is opened with Pillow, corrected according to its embedded orientation through an EXIF transpose operation, and converted to a three-channel RGB representation before architecture-specific preprocessing is applied.

All inputs are resized deterministically to 224×224 pixels. The training transformation applies:

1. resizing to 224×224 pixels;
2. random horizontal flipping with probability 0.5;
3. conversion to a PyTorch tensor;
4. architecture-specific channel normalization.

The validation and evaluation transformations apply the same resizing, tensor-conversion, and normalization operations without random horizontal flipping. The implementation does not introduce random cropping, rotation, color augmentation, or additional geometric distortion.

Table 5.11 reports the channel normalization values used by the implemented models.

Table 5.11: Input normalization used by the proxy models

Models	Channel means	Channel standard deviations
ResNet18 and EfficientNet-B0	(0.485, 0.456, 0.406)	(0.229, 0.224, 0.225)
CLIP ViT-B/32	(0.48145466, 0.45782750, 0.40821073)	(0.26862954, 0.26130258, 0.27577711)

The same architecture-specific normalization is reused for validation and evaluation. A shared normalization scheme is not applied because the ImageNet-initialized convolutional models and the CLIP visual encoder were pretrained under different input-normalization conventions.

Each proxy exposes two output logits corresponding to the fixed binary mapping. Prediction probabilities are derived from these logits by the evaluation pipeline and are not stored in the dataset or preprocessing manifests. Training, held-out-fold handling, checkpoint selection, and model-integrity controls are described in Section 5.3.2.

5.3.2 Training Configuration and Fold-Aware Checkpoints

The proxy models were trained through a fold-aware leave-one-fold-out procedure. For each architecture and target fold, the target fold was reserved as the held-out evaluation partition, while the remaining four folds were used to construct the training and internal validation sets. No checkpoint was therefore trained on the fold on which it was subsequently evaluated or used for fold-matched model-dependent adversarial generation.

For each architecture–fold run, the four non-held-out folds contained 800 images. A deterministic stratified split reserved 15% of these records for internal validation, independently within each binary class. The resulting partitions contained:

- 680 images for parameter optimization;
- 120 images for internal validation;
- 200 images in the untouched held-out fold.

The training–validation split was generated using seed 42. Records were separated by class, shuffled deterministically within each class, and recombined, preserving equal representation of **weapon** and **non_weapon**. The held-out fold was not used for parameter updates or checkpoint selection.

Table 5.12 summarizes the training configuration implemented by `models/scripts/12_train_proxy_models.py`.

Table 5.12: Proxy-model training configuration

Parameter	Value
Training epochs	10
Batch-size configuration	16 as the command-line default; 32 in the dedicated interactive CLIP-only profile. The effective value used for each run is preserved in the checkpoint training configuration.
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Loss function	Unweighted cross-entropy loss
Internal validation ratio	0.15
Random seed	42
Input resolution	224×224 pixels
Data-loader workers	2
Checkpoint-selection criterion	Highest internal validation accuracy; the latest epoch is retained when the same maximum value occurs more than once.

ResNet18 and EfficientNet-B0 were fine-tuned from their ImageNet-initialized weights. Both the convolutional backbones and the newly introduced binary classification layers were optimized. The script also exposes a `-freeze-backbone` option for alternative or diagnostic runs, but this option is not part of the frozen CNN configuration.

For the CLIP-based proxy, the pretrained visual encoder remained frozen and only the binary linear head was optimized. The effective training configuration, including the batch size and backbone-freezing state, is stored inside each fold-specific checkpoint.

The objective function was implemented using unweighted cross-entropy loss. Class weighting was not applied because the training and validation partitions remained balanced between the two supervised classes. Parameter updates used AdamW with the learning-rate and weight-decay values reported in Table 5.12.

Before training, the script validates the complete clean-fold manifest by checking:

- the presence of the required columns;
- the exclusive use of the two accepted binary assignments;
- the presence of all five expected folds;
- the exact count of 100 images per class and fold;
- the uniqueness of the image identifiers;
- the availability of every referenced image file;
- the correspondence between each file and its recorded SHA-256 digest.

Execution terminates when an image is missing or its digest does not match the split manifest, preventing a checkpoint from being produced from a dataset state inconsistent with the frozen partition definition.

The fixed seed is propagated to the Python standard-library pseudo-random generator, NumPy, PyTorch, and all available CUDA devices. The implementation disables cuDNN benchmark-based algorithm selection and requests deterministic cuDNN behavior. These

controls reduce variation from the principal pseudo-random sources, although exact numerical reproduction may still depend on the software and hardware environment.

The execution device may be selected as `cpu` or `cuda`, or resolved automatically according to CUDA availability. Device selection does not modify the frozen folds or declared optimization parameters. The preserved complete-suite execution log records CPU-based training for the reported checkpoints.

During each epoch, the script records training loss, training accuracy, validation loss, and validation accuracy. All 10 epochs are executed; no early stopping is applied. The checkpoint is overwritten whenever the current validation accuracy is greater than or equal to the best value observed during the preceding epochs. When the same maximum accuracy occurs in multiple epochs, the latest qualifying checkpoint is retained. The held-out fold remains excluded from this selection procedure.

Fold-specific checkpoints are stored according to the pattern `models/checkpoints/<model_name>/<fold>.pt`. The complete suite contains:

$$3 \text{ proxy architectures} \times 5 \text{ held-out folds} = 15 \text{ checkpoints.} \quad (5.1)$$

For ResNet18 and EfficientNet-B0, each checkpoint contains the trained model state, class mapping, held-out-fold identifier, input size, training configuration, creation timestamp, and metric history available when the checkpoint was saved. For the CLIP-based proxy, the checkpoint stores the trained binary-head parameters and identifies the external ViT-B/32 architecture and pretrained weights required to reconstruct the frozen visual encoder.

Table 5.13 summarizes the principal outputs of the training stage.

Table 5.13: Outputs produced by the proxy-model training stage

Output	Role
<code>models/checkpoints/resnet18/</code>	Five fold-specific ResNet18 checkpoints.
<code>models/checkpoints/efficientnet_b0/</code>	Five fold-specific EfficientNet-B0 checkpoints.
<code>models/checkpoints/clip/</code>	Five fold-specific binary-head checkpoints for the frozen CLIP visual encoder.
<code>models/reports/proxy_model_training_summary.csv</code>	Consolidated record of checkpoint paths and digests, training, validation, and held-out sample counts, best validation accuracy, final-epoch statistics, and creation timestamps.
<code>models/model_registry.json</code>	Canonical registry of architecture definitions, initialization weights, preprocessing conventions, checkpoint patterns, external-weight requirements, and fold-specific SHA-256 digests.
<code>models/reports/train_all_proxy_models.log</code>	Raw execution log containing the epoch-level training and validation progression of the complete proxy-model suite.

The training summary contains one record for each of the 15 architecture–fold combinations. Every record reports 680 training images, 120 internal-validation images, and 200 held-out images. The report is maintained through an upsert operation: rerunning one architecture–fold combination replaces its corresponding row while preserving the remaining records.

After each checkpoint is written, its SHA-256 digest is computed and stored in both the training summary and `models/model_registry.json`. The registry therefore provides the canonical association among architecture, held-out fold, checkpoint path, initialization convention, preprocessing scheme, external-weight requirements, and checkpoint digest. The checkpoint files are distributed through Git LFS, while their cryptographic identifiers remain directly available in the repository metadata.

This fold-aware implementation ensures that every clean binary sample is evaluated by a checkpoint trained without its assigned fold. The same association is preserved during model-dependent adversarial generation, preventing an artifact from being generated against a checkpoint trained on its corresponding clean sample. The concrete perturbation implementations are described in Section 5.4.

5.4 Perturbation Implementation

5.4.1 Adversarial Attack Implementation

Adversarial generation was implemented through `datasets/scripts/attacks/14_generate_adversarial_attacks.py`, using `datasets/splits/manifests/clean_folds_manifest.csv` as the canonical input. The script generates one perturbed artifact from each of the 1 000 clean binary images for every selected condition.

The frozen suite comprises four model-dependent attacks—FGSM, One Pixel, Sigma-Zero, and SuperDeepFool—and the model-agnostic Color Shift condition. FGSM, Sigma-Zero, and SuperDeepFool require white-box access to gradients or decision-boundary information. One Pixel uses score-based access to the output probabilities of the target proxy model. Color Shift does not query a model and is therefore recorded as an adversarial-style condition rather than as a white-box or score-based attack.

EfficientNet-B0 was used as the primary proxy target for all model-dependent conditions. For each source image, the generator reads the fold identifier from the clean manifest and loads the corresponding checkpoint from `models/checkpoints/efficientnet_b0/<fold>.pt`. Because that checkpoint was trained while excluding the associated fold, the source image was not part of the checkpoint’s training data.

Before generation, the script verifies the required manifest fields, accepted binary assignments, uniqueness of image identifiers, availability of all clean inputs, and correspondence of their SHA-256 and MD5 values with the clean-fold manifest. Model-dependent conditions additionally require the corresponding fold-specific checkpoint. Missing files or integrity mismatches terminate execution.

Table 5.14 reports the parameters used in the frozen generation runs.

Table 5.14: Frozen adversarial and adversarial-style generation parameters

Condition	Dependency	Frozen parameters	Format
FGSM	EfficientNet-B0, fold-aware white-box checkpoint	Untargeted perturbation with $\epsilon = 8/255 \approx 0.03137$, applied in pixel space $[0, 1]$ using the sign of the loss gradient [13].	PNG
One Pixel	EfficientNet-B0, fold-aware score-based checkpoint access	Differential Evolution over five variables (x, y, R, G, B) ; 50 maximum iterations; population-size multiplier 20, corresponding to an effective population of approximately 100 candidates; deterministic base seed 42; early termination after misclassification [52].	PNG
Sigma-Zero	EfficientNet-B0, fold-aware white-box checkpoint	Reference untargeted implementation; 1 000 optimization steps; $\eta_0 = 1.0$, $\sigma = 0.001$, $\tau_0 = 0.3$, threshold-adjustment factor 0.01, and infinity-norm gradient normalization [56].	PNG
SuperDeepFool	EfficientNet-B0, fold-aware white-box checkpoint	Internal paper-based implementation of $\text{SDF}(\infty, 1)$; 20 maximum outer iterations; 50 maximum DeepFool iterations; one projection step; all output classes considered as candidates [54].	PNG
Color Shift	Model-agnostic	Red-channel shift +12, green-channel shift 0, blue-channel shift -12, saturation factor 1.10, contrast factor 1.00, and JPEG quality 95.	JPEG

FGSM. The FGSM implementation preprocesses the clean image according to the EfficientNet-B0 normalization and computes the loss gradient with respect to the reference operational assignment. The normalized gradient is converted to pixel space before applying the perturbation:

$$\mathbf{x}_{\text{adv}} = \text{clip}_{[0,1]}(\mathbf{x} + \epsilon \text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, y))), \quad (5.2)$$

where y is the encoded reference operational assignment and $\epsilon = 8/255$. The adversarial tensor is converted to RGB and stored as a lossless PNG file. The generation record includes clean

and adversarial predictions, maximum predicted-class probability, reference-class probability, and pixel-space perturbation measurements.

One Pixel. The One Pixel implementation uses Differential Evolution to optimize the coordinates and RGB value of one modified pixel. Each candidate is represented by $\theta = (u, v, r, g, b)$, where (u, v) identifies the pixel and (r, g, b) specifies its replacement value. Let $T(\mathbf{x}; \theta)$ denote the corresponding transformed image. The implemented score-based objective is:

$$\begin{aligned} \min_{\theta} \quad & p_y(T(\mathbf{x}; \theta)) \\ \text{subject to} \quad & |\{(i, j) : T(\mathbf{x}; \theta)_{ij} \neq \mathbf{x}_{ij}\}| \leq 1. \end{aligned}$$

where $p_y(\cdot)$ is the probability assigned by the fold-specific target model to the reference class y . The constraint applies to pixel locations, since all three channels of the selected pixel may change jointly.

A deterministic image-specific seed is derived from the base seed, condition name, and stable image identifier through SHA-256. Optimization terminates early when a candidate changes the prediction relative to the reference assignment. Otherwise, the best candidate identified before the iteration limit is exported.

Sigma-Zero. Sigma-Zero was integrated through the reference implementation exposed by the `adversarial-library` project, pinned to commit `b14f81a3e1c414a573b969b402c99e65bfe2ca33`. The FAIRLab adapter invokes the reference function directly rather than reimplementing it.

The reference function operates on unnormalized tensors in $[0, 1]$. The adapter therefore converts the model-normalized input to pixel space and wraps the EfficientNet-B0 checkpoint so that model normalization is reapplied before logit computation. The untargeted sparse-perturbation objective can be summarized as:

$$\begin{aligned} \min_{\mathbf{x}_{\text{adv}} \in [0, 1]^d} \quad & \|\mathbf{x}_{\text{adv}} - \mathbf{x}\|_0 \\ \text{subject to} \quad & \arg \max_k f_k(\mathbf{x}_{\text{adv}}) \neq y. \end{aligned}$$

where $f_k(\cdot)$ is the logit associated with class k . The discrete objective is approximated through the iterative optimization and adaptive threshold procedure configured in Table 5.14.

SuperDeepFool. SuperDeepFool was implemented internally from the published $\text{SDF}(\infty, 1)$ procedure and does not vendor or copy the official repository. The implementation first performs DeepFool-style iterations to identify an adversarial boundary point and then applies a projection based on the gradient difference between the boundary and reference classes.

The frozen untargeted objective is:

$$\begin{aligned} \min_{\delta} \quad & \|\delta\|_{\infty} \\ \text{subject to} \quad & \arg \max_k f_k(\mathbf{x} + \delta) \neq y, \\ & \mathbf{x} + \delta \in [0, 1]^d. \end{aligned}$$

At a boundary point $\mathbf{x}_b$, the projection uses:

$$\mathbf{w}_b = \nabla_{\mathbf{x}} f_{k_b}(\mathbf{x}_b) - \nabla_{\mathbf{x}} f_y(\mathbf{x}_b), \quad (5.3)$$

where k_b is the boundary class. If the projected point remains adversarial, it is retained; otherwise, the implementation falls back to the adversarial boundary point. The manifest records iteration counts, convergence status, and whether the projection preserved adversariality.

Color Shift. Color Shift modifies the RGB channels in the original image space and then applies the configured saturation and contrast factors. No model is queried during generation, and the generation-time attack-success field is marked as not applicable. JPEG quality 95 is used because controlled recompression forms part of the condition. The target-model field is set to `model_agnostic`, and checkpoint-related fields are left empty or marked as not applicable.

Model-dependent outputs follow the pattern `attacks/adversarial/<attack_name>/<target_model>/<fold>/<label>/`. Color Shift uses the same structure with `target_model = model_agnostic`. Generated filenames follow the pattern `<image_id>__<attack_name>__<target_model>.<extension>`.

Each condition produces 1 000 artifacts, comprising 500 `weapon` and 500 `non_weapon` samples, with 200 artifacts per fold. The complete stage therefore produces:

$$5 \text{ conditions} \times 1\,000 \text{ binary images} = 5\,000 \text{ perturbed artifacts.} \quad (5.4)$$

Table 5.15 lists the canonical generation records.

Table 5.15: Canonical adversarial-generation manifests and summaries

Record	Role
<code>attacks/manifests/adversarial_fgsm_efficientnet_b0_manifest.csv</code>	Sample-level traceability for the 1 000 FGSM outputs.
<code>attacks/manifests/adversarial_one_pixel_efficientnet_b0_manifest.csv</code>	Sample-level traceability for the 1 000 One Pixel outputs.
<code>attacks/manifests/adversarial_sigma_zero_efficientnet_b0_manifest.csv</code>	Sample-level traceability for the 1 000 Sigma-Zero outputs.
<code>attacks/manifests/adversarial_superdeepfool_efficientnet_b0_manifest.csv</code>	Sample-level traceability for the 1 000 SuperDeepFool outputs.
<code>attacks/manifests/adversarial_color_shift_manifest.csv</code>	Sample-level traceability for the 1 000 model-agnostic Color Shift outputs.
Associated <code>*_summary.json</code> files	Run-level parameters, generated counts, fold and assignment distributions, target-model information, generation-time success accounting where applicable, and consistency checks.

Each manifest maps the generated artifact to its source through `original_image_id` and assigns a stable `generated_image_id`. It also records the fold, reference assignment, source provenance, clean and perturbed paths, attack family, condition name, serialized parameters, model dependency, target model, checkpoint path and digest, original and perturbed hashes, output size, extension, and creation timestamp.

For model-dependent conditions, the manifest additionally records clean and adversarial predictions, diagnostic probabilities, correctness states, perturbation norms, convergence information, and a generation-time `attack_success` field. This field is true when the perturbed prediction differs from the reference assignment, even when the clean input was already misclassified.

The generation-time indicator is therefore distinct from the comparative attack success rate computed during evaluation. The latter uses only samples correctly classified by the same checkpoint under the clean condition and measures the proportion rendered incorrect by the perturbation. Color Shift is instead analyzed through induced errors and prediction transitions under a shared model-agnostic condition.

The run summaries confirm that each frozen condition generated the expected 1 000 outputs, with 200 artifacts per fold and 500 per reference assignment. They also verify uniqueness of

the generated identifiers and perturbed SHA-256 values. Existing attack directories, manifests, and summaries are not overwritten unless the explicit `-force` option is supplied.

The model-agnostic anti-forensic transformations are described in Section 5.4.2.

5.4.2 Anti-Forensic Transformation Implementation

Anti-forensic artifact generation was implemented through `datasets/scripts/attacks/13_generate_anti_forensic_attacks.py`, using `datasets/splits/manifests/clean_folds_s_manifest.csv` as the canonical input. Each transformation was applied to all 1000 clean binary images, comprising 500 `weapon` and 500 `non_weapon` samples. The 500 clean OOD inputs remained outside this stage.

The frozen suite comprises five deterministic and model-agnostic transformations: JPEG recompression, Resample + Resize, Gaussian Blur, Histogram Modification, and Contrast Stretching. These conditions do not access model gradients, logits, probabilities, or checkpoints. The reference operational assignment of each source image is propagated unchanged to the corresponding transformed artifact.

Before generation, the script verifies the required manifest fields, accepted binary assignments, and uniqueness of image identifiers. It also checks the existence of every clean input and recomputes its SHA-256 and MD5 digests. Missing files or mismatches between the recorded and computed values terminate execution. SHA-256 remains the primary integrity control, whereas MD5 is retained as an auxiliary interoperability identifier.

Each source image is opened with Pillow, corrected according to its embedded orientation through an EXIF transpose operation, and converted to RGB before the transformation-specific operation is applied. All generated artifacts are stored as JPEG files. The dedicated recompression condition uses quality 70, while the remaining transformations use quality 95 to limit additional encoding effects beyond the principal operation.

Table 5.16 reports the frozen configuration of the five transformations.

Table 5.16: Frozen anti-forensic transformation parameters

Transformation	Implementation and frozen parameters	Output encoding
JPEG recompression	The RGB image is re-encoded without changing its spatial dimensions, using JPEG quality 70.	JPEG, quality 70
Resample + Resize	The image is downsampled to 50% of its original width and height using bicubic interpolation and then restored to its original dimensions using the same interpolation method.	JPEG, quality 95
Gaussian Blur	A Gaussian blur filter with radius 1.5 is applied to the complete image.	JPEG, quality 95
Histogram Modification	Per-channel histogram equalization is applied through the Pillow <code>ImageOps.equalize</code> operation.	JPEG, quality 95
Contrast Stretching	Automatic contrast normalization is applied through <code>ImageOps.autocontrast</code> , using a cutoff of 1%.	JPEG, quality 95

JPEG recompression. The recompression condition does not apply an explicit spatial or tonal operation before serialization. The RGB image is re-encoded at JPEG quality 70, introducing a controlled lossy-compression condition through quantization and block-based encoding effects.

Resample + Resize. For an image with original dimensions $W \times H$, the intermediate dimensions are computed as:

$$W_{\text{down}} = \max(1, \text{round}(0.5W)), \quad H_{\text{down}} = \max(1, \text{round}(0.5H)). \quad (5.5)$$

The image is first resized to $W_{\text{down}} \times H_{\text{down}}$ and then restored to $W \times H$. Both operations use bicubic interpolation. The output therefore retains the original spatial dimensions

while incorporating the information loss introduced by the intermediate downsampling and reconstruction.

Gaussian Blur. The Gaussian Blur condition applies a radius of 1.5 to suppress high-frequency spatial detail. Image dimensions remain unchanged before final JPEG serialization.

Histogram Modification. Histogram Modification is implemented through `ImageOps.equalize`, which performs histogram equalization on the RGB image channels. The same deterministic operation is applied to every sample, without image-specific thresholds or model-derived information.

Contrast Stretching. Contrast Stretching is implemented through `ImageOps.autocontrast` with a cutoff value of 1.0. The operation remaps the remaining intensity range after excluding the configured proportion of extreme values, modifying global contrast while retaining the original image dimensions.

Generated files follow the directory structure `attacks/anti_forensic/<transformation>/<fold>/<label>/` and the filename pattern:

`<image_id>__<transformation>.jpg`.

The hierarchy preserves the deterministic fold and binary reference assignment of the source image. The transformation name remains separate from the reference assignment, and no relabeling is performed during generation.

Each transformation produces 1 000 files, comprising 500 `weapon` and 500 `non_weapon` artifacts, with 200 files per fold. The complete stage therefore produces:

$$5 \text{ transformations} \times 1\,000 \text{ binary images} = 5\,000 \text{ transformed artifacts.} \quad (5.6)$$

Table 5.17 summarizes the canonical records produced by the generation stage.

Table 5.17: Canonical records produced by anti-forensic transformation generation

Record	Role
<code>attacks/manifests/anti_forensic_attacks_manifest.csv</code>	Sample-level mapping among each transformed artifact, its source image, reference assignment, fold, provenance, transformation parameters, paths, digests, size, extension, and creation timestamp.
<code>attacks/manifests/anti_forensic_generation_summary.json</code>	Run-level record of the input manifest, selected transformations, frozen parameters, output counts, fold and assignment distributions, and consistency checks.

For each artifact, the sample-level manifest records the generated and original image identifiers, fold, reference assignment, source provenance, clean and transformed paths, transformation name, serialized parameters, original and transformed SHA-256 and MD5 values, file size, extension, and creation timestamp.

The generation summary confirms that all five transformations produced the expected 1 000 outputs, for a total of 5 000 artifacts. It also confirms 200 outputs per fold and 500 outputs per binary reference assignment for every condition, together with uniqueness of the generated identifiers and transformed SHA-256 values.

The script supports optional immediate evaluation of generated artifacts against an EfficientNet-B0 checkpoint and an evaluation-only mode for an existing anti-forensic manifest. These modes are auxiliary to the generation stage. The canonical comparative proxy-model evaluation is performed separately through the common evaluation pipeline described in Section 5.7.1.

Existing selected transformation directories are not overwritten by default. When `-force` is supplied, only the selected directories are removed and regenerated. This behavior reduces the risk of combining artifacts produced under different configurations.

The transformed artifacts are subsequently incorporated into the forensic evaluation bundle described in Section 5.5.

5.5 Forensic Evaluation Bundle Construction

The forensic evaluation bundle was constructed through `datasets/scripts/bundle/16_build_forensic_evaluation_bundle.py`. It constitutes the operational interface between the controlled experimental pipeline and the black-box evaluation of the selected software systems. The builder consolidates clean binary inputs, clean OOD inputs, adversarial artifacts, and anti-forensic artifacts while preserving the identity, provenance, experimental condition, and cryptographic integrity of every logical item.

The builder consumes the following canonical inputs:

- `datasets/splits/manifests/clean_folds_manifest.csv`;
- `datasets/splits/manifests/ood_eval_manifest.csv`;
- the five canonical adversarial-generation manifests stored under `attacks/manifests/`;
- `attacks/manifests/anti_forensic_attacks_manifest.csv`.

The frozen execution produced the 11 500-item logical corpus reported in Table 5.18.

Table 5.18: Composition of the forensic evaluation bundle

Bundle component	Files	Composition
Clean binary inputs	1 000	500 weapon and 500 non_weapon images from the deterministic clean folds.
Clean OOD inputs	500	Separate out-of-distribution evaluation inputs without experimentally applied adversarial or anti-forensic processing.
Adversarial inputs	5 000	Five conditions applied to the 1 000-image binary subset: FGSM, One Pixel, Sigma-Zero, SuperDeepFool, and Color Shift.
Anti-forensic inputs	5 000	Five transformations applied to the 1 000-image binary subset: JPEG recompression, Resample + Resize, Gaussian Blur, Histogram Modification, and Contrast Stretching.
Total	11 500	Complete logical corpus used for black-box software evaluation.

Across the complete bundle, the experimental-group distribution is:

$$5\,500\text{ weapon} + 5\,500\text{ non_weapon} + 500\text{ ood} = 11\,500\text{ items.} \quad (5.7)$$

For each binary source image, the bundle contains the clean input and its ten corresponding perturbed artifacts. Each OOD input appears once because the OOD branch is evaluated in its clean state and is excluded from perturbation generation.

A central implementation requirement was the separation between the files provided to the black-box software systems and the control metadata required for subsequent normalization. The bundle was therefore materialized through the three views reported in Table 5.19.

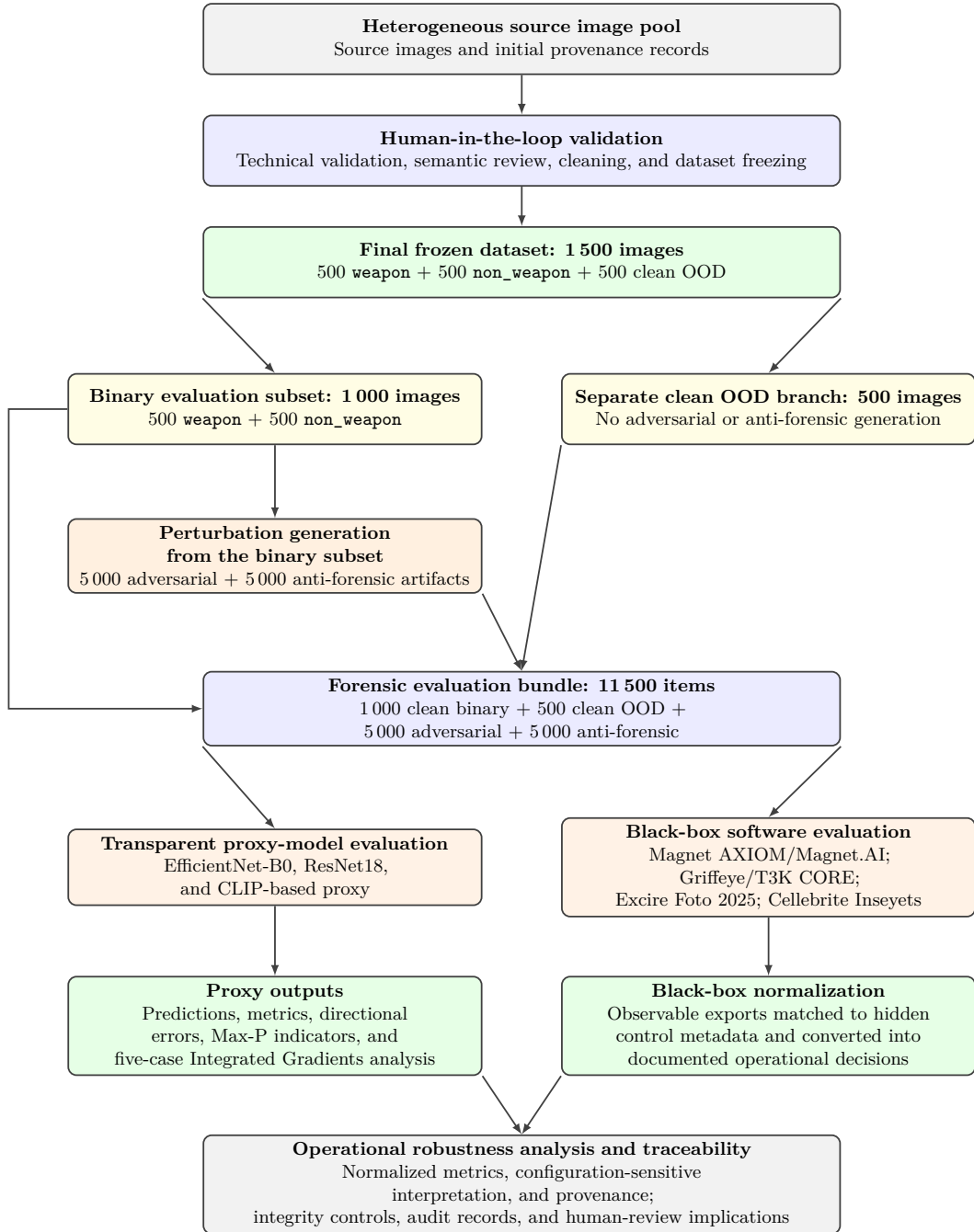


Figure 5.2: Concrete FAIRLab instantiation used in this thesis. The 1 500-image frozen dataset is divided into a 1 000-image binary branch and a separate 500-image clean OOD branch. Adversarial and anti-forensic artifacts are generated only from the binary subset. The resulting 11 500-item bundle supports parallel proxy-model and black-box software evaluation, followed by output normalization, metrics, qualitative explainability for EfficientNet-B0, and traceability controls

Table 5.19: Views materialized within the forensic evaluation bundle

Bundle view	Intended use	Content
<code>datasets/forensic_evaluation_bundle/</code> <code>blind_tool_input/files/</code>	Black-box software input	Flat directory containing neutrally identified files with no explicit reference assignments, perturbation names, fold identifiers, target-model names, or OOD indicators in their paths or file names.
<code>datasets/forensic_evaluation_bundle/</code> <code>structured_audit_view/</code>	Internal inspection and debugging	Structured copy preserving the clean, OOD, adversarial, and anti-forensic hierarchy, together with folds, reference assignments, perturbation names, and target-model information where applicable.
<code>datasets/forensic_evaluation_bundle/</code> <code>metadata/</code>	Post-processing and traceability	Manifests, hash mappings, bundle summary, and embedded-metadata audit records used to reconstruct reference assignments and experimental conditions after the black-box exports have been obtained.

Only `datasets/forensic_evaluation_bundle/blind_tool_input/files/` was imported into the selected black-box software systems. The metadata and structured audit views were excluded from the imported corpus.

This separation prevents experiment-specific descriptors from being disclosed directly through the imported directory hierarchy or file names and limits the analyst’s access to the hidden control mapping during the initial evaluation. It does not conceal the visual content of an image, perceptible perturbation effects, or information that may remain embedded within the image file itself.

Every logical item was assigned a sequential identifier from `bundle_000001` through `bundle_011500`. Blind file names contain only this identifier and the source file’s format extension, as in `bundle_000001.jpg` or `bundle_000002.png`.

The identifiers are neutral and deterministic rather than semantically descriptive. The bundle builder does not claim to randomize the logical item ordering; blindness is achieved through neutral naming, flat-path materialization, exclusion of the internal mapping, and separation of control metadata from the imported directory.

The source file format is retained, with the corresponding extension normalized to lowercase. The builder performs no conversion, resizing, recompression, or metadata sanitization. Files are copied without modifying their byte content.

Blind file names are checked against a list of forbidden semantic and experimental tokens, including binary assignment values, OOD indicators, perturbation names, fold identifiers, and proxy-model names. The file stem must also correspond exactly to the assigned bundle identifier.

The structured audit view reproduces the experimental hierarchy separately. Clean inputs are organized by fold and reference assignment; OOD inputs are maintained under the dedicated evaluation branch; adversarial artifacts are organized by condition, target model, fold, and reference assignment; and anti-forensic artifacts are organized by transformation, fold, and reference assignment. This view supports internal verification and must not be imported into the black-box software systems.

The canonical mapping is stored in `datasets/forensic_evaluation_bundle/metadata/bundle_manifest.csv`. For each logical item, it records the assigned `bundle_id`, the blind input file name, the sample type, the perturbation family and name, the target proxy model when applicable, the experimental fold, the reference operational assignment, and the source dataset.

The manifest also preserves the original and generated image identifiers, the originating manifest and source file path, and the corresponding blind and structured bundle paths. In addition, it records the original, manifest-reported, and materialized-file hashes, the result of their comparison, and the relevant file metadata, including size, extension, layout, and creation timestamp.

This mapping reconnects a black-box result associated with a neutral file name to its source image, reference assignment, fold, and experimental condition. It is withheld during the blind evaluation and used during output normalization as described in Section 5.6.5.

A reduced hash-oriented mapping is stored in `datasets/forensic_evaluation_bundle/metadata/bundle_hashes_sha256.csv`. Despite its name, the file contains both SHA-256 and MD5 values. It associates these digests with the bundle identifier, blind file name, sample type, perturbation information, reference assignment, and source identifiers. SHA-256 remains the primary integrity identifier, whereas MD5 supports matching against software exports that expose MD5-oriented fields.

During materialization, the copied file is hashed and compared with the value recorded in its originating manifest. The frozen summary confirms that all available manifest-recorded SHA-256 values correspond to the materialized files. It also confirms uniqueness of the 11 500 bundle identifiers and materialized-file SHA-256 values.

The principal run-level record is `datasets/forensic_evaluation_bundle/metadata/bundle_summary.json`. It records the selected layout, output locations, and item counts by sample type, perturbation family, perturbation name, and reference assignment.

The file also reports the outcome of four validation checks: uniqueness of the bundle identifiers, uniqueness of the materialized-file SHA-256 values, agreement between the manifest-recorded and materialized-file SHA-256 values whenever a manifest digest is available, and the absence of forbidden semantic or experimental tokens from blind paths.

The summary additionally records the structural condition that the metadata directory is separate from the black-box input directory. All computed checks were satisfied by the frozen bundle, whose layout was materialized in combined mode.

An additional non-destructive audit examines metadata embedded within the blind input files. The audit produces:

- `datasets/forensic_evaluation_bundle/metadata/embedded_metadata_audit.csv`;
- `datasets/forensic_evaluation_bundle/metadata/embedded_metadata_audit_summary.json`.

For each blind input, the audit inspects readable image-information fields and EXIF entries and searches their keys and values for terms that may reveal reference assignments, perturbation conditions, fold identifiers, model names, weapon-related descriptors, or other potentially informative content.

The audit does not remove or rewrite metadata, because doing so would alter the files supplied to the selected black-box systems. Its purpose is to document a potential residual information channel and support the separate sensitivity analysis reported in Section 6.6.5. The presence of a matched term does not by itself demonstrate that a black-box system accessed, interpreted, or used the corresponding metadata.

The bundle builder supports blind-only, structured-only, and combined layouts. The frozen execution used the combined layout and materialized the same 11 500 logical items in both operational views, producing 11 500 blind files and 11 500 structured audit copies. Rebuilding an existing bundle requires the explicit `-force` option, which removes the existing bundle directory before materialization.

The completed bundle provides a stable and auditable input population for the selected black-box software systems. Their concrete versions, import procedures, classification categories, retrieval configurations, and export formats are documented in Section 5.6.

5.6 Black-Box Software Configuration and Output Normalization

All four software systems were evaluated using only `datasets/forensic_evaluation_bundle/blind_tool_input/files/`. Neither the bundle metadata nor the structured audit view was included in the software input. The evaluated systems therefore received neutrally identified files following the `bundle_XXXXXX` convention, without reference operational assignments, perturbation families or conditions, fold identifiers, proxy-model information, source provenance, or explicit OOD indicators in the imported paths or file names.

The same interpretive and distribution boundary applies to all four systems. They were treated as black boxes and assessed only through decision-bearing fields observable in their frozen exports; internal architectures, training data, representations, thresholds, and calibration procedures were not available for inspection. The mappings below are therefore operational recordings for the common binary task, not reconstructions of internal decision rules, and software-specific numerical values are not treated as calibrated probabilities comparable across systems. Complete raw exports remain controlled-access artifacts, whereas minimized decision-bearing extracts are retained in the public repository. The shared matching, completeness, deduplication, and public-equivalence controls are consolidated in Section 5.6.5; the following subsections focus on each system’s configuration and observable mapping.

5.6.1 Magnet AXIOM and Magnet.AI

Magnet AXIOM was evaluated through the image-categorization functionality provided by Magnet.AI, using categories and tags observable through the user interface and exported case records.

The frozen software and case configuration is reported in Table 5.20.

Table 5.20: Frozen Magnet AXIOM and Magnet.AI configuration

Configuration field	Frozen value
Platform	Magnet AXIOM with Magnet.AI media analysis
Software version	10.1.0.48673
Run identifier	FAIRLAB_AXIOM_RUN_02
Imported directory	<code>datasets/forensic_evaluation_bundle/blind_tool_input/files/</code>
Imported files	11 500 neutrally identified bundle images
Processing mode	Full search with Magnet.AI media analysis enabled
Enabled media-analysis category	<code>Weapons</code>
Prediction-bearing export	<code>Pictures.csv</code>
Observable decision field	<code>Tags</code>

Figure 5.3 documents the concrete media-analysis configuration adopted for the frozen AXIOM run.

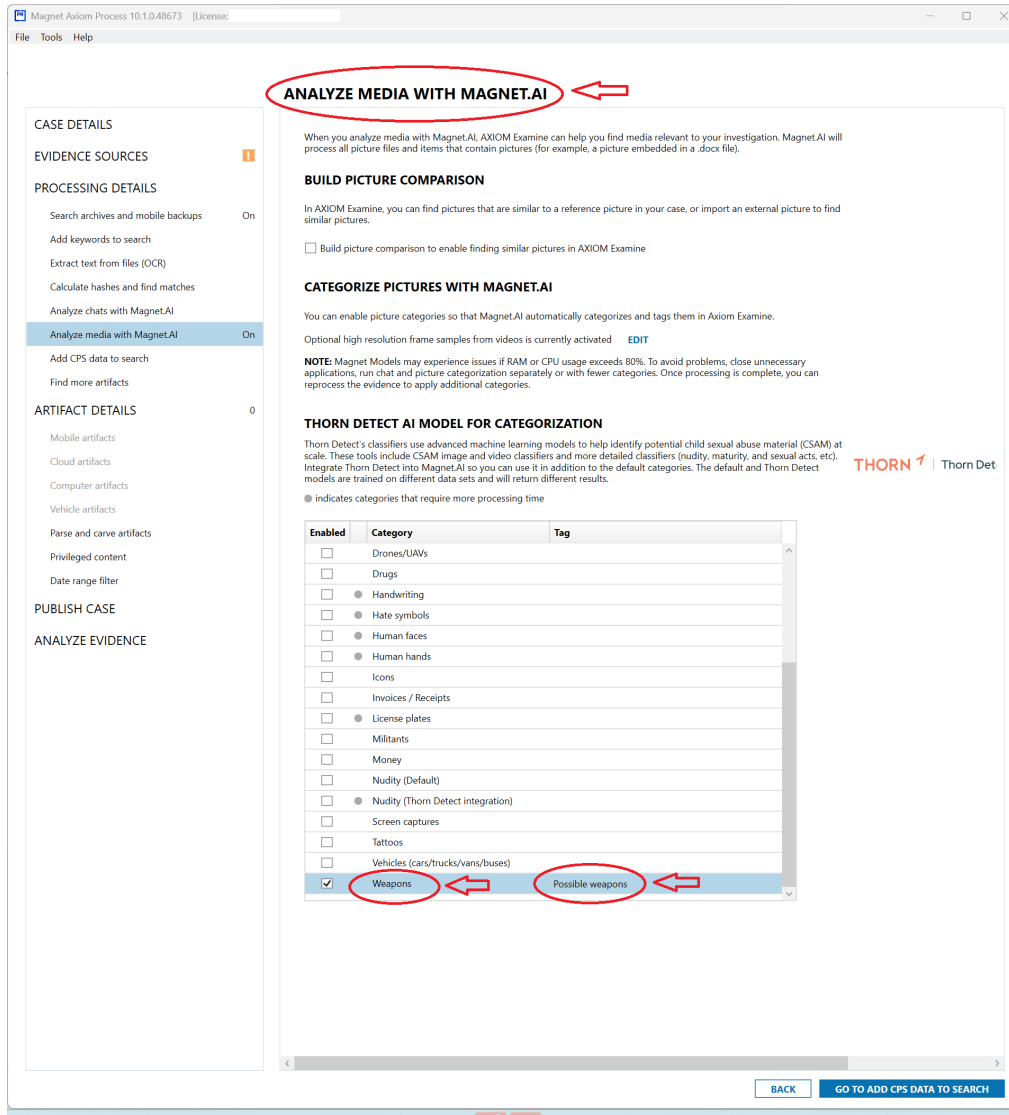


Figure 5.3: Operational view of the Magnet AXIOM/Magnet.AI media-categorization setup used in the black-box evaluation. The screenshot documents activation of the **Weapons** category and the corresponding observable **Possible weapons** tag

A full search was executed with Magnet.AI media analysis enabled and the **Weapons** category selected. Image-level observable decisions were obtained from the **Tags** field of the exported **Pictures.csv** file.

The canonical positive output was the **Possible weapons** tag. The normalization implementation also includes a compatibility rule for nonempty tag values containing the terms **weapon** or **weapons**. Empty tag fields are treated as unflagged images, whereas nonempty tag values that do not satisfy a weapon-oriented rule are preserved as unmapped outputs:

$$\text{weapon_detected} = \begin{cases} \text{true}, & \text{if Tags satisfies the weapon-oriented rule,} \\ \text{false}, & \text{if Tags is empty,} \\ \text{unknown}, & \text{otherwise.} \end{cases}$$

Within the common normalized output space, **weapon_detected = true** is mapped to **weapon**, **weapon_detected = false** is mapped to **non_weapon**, and an unmapped value remains

unknown. The defensive unknown state prevents an unrelated or previously unseen nonempty tag from being interpreted automatically as evidence of the absence of a weapon.

The case export produced five principal files:

- `Pictures.csv`;
- `ExportSummary.json`;
- `File System Information.csv`;
- `Identifiers - Device.csv`;
- `run_metadata.json`.

Only `Pictures.csv` was treated as a prediction-bearing record. The remaining files were retained for export auditing, case identification, software-version documentation, and execution traceability, but did not contribute additional image-classification decisions.

The prediction-bearing export contained exactly 11 500 rows. Every row was associated with one blind-bundle entry, providing complete output coverage without implying correctness or semantic completeness.

The raw AXIOM export includes file-system, hash, EXIF, device, and case metadata beyond the experimental mapping and is therefore controlled at `forensic_tools/magnet_axiom/raw_exports/FAIRLAB_AXIOM_RUN_02/`. The corresponding public decision extract is:

```
forensic_tools/magnet_axiom/public_extracts/magnet_axiom_predictions_extract.csv.
```

It contains one sanitized decision record for each bundle item.

The exported file name, available hash fields, and observable tag were matched against the canonical bundle metadata through the common normalization procedure described in Section 5.6.5. The quantitative results obtained from the normalized Magnet AXIOM output are reported in Section 6.6.1.

5.6.2 Magnet Griffeye and T3K CORE

Magnet Griffeye was evaluated together with the T3K CORE semantic-analysis module through automatically generated semantic bookmarks in the software export. No manual bookmark addition, removal, correction, or reassignment was permitted during the frozen evaluation.

The frozen software and case configuration is reported in Table 5.21.

Table 5.21: Frozen Magnet Griffeye and T3K CORE configuration

Configuration field	Frozen value
Forensic platform	Magnet Griffeye x64
Platform version	26.2.108
Semantic-analysis module	T3K CORE
Module version	1.18.0
Run identifier	FAIRLAB_GRIFFEYE_T3_RUN_01
Imported directory	<code>datasets/forensic_evaluation_bundle/blind_tool_input/files/</code>
Imported bundle files	11 500 neutrally identified images
Analysis functionality	Automatic T3K CORE semantic-bookmark generation
Prediction-bearing export	CSV media export
Observable decision field	Bookmarks

Figure 5.4 documents the concrete post-import configuration adopted for the frozen Griffeye run.

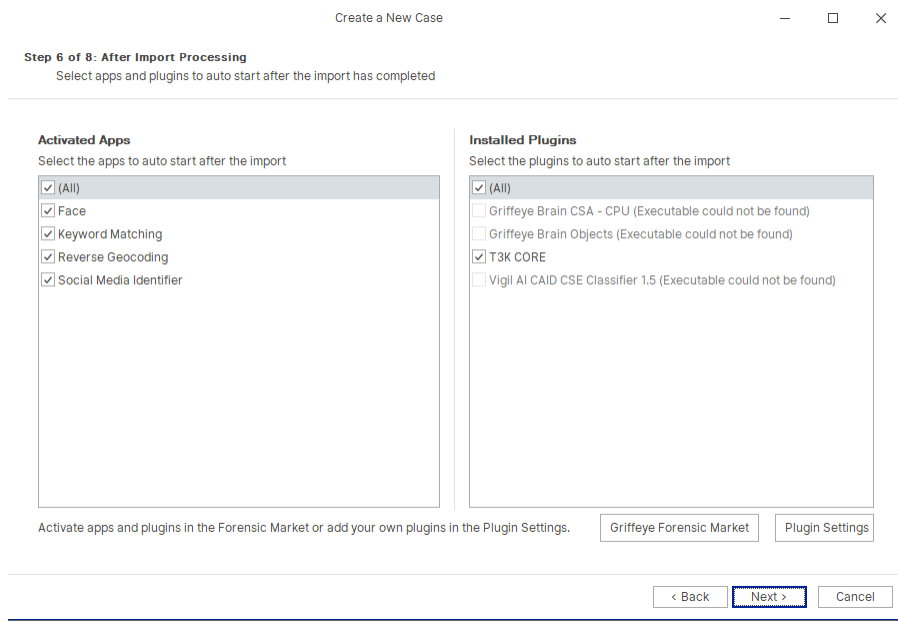


Figure 5.4: Operational setup of Magnet Griffeye/T3K CORE used during the black-box evaluation. The screenshot documents selection of the T3K CORE plugin for automatic post-import processing

After media ingestion, T3K CORE automatically generated semantic bookmarks. The exported **Bookmarks** field could contain one or more hierarchical labels assigned by the module. The primary operational rule adopted in this thesis was deliberately restricted to the firearm-specific **Firearm** bookmark:

$$\text{weapon_detected} = \begin{cases} \text{true}, & \text{if CORE/Violence/Firearm is present,} \\ \text{false}, & \text{otherwise.} \end{cases}$$

Within the common normalized output space, `weapon_detected = true` is mapped to `weapon`, whereas `weapon_detected = false` is mapped to `non_weapon`.

The following automatically generated bookmarks were preserved as secondary semantic indicators:

- CORE/Violence/Explosive Weapon;
- CORE/Violence/Bladed Weapon;
- CORE/Violence/Archery Weapon;
- CORE/Military/Military Equipment.

These bookmarks were intentionally excluded from the primary binary mapping. The positive operational class used in this study concerns firearm-oriented imagery. Combining firearm, bladed-weapon, explosive-weapon, archery, and general military categories into a single positive state would alter the experimental task. The secondary bookmarks were therefore retained for audit and qualitative analysis without contributing to the primary firearm-detection metrics.

The complete CSV export contained 12053 rows. Griffeye case exports may include records that do not correspond to images in the blind input corpus, such as evidence-workflow artifacts, unallocated-space entries, and system-generated or file-system records.

Before matching, the normalization implementation retained only records containing an identifier consistent with the `bundle_XXXXXX` naming convention. Records associated with known non-bundle artifacts, including unallocated-space and system-file entries, were excluded before hash- or file-name-based matching. Applying this filter before matching prevents an unrelated case record from being associated with a genuine bundle item through an incidental hash or path value.

This preliminary procedure excluded 553 non-bundle rows. The remaining 11 500 records were associated with the complete forensic evaluation bundle, and no bundle item remained without an observable exported decision state. This establishes complete matched-output coverage, not semantic correctness.

Only the automatically generated `Bookmarks` field was used to derive the prediction. Other case-management fields, manually assigned categories, and descriptive columns were not treated as model outputs. This restriction prevents analyst-generated annotations from being conflated with the observable behavior of T3K CORE.

The full export contains hashes, PhotoDNA-related values, paths, timestamps, EXIF, face-detection, case-management, and file-system fields and is controlled at `forensic_tools/griffeye/raw_exports/FAIRLAB_GRIFFEYE_T3_RUN_01/`.

The public decision extract is `forensic_tools/griffeye/public_extracts/griffeye_bookmarks_extract.csv`. It contains the neutral bundle identifiers, the primary firearm bookmark, selected secondary weapon-related bookmarks, and the corresponding normalized decision fields for all 11 500 bundle items.

The common normalization and output-matching implementation is described in Section 5.6.5. The experimental behavior observed for Magnet Griffeye and T3K CORE is reported in Section 6.6.2.

5.6.3 Excire Foto 2025

Excire Foto 2025 was evaluated as a standalone, general-purpose, AI-assisted semantic image-retrieval system used within a controlled forensic evaluation context. It was not treated as a native forensic suite or as an internally binary firearm classifier. Its observable behavior was assessed through the images returned by controlled text-based semantic queries.

The frozen software and retrieval configurations are summarized in Table 5.22.

Table 5.22: Frozen Excire Foto 2025 configurations

Setting	Run identifier	Raw hits	Bundle rows	Normalized configuration
D20	FAIRLAB_EXCIRE_D20_FIREARM_PROMPTS	20 329	11 500	excire_foto_2025_d20
D50	FAIRLAB_EXCIRE_D50_FIREARM_PROMPTS	32 076	11 500	excire_foto_2025_d50
D80	FAIRLAB_EXCIRE_D80_FIREARM_PROMPTS	43 736	11 500	excire_foto_2025_d80

Note. All configurations were executed with Excire Foto 2025 version 4.1.5. Raw hits are prompt-level memberships summed across the eight exported hit lists and are not counts of unique retrieved images.

Figure 5.5 documents the semantic retrieval interface used during the frozen Excire runs.

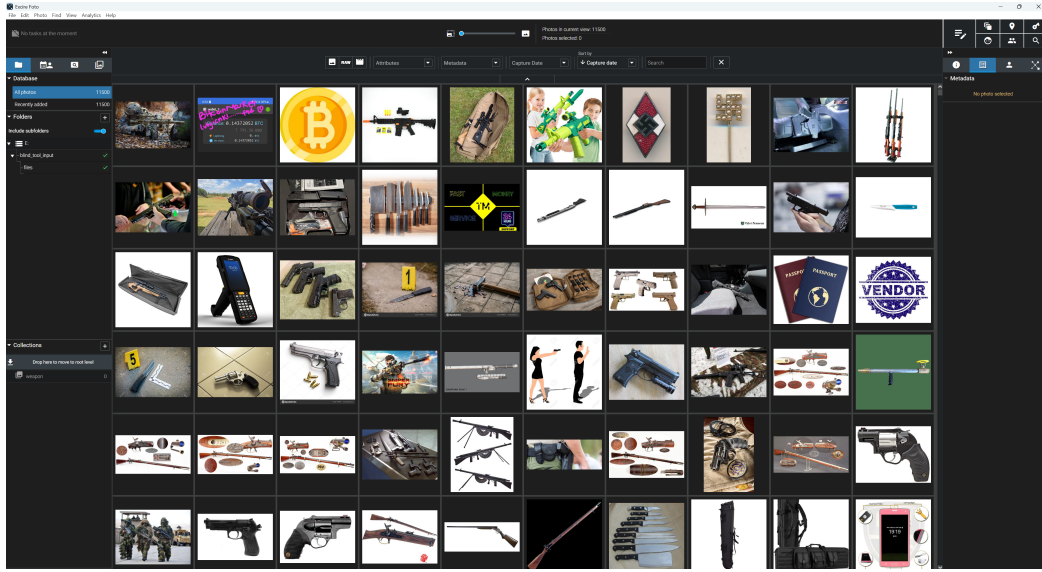


Figure 5.5: Operational view of Excire Foto 2025 during the black-box evaluation. The screenshot documents the blind image collection loaded into the software and the text-prompt retrieval interface used to obtain the observable outputs

The same frozen set of eight firearm-oriented English prompts was used for each semantic-distance configuration:

1. `firearm;`
2. `gun;`
3. `pistol;`
4. `handgun;`
5. `revolver;`
6. `rifle;`
7. `shotgun;`
8. `assault rifle.`

The prompt set was defined before the frozen evaluations and was not modified on the basis of the observed retrieval results. The broader term **weapon** was intentionally excluded because the experimental positive class was firearm-oriented rather than a general category encompassing all types of weapons.

Each prompt was executed separately under the semantic-distance limits 20, 50, and 80. This procedure produced:

$$8 \text{ fixed prompts} \times 3 \text{ distance settings} = 24 \text{ prompt-level export files.} \quad (5.8)$$

The export names follow the pattern `excire_<prompt>_distance<limit>.csv`. For file-system compatibility, the multiword prompt **assault rifle** is represented as **assault_rifle** in the corresponding file name. The prompt and distance configuration can therefore be reconstructed deterministically from each export name.

The Excire outputs are retrieval hit lists rather than complete classification tables. Each file contains the paths of the images returned for one prompt under one distance setting, generally as a headerless list with one path per line. An image may occur in multiple files when it is

retrieved by more than one prompt. Consequently, the raw hit totals reported in Table 5.22 represent prompt-level memberships and may exceed the number of unique positive images.

The normalizer reads each hit list deterministically, removes surrounding quotation marks, ignores an optional path-oriented header, extracts the file name from each exported path, and matches it against the neutral file names in the canonical bundle manifest. The original local directory component does not contribute to the experimental decision.

For a bundle image x , semantic-distance setting d , and frozen prompt set $\mathcal{P}$, the operational rule is:

$$\text{weapon_detected}(x; d) = \begin{cases} \text{true}, & \text{if } \exists p \in \mathcal{P} : \text{retrieved}(x, p, d), \\ \text{false}, & \text{otherwise.} \end{cases}$$

The corresponding common normalized prediction is **weapon** when **weapon_detected** = **true** and **non_weapon** otherwise. Images absent from all eight hit lists under the same distance setting are completed against the canonical 11 500-item bundle as operational negative decisions.

The distance limits are software-specific retrieval settings, not calibrated probabilities or thresholds comparable with those of another system.

The D20, D50, and D80 settings were retained as separate system configurations throughout normalization and metric computation. Their retrieval results were not combined into a single Excire decision. Each setting produced one complete table containing exactly 11 500 normalized predictions.

For auditability, each normalized record preserves the semantic-distance limit, the complete frozen prompt set, and the number and names of the prompts that retrieved the image. It also includes one binary membership indicator for each of the eight prompts, together with the corresponding bundle identifier, neutral input file name, and SHA-256 and MD5 values.

The record further retains the sample type, perturbation family, experimental condition, fold, reference operational assignment, source dataset, and source-image identifiers, as well as the resulting **weapon_detected** and normalized-prediction fields.

Preserving the individual prompt memberships prevents the union operation from discarding relevant retrieval information. Two images may receive the same positive normalized decision even when one is retrieved by a single prompt and the other by several semantically related prompts.

After union and bundle completion, the three configurations each contained one prediction per bundle item, producing 34 500 normalized Excire records with no unmatched item.

Because the 24 original CSV files contain local Windows paths, they are controlled at `forensic_tools/excire_foto_2025/raw_exports/`.

The public decision extract is `forensic_tools/excire_foto_2025/public_extracts/excire_prompt_hits_extract.csv`. It contains the bundle identifiers, semantic-distance configurations, prompt-membership information, and normalized decision fields for the three complete 11 500-item configurations.

The common normalization procedure is described in Section 5.6.5. The sensitivity of the observed retrieval behavior to the three semantic-distance settings is reported in Section 6.6.3.

5.6.4 Cellebrite Inseyets

Cellebrite Inseyets was evaluated through the classifications observable in the report exported by Cellebrite Physical Analyzer.

The frozen software and case configuration is reported in Table 5.23.

Table 5.23: Frozen Cellebrite Inseyets and Physical Analyzer configuration

Configuration field	Frozen value
Media-analysis platform	Cellebrite Inseyets
Platform version	10.9
Physical Analyzer version	10.9.0.3029
UFED component version	10.9.0.284
Run identifier	FAIRLAB_CELLEBRITE_INS EYETS_RUN_01
Imported directory	datasets/forensic_evaluation_bundle/blind_to ol_input/files/
Imported bundle files	11 500 neutrally identified images
Analysis functionality	Media and image classifica- tions
Prediction-bearing export	Physical Analyzer XLSX re- port
Prediction-bearing worksheet	Immagini
Observable decision field	Classifications
Raw report rows	11 829
Matched bundle rows	11 500
Non-bundle rows	329

The prediction-bearing record was an XLSX report exported through Physical Analyzer. The normalization procedure reads the **Immagini** worksheet and uses the exported **Classifications** field as the sole source of the operational prediction. Other report fields are retained only for matching or audit purposes and are not interpreted as additional model decisions.

A single image may be associated with multiple exported classifications, with each label optionally followed by a percentage value. Representative field values include **Armi (100%)**, **Pistola (100%)**; **Armi (100%)**, and **Fucile (91%) | Oggetto tenuto in mano (78%)**.

The parser separates labels expressed through semicolons, vertical bars, commas, or line breaks. Parenthesized percentage values are removed before the operational mapping is applied. The original observable classification string, including its percentages, is retained for auditability. These values are not treated as calibrated probabilities and do not influence the binary decision.

Let $\mathcal{C}(x)$ denote the set of normalized classification labels exported for image x , and let:

$$\mathcal{W}_{\text{Cellebrite}} = \{\text{Armi}, \text{Pistola}, \text{Fucile}\}.$$

The frozen operational mapping is:

$$\text{weapon_detected}(x) = \begin{cases} \text{true}, & \text{if } \mathcal{W}_{\text{Cellebrite}} \cap \mathcal{C}(x) \neq \emptyset, \\ \text{false}, & \text{otherwise.} \end{cases}$$

Within the common normalized output space, **weapon_detected = true** is mapped to **weapon**, whereas **weapon_detected = false** is mapped to **non_weapon**.

The inclusion of **Pistola** and **Fucile** alongside the broader **Armi** label defines the extended mapping used for the frozen evaluation. This rule prevents firearm-specific positive classifications from being discarded when the broader **Armi** category is absent from the same exported field.

The mapping does not assume that the exported labels are mutually exclusive or that Celebrite internally implements the same binary taxonomy.

The original report contained 11 829 image rows. Of these, 11 500 were matched to the complete forensic evaluation bundle, whereas 329 did not correspond to bundle items. The non-bundle records were preserved in the local unmatched-output artifact and summarized by tool in the public normalization report. They were excluded from the canonical bundle-level prediction table and from the quantitative robustness analysis.

The resulting canonical output contains one matched decision for each bundle item; the shared normalized schema preserves the original `Classifications` value and the mapping reason.

Because the complete XLSX report also contains file-system, evidence, and case metadata, it is controlled at `forensic_tools/cellebrite_inseyets/raw_exports/FAIRLAB_CELLEBRIT E_INSEYETS_RUN_01/`.

The public decision extract is `forensic_tools/cellebrite_inseyets/public_extracts /cellebrite_classifications_extract.csv`. It contains the neutral bundle identifiers, experimental-condition fields, observable classification values, and normalized decision fields for all 11 500 matched bundle items. Local paths and the 329 non-bundle report rows are not included in this public extract.

The common matching, normalization, completeness, and deduplication procedures applied across the selected black-box software systems are described in Section 5.6.5. The resulting behavior of Celebrite Inseyets is reported in Section 6.6.4.

5.6.5 Output Matching and Normalization

The heterogeneous outputs produced by the selected black-box software systems were converted into a common sample-level representation through the canonical entry point `evaluation/scripts/19_normalize_forensic_ai_tool_predictions.py`. The entry point acts as a controlled-output wrapper around the preserved normalization implementation. It adds overwrite protection, field minimization, explicit preservation of unmatched records, complete version metadata, strict JSON serialization, and optional generation of separate per-configuration prediction files.

The normalization implementation reads the controlled-access exports associated with the frozen runs and associates their observable records with the forensic evaluation bundle. Its primary bundle inputs are `datasets/forensic_evaluation_bundle/metadata/bundle_manifest.csv` and `datasets/forensic_evaluation_bundle/metadata/bundle_hashes_sha256.csv`.

These records provide the association among neutrally identified software-input files, cryptographic hashes, reference operational assignments, folds, sample types, perturbation conditions, source-image identifiers, and generated-image identifiers. Reference assignments and experimental metadata are joined to the observable software outputs only after export and were not supplied to the evaluated systems during their analysis.

The normalization pipeline processes four software families while retaining the three Excire Foto 2025 semantic-distance settings as distinct operational configurations:

- `magnet_axiom;`
- `griffeye;`
- `excire_foto_2025_d20;`
- `excire_foto_2025_d50;`
- `excire_foto_2025_d80;`
- `cellebrite_inseyets.`

Each configuration is evaluated against the same 11 500-item bundle. The complete normalized population therefore contains:

$$6 \text{ configurations} \times 11\,500 \text{ bundle items} = 69\,000 \text{ matched decisions.} \quad (5.9)$$

Bundle indexing and record matching. When the bundle metadata is loaded, the implementation constructs three lookup indexes:

1. a SHA-256-based index;
2. an MD5-based index;
3. a neutral file-name index.

For conventional row-oriented exports, each parsed software record is matched according to the following priority:

```
SHA-256 > MD5
    > exact file name
    > file name embedded in an exported path or text field.
```

The selected matching method is retained in the normalized record. This permits subsequent verification of whether an association was established through the primary cryptographic identifier or through a fallback identifier exposed by the software export.

Hash values are normalized before comparison by removing separators and non-hexadecimal formatting characters. Exported paths are reduced to their base file names before lookup so that differences between Windows paths and repository-relative paths do not prevent association with the neutral bundle files.

The Griffeye export is filtered before matching to exclude unallocated-space, system-file, and other non-bundle records. The Excire workflow instead requires a bundle-completion procedure because its exports contain only the images retrieved by each semantic query. Neutral file names found in the eight prompt-level hit lists are associated with bundle items, and the union of the retrieval results is subsequently completed against the full bundle manifest. This produces an explicit operational decision for every bundle item, including images absent from all prompt exports.

Software-specific observable recoding. The observable fields exposed by the software systems are converted into a common `weapon_detected` state according to the frozen rules summarized in Table 5.24.

Table 5.24: Frozen mappings from observable software outputs to normalized decisions

Configuration	Observable source	Operational mapping
Magnet AXIOM/Magnet.AI	Pictures.csv: Tags	Possible weapons, or a compatible nonempty weapon-oriented tag, is mapped to <code>true</code> . An empty tag is mapped to <code>false</code> ; an unrecognized nonempty tag is mapped to <code>unknown</code> .
Magnet CORE	Griffeye/T3K CSV: Bookmarks	The automatically generated bookmark <code>CORE/Violence/Firearm</code> is mapped to <code>true</code> ; its absence is mapped to <code>false</code> . Other weapon-related bookmarks remain secondary semantic indicators.
Excire Foto 2025 D20, D50, and D80	Eight prompt-level re- trieval hit lists per config- uration	Retrieval by at least one frozen firearm-oriented prompt under the corresponding distance setting is mapped to <code>true</code> ; absence from all eight lists is mapped to <code>false</code> .
Cellebrite Inseyets	XLSX, worksheet Immagini: Classifications	A classification set containing at least one among <code>Armi</code> , <code>Pistola</code> , or <code>Fucile</code> is mapped to <code>true</code> ; all other matched bundle records are mapped to <code>false</code> .

The resulting common prediction is assigned as:

$$\text{normalized_prediction} = \begin{cases} \text{weapon}, & \text{if } \text{weapon_detected} = \text{true}, \\ \text{non_weapon}, & \text{if } \text{weapon_detected} = \text{false}, \\ \text{unknown}, & \text{otherwise.} \end{cases}$$

The **unknown** state prevents an unsupported or unrecognized observable value from being interpreted silently as a negative result. **Unmapped** describes the outcome of a software-specific parsing rule, whereas **unknown** is the corresponding value in the common normalized schema. The frozen official configurations produced no unknown decisions in the final 69 000-row population.

Controlled normalized schema. During controlled local execution, the normalized representation separates software-observable information from the reference assignments and experimental metadata recovered from the bundle. Its principal fields include **tool_name**, **bundle_id**, **matched**, and **match_method**, together with **sample_type**, **attack_family**, **attack_name**, and **attack_target_model**. The record also preserves the experimental **fold**, **final_label**, **source_dataset**, **original_image_id**, and **generated_image_id**, as well as the neutral input file name and the available SHA-256 and MD5 values.

The original software-observable tag, classification, bookmark, or prompt-membership information is retained alongside **weapon_detected**, **normalized_prediction**, and **mapping_reason**. When the reference operational assignment is **weapon** or **non_weapon**, the normalized representation also includes binary correctness, false-negative, and false-positive indicators.

The literal field name **final_label** is retained because it belongs to the implemented schema. Semantically, its value represents the reference operational assignment inherited from the frozen bundle.

Software-specific information is preserved in dedicated fields rather than being discarded after binary recoding. These fields include the original AXIOM tags, Cellebrite classifications, Excire distance limit and per-prompt membership indicators, and the primary and secondary Griffeye bookmarks.

For records whose reference operational assignment is **ood**, the binary correctness, false-negative, and false-positive fields remain undefined. An OOD input is not treated as either a positive or negative binary reference instance. Its normalized detection state is preserved separately for calculation of the OOD weapon-flag rate.

Deduplication and completeness. The normalizer supports deterministic consolidation to a single row for each normalized-configuration and bundle-identifier pair. When multiple matched records refer to the same pair, the consolidated decision follows a fixed precedence rule: it is set to **true** if at least one row contains a positive decision; otherwise, it is set to **false** if at least one row contains an explicit negative decision. If neither condition is satisfied, the consolidated decision is assigned as **unknown**.

The consolidation procedure preserves the combined export references, row numbers, observable labels, mapping reasons, and contributing-row count. When multiple numerical values are exposed by the same software for the consolidated records, the maximum value may be retained as part of the original observable information. It is not converted into a cross-system calibrated confidence measure, and no numerical confidence is fabricated for systems that do not expose one.

Unmatched records are excluded from the canonical matched population because they do not correspond to official bundle items. They are preserved separately in the controlled local output and summarized in the public normalization report.

After the preliminary exclusion of the 553 non-bundle Griffeye records, the normalization workspace contained 69 329 candidate records. This population comprised 69 000 matched bundle-level decisions and 329 unmatched Cellebrite report records:

$$69\,329 - 329 = 69\,000 \text{ matched decisions.} \quad (5.10)$$

Deduplication was enabled in the frozen execution. The matched population already contained one decision for every configuration–bundle pair, and its size therefore remained 69 000 rows after consolidation. Every configuration covered all 11 500 bundle items, and no matched decision remained in the `unknown` state.

Complete coverage establishes the availability of an observable and matched decision for every bundle item. It does not establish classification correctness.

Controlled and public output records. The controlled normalization execution produces the complete normalized prediction workspace, a separate table of unmatched records, the export audit, the software-version log, and the normalization summary. It also generates the per-configuration and aggregate metric tables.

Separate files following the pattern `evaluation/forensic_tools/<tool_name>_normalized_predictions.csv` are optional and are generated only when the `-write-per-tool-files` option is supplied. They are not produced by the default canonical execution.

For public distribution, four minimized software-specific extracts are built from the validated local normalization output. These extracts retain neutral bundle identifiers, experimental-condition fields, the observable signal needed to audit each software-specific mapping, and the resulting normalized decision. They exclude raw local paths, unrelated case fields, device identifiers, and other proprietary-export-derived metadata.

The repository-facing canonical table `evaluation/forensic_tools/normalized_predictions.csv` is subsequently reconstructed from these four validated extracts through `forensic_tools/scripts/build_canonical_normalized_predictions.py`. It contains all 69 000 normalized decisions but does not reproduce the complete controlled local workspace.

Table 5.25 summarizes the principal normalization, audit, and public-validation records.

Table 5.25: Principal normalization, audit, and validation records

Record	Role
<code>evaluation/forensic_tools/normalized_predictions.csv</code>	Canonical sanitized table containing 69 000 matched decisions reconstructed from the four validated public extracts.
<code>evaluation/forensic_tools/unmatched_predictions.csv</code>	Controlled local record of export rows that could not be associated with official bundle items; the frozen run contains 329 Cellebrite records.
<code>evaluation/forensic_tools/tool_export_audit.csv</code>	Documents inspected export files, prediction-bearing status, parsed-row counts, excluded non-bundle records, and parsing errors.
<code>evaluation/forensic_tools/tool_version_log.csv</code>	Records software versions, run identifiers, selected export directories, enabled functionality, import path, export status, and normalization notes.
<code>evaluation/forensic_tools/normalization_summary.json</code>	Records selected runs, raw and normalized row counts, unmatched-record counts, output configurations, deduplication state, and metric-output validation.
<code>evaluation/forensic_tools/normalized_predictions_public_summary.json</code>	Documents the canonical sanitized table, its row count, schema, source extracts, and SHA-256 digest.
Four software-specific files under <code>forensic_tools/*/public_extracts/</code>	Minimized observable-signal and decision records used to reconstruct the canonical sanitized table.
<code>forensic_tools/public_extracts_validation.json</code>	Verifies decision-level and metric-level equivalence between the sanitized public extracts and the controlled local normalization results.
<code>forensic_tools/run_registry.json</code>	Provides the canonical registry of software versions, run identifiers, observable fields, mappings, controlled raw-export locations, and public extracts.

The public schemas `evaluation/forensic_tools/normalized_predictions.schema.csv` and `evaluation/forensic_tools/unmatched_predictions.schema.csv` document the prediction-level formats independently of the controlled local records.

The validation process confirms that the public extracts preserve all 69 000 normalized decisions and reproduce all 186 frozen black-box software metric rows without differences. The corresponding extract hashes are recorded in the validation artifact.

Canonical outputs are protected against accidental replacement. Existing normalization predictions, unmatched records, and metric outputs require the explicit `-force` option before regeneration. A partial software selection cannot write to the canonical output directories; partial or diagnostic runs must use alternative output locations.

This distribution boundary preserves auditability and reproducibility while avoiding publication of complete proprietary software exports, unrelated forensic fields, local execution paths, and controlled case metadata. The computation of the corresponding binary and OOD metrics is described in Section 5.7.1.

5.7 Metric Computation and Explainability Implementation

5.7.1 Metric Computation and Evaluation Artifacts

Proxy-model outputs and normalized black-box software decisions were processed through two separate computational pipelines. This separation preserves the distinction between predictions obtained from fully accessible models and operational decisions reconstructed from observable commercial-software exports.

Proxy-model predictions and metrics were generated through `evaluation/scripts/15_evaluate_proxy_models.py`. Black-box software metrics were generated as part of the normalization pipeline implemented through `evaluation/scripts/19_normalize_forensic_ai_tool_predictions.py`.

Both pipelines use the same binary reference classes:

positive class = `weapon`, negative class = `non_weapon`.

The OOD branch is handled separately and does not contribute to the denominators of binary accuracy, precision, recall, F1-score, False Positive Rate (FPR), or False Negative Rate (FNR).

Proxy-model prediction generation. The proxy-model evaluation pipeline consumes the clean-fold manifest, the separate OOD evaluation manifest, the five canonical adversarial-generation manifests, the anti-forensic transformation manifest, and the 15 fold-specific proxy-model checkpoints.

The public evaluation entry point wraps the preserved implementation with additional integrity controls. Before inference, it verifies that the loaded population contains exactly 1 000 clean binary inputs, 500 OOD inputs, 5 000 adversarial artifacts, and 5 000 anti-forensic artifacts. Manifest discovery is restricted to the five frozen adversarial conditions rather than accepting every file matching a generic naming pattern.

The wrapper verifies the existence and manifest-recorded SHA-256 value of every evaluated image. It also recomputes and validates the SHA-256 digest of each of the 15 checkpoints against `models/model_registry.json`. Missing inputs, digest mismatches, or inference errors terminate the canonical execution.

A run using fewer than all three architectures, a limited input population, or a single-fold OOD mode cannot write to the canonical output locations. Such partial executions require an explicit diagnostic output directory.

Each clean or perturbed binary sample is evaluated using the checkpoint associated with its assigned fold. Consequently, the checkpoint used for prediction was trained while excluding the fold containing the corresponding source image.

For each evaluated input, the implementation records the evaluated architecture and fold-specific checkpoint, together with whether the sample is clean, OOD, or perturbed. It also preserves the perturbation family, condition, and attack target model, as well as the original and generated image identifiers, the assigned fold, the reference operational assignment, the image path, and the recomputed SHA-256 and MD5 values.

The resulting evaluation record includes the predicted class, the maximum predicted-class probability, and the probabilities assigned to `weapon` and `non_weapon`. For binary samples, it additionally stores the probability assigned to the reference class, together with the correctness state and any error encountered during evaluation.

The complete prediction-level record is stored in `evaluation/proxy_models/proxy_model_predictions.csv`. Before inference, every image is opened through the EXIF-aware RGB-loading procedure used elsewhere in the pipeline, and the fold-aware adapter applies the architecture-specific preprocessing defined by the model implementation.

The evaluation population contains 11 500 distinct experimental inputs:

$$\begin{aligned}
 &1\,000 \text{ clean binary inputs} + 500 \text{ OOD inputs} \\
 &\quad + 5\,000 \text{ adversarial artifacts} \\
 &\quad + 5\,000 \text{ anti-forensic artifacts} \\
 &= 11\,500 \text{ inputs.}
 \end{aligned} \tag{5.11}$$

Clean and perturbed binary samples are evaluated once by each of the three proxy architectures, using the checkpoint associated with their assigned fold:

$$1\,000 \times 3 = 3\,000 \text{ clean prediction rows,} \tag{5.12}$$

and:

$$10\,000 \times 3 = 30\,000 \text{ perturbed prediction rows.} \tag{5.13}$$

The OOD branch follows a different protocol. Each of the 500 unique OOD inputs is evaluated by all five fold-specific checkpoints of each architecture:

$$500 \times 5 \times 3 = 7\,500 \text{ OOD prediction rows.} \tag{5.14}$$

The complete proxy-model prediction table therefore contains:

$$3\,000 + 30\,000 + 7\,500 = 40\,500 \text{ prediction rows.} \tag{5.15}$$

The 2 500 OOD responses associated with each architecture correspond to 500 unique images processed by five different fold-specific checkpoints. They must not be interpreted as 2 500 distinct or statistically independent out-of-distribution samples.

Binary metric computation. For each binary evaluation group, the confusion matrix uses the fixed layout:

$$\mathbf{C} = \begin{bmatrix} \text{TN} & \text{FP} \\ \text{FN} & \text{TP} \end{bmatrix}. \tag{5.16}$$

The resulting metric records contain accuracy and balanced accuracy, together with class-specific precision, recall, and F1-score and their corresponding macro-averaged values. They also report the True Positive (TP), True Negative (TN), False Positive (FP), and False Negative

(FN) counts, the FPR and FNR, and the overall misclassification rate. Since all evaluated binary groups contain one interpretable prediction for every sample, the misclassification rate is equal to $1 - \text{accuracy}$. It is retained in the consolidated records for schema completeness and automated validation but is not reported separately in Chapter 6, where accuracy and the direction-specific error measures provide the same information in a more operationally interpretable form.

In addition, the records preserve the directional **weapon-to-non_weapon** and **non_weapon-to-weapon** error counts, as well as the mean maximum predicted-class probability.

The FPR and FNR are implemented as:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad \text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}}. \quad (5.17)$$

Clean metrics are grouped by proxy architecture. Adversarial-stage metrics are grouped by evaluated architecture, perturbation condition, and attack target model. Anti-forensic metrics are grouped by evaluated architecture and transformation condition.

The attack-target field is retained because the model-dependent adversarial artifacts were generated against fold-aware EfficientNet-B0 checkpoints but were subsequently evaluated by all three proxy architectures. This distinction supports separate analysis of direct target-model behavior and cross-model transfer. Color Shift remains marked as model-agnostic and is not interpreted as a transfer attack.

The maximum predicted-class probabilities stored by the evaluation pipeline are used only as intra-model diagnostic values. They are not assumed to be calibrated across architectures and are not used as directly comparable confidence measures between ResNet18, EfficientNet-B0, and the CLIP-based proxy.

Clean-to-perturbed comparison. Each perturbed prediction is associated with its corresponding clean prediction through the tuple:

`(evaluated_model, evaluation_fold, original_image_id).`

The perturbed prediction record is enriched with the clean prediction, clean correctness state, clean class probabilities, clean maximum predicted-class probability, and probability difference relative to the perturbed input.

For each proxy architecture and perturbation condition, the comparative layer records clean and perturbed accuracy, the schema field **robust_accuracy**, and the corresponding accuracy drop. The stored value of **robust_accuracy** is identical to the perturbed-condition accuracy. It is retained for schema compatibility with the evaluation pipeline and does not represent an additional robustness metric; consequently, it is not reported separately in Chapter 6.

The comparative layer also computes the clean and perturbed macro F1-scores and their difference. The macro-F1 difference is preserved as a secondary descriptive indicator in the consolidated records. The results chapter prioritizes accuracy degradation, false-negative and false-positive behavior, and clean-correct decision transitions because these quantities provide a more direct operational interpretation for forensic triage. The macro-F1 difference is therefore not reported as a separate primary robustness measure.

The resulting comparative record further includes the induced-error count and rate, the schema field **attack_success_rate**, the shift in mean maximum predicted-class probability, and both the counts and rates of the directional **weapon-to-non_weapon** and **non_weapon-to-weapon** transitions.

Chapter 6 reports the directional transition counts together with the overall induced-error rate or attack success rate. The corresponding direction-specific rates remain available in the

consolidated records but are not tabulated separately because the frozen binary evaluation groups are class-balanced and have fixed denominators. The counts therefore provide an immediate operational interpretation of newly missed target images and additional false-positive review workload. The normalized directional rates remain relevant for machine-readable comparison and for possible future evaluations with unequal class distributions.

Let $\mathcal{C}_{\text{clean}}$ denote the set of samples correctly classified under the corresponding clean condition. The induced-error rate for perturbation condition p is:

$$\text{IER}_p = \frac{\left| \left\{ i \in \mathcal{C}_{\text{clean}} : \hat{y}_i^{(p)} \neq y_i \right\} \right|}{|\mathcal{C}_{\text{clean}}|}. \quad (5.18)$$

Samples already misclassified in the clean condition are excluded from the denominator because they cannot represent newly induced errors.

The implementation stores this clean-correct transition rate in the `attack_success_rate` schema field for every perturbation group. It is interpreted as attack success rate only for FGSM, One Pixel, Sigma-Zero, and SuperDeepFool. For Color Shift and the five anti-forensic transformations, the same numerical transition is reported and interpreted as an induced-error rate, not as adversarial or transfer success.

The maximum-probability shift is computed as:

$$\Delta c = c_{\text{perturbed}} - c_{\text{clean}}, \quad (5.19)$$

and averaged within each architecture-condition group. This quantity compares the maximum probability assigned to the predicted class before and after perturbation. If the predicted class changes, it does not compare the probability assigned to the same semantic class. It is therefore retained only as an intra-model diagnostic.

Proxy-model OOD metrics. The OOD branch is summarized independently for each proxy architecture. The resulting records report the fold-aggregated number of checkpoint responses, the number and rate of predictions assigned to `weapon`, and the number assigned to `non_weapon`. They also preserve the mean maximum predicted-class probability, the adopted high-Max-P threshold, and the number and rate of responses whose maximum predicted-class probability meets or exceeds that threshold.

The frozen threshold is:

$$\tau_{\text{high}} = 0.90. \quad (5.20)$$

The high-Max-P count includes responses assigned to either binary class and characterizes closed-set assignments whose maximum predicted-class probability meets or exceeds 0.90. The predicted-weapon rate instead characterizes the tendency to flag OOD content as firearm-related.

Neither quantity represents supervised correctness because the OOD branch has no positive or negative binary reference assignment. The associated probabilities remain architecture-specific diagnostic outputs and are not treated as calibrated across proxy models.

Proxy-model evaluation artifacts. Table 5.26 lists the principal prediction and metric records generated by the proxy-model evaluation stage.

Table 5.26: Proxy-model prediction records and metric outputs

Output	Role
evaluation/proxy_models/proxy_model_predictions.csv	Sample-level predictions, probabilities, fold associations, integrity values, clean-to-perturbed links, and evaluation errors.
results/metrics/proxy_model_clean_metrics.csv	Architecture-level clean binary metrics.
results/metrics/proxy_model_adversarial_metrics.csv	Binary metrics grouped by evaluated architecture, adversarial-stage condition, and attack target model.
results/metrics/proxy_model_anti_forensic_metrics.csv	Binary metrics grouped by evaluated architecture and anti-forensic transformation.
results/metrics/proxy_model_ood_metrics.csv	Fold-aggregated OOD prediction, weapon-rate, mean-maximum-probability, and high-Max-P statistics.
results/metrics/proxy_model_comparative_metrics.csv	Clean-to-perturbed accuracy, F1-score, induced-error, directional-error, and maximum-probability-shift comparisons.
results/metrics/proxy_model_evaluation_summary.json	Models, output paths, row counts, evaluation-error count, metric-table counts, and schema notes.

The stage also generates four schema-stable reporting tables:

- results/metrics/final_core_metrics.csv;
- results/metrics/final_robustness_metrics.csv;
- results/metrics/final_confusion_matrices.csv;
- results/metrics/final_ood_metrics.csv.

The core table contains 33 binary-condition rows:

$$3 \text{ clean architecture rows} + 15 \text{ adversarial-stage rows} + 15 \text{ anti-forensic rows} = 33 \text{ rows. (5.21)}$$

The 15 adversarial-stage rows comprise three evaluated architectures under the four model-dependent attacks and the model-agnostic Color Shift condition.

The robustness table contains:

$$3 \text{ architectures} \times 10 \text{ perturbation conditions} = 30 \text{ clean-to-perturbed comparison rows. (5.22)}$$

The confusion-matrix table contains one record for each of the 33 binary condition rows, while the final OOD table contains one record for each of the three proxy architectures. The frozen evaluation completed all 40 500 prediction operations without recorded inference errors.

Black-box software metric computation. Black-box software metrics are computed from the normalized decisions described in Section 5.6.5. Only matched binary records with an interpretable **true** or **false weapon_detected** value contribute to the binary confusion matrix. Records assigned to **unknown** are excluded from binary denominators and reported separately.

The black-box metric layer computes accuracy and balanced accuracy, together with precision, recall, and F1-score for the **weapon** class and specificity for the **non_weapon** class. It also reports the FPR, FNR, and the corresponding TP, TN, FP, and FN counts.

The evaluation further records the number and rate of **unknown** decisions. For the separate OOD branch, it reports the number and rate of inputs flagged as **weapon**, together with the number assigned to **non_weapon** or **unknown**.

Binary metrics are computed only for records whose reference operational assignment is **weapon** or **non_weapon**. The OOD weapon-flag rate uses the complete matched OOD population of the corresponding configuration as its denominator. This rate characterizes observable flagging behavior and is not interpreted as supervised accuracy or error.

Metrics are generated at multiple aggregation scopes, including the complete normalized configuration, the sample type, the perturbation family, the individual perturbation condition, and the combined sample-type and perturbation-condition level.

Each of the six normalized configurations produces 31 metric rows:

$$6 \times 31 = 186 \text{ black-box software metric rows.} \quad (5.23)$$

The canonical consolidated record is `results/metrics/forensic_tools_metrics.csv`. Corresponding per-configuration files follow the pattern `results/metrics/<tool_name>_metrics.csv`.

The normalizer verifies that every per-configuration file contains the expected number of rows and that the consolidated record was generated from the same in-memory metric population. The frozen validation confirmed consistency across all 186 rows.

Canonical reporting layer. CSV and JSON metric records remain the canonical sources for numerical reporting. Figures and thesis-ready tabular extracts are generated from these records rather than edited manually.

The downstream reporting entry point is `results/scripts/20_generate_experimental_reporting_assets.py`. It reads the consolidated proxy-model and black-box metric files and generates presentation-layer assets without recreating datasets, perturbations, predictions, or metrics.

The frozen reporting execution generated PDF and PNG outputs at 300 dpi, together with a manifest and summary under `results/figures/chapter_5/`. It produced 24 unique figure or table identifiers and 41 materialized files.

The reporting layer intentionally contains a broader set of presentation assets than those selected for inclusion in Chapter 6. Some generated figures and tabular extracts provide alternative views of the same consolidated metrics and are retained as auditable reporting artifacts without being referenced by the final thesis text. Their absence from the chapter therefore does not indicate missing experimental results. The reporting-asset audit distinguishes assets referenced by the authoritative L^AT_EX source from generated but unselected assets and verifies the availability and integrity of the corresponding files.

The directory name `chapter_5` and the corresponding manifest identifiers are historical frozen names established before the final thesis reorganization. They are preserved for artifact identity and reproducibility, although the final experimental results are reported in Chapter 6.

The complete metric layer therefore preserves three distinct levels:

1. sample-level proxy-model predictions or normalized black-box software decisions;
2. condition-level and comparative metric records;
3. presentation-layer figures and thesis-ready tables.

Maintaining this separation allows each value reported in Chapter 6 to be traced back to its corresponding prediction record, frozen input manifest, fold-specific checkpoint, or observable black-box software output.

5.7.2 Integrated Gradients Implementation

The explainability layer was implemented using IG, following the method introduced by Sundararajan et al. [66]. The canonical entry point is `explainability/scripts/17_generate_integrated_gradients_case_studies.py`, which wraps the preserved implementation in `explainability/scripts/_17_generate_integrated_gradients_case_studies_impl.py`.

The wrapper adds headless-safe execution, checkpoint and input-integrity verification, adaptive convergence diagnostics, pseudonymous reviewer identifiers, and run-specific output cleanup.

Interactive manual review uses a graphical backend, whereas noninteractive attribution generation uses a headless-safe backend.

The xAI workflow is restricted to transparent proxy models. It is not used to explain Magnet.AI, T3K CORE, Excire Foto 2025, or Celebrite Inseyets, because the gradients, internal representations, and model parameters required by IG are not accessible for these black-box software systems.

The prediction-level source used for case selection is `evaluation/proxy_models/proxy_model_predictions.csv`. This record provides the prediction, maximum predicted-class probability, fold, reference operational assignment, perturbation condition, clean-to-perturbed association, and input path required to identify diagnostically relevant cases.

The frozen thesis workflow applies IG exclusively to EfficientNet-B0, which was also the primary proxy target used for model-dependent adversarial generation. The corresponding attribution maps must therefore be interpreted as evidence concerning the transparent EfficientNet-B0 proxy rather than as explanations of the selected black-box software systems.

For each selected case, the implementation loads the checkpoint associated with the input's assigned fold. The hardened entry point computes the checkpoint SHA-256 digest and compares it with the canonical value recorded in `models/model_registry.json`. Generation terminates when the checkpoint is missing, when no expected digest is available, or when the computed and recorded digests differ.

The input-image SHA-256 value is also verified whenever an expected digest is available in the prediction or perturbation metadata. The image is then opened through the common EXIF-aware RGB-loading procedure and processed using the same EfficientNet-B0 transformation applied during proxy-model evaluation.

The resulting model-input tensor is passed to the Captum implementation of IG. The baseline is generated as a tensor of zeros having the same shape as the preprocessed input:

$$\mathbf{x}' = \mathbf{0} \quad \text{in normalized model-input space.}$$

This baseline does not represent an independently stored reference image and must not be interpreted as a black RGB image in the original pixel space.

The frozen thesis cases use the proxy model's predicted class as the attribution target. This configuration produces an attribution map for the output actually selected by the model, including incorrect binary predictions and OOD assignments. The implementation also supports attribution to the reference binary class or to both binary outputs, but these alternatives were not used for the retained thesis assets.

Table 5.27 summarizes the frozen configuration used to generate the historical thesis figures.

Table 5.27: Frozen Integrated Gradients configuration used for the thesis assets

Configuration field	Frozen value
Proxy architecture	EfficientNet-B0
Selection strategy	Thesis-oriented diagnostic case selection
Attribution target	Predicted binary class
Baseline	Zero tensor in the normalized, preprocessed input space
Historical integration steps	32
Input resolution	224×224 pixels
High-Max-P selection threshold	0.90
Attribution-mask percentile	90th percentile, corresponding approximately to the image locations with the highest 10% of attribution magnitudes
Reviewer identifier	Pseudonymous identifier <code>reviewer_01</code>
Historical convergence-validation status	<code>pending_adaptive_regeneration</code>

The repository path `explainability/manifests/chapter5/` retains a historical directory name created before the final reorganization of the thesis chapters. The path identifies the frozen selection and provenance records and does not indicate the chapter in which the qualitative results are currently presented. The resulting attribution analyses are discussed in Section 6.8.

Case-selection procedure. The thesis-oriented selection procedure organizes the prediction records into five diagnostic categories. These comprise a correctly classified clean **weapon** image, a clean **weapon** false negative, and a high-Max-P OOD input assigned to **weapon**. The remaining categories include an image that is correctly classified in its clean state but rendered incorrect by an anti-forensic transformation, and an image that is correctly classified in its clean state but rendered incorrect with a high maximum predicted-class probability under an adversarial condition.

Candidates are ordered deterministically using the maximum predicted-class probability, fold, and image path. For perturbation-induced failures, the ranking prioritizes **weapon-to-non_weapon** transitions, followed by **non_weapon-to-weapon** transitions and other clean-correct-to-incorrect transitions.

The frozen selection summary records 1 175 candidate records across the five buckets. Human review retained three cases per bucket, producing a 15-case intermediate selection. The reviewer inspected the image together with its fold, reference operational assignment, prediction, maximum predicted-class probability, perturbation condition, and selection rationale.

Selection and deselection actions were logged together with timestamps, ranks, case identifiers, and the pseudonymous reviewer identifier. A subsequent thesis-level reduction purposively retained one case from each diagnostic bucket.

Table 5.28 summarizes both the candidate population and the final five-case selection.

Table 5.28: Canonical Integrated Gradients case-selection process

Diagnostic bucket	Candidates	Reviewed selection	Final thesis case and role
Clean correct weapon	492	3	<code>xai_case_0001</code> : correctly classified clean <code>weapon</code> reference case.
Clean weapon false negative	8	3	<code>xai_case_0006</code> : clean <code>weapon</code> input classified as <code>non_weapon</code> .
High-Max-P OOD as weapon	139	3	<code>xai_case_0009</code> : high-Max-P OOD assignment to <code>weapon</code> .
Anti-forensic failure	36	3	<code>xai_case_0010</code> : clean-correct image rendered incorrect by Histogram Modification.
Adversarial high-Max-P failure	500	3	<code>xai_case_0015</code> : high-Max-P induced failure associated with Sigma-Zero.
Total	1 175	15	Five final cases, one per diagnostic bucket.

These categories define the intended diagnostic roles of the cases. Their visual interpretation and the corresponding attribution patterns are reported in Section 6.8.

Attribution processing and visual outputs. For each target, the channel-wise attribution tensor is converted into a two-dimensional map by summing the absolute attribution magnitudes across the three input channels. The resulting map is min-max normalized to the interval $[0, 1]$.

Because absolute values are used, the visualization represents attribution magnitude rather than contribution direction. It identifies regions associated with larger attribution magnitudes for the selected model output along the baseline-to-input path, but it does not distinguish positive evidence from negative evidence.

For each selected case, the implementation produces a resized RGB input image, a colorized attribution heatmap, and an overlay combining the input with the heatmap. It also generates a normalized grayscale attribution mask, a binary high-attribution mask, a side-by-side diagnostic comparison, and a histogram of the normalized attribution values.

The heatmap uses the `inferno` color map. The overlay combines the resized input and colorized map using an attribution opacity of 0.45. The binary high-attribution mask retains pixels whose normalized magnitude is at or above the 90th-percentile threshold.

This threshold identifies approximately the 10% of image locations with the largest attribution magnitudes. The exact proportion may differ when multiple locations share the percentile value.

For each selected case, the thesis retains four visual assets: the input image, the standalone attribution heatmap, the heatmap overlay, and the top-10% attribution mask.

The resulting thesis-level visual corpus contains:

$$5 \text{ cases} \times 4 \text{ visual assets} = 20 \text{ images.} \quad (5.24)$$

The thesis-ready files are stored under `docs/LatexThesis/images/`. The complete generated diagnostic tree under `explainability/outputs/integrated_gradients/` remains a local output and is not distributed through the public repository.

Convergence diagnostics. The historical thesis assets were generated using 32 integration steps. Their raw Captum convergence deltas are retained in `explainability/manifests/chapter5/thesis_selection.csv`. These historical outputs were not generated under the adaptive acceptance rule subsequently introduced by the hardened entry point.

The hardened procedure evaluates the normalized convergence delta:

$$\delta_{\text{norm}} = \frac{|\delta_{\text{IG}}|}{\max(|f_t(\mathbf{x}) - f_t(\mathbf{x}')|, 10^{-12})}, \quad (5.25)$$

where f_t is the target-class output, $\mathbf{x}$ is the preprocessed input, and $\mathbf{x}'$ is the baseline.

Starting from the requested integration resolution, the procedure doubles the number of steps until either:

$$\delta_{\text{norm}} \leq 0.05,$$

or the maximum of 256 integration steps is reached. A regenerated case is marked as **passed** only when the threshold is satisfied; otherwise, the state **threshold_not_met** is retained.

Adaptive regeneration had not been completed for the five historical thesis assets at the time of artifact freezing. Their validation state is therefore recorded as **pending_adaptive_regeneration**. The figures are interpreted exclusively as qualitative diagnostic visualizations, and no strict numerical convergence or completeness claim is made for them.

Selection and provenance records. Table 5.29 lists the canonical records retained for the explainability workflow.

Table 5.29: Canonical Integrated Gradients selection and provenance records

Record	Role
<code>explainability/manifests/chapter5/thesis_selection.csv</code>	Canonical mapping of the five cases, prediction metadata, historical attribution configuration, thesis image paths, raw convergence deltas, validation status, and selection rationales.
<code>explainability/manifests/chapter5/candidate_selection_summary.json</code>	Candidate counts, reviewed selections, bucket distribution, and final five-case mapping.
<code>explainability/manifests/chapter5/run_summary.json</code>	Frozen model, strategy, baseline, attribution target, historical integration steps, percentile, selection counts, and public-distribution policy.
<code>explainability/manifests/chapter5/run_provenance.json</code>	Entry point, preserved implementation, prediction source, historical snapshot, model registry, integrity controls, and adaptive-convergence policy.
<code>explainability/logs/chapter5_manual_selection.csv</code>	Human-in-the-loop case-selection actions recorded under a pseudonymous reviewer identifier.

The explainability layer is intentionally qualitative and diagnostic. The generated maps are not semantic segmentations, causal demonstrations, calibrated reliability estimates, or explanations of the selected black-box software systems. They provide evidence concerning the attribution behavior of the transparent EfficientNet-B0 proxy for the selected cases and complement, rather than replace, the quantitative robustness metrics.

The integrity and provenance controls shared across the complete experimental pipeline are consolidated in Section 5.8.

5.8 Traceability, Integrity, and Reproducibility Controls

Traceability, integrity, and controlled reproducibility were implemented as cross-cutting properties of the experimental workflow rather than as controls introduced only after evaluation. The resulting protocol links the frozen dataset, deterministic partitions, proxy-model checkpoints, derived perturbations, black-box inputs, normalized software outputs, metric tables, explainability cases, and reporting assets through stable identifiers, cryptographic digests, manifests, registries, and version-controlled metadata.

The implementation distinguishes three related but non-equivalent objectives. *Traceability* refers to the ability to reconstruct the provenance and processing history of an experimental input, derived artifact, model, or result record. *Integrity* concerns the ability to verify that an input, artifact, model, or record has not changed relative to its canonical recorded state. *Controlled reproducibility* refers to the extent to which the workflow can be inspected or

rerun under the constraints imposed by data-access restrictions, licensed-software requirements, human-review stages, and the available execution environment.

This distinction is necessary because public availability of code, manifests, and sanitized results does not imply unrestricted redistribution of the source images or availability of the proprietary software required to repeat the black-box processing.

Stable identity and provenance chain. Stable identifiers are assigned at successive processing stages. Prepared source images receive an `image_id`; generated perturbations receive a `generated_image_id`; black-box inputs receive a semantically neutral `bundle_id`; and selected explainability cases receive a `case_id`. Proxy-model checkpoints are identified through the combination of architecture and held-out fold.

The principal provenance chain is:

```
image_id → generated_image_id → bundle_id
          → normalized_prediction → metric_row
          → reporting_asset.
```

Original and generated identifiers coincide when no derived artifact exists; otherwise, the generated identifier retains the source association and the perturbation condition or target model. Canonical manifests use repository-relative paths, while run summaries retain the configuration, counts, folds, timestamps, and output locations needed to reconstruct a stage.

Table 5.30 summarizes the principal identity controls.

Table 5.30: Stable identifiers used throughout the experimental pipeline

Identifier	Canonical context	Traceability role
<code>image_id</code>	Prepared and frozen dataset manifests	Identifies the reviewed source image across dataset preparation, freezing, partitioning, and clean evaluation.
<code>generated_image_id</code>	Adversarial and anti-forensic manifests	Identifies a derived image artifact while preserving its association with the source image and perturbation condition.
<code>bundle_id</code>	Forensic evaluation bundle manifest	Provides a semantically neutral identifier for black-box import and subsequent reconnection to the reference operational assignment and experimental condition.
Architecture and fold	Model registry and checkpoint paths	Identifies the fold-aware checkpoint used for training, inference, model-dependent perturbation generation, and explainability.
<code>case_id</code>	Integrated Gradients selection manifest	Identifies a manually selected qualitative diagnostic case and its associated thesis assets.

Cryptographic integrity controls. SHA-256 is the primary integrity digest throughout the pipeline. MD5 is retained only as an auxiliary interoperability identifier because some forensic software exports expose MD5-oriented fields or matching workflows. It is not treated as the primary integrity mechanism.

Per-file SHA-256 values are recorded or verified at the principal transformation and transfer boundaries, covering source and derived images, partitions, bundle copies, checkpoints, sanitized black-box extracts, canonical predictions, and metric records. Copy operations recompute the destination digest and fail on mismatch. Thesis-level explainability images are checked through their manifest paths, case metadata, and asset audits.

The checkpoint digests stored in `models/model_registry.json` bind every proxy architecture and held-out fold to its serialized checkpoint. They are checked before canonical proxy evaluation, model-dependent adversarial generation, and hardened explainability generation.

Archive-level integrity is maintained separately from per-file integrity. The canonical complete-archive digests are stored in `docs/artifact/CONTROLLED_ARTIFACT_CHECKSUMS.sha256`. This file identifies:

- `00_raw_datasets_bundle.zip`;
- `16_frozen_forensic_evaluation_bundle.zip`.

The archive digest authenticates the complete downloaded ZIP file. Restoration of the frozen forensic evaluation bundle additionally uses `datasets/forensic_evaluation_bundle/metadata/bundle_hashes_sha256.csv` to verify the expected file-name population and the SHA-256 value of each of the 11 500 blind inputs.

The restoration entry point rejects malformed or corrupt ZIP members, symbolic links, path traversal, unexpected roots, population differences, and content hash mismatches; repository metadata remains authoritative. Archive and post-extraction verification are therefore complementary controls.

Deterministic configuration controls. A fixed seed of 42 is used for the principal stochastic procedures, including fold construction, internal training-validation partitioning, and image-specific randomized-search initialization where applicable. The seed is recorded in the corresponding machine-readable configuration and summary records.

The fixed seed and supported PyTorch/cuDNN deterministic options improve repeatability but do not guarantee bit-identical results across operating systems, hardware, libraries, or CPU/GPU implementations.

Machine-readable CSV and JSON configurations preserve population counts, partitions, preprocessing, training and checkpoint rules, perturbation parameters, black-box versions and mappings, metric definitions, and explainability settings.

Overwrite and rerun controls. Generation and evaluation scripts fail rather than silently replacing canonical outputs. A rebuild requires an explicit `-force` option scoped to the selected run; partial diagnostics must use separate locations. Regenerated outputs remain subject to the recorded counts, identifiers, digests, configurations, and downstream validations.

Machine-readable invariants and completeness checks. Each major stage produces sample-level records and a run-level summary. Summary files record input manifests, parameters, output paths, expected and actual counts, validation states, and creation metadata. These records permit structural checks without requiring manual inspection of every image or prediction.

The principal frozen invariants comprise 1 500 records in the final dataset, including 1 000 binary samples and 500 OOD inputs. The binary subset is partitioned into five clean folds containing 200 images each and is associated with 15 fold-aware proxy-model checkpoints. The experimental pipeline also produces 5 000 adversarial and 5 000 anti-forensic artifacts, resulting in 11 500 unique items in the forensic evaluation bundle.

At the evaluation level, the frozen outputs include 40 500 proxy-model prediction rows, 69 000 canonical sanitized black-box software decisions, and 186 black-box software metric rows. The explainability branch retains five canonical thesis-level cases and 20 associated visual assets.

These counts are implementation invariants and completeness checks. Their satisfaction does not establish model or software robustness. The quantitative behavior of the evaluated systems is analyzed separately in Chapter 6.

Black-box output correspondence and equivalence controls. Complete commercial exports remain controlled because they include proprietary or case-specific fields. The four minimized extracts reconstruct the 69 000 sanitized decisions in `evaluation/forensic_tools/normalized_predictions.csv` through `forensic_tools/scripts/build_canon`

`ical_normalized_predictions.py`. Decision- and metric-level equivalence is checked by `forensic_tools/scripts/validate_public_extract_equivalence.py`, which verifies source and output digests and reproduction of all 186 metric rows. This establishes equivalence of the normalized decision population and derived metrics, not field-level equivalence with every raw-export column.

Result, explainability, reporting, and LaTeX validation. The repository provides dedicated read-only or fail-visible validation utilities, consistent with the principle that forensic methods and tools should undergo documented validation before their operational results are relied upon [100].

Table 5.31 summarizes their principal roles.

Table 5.31: Validation utilities applied to the frozen research artifact

Validation entry point	Principal checks
<code>forensic_tools/scripts/validate_public_extract_equivalence.py</code>	Decision-level equivalence across the 69 000 sanitized black-box software decisions and metric-level equivalence across the 186 black-box metric rows, together with source and extract digests.
<code>results/scripts/23_validate_results_artifacts.py</code>	Canonical commercial prediction schema and counts, commercial metric counts and equivalence status, proxy-evaluation summary counts, fold-aggregated OOD accounting, reporting-manifest counts, generated-output availability, and metadata-sensitivity counts.
<code>results/scripts/24_audit_reporting_asset_usage.py</code>	Consistency among the reporting manifest, generated reporting assets, LaTeX references, and existing thesis-ready copies, including byte-level comparison through SHA-256.
<code>explainability/scripts/validate_chapter5_xai_artifacts.py</code>	Five canonical case identifiers and buckets, prediction values displayed in the results chapter, availability of the 20 thesis assets, absence of local path leakage, and optional adaptive-convergence and integrity checks for regenerated manifests.
<code>tools/latex/audit_latex_images_used.py</code>	Missing, ambiguous, unused, and duplicate image references within the authoritative LaTeX project.
<code>tools/tasks.shaudit-all</code>	Repository status, JSON parsing, Python syntax, repository-layout and selected text-guard checks, XAI validation, result validation, reporting-asset audit, LaTeX-image audit, and thesis-log inspection when a compilation log is available.

The metric CSV and JSON files remain the canonical numerical records. PDF and PNG figures are presentation-layer outputs generated from those frozen values and must not be edited manually to alter numerical content.

The authoritative LaTeX entry point is `docs/LaTeXThesis/main.tex`. The final compilation audit checks bibliography resolution, cross-references, glossary entries, image availability, and unresolved LaTeX errors. Generated auxiliary files and the local PDF are not treated as canonical source files.

Public, controlled, and licensed reproducibility layers. The research artifact defines the three access layers summarized in Table 5.32.

Table 5.32: Reproducibility boundary of the research artifact

Layer	Supported activities
Public	Inspection of source code, manifests, proxy checkpoints, prediction and metric tables, sanitized black-box decisions, explainability metadata, thesis-ready assets, validation utilities, documentation, and LaTeX sources; reconstruction of the canonical normalized black-box prediction table; and recomputation of metrics from publicly retained decisions.
Controlled	Restoration of the heterogeneous source corpora; regeneration of the prepared and frozen image pipeline; restoration of the exact 11 500-file black-box input bundle; proxy-model retraining; perturbation regeneration; and selected image-level validation, subject to approved data access and the applicable source restrictions.
Licensed	Reprocessing of the blind bundle through the proprietary software versions and compatible import/export environments used during the frozen evaluation.

The two authenticated archives listed under the integrity controls support the controlled layer. Their availability reflects legal, ethical, contractual, and licensing restrictions and does not grant redistribution rights or access to proprietary model internals.

Environment and public-distribution boundary. The pipeline uses Python environments across Windows and Linux, with CPU or CUDA execution selected by stage; black-box evaluation additionally requires compatible licensed software. Recorded dependencies support reconstruction but do not claim a byte-identical operating environment, and numerical reproduction may depend on hardware, drivers, libraries, and commercial-software behavior.

The public release excludes source corpora, complete commercial exports, proprietary case material, credentials, licence data, private URLs, usernames, unnecessary absolute paths, and generated LaTeX outputs. Tracked records are audited before release; sanitization changes the distribution layer, not the byte-identical images used in the frozen evaluation.

Version control, historical provenance, and artifact freezing. Source code, manifests, registries, sanitized decisions, metric tables, documentation, and thesis sources are maintained through Git version control. Large proxy-model checkpoints are managed through Git LFS, while their SHA-256 values remain recorded independently in the model registry.

For historical provenance, the repository preserves the pre-cleanup state through the protected branch `archive/pre-commission-cleanup-2026-07-16` and the protected tag `snapshot/pre-commission-cleanup-2026-07-16`. Both references identify the same historical commit. This snapshot supports provenance and repository audit but is not the current source of truth for final documentation or experimental reporting.

The curated `main` branch remains the authoritative current artifact. Substantive changes to the data, perturbations, checkpoints, normalized decisions, metrics, or thesis conclusions require a new versioned release rather than silent replacement of the frozen experimental state. Documentation corrections, security fixes, citation updates, and non-substantive archival improvements must remain clearly distinguishable from changes to the experimental evidence.

Taken together, these controls provide a traceable connection between the implemented protocol and the results reported in the following chapter. They do not eliminate limitations associated with controlled data access, execution environment variation, human review, historical explainability convergence, or proprietary black-box systems. Instead, they make these boundaries explicit and preserve the information required to audit the evaluation within them.

Chapter 6 presents the quantitative outcomes obtained from the frozen implementation and analyzes their implications for the operational robustness of the evaluated proxy models and selected black-box software systems.

Chapter 6

Experimental Results and Operational Robustness Analysis

This chapter presents the quantitative outcomes obtained from the implementation documented in Chapter 5 and interprets them according to the methodological protocol defined in Chapter 4. The analysis focuses on the operational robustness of AI-based systems for automated image classification and triage in forensic contexts.

A central feature of the evaluation is the parallel analysis of two complementary system levels. The first comprises the transparent proxy models defined in Section 5.3.1, for which predictions, maximum predicted-class probabilities, confusion matrices, and internal attribution information are available. The second comprises the selected black-box software systems, whose behavior is assessed exclusively through observable and normalized outputs.

Both evaluation levels operate on the same frozen source population. The 1 000 clean binary images and 500 clean OOD images constitute the common source corpus, whereas 10 000 adversarial and anti-forensic artifacts are derived exclusively from the binary subset. The resulting comparison is therefore based on the same underlying samples and experimental conditions, while preserving the distinction between transparent model evaluation and operational black-box assessment.

The purpose of the chapter is not to construct an absolute ranking among heterogeneous systems or to reduce the study to a conventional Adv-ML benchmark. The systems differ in architecture, output semantics, configuration, operational scope, and degree of transparency. Their results are therefore interpreted from a forensic perspective, with particular attention to false negatives on the firearm-oriented target category, false positives on `non_weapon` content, closed-set assignments of OOD inputs, degradation under manipulated conditions, and the continued need for human review [1, 2, 3].

6.1 Experimental Evaluation Scope and Completeness

The experimental results cover the complete population generated through the frozen implementation described in Chapter 5. The common corpus contains 11 500 logical experimental inputs, comprising clean images and derived perturbed artifacts:

$$\begin{aligned} N_{\text{bundle}} &= 1\,000 \text{ clean binary inputs} + 500 \text{ OOD inputs} \\ &\quad + 5\,000 \text{ adversarial artifacts} + 5\,000 \text{ anti-forensic artifacts} \\ &= 11\,500. \end{aligned} \tag{6.1}$$

The proxy-model and black-box evaluations use different output representations but cover the

same experimental conditions. Each clean binary input or perturbed binary artifact is evaluated by all three proxy architectures through the checkpoint associated with its assigned fold.

The OOD branch follows a different proxy-model evaluation protocol. Each of the 500 unique OOD images is evaluated by all five fold-specific checkpoints of each architecture. The resulting 7 500 prediction rows therefore represent repeated checkpoint responses to 500 unique images and must not be interpreted as 7 500 independent OOD samples.

The black-box evaluation comprises four software families and six normalized operational configurations. Magnet AXIOM/Magnet.AI and Griffeye/T3K CORE each contribute one configuration; the D20, D50, and D80 Excire Foto 2025 settings are retained as three separate configurations; and Cellebrite Inseyets contributes one configuration. Each configuration produces one normalized decision for every item in the 11 500-file forensic evaluation bundle.

Table 6.1 summarizes the completeness of the result population.

Table 6.1: Completeness of the experimental result population

Result layer	Records	Coverage
Proxy-model predictions	40 500	3 000 clean predictions, 7 500 fold-specific OOD responses, and 30 000 predictions on adversarial and anti-forensic artifacts; no recorded inference errors.
Normalized black-box decisions	69 000	Six normalized configurations evaluated on all 11 500 bundle items, with no missing bundle-level decision.
Proxy binary-condition metrics	33	Three clean architecture-level rows and 30 architecture-perturbation combinations.
Proxy robustness comparisons	30	Clean-to-perturbed comparisons for three architectures and ten perturbation conditions.
Proxy OOD summaries	3	One fold-aggregated OOD record for each proxy architecture.
Black-box software metrics	186	Thirty-one aggregation records for each of the six normalized black-box configurations.
Explainability cases	5	One purposively selected case for each predefined diagnostic category, associated with 20 thesis-level visual assets.

Binary accuracy, precision, recall, F1-score, FPR, and FNR are computed exclusively on samples whose reference operational assignment is **weapon** or **non_weapon**. The OOD branch remains separate and is analyzed through the proportion of inputs assigned to the **weapon** class and, for the proxy models, through architecture-specific maximum-probability indicators.

Accuracy drop, induced-error counts, directional error transitions, and the confusion-matrix layout follow the definitions established in Section 4.13. Positive accuracy-drop values indicate degradation relative to the clean baseline, whereas negative values indicate a condition-specific increase in accuracy that must not be interpreted as evidence of general robustness.

The clean-correct-to-incorrect transition rate is interpreted as attack success rate only for the four model-dependent adversarial attacks: FGSM, One Pixel Attack, Sigma Zero Attack, and SuperDeepFool. For Color Shift and the five anti-forensic transformations, the same numerical transition is interpreted as an induced-error rate rather than as adversarial or transfer success.

Maximum predicted-class probabilities are available only for the transparent proxy models. They are retained as intra-model diagnostic indicators and are not interpreted as calibrated probabilities directly comparable across architectures. The selected black-box software systems do not expose homogeneous numerical uncertainty measures; consequently, their comparison is based on normalized observable decisions rather than cross-system probability or uncertainty scores.

Unless a subsection explicitly states otherwise, these population, accuracy-drop, attack-transfer, OOD, and probability-interpretation rules apply throughout the chapter.

6.2 Clean Baseline Performance

Evaluation on clean inputs establishes the nominal baseline against which behavior under OOD, adversarial, and anti-forensic conditions is subsequently assessed. The three proxy architectures were evaluated on the frozen binary subset of 1 000 images, comprising 500 **weapon** and 500 **non_weapon** samples.

The positive class is **weapon**, and the negative class is **non_weapon**. A false negative therefore corresponds to a **weapon** image classified as **non_weapon**, whereas a false positive corresponds to a **non_weapon** image classified as **weapon**.

Table 6.2 reports the principal clean binary metrics, including the architecture-specific mean maximum predicted-class probability defined in Section 6.1.

Table 6.2: Clean performance of the proxy models on the binary subset

Model	Acc.	Bal.	M-P	M-R	M-F1	P _w	R _w	F1 _w	Max-P	FPR	FNR
EfficientNet-B0	0.978	0.978	0.978	0.978	0.978	0.972	0.984	0.978	0.979	0.028	0.016
ResNet18	0.964	0.964	0.964	0.964	0.964	0.958	0.970	0.964	0.974	0.042	0.030
CLIP	0.927	0.927	0.933	0.927	0.927	0.881	0.988	0.931	0.559	0.134	0.012

Note. *Bal.* indicates balanced accuracy; *M-P*, *M-R*, and *M-F1* indicate macro-precision, macro-recall, and macro-F1, respectively; *P_w*, *R_w*, and *F1_w* indicate precision, recall, and F1-score for the **weapon** class. *Max-P* indicates the mean maximum predicted-class probability and is reported only as an intra-model diagnostic indicator.

Among the two fully fine-tuned CNNs, EfficientNet-B0 provides the strongest overall balance, with 0.978 accuracy and macro-F1, an FNR of 0.016, and an FPR of 0.028. ResNet18 follows at 0.964 accuracy and macro-F1, with an FNR of 0.030 and an FPR of 0.042. The CLIP-based proxy has lower overall accuracy (0.927) but the highest weapon recall (0.988) and lowest FNR (0.012), at the cost of the highest FPR (0.134). Its lower model-specific Max-P value (0.559) accompanies this positive-class tendency but is not a cross-architecture calibration result.

This behavior is not the result of zero-shot classification. In the adopted implementation, CLIP serves as a pretrained and frozen visual encoder connected to a supervised binary classification head trained on the same frozen binary task. Its error profile must therefore be attributed to the combination of the pretrained visual representation and the learned binary decision head rather than to the original image–text similarity interface.

Table 6.3 reports the aggregate confusion-matrix components for the three proxy models. The layout is:

$$\mathbf{C} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix}.$$

Table 6.3: Clean confusion matrices of the proxy models

Model	TN	FP	FN	TP
EfficientNet-B0	486	14	8	492
ResNet18	479	21	15	485
CLIP	433	67	6	494

The same error distributions are visualized in Figure 6.1.

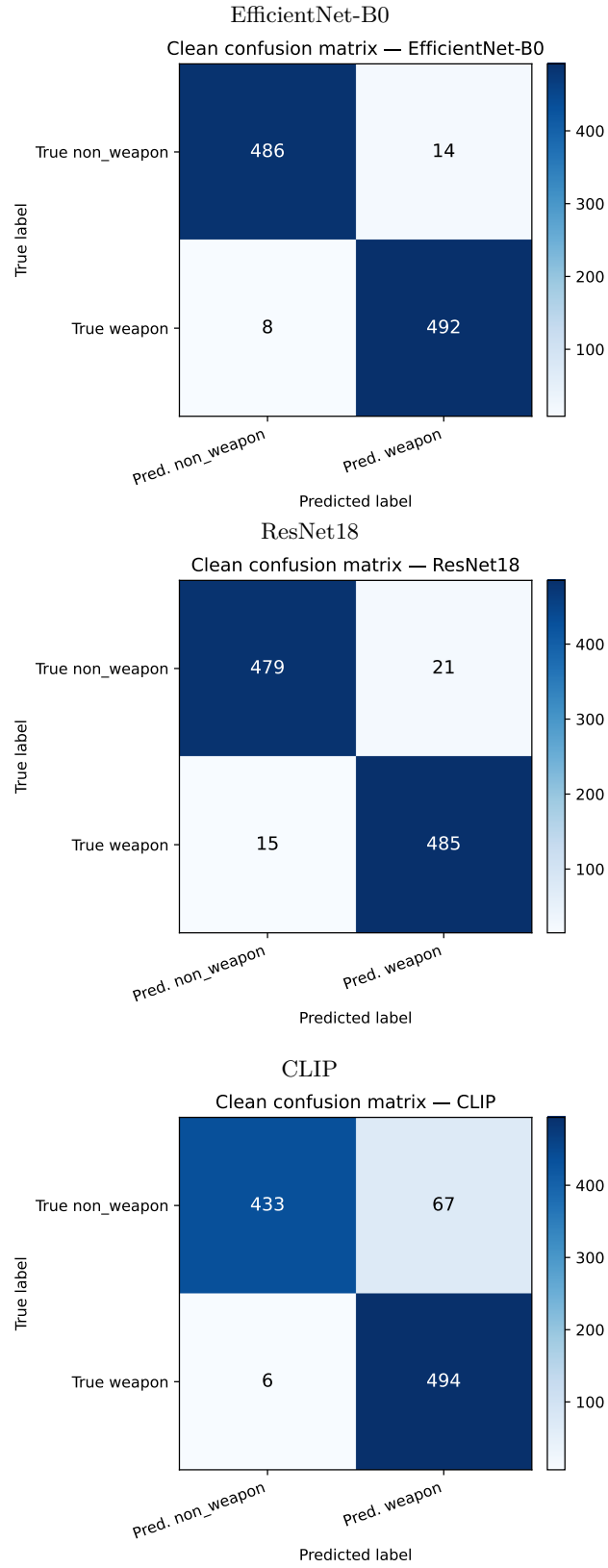


Figure 6.1: Clean confusion matrices of the three proxy models on the binary subset

The direction of error is operationally decisive: the three models respectively produce 8/14, 15/21, and 6/67 false negatives/false positives, for 22, 36, and 73 errors in total. Thus, the CLIP-based proxy minimizes missed targets while substantially increasing review workload, whereas EfficientNet-B0 offers the strongest balance. These distinct clean profiles form the baseline for the following robustness analyses.

6.3 OOD Behavior

The OOD evaluation examines the behavior of the three proxy architectures on the separate 500-image OOD branch. As established in Chapter 4, these images are not treated as a third supervised class and do not contribute to binary accuracy, precision, recall, or confusion-matrix counts.

The analysis instead characterizes how inputs outside the frozen binary distribution are forced into one of the two available closed-set operational classes [20, 47, 101].

Each of the 500 unique OOD images was evaluated by all five fold-specific checkpoints of each proxy architecture:

$$500 \text{ unique OOD images} \times 5 \text{ fold-specific checkpoints} = 2\,500 \text{ responses per architecture. (6.2)}$$

The resulting responses are not independent images. They represent five fold-specific model outputs for each unique OOD input. The reported counts and rates therefore characterize checkpoint-response behavior rather than unique-image-level classification frequency.

Two principal indicators are considered. The *weapon assignment rate* corresponds to the proportion of the 2 500 responses assigned to **weapon**. The *high-maximum-probability rate* instead represents the proportion of responses whose maximum predicted-class probability is at least 0.90.

Neither indicator represents supervised correctness because the OOD inputs have no binary reference assignment.

Table 6.4 reports the fold-aggregated results.

Table 6.4: Fold-aggregated behavior of the proxy models on OOD inputs

Model	Responses	Weapon responses	Non-weapon responses	W-rate	Max-P	HP-count	HP-rate
EfficientNet-B0	2 500	924	1 576	0.3696	0.8505	1 252	0.5008
ResNet18	2 500	1 094	1 406	0.4376	0.8896	1 617	0.6468
CLIP	2 500	2 089	411	0.8356	0.5358	0	0.0000

Note. *W-rate* indicates the proportion of fold-specific responses assigned to **weapon**; *Max-P* indicates the mean maximum predicted-class probability; *HP-count* indicates the number of responses whose maximum probability is at least 0.90; and *HP-rate* indicates the corresponding proportion over the 2 500 responses. Each total comprises five checkpoint responses for each of the 500 unique OOD images.

The CLIP-based proxy assigns 0.8356 of the responses to **weapon**, far more than EfficientNet-B0 (0.3696) or ResNet18 (0.4376), yet none of its responses reaches Max-P 0.90. Conversely, EfficientNet-B0 and ResNet18 produce high-Max-P rates of 0.5008 and 0.6468. Assignment direction and output concentration are therefore distinct diagnostics.

Figure 6.2 compares the weapon assignment rate, mean maximum predicted-class probability, and high-maximum-probability rate.

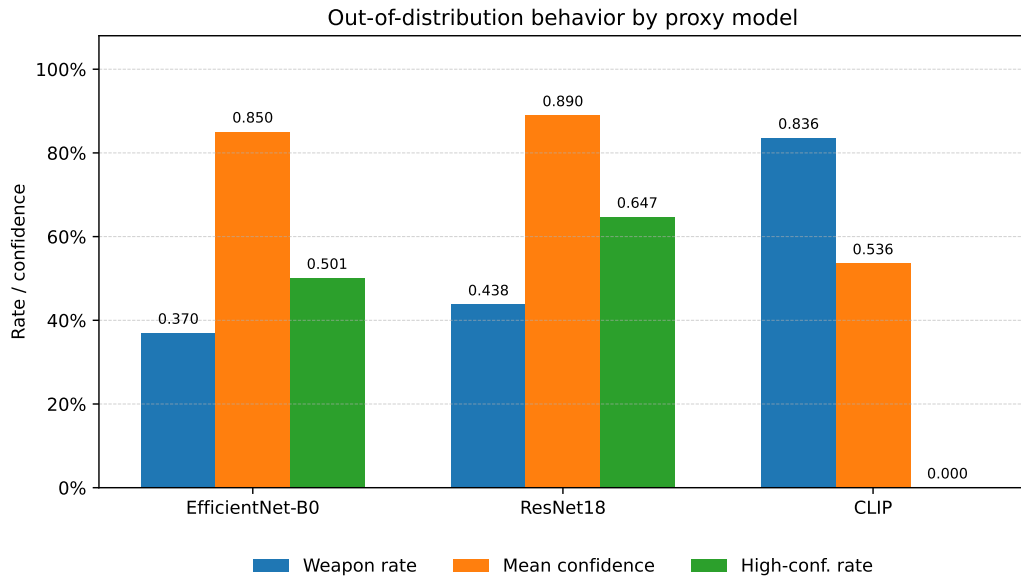


Figure 6.2: Weapon assignment rate and maximum predicted-class probability indicators on OOD inputs

The comparison confirms the separation between assignment frequency and architecture-specific Max-P behavior; as established in Section 6.1, it is not a cross-model uncertainty ranking.

Figure 6.3 presents the complete distribution of maximum predicted-class probabilities for the three architectures.

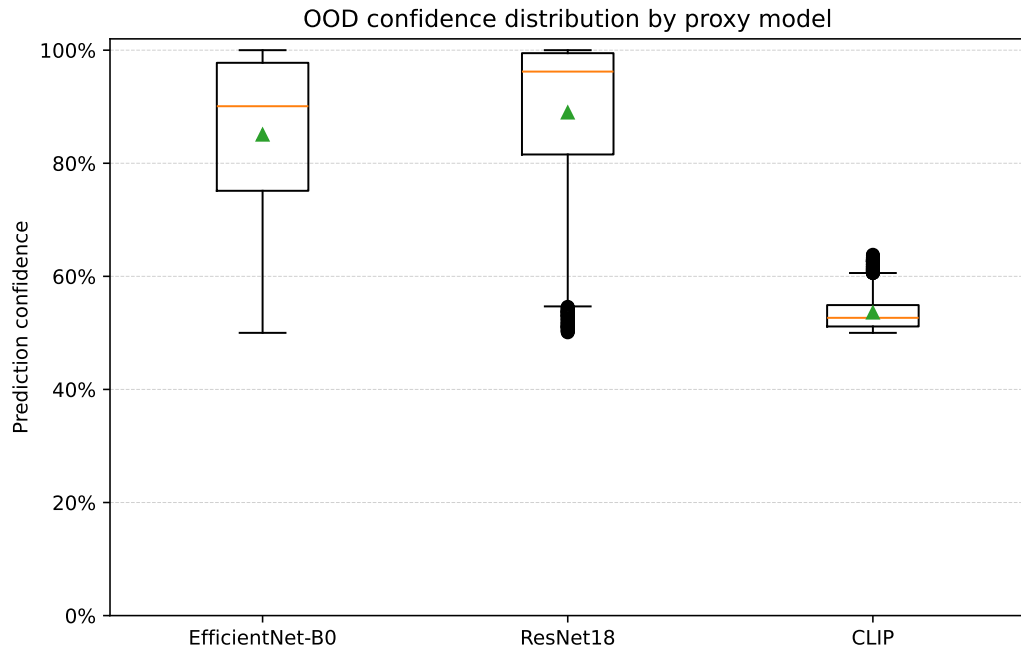


Figure 6.3: Distribution of maximum predicted-class probabilities on OOD inputs for each proxy architecture

The full distributions confirm these architecture-specific score profiles. Operationally, a positive OOD assignment can increase unrelated review volume, while a high Max-P indicates only a

peaked closed-set output. Neither is supervised correctness or stand-alone evidence, and both require human review [102, 103].

6.4 Anti-Forensic Robustness

The anti-forensic evaluation examines the stability of the proxy models under five deterministic, model-agnostic image transformations. These conditions do not require access to model parameters, gradients, logits, or decision thresholds. They represent operations compatible with ordinary image circulation and processing, although the same transformations may also be applied intentionally to alter observable image characteristics [12, 21, 59].

The evaluated conditions comprise JPEG recompression, downsampling followed by resizing to the original dimensions, Gaussian blurring, histogram modification, and contrast stretching. Their operational relevance derives from the fact that they may occur during compression, platform-mediated sharing, editing, conversion, or deliberate anti-forensic manipulation.

For each architecture and transformation, performance is compared with the corresponding clean baseline. Accuracy drop is defined as:

$$\Delta_{\text{acc}} = \text{Acc}_{\text{clean}} - \text{Acc}_{\text{transformed}}. \quad (6.3)$$

Positive values indicate lower aggregate accuracy under the transformed condition. A value of zero indicates unchanged aggregate accuracy, whereas a negative value indicates higher accuracy on the specific transformed population.

A zero or negative accuracy drop must not be interpreted as evidence that the transformation introduced no adverse decisions. Aggregate accuracy may remain unchanged or increase when some clean-condition errors are corrected while other initially correct samples become incorrect. For this reason, accuracy drop is examined together with the induced-error count, induced-error rate, and direction of the clean-correct-to-incorrect transitions.

Table 6.5 reports transformed accuracy and accuracy drop for all architecture–transformation combinations.

Table 6.5: Accuracy and accuracy drop under anti-forensic transformations

Transformation	CLIP acc.	Drop	EffNet acc.	Drop	ResNet acc.	Drop
JPEG recompression	0.925	0.002	0.976	0.002	0.964	0.000
Resample + Resize	0.929	-0.002	0.976	0.002	0.963	0.001
Gaussian Blur	0.934	-0.007	0.965	0.013	0.959	0.005
Histogram Modification	0.925	0.002	0.973	0.005	0.934	0.030
Contrast Stretching	0.927	0.000	0.974	0.004	0.963	0.001

Note. Drop is computed as clean accuracy minus transformed accuracy. Positive values indicate aggregate degradation, zero indicates unchanged aggregate accuracy, and negative values indicate a condition-specific increase. None of these values alone establishes the absence or presence of newly induced errors.

Figure 6.4 presents the same comparison as a heatmap.

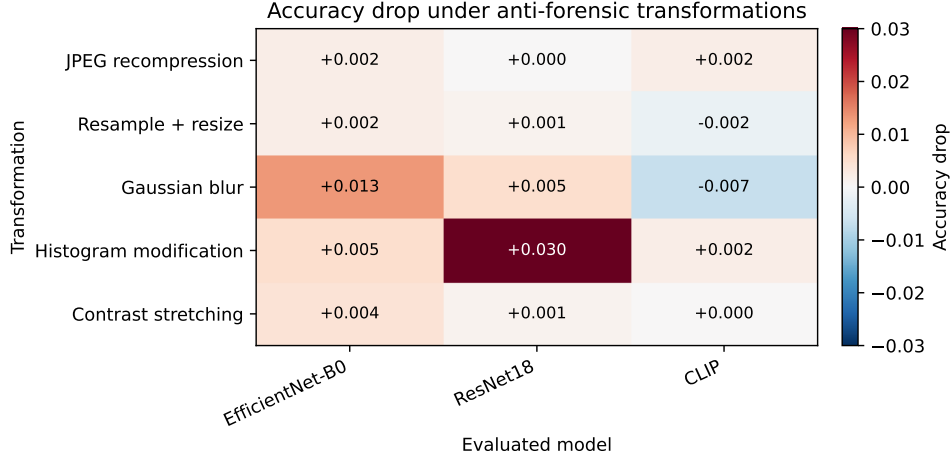


Figure 6.4: Accuracy drop of the proxy models under anti-forensic transformations

The aggregate results indicate that the five transformations produce smaller accuracy variations than the strongest model-dependent adversarial attacks examined in Section 6.5. This relative difference does not make the anti-forensic effects operationally negligible. Even limited aggregate degradation may correspond to newly missed target-category images.

EfficientNet-B0 remains above 0.965 accuracy under every transformation. Gaussian Blur produces its largest degradation, reducing accuracy from 0.978 to 0.965 and yielding an accuracy drop of 0.013. Histogram Modification is the second most influential condition for this model, with an accuracy of 0.973 and a drop of 0.005.

ResNet18 is most affected by Histogram Modification. Its accuracy decreases from 0.964 to 0.934, corresponding to the largest anti-forensic accuracy drop observed across the three architectures, at 0.030. Gaussian Blur produces a smaller reduction to 0.959, corresponding to a drop of 0.005. JPEG Recompression leaves aggregate accuracy unchanged at 0.964.

The CLIP-based proxy exhibits small aggregate variations across the five conditions. JPEG Recompression and Histogram Modification reduce accuracy by 0.002, Contrast Stretching leaves it unchanged, and Resample Resize increases it by 0.002. Gaussian Blur increases aggregate accuracy from 0.927 to 0.934.

Table 6.6 reports the most operationally relevant condition for each architecture. The selected conditions capture the largest aggregate degradation or, where aggregate drops are tied or limited, the greatest newly induced error burden.

Table 6.6: Operational indicators for representative anti-forensic conditions

Model	Transformation	Induced	IER	Max-P shift	W→NW	NW→W
EfficientNet-B0	Gaussian Blur	17	0.0174	-0.0099	17	0
ResNet18	Histogram Modification	39	0.0405	-0.0157	23	16
CLIP	Histogram Modification	13	0.0140	-0.0037	6	7

Note. *Induced* is the number of clean-correct samples rendered incorrect by the transformation. *IER* is the induced-error rate over the clean-correct denominator. *Max-P shift* is the mean transformed maximum predicted-class probability minus the corresponding clean value. W→NW denotes **weapon-to-non_weapon** transitions; NW→W denotes the opposite direction.

For EfficientNet-B0, Gaussian Blur introduces 17 errors among samples that were correctly classified in their clean form. All 17 transitions occur from **weapon** to **non_weapon**, producing an induced-error rate of 0.0174. The direction of these transitions is operationally more consequential than the aggregate drop alone because every newly induced error corresponds to a missed firearm-oriented sample.

For ResNet18, Histogram Modification introduces 39 errors and produces the highest anti-forensic induced-error rate, at 0.0405. Of these transitions, 23 occur from **weapon** to **non_weapon**, while 16 occur from **non_weapon** to **weapon**. The condition therefore increases both missed-target risk and false-positive review workload.

For the CLIP-based proxy, Histogram Modification introduces 13 errors among clean-correct samples, corresponding to an induced-error rate of 0.0140. Six transitions occur from **weapon** to **non_weapon**, and seven occur in the opposite direction. Although the aggregate accuracy drop is only 0.002, the transition analysis confirms that the transformed condition is not behaviorally identical to the clean baseline.

The model-specific Max-P shifts are small and negative. Gaussian Blur also shows why aggregate and transition metrics remain separate: CLIP accuracy increases by 0.007 while four clean-correct samples become incorrect.

From an operational perspective, the anti-forensic results identify architecture- and transformation-specific failure modes rather than a uniform degradation pattern. Gaussian Blur primarily affects EfficientNet-B0 through new **weapon-to-non_weapon** transitions, whereas Histogram Modification produces the largest effects for ResNet18 and the most relevant transition burden for the CLIP-based proxy.

These findings are important because the evaluated transformations do not require model access and may arise during routine handling, conversion, compression, editing, or re-sharing. In a forensic workflow, even modest aggregate variation may become consequential when it produces missed target-category images, increases irrelevant review volume, or interacts with low visual quality and ambiguous context.

Thus, comparatively stable aggregate accuracy under the frozen transformations coexists with architecture-specific decision changes and cannot support a general robustness claim.

6.5 Adversarial Robustness

The adversarial evaluation examines the response of the transparent proxy models to perturbations generated to alter classifier behavior [13, 15, 48]. These conditions are used to assess operational robustness in a controlled forensic evaluation rather than to construct an independent Adv-ML benchmark.

Four model-dependent attacks were generated against the fold-specific EfficientNet-B0 checkpoints. FGSM, Sigma Zero Attack, and SuperDeepFool used white-box checkpoint access, whereas One Pixel Attack used score-based access to the same target architecture. Evaluation of these artifacts on EfficientNet-B0 constitutes direct-target assessment.

The same artifacts were subsequently evaluated on ResNet18 and the CLIP-based proxy to measure empirical transfer behavior. Their effects on these architectures must therefore not be interpreted as white-box attacks against ResNet18 or CLIP [15, 16, 60].

Color Shift is treated separately as a deterministic, model-agnostic adversarial-style condition. It does not use model parameters, gradients, scores, or checkpoints and is therefore not associated with a target-model attack success rate.

For every architecture and condition, accuracy drop is defined as:

$$\Delta_{\text{acc}} = \text{Acc}_{\text{clean}} - \text{Acc}_{\text{perturbed}}. \quad (6.4)$$

Positive values indicate lower aggregate accuracy under the perturbed condition. Zero indicates unchanged aggregate accuracy, whereas negative values indicate a condition-specific increase.

As in the anti-forensic analysis, zero or negative accuracy drop does not imply that no new errors were introduced. Newly incorrect decisions may be offset by clean-condition errors that become correct under the perturbed distribution.

For the four model-dependent attacks, attack success rate is computed over the clean-correct population:

$$\text{ASR} = \frac{N(\hat{y}_{\text{clean}} = y \wedge \hat{y}_{\text{perturbed}} \neq y)}{N(\hat{y}_{\text{clean}} = y)}. \quad (6.5)$$

This definition differs from the generation-time success indicator recorded while creating the artifacts. The results reported here use matched clean and perturbed predictions and exclude samples already misclassified in their clean form from the ASR denominator.

Table 6.7 reports perturbed accuracy and accuracy drop for the five evaluated conditions.

Table 6.7: Accuracy and accuracy drop under adversarial and adversarial-style conditions

Condition	CLIP acc.	Drop	EffNet acc.	Drop	ResNet acc.	Drop
FGSM	0.914	0.013	0.575	0.403	0.918	0.046
SuperDeepFool	0.926	0.001	0.931	0.047	0.964	0.000
Sigma Zero Attack	0.944	-0.017	0.015	0.963	0.968	-0.004
One Pixel Attack	0.928	-0.001	0.976	0.002	0.963	0.001
Color Shift	0.925	0.002	0.977	0.001	0.963	0.001

Note. The first four conditions were generated against fold-specific EfficientNet-B0 checkpoints. Their evaluation on ResNet18 and the CLIP-based proxy measures transfer behavior. Color Shift is model-agnostic. Drop is computed as clean accuracy minus perturbed accuracy; negative values indicate a condition-specific aggregate increase and do not establish the absence of newly induced errors.

Figure 6.5 visualizes the accuracy-drop distribution. The concentration of the largest values in the EfficientNet-B0 column reflects the fact that the four model-dependent attacks were generated against this architecture.

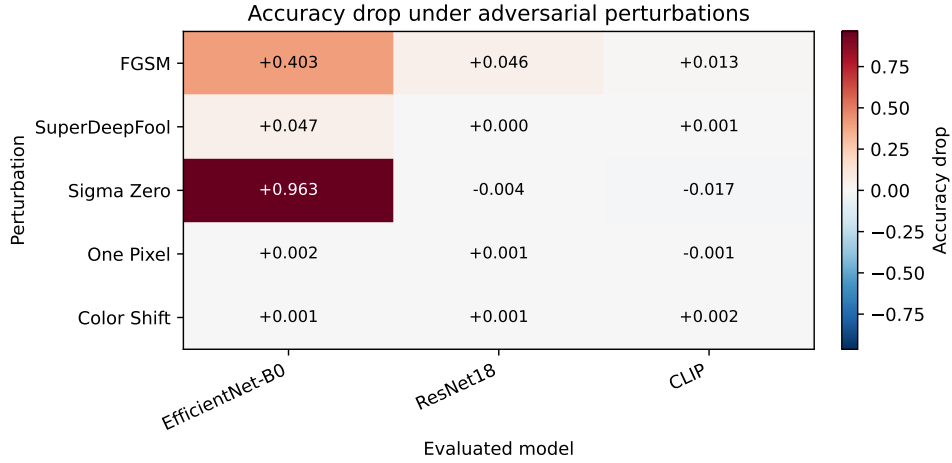


Figure 6.5: Accuracy drop of the proxy models under adversarial and adversarial-style conditions

The most severe result is observed for EfficientNet-B0 under Sigma Zero Attack. Accuracy decreases from 0.978 to 0.015, corresponding to a drop of 0.963. Among the 978 samples classified correctly in their clean form, 963 become incorrect, yielding an ASR of 0.9847.

The directional transitions are nearly balanced: 489 clean-correct **weapon** samples become **non_weapon**, while 474 clean-correct **non_weapon** samples become **weapon**. The result therefore represents a near-total collapse of the direct-target classifier rather than a degradation confined to one class.

The operational consequence is twofold. The 489 **weapon-to-non_weapon** transitions correspond to newly missed firearm-oriented images, while the 474 transitions in the opposite direction substantially increase irrelevant review workload.

FGSM also produces severe direct-target degradation. EfficientNet-B0 accuracy decreases to 0.575, corresponding to a drop of 0.403. The attack induces 406 errors among clean-correct samples and reaches an ASR of 0.4151.

In this case, the directional distribution is strongly asymmetric. Of the 406 induced errors, 321 are **weapon-to-non_weapon** transitions, whereas 85 occur in the opposite direction. FGSM therefore produces a particularly relevant missed-target risk in addition to its aggregate performance degradation.

SuperDeepFool has a smaller direct-target effect. EfficientNet-B0 accuracy decreases to 0.931, producing a drop of 0.047 and an ASR of 0.0481. All 47 induced errors occur from **non_weapon** to **weapon**; no clean-correct **weapon** image becomes **non_weapon** under this condition.

This distinction is operationally important. The aggregate accuracy drop indicates reduced correctness, but the observed direct-target failure mode increases false-positive review workload without introducing additional missed firearm-oriented images among the clean-correct target samples.

One Pixel Attack produces only two direct-target induced errors on EfficientNet-B0, one in each direction, corresponding to an ASR of 0.0020. Its perturbed accuracy is 0.976, only 0.002 below the clean baseline. Under the frozen parameters, this score-based condition is therefore substantially less effective against the target architecture than Sigma-Zero, FGSM, or SuperDeepFool.

Table 6.8 summarizes the most relevant direct-target and transferred model-dependent conditions.

Table 6.8: Operational indicators for selected model-dependent adversarial conditions

Model	Attack	Induced	ASR	Max-P shift	W→NW	NW→W
EfficientNet-B0	Sigma Zero Attack	963	0.9847	-0.2578	489	474
EfficientNet-B0	FGSM	406	0.4151	-0.1246	321	85
EfficientNet-B0	SuperDeepFool	47	0.0481	-0.3063	0	47
ResNet18	FGSM	47	0.0488	-0.0230	27	20
CLIP	FGSM	23	0.0248	-0.0043	8	15

Note. *Induced* is the number of clean-correct samples rendered incorrect by the perturbed condition. *ASR* uses the clean-correct denominator. *Max-P shift* is the mean perturbed maximum predicted-class probability minus the corresponding clean value. W→NW denotes **weapon-to-non_weapon** transitions; NW→W denotes the opposite direction.

The maximum-probability shift must be interpreted independently of the direction and number of induced errors. For example, SuperDeepFool produces the largest negative shift among the selected EfficientNet-B0 conditions, at -0.3063 , but induces substantially fewer errors than Sigma-Zero or FGSM. Moreover, all of its induced errors occur in the false-positive direction.

The shift is computed from the maximum predicted-class probability before and after perturbation. The maximizing class may differ between the clean and perturbed predictions. Consequently, the value does not measure the probability change of a fixed class and does not constitute a calibrated uncertainty estimate.

The transfer assessment shows that perturbations optimized against EfficientNet-B0 do not produce equivalent effects on the other proxy architectures. The largest transferred degradation occurs under FGSM. ResNet18 accuracy decreases from 0.964 to 0.918, corresponding to a drop of 0.046, 47 induced errors, and an ASR of 0.0488. Twenty-seven of these errors are **weapon-to-non_weapon** transitions.

For the CLIP-based proxy, FGSM decreases accuracy from 0.927 to 0.914, corresponding to a drop of 0.013. It induces 23 errors and reaches an ASR of 0.0248. Eight transitions occur from **weapon** to **non_weapon**, while 15 occur in the opposite direction.

Transferred Sigma-Zero artifacts do not reproduce the direct-target collapse. ResNet18 accuracy increases by 0.004 and the CLIP-based proxy accuracy increases by 0.017. These aggregate increases nevertheless coexist with three induced errors for ResNet18 and eight for CLIP. The result again confirms that aggregate improvement does not establish sample-level invariance.

Transferred SuperDeepFool and One Pixel artifacts produce limited effects. SuperDeepFool induces no errors among clean-correct ResNet18 samples and three among clean-correct CLIP samples. One Pixel induces one error for ResNet18 and none for CLIP under the frozen evaluation.

Color Shift produces only marginal aggregate changes across the three architectures. Because it is model-agnostic, its clean-correct-to-incorrect transitions are interpreted as induced errors rather than attack or transfer success. Its role is to provide a common adversarial-style comparison condition rather than evidence of white-box vulnerability.

The adversarial results therefore identify a strong dependence on the relationship between the perturbation-generation model and the evaluated architecture. EfficientNet-B0 achieves the highest clean accuracy but undergoes the most severe degradation when the attacks are optimized directly against its fold-specific checkpoints. The same artifacts transfer only partially to ResNet18 and the CLIP-based proxy.

This limited transferability must not be interpreted as general robustness of the non-target architectures. It establishes only that the frozen EfficientNet-B0-oriented perturbations were less effective against the other proxy models. A direct attack generated against ResNet18 or the CLIP-based proxy could produce a different result and was outside the experimental scope.

From a forensic perspective, the findings demonstrate that high clean accuracy does not ensure operational stability under deliberately manipulated inputs. They also show that aggregate degradation alone is insufficient: the same accuracy drop may represent newly missed target images, increased false-positive workload, or a combination of both.

The output of an AI-based classification model must therefore remain a triage indicator rather than autonomous evidence. Its use requires systematic human verification, traceable processing, explicit knowledge of the evaluated threat model, and awareness that robustness conclusions are bounded to the tested architectures, perturbations, parameters, and access assumptions.

6.6 Black-Box Software Results

The implementation-specific configurations, observable output fields, operational mappings, and normalization procedures adopted for the selected black-box software systems are documented in Section 5.6. The present section reports the resulting normalized behavior without making assumptions about inaccessible internal architectures, training data, thresholds, or confidence-calibration procedures.

The analysis comprises four software families and six normalized operational configurations. These include Magnet AXIOM/Magnet.AI, Magnet Griffeye/T3K CORE, Excire Foto 2025 under the D20, D50, and D80 semantic-distance settings, and Cellebrite Inseyets.

Each configuration produced one normalized decision for every item in the 11 500-file forensic evaluation bundle. The complete black-box population therefore contains:

$$6 \text{ normalized configurations} \times 11\,500 \text{ bundle items} = 69\,000 \text{ matched decisions.} \quad (6.6)$$

No bundle item remained without a matched normalized decision, and the canonical normalized result population contains no **unknown** outputs. Non-bundle records present in some controlled

raw exports were excluded during normalization, as described in Section 5.6.5.

The 11 500 matched decisions associated with each configuration comprise two analytically distinct populations. The first consists of 11 000 binary items with a reference operational assignment of **weapon** or **non_weapon**. The second consists of 500 OOD inputs, which are analyzed separately through their observable weapon assignment rate.

Accuracy, balanced accuracy, precision, recall, FNR, and FPR are computed only on the 11 000 interpretable binary items. The OOD inputs are excluded from those metrics because they have no binary reference assignment.

The binary aggregate is itself bundle-weighted rather than condition-balanced. It contains 1 000 clean inputs and 10 000 perturbed artifacts:

$$N_{\text{binary}} = 1\,000 \text{ clean inputs} + 5\,000 \text{ adversarial artifacts} + 5\,000 \text{ anti-forensic artifacts} = 11\,000. \quad (6.7)$$

Consequently, the bundle-level metrics are influenced primarily by behavior on the perturbed population. They must not be interpreted as the arithmetic mean of clean, adversarial, and anti-forensic performance, nor as estimates of the frequency with which these conditions would occur in an operational case.

The four model-dependent adversarial conditions were generated against fold-specific EfficientNet-B0 checkpoints. Their evaluation through the selected black-box software systems therefore constitutes an empirical transfer assessment. It does not constitute a white-box attack against any commercial system. Color Shift instead provides a deterministic model-agnostic comparison condition.

Table 6.9 summarizes the normalized results for the six configurations.

Table 6.9: Bundle-weighted summary of the selected black-box software configurations

Configuration	Matched	Clean FNR	Bundle Acc.	Bundle Bal.	Bundle R_w	Bundle FNR	Bundle FPR	OOD W-rate
Magnet AXIOM/Magnet.AI	11 500	0.070	0.933	0.933	0.901	0.099	0.035	0.360
Magnet Griffeye/T3K CORE	11 500	0.036	0.972	0.972	0.951	0.049	0.007	0.260
Excire Foto 2025 D20	11 500	0.120	0.911	0.911	0.857	0.143	0.036	0.238
Excire Foto 2025 D50	11 500	0.042	0.925	0.925	0.949	0.051	0.099	0.340
Excire Foto 2025 D80	11 500	0.014	0.887	0.887	0.981	0.019	0.207	0.522
Cellebrite Inseyets	11 500	0.032	0.958	0.958	0.958	0.042	0.042	0.292

Note. *Matched* indicates the complete 11 500-item bundle population associated with each normalized configuration. *Clean FNR* is computed on the 1 000 unperturbed binary inputs. *Bundle accuracy*, *balanced accuracy*, *weapon recall*, *FNR*, and *FPR* are computed on the 11 000-item binary population, comprising clean, adversarial, and anti-forensic samples and excluding OOD inputs. *OOD W-rate* indicates the proportion of the 500 OOD inputs assigned to **weapon**. The bundle-level values reflect the frozen experimental composition and do not constitute general validation or an absolute ranking of the selected software systems.

Figure 6.6 visualizes the same bundle-weighted metrics.

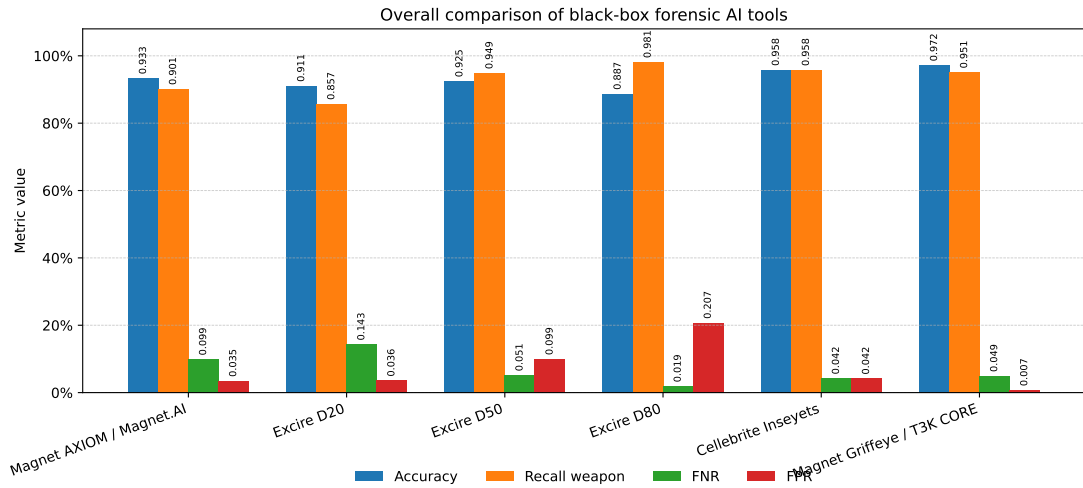


Figure 6.6: Bundle-weighted normalized metrics of the selected black-box software configurations

The aggregate comparison identifies distinct operational profiles, but it must not be interpreted as a direct ranking of intrinsic system quality. The evaluated software systems differ in intended use, internal taxonomy, observable output semantics, configuration, and decision mechanism. Furthermore, the binary bundle contains repeated manipulated versions derived from the same 1 000 source images rather than 11 000 independent real-world cases.

The table exposes three distinct profiles that the software-specific subsections examine in detail. Griffeye/T3K CORE has the highest bundle accuracy and lowest FPR; Cellebrite is comparatively balanced; and Magnet.AI has a larger perturbed missed-target burden than its clean result. Excire most clearly demonstrates configuration sensitivity: moving from D20 to D80 reduces the FNR while increasing both the FPR and OOD weapon-assignment rate. Greater retrieval coverage therefore exchanges missed-target risk for additional review workload.

Figure 6.7 reports the observable weapon assignment rate on the separate OOD branch.

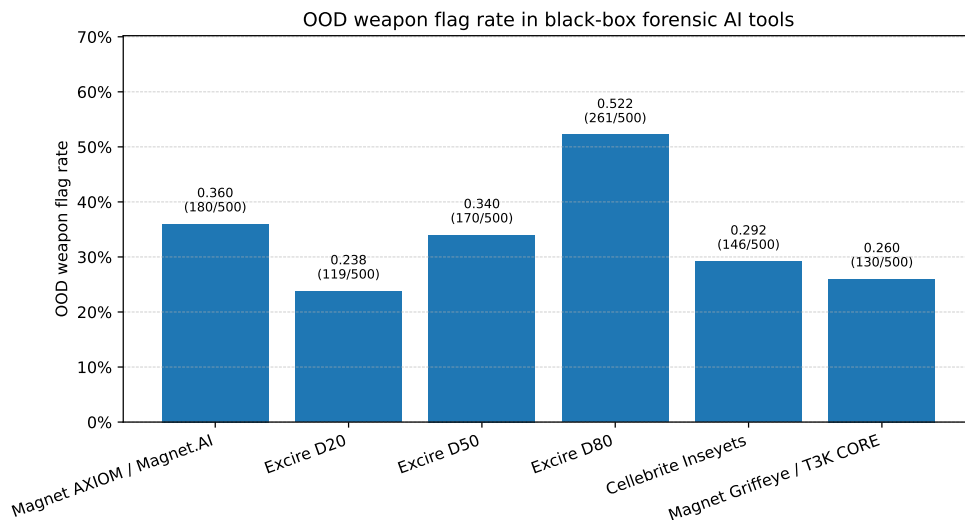


Figure 6.7: Weapon assignment rate of the selected black-box software configurations on OOD inputs

All configurations assign some OOD inputs to `weapon`, with rates ranging from 0.238 (Excire

D20) to 0.522 (Excire D80). These are not supervised false positives, but they can increase unrelated review volume. No configuration simultaneously minimizes missed targets, false positives, and OOD assignments; suitability therefore depends on the intended workflow and the consequence of each error direction.

6.6.1 Magnet AXIOM/Magnet.AI Results

Under the mapping documented in Section 5.6.1, the exported `Possible weapons` tag is the positive observable output; its absence is mapped to `non_weapon`. No unsupported `unknown` state occurs in the canonical population.

Table 6.10 reports the bundle-weighted binary metrics.

Table 6.10: Bundle-weighted performance of Magnet.AI on the binary `weapon/non_weapon` task

Metric	Value
Accuracy	0.933
Balanced accuracy	0.933
Precision <code>weapon</code>	0.963
Recall <code>weapon</code>	0.901
F1-score <code>weapon</code>	0.931
False negative rate	0.099
False positive rate	0.035
True positives	4 958
True negatives	5 309
False positives	191
False negatives	542

Note. The metrics are computed on the 11 000-item binary bundle, comprising 1 000 clean inputs and 10 000 perturbed artifacts. The values are therefore bundle-weighted and do not represent an arithmetic mean across experimental conditions.

Within the frozen binary bundle, Magnet.AI achieves an accuracy and balanced accuracy of 0.933. Precision for the `weapon` class is 0.963, indicating that most files assigned the positive operational state are reference `weapon` samples.

Weapon recall is lower, at 0.901, corresponding to an FNR of 0.099. Across the 5 500 reference `weapon` items in the binary bundle, 542 do not receive the positive observable tag. The FPR is 0.035, corresponding to 191 `non_weapon` items assigned to the operational `weapon` class.

These aggregate values must be interpreted in relation to the composition of the bundle. Perturbed artifacts outnumber clean inputs by ten to one. Consequently, the overall metrics primarily describe behavior across the combined perturbed population rather than nominal clean-condition performance.

Table 6.11 separates the clean, aggregate perturbed, adversarial-family, and anti-forensic-family binary results. The separate OOD result is discussed below.

Table 6.11: Behavior of Magnet.AI by experimental condition

Condition	Accuracy	Weapon recall	FNR	FPR
Clean	0.950	0.930	0.070	0.030
Perturbed	0.932	0.899	0.101	0.035
Adversarial family	0.923	0.884	0.116	0.038
Anti-forensic family	0.940	0.913	0.087	0.032

The clean baseline yields an accuracy of 0.950, a weapon recall of 0.930, an FNR of 0.070, and an

FPR of 0.030. This corresponds to 35 missed **weapon** images and 15 false-positive assignments among the 1 000 unperturbed inputs.

Across the complete perturbed population, accuracy decreases to 0.932 and weapon recall decreases to 0.899. The FNR consequently increases to 0.101, while the FPR increases more moderately to 0.035. The principal aggregate change therefore concerns a greater proportion of missed firearm-oriented inputs.

The adversarial-family population produces a stronger effect than the anti-forensic population under the frozen protocol. Its accuracy is 0.923, weapon recall is 0.884, and FNR is 0.116. The anti-forensic population retains an accuracy of 0.940, weapon recall of 0.913, and FNR of 0.087.

As defined in Section 6.1, the adversarial-family results measure transfer from EfficientNet-B0; Color Shift and the anti-forensic transformations are model-agnostic.

On the separate OOD branch, 180 of 500 inputs receive the positive tag (0.360), an operational review-load indicator rather than a supervised false positive. Magnet.AI exposes no comparable numerical uncertainty output for this mapping.

Table 6.12 reports the condition-level behavior for all ten perturbed populations.

Table 6.12: Performance of Magnet.AI under adversarial, adversarial-style, and anti-forensic conditions

Family	Condition	Accuracy	Recall	FNR	FPR	FN	FP
Adversarial-style	Color Shift	0.951	0.934	0.066	0.032	33	16
Adversarial transfer	FGSM	0.867	0.808	0.192	0.074	96	37
Adversarial transfer	One Pixel Attack	0.950	0.930	0.070	0.030	35	15
Adversarial transfer	Sigma Zero Attack	0.916	0.856	0.144	0.024	72	12
Adversarial transfer	SuperDeepFool	0.932	0.894	0.106	0.030	53	15
Anti-forensic	Contrast Stretching	0.951	0.930	0.070	0.028	35	14
Anti-forensic	Gaussian Blur	0.927	0.880	0.120	0.026	60	13
Anti-forensic	Histogram Modification	0.936	0.920	0.080	0.048	40	24
Anti-forensic	JPEG recompression	0.944	0.928	0.072	0.040	36	20
Anti-forensic	Resample + Resize	0.943	0.906	0.094	0.020	47	10

Among the transferred model-dependent adversarial conditions, FGSM produces the most substantial degradation. Accuracy decreases to 0.867 and weapon recall to 0.808, while the FNR increases to 0.192. This corresponds to 96 missed **weapon** items and 37 false-positive assignments in the 1 000-item FGSM population.

Transferred Sigma-Zero artifacts produce the second-highest missed-target burden. Accuracy is 0.916, weapon recall is 0.856, and the FNR is 0.144, corresponding to 72 false negatives. SuperDeepFool produces an FNR of 0.106 and 53 false negatives.

One Pixel produces the same aggregate confusion-matrix values as the clean baseline: accuracy 0.950, FNR 0.070, and FPR 0.030. This result is specific to the frozen transferred artifacts and does not establish general invariance to score-based attacks generated directly against Magnet.AI.

The model-agnostic Color Shift condition produces an accuracy of 0.951 and an FNR of 0.066, values close to the clean baseline. Because it was not optimized against a model, its effect is interpreted as condition-specific behavior rather than adversarial transfer success.

Among the anti-forensic transformations, Gaussian Blur produces the largest missed-target effect. Weapon recall decreases to 0.880 and the FNR increases to 0.120, corresponding to 60 false negatives. Histogram Modification produces a lower FNR, at 0.080, but the highest anti-forensic FPR, at 0.048, corresponding to 24 false positives.

The remaining transformations produce smaller changes but do not preserve the same operational profile. JPEG Recompression yields 36 false negatives and 20 false positives, while

Resample Resize yields 47 false negatives and 10 false positives. Contrast Stretching remains close to the clean baseline.

The results therefore identify failure modes that are not captured by overall accuracy alone. FGSM transfer and Gaussian Blur primarily increase the proportion of firearm-oriented inputs not receiving the positive tag, whereas Histogram Modification produces a comparatively greater increase in false-positive review workload.

These results characterize only the frozen Magnet.AI version, category, configuration, population, and mapping. The tag remains a traceable triage aid requiring human verification, not an evidentiary conclusion.

6.6.2 Magnet Griffeye/T3K CORE Results

Under the frozen operational mapping, only the **CORE/Violence/Firearm** bookmark is treated as a positive **weapon** output. Other bookmarks related to weapons, violence, military equipment, or adjacent semantic categories are preserved in the controlled raw output but are not included as primary positives, as documented in Section 5.6.2.

This mapping is deliberately conservative. Consequently, the observed false positive rate must be interpreted jointly with the selected bookmark rule. A more permissive mapping incorporating secondary weapon-related bookmarks would define a different operational task and could alter both target recall and false-positive review workload. That alternative was not evaluated in the frozen protocol.

Table 6.13 reports the bundle-weighted binary results.

Table 6.13: Bundle-weighted performance of Magnet Griffeye/T3K CORE on the binary **weapon/non_weapon** task

Metric	Value
Accuracy	0.972
Balanced accuracy	0.972
Precision weapon	0.992
Recall weapon	0.951
F1-score weapon	0.971
False negative rate	0.049
False positive rate	0.007
True positives	5 229
True negatives	5 460
False positives	40
False negatives	271

Note. The metrics are computed on the 11 000-item binary bundle, comprising 1 000 clean inputs and 10 000 perturbed artifacts. The 500 OOD inputs are excluded from binary correctness metrics. The low FPR must also be interpreted in relation to the conservative positive mapping based exclusively on the **CORE/Violence/Firearm** bookmark.

Within the frozen binary bundle, Magnet Griffeye/T3K CORE achieves an accuracy and balanced accuracy of 0.972. Precision for the **weapon** class is 0.992, corresponding to only 40 false-positive assignments among the 5 500 reference **non_weapon** items.

Weapon recall is 0.951, corresponding to an FNR of 0.049. A total of 271 reference **weapon** items are therefore not associated with the primary firearm bookmark.

The result represents the highest bundle-level accuracy and the lowest bundle-level FPR among the six normalized configurations under the frozen protocol. It must not be interpreted as evidence of intrinsically superior system quality because it is conditional on the evaluated software version, the selected category, the imported population, and the conservative normalization rule.

Table 6.14 separates the clean, aggregate perturbed, adversarial-family, and anti-forensic-family binary results. The separate OOD result is discussed below.

Table 6.14: Behavior of Magnet Griffey/T3K CORE by experimental condition

Condition	Accuracy	Weapon recall	FNR	FPR
Clean	0.978	0.964	0.036	0.008
Perturbed	0.971	0.949	0.051	0.007
Adversarial family	0.969	0.945	0.055	0.008
Anti-forensic family	0.974	0.954	0.046	0.006

The clean baseline produces an accuracy of 0.978, weapon recall of 0.964, FNR of 0.036, and FPR of 0.008. This corresponds to 18 missed **weapon** inputs and four false-positive assignments among the 1 000 clean images.

Across the 10 000 perturbed artifacts, accuracy decreases to 0.971 and weapon recall decreases to 0.949. The FNR increases to 0.051, whereas the FPR remains slightly lower than the clean value, at 0.007. The primary aggregate change therefore concerns a moderate increase in missed firearm-oriented inputs rather than increased false-positive workload.

The adversarial-family population produces an accuracy of 0.969, weapon recall of 0.945, and FNR of 0.055. The anti-forensic population exhibits an accuracy of 0.974, weapon recall of 0.954, and FNR of 0.046.

The difference between the two families is limited but directionally consistent: under the frozen protocol, the transferred adversarial artifacts produce a somewhat larger missed-target burden than the model-agnostic anti-forensic transformations.

The adversarial-family values are transfer results under the common scope. On the separate OOD branch, 130 of 500 inputs receive the firearm bookmark (0.260), which can affect review workload but is not a supervised false positive.

Table 6.15 reports the ten individual perturbed conditions.

Table 6.15: Performance of Magnet Griffey/T3K CORE under adversarial, adversarial-style, and anti-forensic conditions

Family	Condition	Accuracy	Recall	FNR	FPR	FN	FP
Adversarial-style	Color Shift	0.977	0.960	0.040	0.006	20	3
Adversarial transfer	FGSM	0.960	0.928	0.072	0.008	36	4
Adversarial transfer	One Pixel Attack	0.978	0.964	0.036	0.008	18	4
Adversarial transfer	Sigma Zero Attack	0.959	0.926	0.074	0.008	37	4
Adversarial transfer	SuperDeepFool	0.969	0.948	0.052	0.010	26	5
Anti-forensic	Contrast Stretching	0.977	0.964	0.036	0.010	18	5
Anti-forensic	Gaussian Blur	0.972	0.948	0.052	0.004	26	2
Anti-forensic	Histogram Modification	0.963	0.932	0.068	0.006	34	3
Anti-forensic	JPEG recompression	0.976	0.960	0.040	0.008	20	4
Anti-forensic	Resample + Resize	0.980	0.964	0.036	0.004	18	2

Among the transferred model-dependent conditions, Sigma-Zero produces the largest missed-target burden. Accuracy decreases to 0.959, weapon recall to 0.926, and the FNR increases to 0.074, corresponding to 37 false negatives and four false positives.

FGSM produces a closely related profile, with an accuracy of 0.960, weapon recall of 0.928, and FNR of 0.072. The population contains 36 false negatives and four false positives.

The transferred Sigma-Zero artifacts therefore produce measurable degradation but do not reproduce the near-total collapse observed for their direct EfficientNet-B0 target. This difference

is evidence of limited transfer under the evaluated conditions, not proof of general robustness against attacks generated directly against Griffeye/T3K CORE.

SuperDeepFool produces an accuracy of 0.969 and an FNR of 0.052. One Pixel preserves the same aggregate confusion-matrix values as the clean baseline, with 18 false negatives and four false positives. This result is specific to the frozen transferred population.

The model-agnostic Color Shift condition remains close to the clean baseline, with an accuracy of 0.977, FNR of 0.040, and FPR of 0.006. Its effect is interpreted as condition-specific behavior rather than adversarial transfer success.

Among the anti-forensic transformations, Histogram Modification produces the largest missed-target effect. Accuracy decreases to 0.963, weapon recall to 0.932, and the FNR increases to 0.068, corresponding to 34 false negatives.

Gaussian Blur produces 26 false negatives but only two false positives. JPEG Recompression produces 20 false negatives and four false positives. Contrast Stretching produces the highest anti-forensic FPR, at 0.010, although this corresponds to only five false-positive assignments.

Resample Resize produces the highest condition-level accuracy, at 0.980, while retaining 18 false negatives and two false positives. The slightly higher accuracy than the clean baseline does not establish sample-level invariance: aggregate improvements may coexist with changed individual decisions.

Overall, the frozen Griffeye/T3K CORE configuration combines high firearm-oriented recall with a particularly low false-positive rate. The profile is operationally selective, but its interpretation is inseparable from the conservative choice to count only **CORE/Violence/Firearm** as a primary positive.

The profile is bounded to the frozen version, modules, population, and bookmark mapping. The automatic bookmark remains a traceable triage signal requiring human verification, not an evidentiary conclusion.

6.6.3 Excire Foto 2025 Results and Sensitivity Analysis

Using the union-based mapping documented in Section 5.6.3, this subsection compares the same eight firearm-oriented queries at D20, D50, and D80. Within the evaluated range, these settings are respectively the most restrictive, intermediate, and most permissive retrieval configurations; the terms describe observable behavior, not the internal scoring mechanism.

Table 6.16 reports clean binary performance and OOD weapon assignment behavior for the three configurations.

Table 6.16: Sensitivity of Excire Foto 2025 to the `Distance limit` setting

Setting	Clean Acc.	Clean R_w	Clean FNR	Clean FPR	OOD W-rate	Operational profile
D20	0.927	0.880	0.120	0.026	0.238	Restrictive profile with the lowest clean false-positive rate and the lowest OOD weapon assignment rate, but the highest clean missed-target rate.
D50	0.944	0.958	0.042	0.070	0.340	Intermediate profile balancing target recall, false-positive workload, and OOD weapon assignments.
D80	0.915	0.986	0.014	0.156	0.522	Permissive profile with the highest clean target recall, accompanied by a substantial increase in false-positive and OOD assignments.

The three settings expose a clear operational tradeoff. Under clean conditions, D20 produces the lowest FPR, at 0.026, but its weapon recall is 0.880, corresponding to an FNR of 0.120.

The configuration therefore limits irrelevant retrievals at the cost of missing a larger proportion of firearm-oriented images.

At D50, weapon recall increases to 0.958 and the FNR decreases to 0.042. The FPR increases to 0.070. This setting provides the highest clean accuracy among the three configurations, at 0.944, but this value does not independently establish the most appropriate operating point for every forensic workflow.

At D80, weapon recall reaches 0.986 and the clean FNR decreases to 0.014. Only seven of the 500 clean reference **weapon** images are not retrieved by any of the eight firearm-oriented queries. This reduction in missed detections is accompanied by 78 false-positive assignments and an FPR of 0.156.

The same progressive expansion is visible on the separate OOD branch. The weapon assignment rate increases from 0.238 at D20, corresponding to 119 of the 500 OOD inputs, to 0.340 at D50, corresponding to 170 inputs, and to 0.522 at D80, corresponding to 261 inputs.

These OOD assignments are not false positives in the supervised binary sense because the images have no **weapon** or **non_weapon** reference assignment. Nevertheless, their increase indicates that the more permissive settings route a larger proportion of material outside the nominal binary task toward the firearm-oriented review category.

The **Distance limit** setting must therefore be treated as an operational configuration parameter rather than a purely technical detail. A restrictive setting reduces false-positive review workload but increases missed-target risk. A permissive setting reduces missed detections but increases the volume of **non_weapon** and OOD material returned for human review.

Table 6.17 reports aggregate behavior for the adversarial and anti-forensic populations.

Table 6.17: Aggregate performance of Excire Foto 2025 on perturbed populations

Setting	Family	Accuracy	Recall weapon	FNR	FPR
D20	Adversarial	0.892	0.834	0.166	0.051
D20	Anti-forensic	0.927	0.875	0.125	0.022
D50	Adversarial	0.904	0.938	0.062	0.130
D50	Anti-forensic	0.941	0.957	0.043	0.074
D80	Adversarial	0.861	0.977	0.023	0.256
D80	Anti-forensic	0.908	0.984	0.016	0.169

Note. The adversarial family comprises four model-dependent artifact populations generated against EfficientNet-B0 and the model-agnostic Color Shift condition. Their evaluation through Excire constitutes transfer or condition-level assessment, not a white-box attack against the software.

The same recall-workload tradeoff persists under perturbed conditions. D20 maintains the lowest FPR in both families, at 0.051 for the adversarial population and 0.022 for the anti-forensic population. However, its FNR reaches 0.166 and 0.125, respectively.

At D50, the adversarial-family FNR decreases to 0.062, while the FPR increases to 0.130. Under anti-forensic conditions, the corresponding values are 0.043 and 0.074.

At D80, the adversarial-family FNR decreases to 0.023, but the FPR reaches 0.256. Under anti-forensic conditions, the FNR is 0.016 and the FPR is 0.169.

For every setting, the adversarial-family population produces lower aggregate accuracy and a higher FPR than the anti-forensic population. The difference is particularly pronounced for D80, where permissive retrieval interacts with the adversarial artifacts to route more than one quarter of the reference **non_weapon** population to the positive operational state.

These results do not demonstrate that the software was attacked directly. The four model-dependent artifact populations were generated against fold-specific EfficientNet-B0 checkpoints and subsequently processed by Excire. Their effects therefore represent empirical

transfer behavior. Color Shift and the five anti-forensic transformations are model-agnostic conditions.

The D50 configuration is retained as the intermediate reference for condition-level analysis. This choice facilitates comparison without implying that D50 is universally optimal. The appropriate operating point depends on the relative cost assigned to missed target images and additional human-review workload.

Table 6.18 reports the ten individual perturbed conditions for D50.

Table 6.18: Performance of Excire Foto 2025 D50 under adversarial, adversarial-style, and anti-forensic conditions

Family	Condition	Accuracy	Recall	FNR	FPR	FN	FP
Adversarial-style	Color Shift	0.949	0.966	0.034	0.068	17	34
Adversarial transfer	FGSM	0.926	0.890	0.110	0.038	55	19
Adversarial transfer	One Pixel Attack	0.946	0.964	0.036	0.072	18	36
Adversarial transfer	Sigma Zero Attack	0.742	0.926	0.074	0.442	37	221
Adversarial transfer	SuperDeepFool	0.956	0.944	0.056	0.032	28	16
Anti-forensic	Contrast Stretching	0.941	0.952	0.048	0.070	24	35
Anti-forensic	Gaussian Blur	0.941	0.960	0.040	0.078	20	39
Anti-forensic	Histogram Modification	0.932	0.952	0.048	0.088	24	44
Anti-forensic	JPEG recompression	0.946	0.958	0.042	0.066	21	33
Anti-forensic	Resample + Resize	0.947	0.964	0.036	0.070	18	35

At D50, FGSM produces the largest missed-target burden among the transferred model-dependent conditions. Weapon recall decreases to 0.890 and the FNR increases to 0.110, corresponding to 55 false negatives. However, its FPR remains comparatively limited, at 0.038.

Sigma-Zero produces a different operational failure mode. Weapon recall remains relatively high, at 0.926, and the FNR is 0.074. Nevertheless, the FPR increases to 0.442: 221 of the 500 reference `non_weapon` images are returned by at least one firearm-oriented query.

The resulting accuracy of 0.742 is therefore driven primarily by an expansion of positive assignments rather than by a predominant increase in missed firearm-oriented content. The operational consequence is a substantial increase in false-positive review workload.

SuperDeepFool produces the highest D50 accuracy among the transferred model-dependent conditions, at 0.956, with 28 false negatives and 16 false positives. One Pixel remains close to the clean D50 profile, with 18 false negatives and 36 false positives.

The model-agnostic Color Shift condition produces 17 false negatives and 34 false positives. Its behavior is interpreted as a common perturbation condition rather than adversarial transfer success.

Among the anti-forensic transformations, Histogram Modification produces the lowest accuracy, at 0.932, and the highest FPR, at 0.088, corresponding to 44 false positives. Contrast Stretching produces 24 false negatives and 35 false positives, whereas Gaussian Blur produces 20 false negatives and 39 false positives.

JPEG Recompression and Resample Resize remain close to the clean D50 operating profile. Their aggregate similarity does not establish sample-level invariance, because individual assignments may differ even when the resulting confusion counts are similar.

Sigma-Zero provides the clearest illustration of the interaction between the perturbed population and the semantic-distance configuration. Table 6.19 compares the three settings.

Table 6.19: Behavior of Excire Foto 2025 on the transferred Sigma Zero Attack population

Setting	Accuracy	Recall	FNR	FPR	FN	FP
D20	0.815	0.820	0.180	0.190	90	95
D50	0.742	0.926	0.074	0.442	37	221
D80	0.657	0.976	0.024	0.662	12	331

As the distance setting becomes more permissive, weapon recall on the Sigma-Zero population increases from 0.820 at D20 to 0.926 at D50 and 0.976 at D80. The number of false negatives consequently decreases from 90 to 37 and then to 12.

At the same time, the FPR increases from 0.190 to 0.442 and then to 0.662. The number of false positives increases from 95 at D20 to 221 at D50 and 331 at D80.

The reduction in aggregate accuracy across the three settings must therefore not be interpreted as a conventional evasion pattern dominated by missed target images. Under D50 and especially D80, the principal effect is the expansion of positive retrievals into the `non_weapon` population.

The observed behavior may derive from the interaction between the perturbed visual content, the eight semantic queries, the internal retrieval representation, and the selected distance limit. Because the software is black-box, the experiment cannot determine the internal mechanism responsible for this expansion.

The result nevertheless has a clear operational implication. A permissive setting may preserve or increase firearm-oriented recall while substantially reducing the selectivity of the automated filter. Analysts may receive fewer missed target images but a much larger volume of irrelevant material requiring manual review.

Overall, Excire Foto 2025 demonstrates that configuration sensitivity is a central component of black-box robustness evaluation. The same software, queries, and image population produce materially different false-negative, false-positive, and OOD assignment profiles when the semantic-distance setting changes.

No single setting simultaneously minimizes missed detections, false-positive workload, and OOD weapon assignments. D20 favors selectivity, D80 favors retrieval coverage, and D50 provides an intermediate operating point within the evaluated range.

The conclusion is bounded to the frozen version, eight-query set, distance settings, population, and union mapping. Excire outputs remain triage indicators requiring documented configuration and human verification.

6.6.4 Celebrite Inseyets Results

Under the frozen mapping, an image is assigned to `weapon` when the exported `Classifications` field contains at least one of the labels `Armi`, `Pistola`, or `Fucile`. In the absence of these labels, the image is assigned to `non_weapon`, as documented in Section 5.6.4.

The percentages associated with individual exported labels are preserved in the auditable source field but are removed before applying the operational mapping. They are not treated as calibrated probabilities and do not influence the normalized binary decision.

Table 6.20 reports the bundle-weighted binary results.

Table 6.20: Bundle-weighted performance of Cellebrite Inseyets on the binary `weapon/non_weapon` task

Metric	Value
Accuracy	0.958
Balanced accuracy	0.958
Precision <code>weapon</code>	0.958
Recall <code>weapon</code>	0.958
F1-score <code>weapon</code>	0.958
False negative rate	0.042
False positive rate	0.042
True positives	5 268
True negatives	5 271
False positives	229
False negatives	232

Note. The metrics are computed on the 11 000-item binary bundle, comprising 1 000 clean inputs and 10 000 perturbed artifacts. The values are therefore bundle-weighted and do not represent an arithmetic mean across experimental conditions. The 500 OOD inputs are excluded from binary correctness metrics.

Within the frozen binary bundle, Cellebrite Inseyets achieves an accuracy and balanced accuracy of 0.958. Precision and recall for the `weapon` class are also approximately 0.958.

The resulting error profile is comparatively symmetrical. A total of 232 reference `weapon` items are assigned to `non_weapon`, whereas 229 reference `non_weapon` items are assigned to `weapon`. The corresponding FNR and FPR are 0.042 and 0.042, respectively.

This approximate balance does not imply that the two error directions have the same operational consequence. False negatives may reduce the priority assigned to firearm-oriented evidence, whereas false positives increase the volume of material requiring analyst review.

Table 6.21 separates the clean, aggregate perturbed, adversarial-family, and anti-forensic-family binary results. The separate OOD result is discussed below.

Table 6.21: Behavior of Cellebrite Inseyets by experimental condition

Condition	Accuracy	Weapon recall	FNR	FPR
Clean	0.964	0.968	0.032	0.040
Perturbed	0.958	0.957	0.043	0.042
Adversarial family	0.955	0.950	0.050	0.040
Anti-forensic family	0.960	0.963	0.037	0.043

The clean baseline yields an accuracy of 0.964, weapon recall of 0.968, FNR of 0.032, and FPR of 0.040. This corresponds to 16 missed `weapon` inputs and 20 false-positive assignments among the 1 000 unperturbed images.

Across the complete perturbed population, accuracy decreases to 0.958 and weapon recall decreases to 0.957. The FNR increases to 0.043, whereas the FPR remains close to the clean value, at 0.042.

The aggregate changes are therefore limited, but the perturbed population still contains a greater proportion of missed firearm-oriented inputs than the clean baseline. Aggregate stability must not be interpreted as invariance of individual decisions.

The adversarial-family population produces an accuracy of 0.955, weapon recall of 0.950, and FNR of 0.050. The anti-forensic population exhibits an accuracy of 0.960, weapon recall of 0.963, and FNR of 0.037.

Under the frozen protocol, the adversarial-family population therefore produces a somewhat larger missed-target burden than the anti-forensic population. The anti-forensic family instead produces a slightly higher FPR, at 0.043, compared with 0.040 for the adversarial family.

The adversarial-family values are transfer results under the common scope. On the separate OOD branch, 146 of 500 inputs are assigned to the positive state (0.292), a review-load indicator rather than a supervised false positive.

Table 6.22 reports the ten individual perturbed conditions.

Table 6.22: Performance of Celebrite Inseyets under adversarial, adversarial-style, and anti-forensic conditions

Family	Condition	Accuracy	Recall	FNR	FPR	FN	FP
Adversarial-style	Color Shift	0.964	0.968	0.032	0.040	16	20
Adversarial transfer	FGSM	0.918	0.894	0.106	0.058	53	29
Adversarial transfer	One Pixel Attack	0.962	0.968	0.032	0.044	16	22
Adversarial transfer	Sigma Zero Attack	0.970	0.958	0.042	0.018	21	9
Adversarial transfer	SuperDeepFool	0.961	0.964	0.036	0.042	18	21
Anti-forensic	Contrast Stretching	0.967	0.972	0.028	0.038	14	19
Anti-forensic	Gaussian Blur	0.954	0.964	0.036	0.056	18	28
Anti-forensic	Histogram Modification	0.949	0.940	0.060	0.042	30	21
Anti-forensic	JPEG recompression	0.966	0.972	0.028	0.040	14	20
Anti-forensic	Resample + Resize	0.964	0.968	0.032	0.040	16	20

Among the transferred model-dependent adversarial conditions, FGSM produces the largest degradation. Accuracy decreases to 0.918 and weapon recall to 0.894, while the FNR increases to 0.106. The population contains 53 false negatives and 29 false positives.

The FGSM result therefore represents the most relevant missed-target failure mode observed for the frozen Celebrite configuration. Its 53 missed **weapon** inputs considerably exceed the 16 false negatives observed on the clean population.

Transferred Sigma-Zero artifacts produce a different profile. Accuracy reaches 0.970, exceeding the clean aggregate value, while the FNR is 0.042 and the FPR is 0.018. The corresponding population contains 21 false negatives and nine false positives.

This result does not demonstrate general robustness to Sigma-Zero. It shows only that artifacts generated against EfficientNet-B0 do not reproduce their direct-target failure mode when processed by the frozen Celebrite configuration.

One Pixel remains close to the clean profile, retaining 16 false negatives but increasing false positives from 20 to 22. SuperDeepFool produces 18 false negatives and 21 false positives.

The model-agnostic Color Shift condition reproduces the same aggregate confusion-matrix values as the clean baseline. This equality does not guarantee sample-level invariance because different individual decisions may produce the same aggregate counts.

Among the anti-forensic transformations, Histogram Modification produces the largest missed-target burden. Weapon recall decreases to 0.940 and the FNR increases to 0.060, corresponding to 30 false negatives and 21 false positives.

Gaussian Blur produces the highest anti-forensic FPR, at 0.056, corresponding to 28 false positives. Its FNR remains comparatively limited, at 0.036, with 18 false negatives.

Contrast Stretching and JPEG Recompression yield the lowest anti-forensic FNR, at 0.028, corresponding to 14 false negatives each. Resample Resize reproduces the same aggregate confusion-matrix values as the clean baseline.

The condition-level results therefore show that the aggregate balance between false negatives and false positives is not preserved uniformly. FGSM transfer primarily increases missed

firearm-oriented content, Histogram Modification produces the largest anti-forensic missed-target burden, and Gaussian Blur increases false-positive review workload.

The observable percentages retained in the original **Classifications** field are not used to interpret these differences. They are not established as calibrated probabilities and are not directly comparable across labels, images, conditions, or software systems.

Overall, the frozen Cellebrite configuration presents a comparatively balanced bundle-level profile and limited aggregate degradation. This observation is bounded to its version, configuration, exported labels, mapping, and population; the classifications remain traceable triage indicators requiring human review.

6.6.5 Embedded Metadata Sensitivity Check

After normalization of the selected black-box outputs, a secondary sensitivity check was performed on the residual metadata embedded in the blind-view image files. Bundle blindness was enforced through opaque file names, paths, and directory structure. Embedded metadata was preserved, however, because some software systems may index information stored within the image bytes.

The purpose of this analysis is limited to measuring how the reported aggregate metrics change when files containing predefined semantically or experimentally sensitive metadata terms are excluded. It does not establish whether any software system accessed those fields or whether metadata causally influenced an individual decision.

The audit covered all 11 500 files in the forensic evaluation bundle. Embedded metadata was detected in 7 640 files, while 15 files contained at least one predefined sensitive-term match. These 15 files comprise 11 clean binary images and four OOD images. No adversarial or anti-forensic artifact contains a sensitive-term match. Consequently, the identified metadata channel does not directly affect the perturbation-family populations used for the principal robustness comparisons.

Table 6.23 summarizes the affected populations and the corresponding leave-out denominators.

Table 6.23: Scope of the residual embedded-metadata sensitivity check

Population	Flagged files	Original denominator	Leave-out denominator	Interpretive scope
Complete bundle	15	11 500	11 485	Describes the prevalence of predefined metadata-term matches in the complete blind-view bundle.
Clean binary subset	11	1 000	989	Supports a leave-out comparison of clean accuracy for each black-box configuration.
Adversarial family	0	5 000	5 000	No sensitive-term match occurs in the adversarial or adversarial-style artifact populations.
Anti-forensic family	0	5 000	5 000	No sensitive-term match occurs in the anti-forensic artifact populations.
OOD branch	4	500	496	Supports a leave-out comparison of the observable OOD weapon assignment rate.

Table 6.24 reports the exact metric changes after excluding the affected files. The clean-accuracy delta is defined as the leave-out value minus the original value. The OOD delta follows the same convention for the weapon assignment rate.

Table 6.24: Leave-out sensitivity results for files containing predefined embedded-metadata term matches

Configuration	Clean original	Clean leave-out	Clean delta	OOD original	OOD leave-out	OOD delta
Magnet AXIOM/Magnet.AI	0.9500	0.9515	+0.0015	0.3600	0.3548	-0.0052
Magnet Griffeye/T3K CORE	0.9780	0.9788	+0.0008	0.2600	0.2601	+0.0001
Excire Foto 2025 D20	0.9270	0.9282	+0.0012	0.2380	0.2379	-0.0001
Excire Foto 2025 D50	0.9440	0.9454	+0.0014	0.3400	0.3347	-0.0053
Excire Foto 2025 D80	0.9150	0.9141	-0.0009	0.5220	0.5181	-0.0039
Cellebrite Inseyets	0.9640	0.9636	-0.0004	0.2920	0.2883	-0.0037

Note. Clean values use 1 000 original and 989 leave-out binary inputs. OOD values use 500 original and 496 leave-out inputs. Deltas are leave-out minus original. The analysis measures metric sensitivity only and does not identify a causal contribution of embedded metadata to any black-box decision.

The clean-accuracy changes range from -0.0009 to $+0.0015$. The largest absolute change in the OOD weapon assignment rate is approximately 0.0053 . These variations do not alter the qualitative ordering or the principal operational conclusions reported for the selected configurations.

The result must nevertheless not be interpreted as proof that embedded metadata was irrelevant to every affected decision. For the four OOD files with sensitive-term matches, the observable outputs do not permit separation of visual and metadata contributions. Similarly, the fact that

some clean **weapon** images remain unflagged despite weapon-related metadata shows only that the presence of such a term is not sufficient to determine every normalized output.

A causal assessment would require a paired metadata-neutralized bundle in which the image pixels remain unchanged while the embedded fields are removed or standardized. Such a paired experiment was not included in the frozen protocol. The identified cases are therefore retained in the corpus and documented as a residual limitation of bundle blindness rather than removed retrospectively.

6.7 Comparative Operational Robustness Analysis

The common frozen source population enables a parallel comparison between the transparent proxy models and the selected black-box software configurations. The comparison is operational rather than architectural: all outputs are interpreted through the common **weapon/non_weapon** decision space, but the mechanisms producing those outputs remain fundamentally different.

The proxy models expose fold-specific checkpoints, predicted classes, maximum predicted-class probabilities, and internal attribution information. They also permit matched clean-to-perturbed transition analysis. The black-box software systems instead expose heterogeneous observable outputs, including tags, semantic retrieval results, exported classifications, and bookmarks. Their internal architectures, training populations, preprocessing stages, thresholds, and decision logic are not available for inspection.

The resulting comparison must therefore not be interpreted as an absolute ranking or as evidence that the evaluated systems implement equivalent classification tasks. It characterizes their observable operating profiles under the same frozen source population, derived perturbation suite, reference operational assignments, and explicit normalization rules.

A further distinction concerns the structure of the evaluated populations. The 1 000 clean binary images are common unique inputs. The 10 000 perturbed artifacts are derived from those same source images and must not be interpreted as independent real-world cases. On the OOD branch, each proxy-model rate is computed over 2 500 fold-specific responses:

$$500 \text{ unique OOD images} \times 5 \text{ checkpoints} = 2\,500 \text{ responses per proxy architecture}, \quad (6.8)$$

whereas each black-box rate is computed over one normalized decision for each of the 500 unique OOD images. The resulting proportions describe parallel operational tendencies but are not based on identical numbers or types of prediction events.

Clean performance and OOD behavior. Table 6.25 compares nominal clean performance with observable behavior on the separate OOD branch. Clean accuracy and clean FNR characterize the supervised binary task. The OOD weapon assignment rate instead measures how frequently inputs without a binary reference assignment are routed to the operational **weapon** state.

Similar clean accuracies do not imply equivalent operating profiles because the systems, mappings, and prediction units differ. The CLIP-based proxy most clearly separates nominal accuracy from OOD behavior: it has only six clean false negatives but assigns 0.836 of fold-specific OOD responses to **weapon**, as discussed in Section 6.3. Black-box reference rates are lower (0.260–0.360), although Excire D80 reaches 0.522. Clean errors and OOD review-load tendency must therefore be read jointly and within configuration.

Robustness under anti-forensic transformations. Table 6.26 compares the aggregate anti-forensic accuracy of each reference system with its clean baseline. The family-level drop is defined as:

Table 6.25: Clean performance and OOD behavior of the proxy models and selected black-box reference configurations

System	Type	Clean acc.	Clean FNR	OOD W-rate	Scoped operational interpretation
EfficientNet-B0	Proxy	0.978	0.016	0.370	Highest clean accuracy among the proxy models, with eight clean false negatives. Its OOD rate represents fold-specific checkpoint responses, not unique-image-level decisions.
ResNet18	Proxy	0.964	0.030	0.438	Intermediate clean profile among the proxy models, with 15 clean false negatives and a higher OOD weapon assignment rate than EfficientNet-B0.
CLIP	Proxy	0.927	0.012	0.836	Lowest clean accuracy but lowest clean FNR among the proxies. The main distinctive behavior is the high proportion of OOD checkpoint responses assigned to weapon .
Magnet AX-IOM/Magnet.AI	Black-box software	0.950	0.070	0.360	High clean accuracy under the adopted Possible weapons mapping, with a higher missed-target rate than the other black-box reference configurations.
Magnet Griffeye/T3K CORE	Black-box software	0.978	0.036	0.260	Highest clean accuracy and lowest OOD weapon assignment rate among the selected reference configurations, conditional on the conservative primary bookmark mapping.
Excire Foto 2025 D50	Black-box software	0.944	0.042	0.340	Intermediate semantic-retrieval operating point between the more restrictive D20 and more permissive D80 configurations.
Cellebrite Inseyets	Black-box software	0.964	0.032	0.292	High clean weapon recall with an OOD weapon assignment rate lower than those of Magnet.AI and Excire D50.

Note. Proxy-model OOD rates use 2 500 fold-specific responses per architecture. Black-box rates use 500 unique-image-level normalized decisions per configuration. The proportions therefore support descriptive operational comparison but do not constitute identical statistical observations.

$$\Delta_{\text{anti}} = \text{Acc}_{\text{clean}} - \text{Acc}_{\text{anti-forensic}}. \quad (6.9)$$

The proxy rows additionally report clean-correct-to-incorrect transitions for the most relevant condition. The black-box rows report condition-level false negative or false positive counts because their normalized evaluation does not provide an equivalent white-box attack-success interpretation.

Table 6.26: Comparative behavior under anti-forensic transformations

System	Family acc. / drop	Most relevant condition	con-	Observed burden	error	Scoped interpretation
EfficientNet-B0	0.9728 0.0052	/ Gaussian Blur; accuracy 0.965	accu-	17 induced all W→NW	errors,	Limited aggregate degradation, but every newly induced error in the selected condition corresponds to a missed target image.
ResNet18	0.9566 0.0074	/ Histogram Modification; accuracy 0.934		39 induced 23 W→NW and 16 NW→W	errors:	The average family drop is limited, but the most critical tonal transformation introduces errors in both operational directions.
CLIP	0.9280 −0.0010	/ Histogram Modification		13 induced 6 W→NW and 7 NW→W	errors:	A small aggregate increase coexists with newly induced errors and therefore does not establish sample-level invariance.
Magnet AX-IOM/Magnet.AI	0.9402 0.0098	/ Gaussian Blur; accuracy 0.927	accu-	60 false negatives; FNR 0.120		The principal anti-forensic concern is an increased missed-target burden under blurring.
Magnet Griffeye/T3K CORE	0.9736 0.0044	/ Histogram Modification; accuracy 0.963		34 false negatives; FNR 0.068		High aggregate accuracy and low false-positive burden under the conservative mapping, with residual missed targets under tonal modification.
Excire Foto 2025 D50	0.9414 0.0026	/ Histogram Modification; accuracy 0.932		44 false positives; FPR 0.088		The most evident D50 change concerns expansion of positive retrievals and additional review workload rather than predominant target concealment.
Cellebrite Inseyets	0.9600 0.0040	/ Histogram Modification; accuracy 0.949		30 false negatives; FNR 0.060		Limited family-level degradation, with a condition-specific increase in missed target images under histogram modification.

Note. Family accuracy is computed over the five anti-forensic conditions. The reported drop is relative to the corresponding clean accuracy. For proxy models, the error burden uses matched induced-error transitions. For black-box configurations, it reports condition-level confusion counts under the adopted normalized mapping.

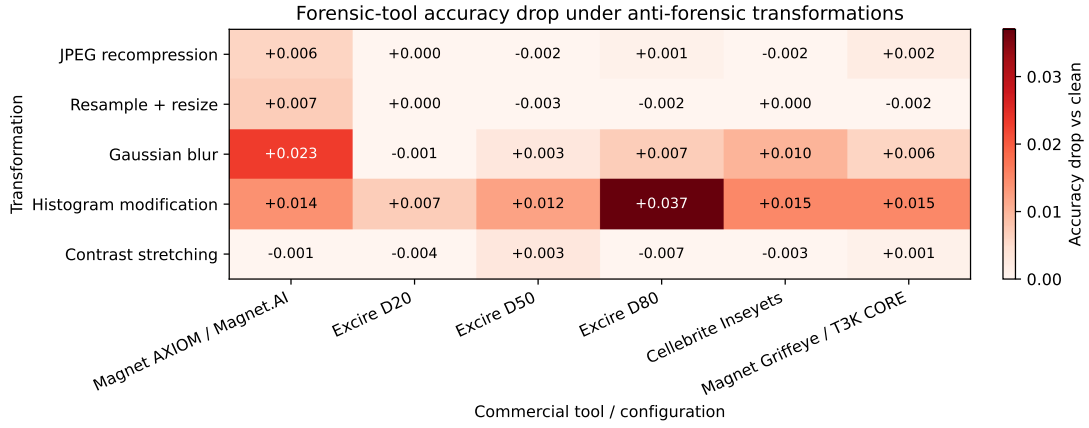


Figure 6.8: Accuracy drop of the selected black-box software configurations under anti-forensic transformations

Figure 6.8 extends the comparison to all Excire settings. Anti-forensic family-level degradation is smaller than the strongest direct-target effects, but the dominant burden differs: blurring increases missed targets for EfficientNet-B0 and Magnet.AI, Histogram Modification produces bidirectional ResNet18 errors, and Excire D50 expands false-positive workload. The CLIP row further shows that a small aggregate improvement can coexist with 13 newly induced errors. These are mapping- and configuration-specific operating points, not intrinsic system properties.

Robustness under adversarial and adversarial-style conditions. The direct-target, transfer, and model-agnostic distinctions defined in Section 6.1 apply to this comparison.

The family-level adversarial drop is:

$$\Delta_{\text{adv}} = \text{Acc}_{\text{clean}} - \text{Acc}_{\text{adversarial}}. \quad (6.10)$$

Table 6.27 reports the resulting operating profiles.

Table 6.27: Comparative behavior under adversarial and adversarial-style conditions

System	Family acc. / drop	Most relevant condition	Observed burden	error	Scoped interpretation
EfficientNet-B0	0.6948 0.2832	/ Sigma Zero Attack; accuracy 0.015	963 induced errors: 489 W→NW and 474 NW→W		Severe direct-target vulnerability under the frozen white-box condition; strong clean performance does not imply adversarial robustness.
ResNet18	0.9552 0.0088	/ FGSM; accuracy 0.918	47 induced errors: 27 W→NW and 20 NW→W		Limited family-level transfer from EfficientNet-B0-oriented artifacts; this is not a direct white-box assessment of ResNet18.
CLIP	0.9274 −0.0004	/ FGSM; accuracy 0.914	23 induced errors: 8 W→NW and 15 NW→W		Aggregate stability coexists with transferred errors and does not establish robustness against attacks optimized directly against the CLIP-based proxy.
Magnet AX-IOM/Magnet.AI	0.9232 0.0268	/ FGSM; accuracy 0.867	96 false negatives; FNR 0.192		The most relevant transferred effect is a substantial increase in missed firearm-oriented inputs under FGSM.
Magnet Griffeye/T3K CORE	0.9686 0.0094	/ Sigma Zero Attack; accuracy 0.959	37 false negatives; FNR 0.074		Highest family-level black-box accuracy among the selected reference configurations, conditional on the conservative bookmark mapping.
Excire Foto 2025 D50	0.9038 0.0402	/ Sigma Zero Attack; accuracy 0.742	221 false positives; FPR 0.442		The principal failure mode is expansion of positive semantic retrievals into the <code>non_weapon</code> population rather than predominant target concealment.
Cellebrite Inseyets	0.9550 0.0090	/ FGSM; accuracy 0.918	53 false negatives; FNR 0.106		Limited family-level transfer overall, with a condition-specific increase in missed target images under FGSM.

Note. The adversarial family contains four EfficientNet-B0-oriented model-dependent artifact populations and the model-agnostic Color Shift condition. Only the EfficientNet-B0 rows represent direct-target evaluation for the model-dependent attacks. Proxy error burdens use matched induced-error transitions; black-box rows use condition-level confusion counts.

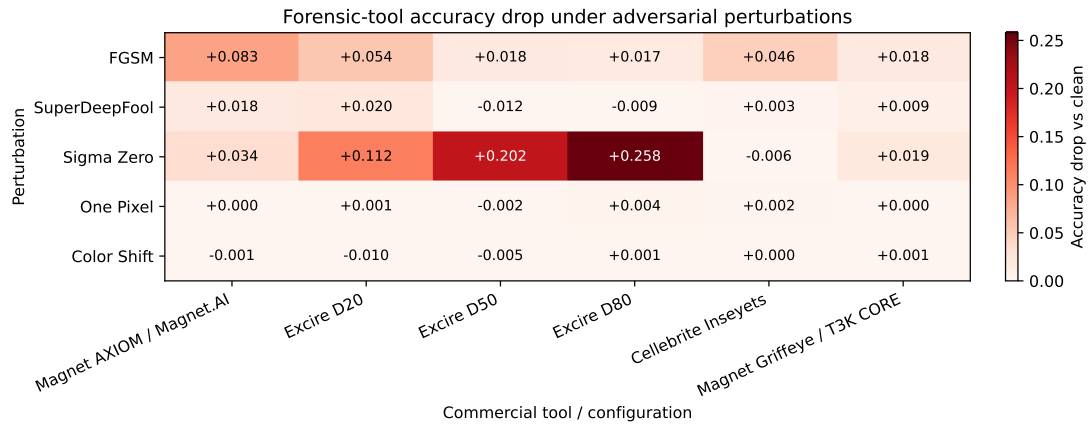


Figure 6.9: Accuracy drop of the selected black-box software configurations under adversarial and adversarial-style conditions

Table 6.27 and Figure 6.9 distinguish a severe direct-target collapse from much smaller transferred effects. Sigma-Zero drives EfficientNet-B0 to 0.015 accuracy with 963 induced errors, while transferred FGSM produces the largest missed-target burden for Magnet.AI and Celebrite (96 and 53 false negatives). Griffeye’s largest burden occurs under Sigma-Zero (37 false negatives), whereas Excire D50 responds chiefly through false-positive expansion (FPR 0.442). The D20/D80 heatmap rows confirm that transfer behavior also depends on the retrieval setting; inaccessible internal causes cannot be inferred from these differences.

Configuration sensitivity and black-box interpretation. The Excire sensitivity analysis provides the clearest demonstration that configuration is part of the evaluated system. Under Sigma-Zero, moving from D20 to D80 decreases the FNR from 0.180 to 0.024 but increases the FPR from 0.190 to 0.662.

The same queries therefore support different priorities: D20 favors selectivity, D80 retrieval coverage, and D50 an intermediate point. The other black-box profiles likewise depend on the frozen Griffeye bookmark, Celebrite label union, and Magnet.AI tag. These reproducible recordings are not semantically equivalent internal taxonomies; interpretation is limited to observable decisions and requires the version, configuration, export, and mapping to remain recorded.

Operational-risk summary. Figure 6.10 summarizes seven quantitative dimensions for the three proxy models and four black-box reference configurations. These comprise the clean FNR, the bundle-level FPR, the OOD weapon-assignment rate, and the adversarial- and anti-forensic-family FNR values. The figure also reports the degradation in accuracy produced by adversarial and anti-forensic conditions relative to the corresponding clean baseline.

The figure does not encode explainability, auditability, or system interpretability as numerical values. The distinction between transparent proxy models and black-box software is a methodological property rather than an experimentally measured risk score.

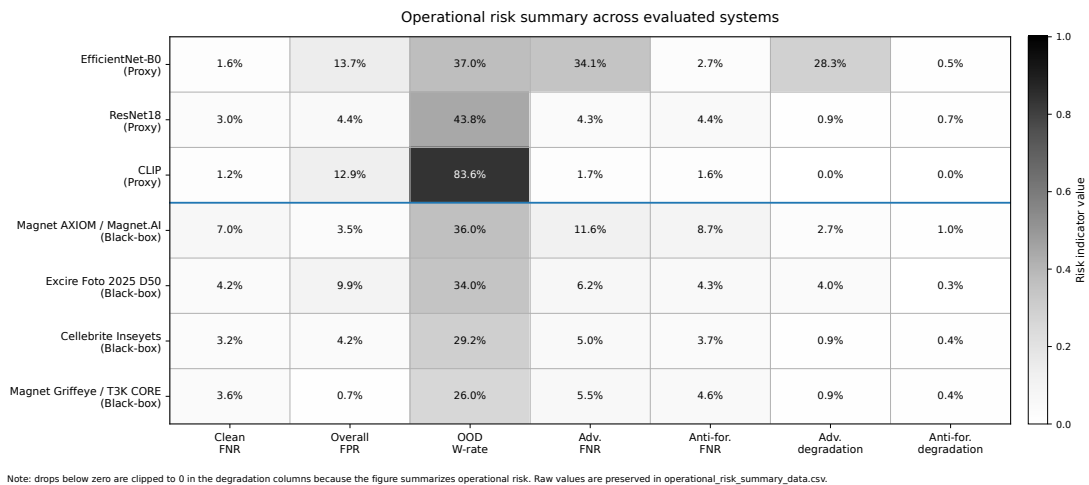


Figure 6.10: Quantitative operational-risk indicators across the proxy models and selected black-box reference configurations

Note. Clean FNR is computed on the 1000 unperturbed binary inputs. Bundle-level FPR is computed on the complete 11 000-item binary population. Adversarial and anti-forensic FNR values are computed over the corresponding 5 000-item family populations. Proxy OOD rates use 2 500 checkpoint responses, whereas black-box rates use 500 unique-image-level decisions. In the two degradation columns, negative raw drops are displayed as zero because the figure represents non-negative operational degradation; the original signed values remain preserved in the underlying metric table.

No row minimizes all seven indicators. Low clean missed-target rates can coexist with adversarial degradation, high OOD routing, or increased false-positive workload; the black-box

reference profiles likewise exchange selectivity, recall, and balance according to their frozen mappings.

Overall operational interpretation. The comparative results support four principal conclusions.

First, clean accuracy is necessary but insufficient for characterizing operational robustness. Systems with similar nominal accuracy may differ substantially in error direction, OOD assignments, perturbation sensitivity, and false-positive review burden.

Second, robustness is conditional on the evaluated threat model. The severe EfficientNet-B0 results describe attacks optimized directly against its fold-specific checkpoints. The corresponding ResNet18, CLIP, and black-box results describe only the transfer of those frozen artifacts.

Third, configuration and output mapping are part of the operating profile. Excire distance settings, Griffeye bookmark selection, Celebrite label mapping, and the Magnet.AI exported tag directly determine how observable outputs are recoded and evaluated.

Fourth, no evaluated system eliminates the need for human verification. False negatives may prevent relevant content from being prioritized, while false positives and OOD assignments may increase review volume or alter attention allocation. Maximum predicted-class probabilities, when available, do not independently establish forensic reliability.

Accordingly, the comparison identifies bounded tradeoffs rather than a universally superior system. Outputs remain traceable triage support associated with their source, version, configuration, mapping, condition, and limitations; final assessment remains human.

6.8 Explainability Analysis and Purposively Selected Cases

The xAI analysis provides a qualitative diagnostic link between the aggregate metrics reported in the preceding sections and a restricted set of individual EfficientNet-B0 outputs. It is not treated as a primary robustness metric, a quantitative validation of model reliability, or an evidentiary interpretation of the analyzed images.

The analysis is limited to the transparent EfficientNet-B0 proxy. It does not explain the behavior of Magnet AXIOM/Magnet.AI, Magnet Griffeye/T3K CORE, Excire Foto 2025, or Celebrite Inseyets. These systems remain black-box within the experimental protocol, and their observable outputs cannot be attributed using the proxy model's internal gradients.

The retained visualizations were generated using IG. For each case, the model-predicted class, recorded as `predicted_label`, was used as the attribution target. The maps therefore represent attribution magnitudes for the output actually produced by EfficientNet-B0.

This distinction is particularly important for erroneous predictions. In a false-negative case, the attribution target is the incorrectly predicted `non_weapon` class. In a false-positive case, the target is the incorrectly predicted `weapon` class. The map does not represent an explanation of the reference operational assignment or indicate which regions the model should have used to produce the correct result.

Each retained case comprises four visual assets: the model input, the IG attribution-magnitude heatmap, the attribution overlay on the input, and a mask retaining the image locations associated with the largest 10% of absolute attribution magnitudes.

The mask is not a semantic segmentation, object annotation, localization ground truth, or causal explanation. It reports only the spatial locations associated with the largest absolute attribution values under the adopted model, target, preprocessing procedure, and baseline path.

The historical thesis assets were generated using 32 integration steps and a zero baseline in the preprocessed input space. The retained manifest does not establish numerical convergence for these figures, and the cases remain pending adaptive regeneration under the hardened convergence-validation procedure described in Section 5.7.2. They are therefore used exclusively as qualitative diagnostic visualizations. No conclusion in this section depends on the numerical magnitude of the attribution values or on a claim of IG convergence.

The case-selection process was also qualitative. The original review considered 1 175 candidate rows and retained 15 cases, comprising three manually selected examples for each of five diagnostic categories. One case from each category was subsequently included in the thesis through a purposive selection process.

The resulting five-case set is purposive rather than random or statistically representative. It illustrates distinct operating behaviors but does not estimate their prevalence in the complete evaluation population.

Table 6.28 summarizes the selected cases.

Table 6.28: Purposively selected EfficientNet-B0 cases for qualitative analysis using IG

Identifier	Scenario	Reference	Prediction	Max-P	Qualitative role
xai_case_0001	clean	weapon	weapon	1.000	Correct clean prediction retained as a reference case for visual comparison with the subsequent failures.
xai_case_0006	clean	weapon	non_weapon	0.692	Clean false negative illustrating that missed-target behavior can occur without adversarial or anti-forensic manipulation.
xai_case_0009	ood	ood	weapon	0.999	Closed-set OOD assignment illustrating that a high maximum probability does not establish binary correctness or forensic relevance.
xai_case_0010	Histogram Modification	weapon	non_weapon	0.852	Anti-forensic false negative illustrating model sensitivity to a model-agnostic tonal transformation.
xai_case_0015	Sigma Zero Attack	non_weapon	weapon	1.000	Direct-target adversarial false positive illustrating that a maximum predicted-class probability of 1.000 does not establish operational reliability.

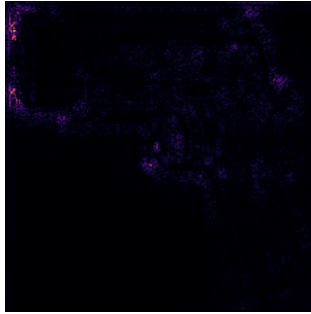
Note. *Max-P* denotes the maximum predicted-class probability produced by EfficientNet-B0. It is not treated as calibrated. For `xai_case_0009`, `ood` denotes the separate OOD evaluation branch and not a third supervised class.



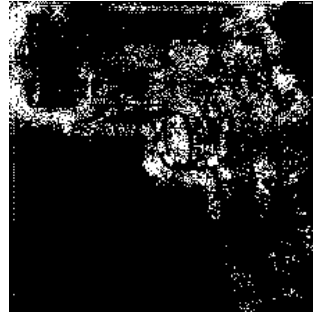
Input image
Original sample.



IG overlay
Attribution over input image.



IG heatmap
Normalized attribution map.



Top-attribution mask
Highest-attribution regions.

Case metadata: **scenario:** clean; **reference state:** weapon; **prediction:** weapon; **Max-P:** 1.000

Figure 6.11: Correct prediction on clean input `xai_case_0001`. EfficientNet-B0 assigns an image with reference operational state `weapon` to `weapon`

Clean correct reference case. Figure 6.11 presents a clean input for which the EfficientNet-B0 prediction agrees with the reference operational assignment. Both are `weapon`, and the model produces a maximum predicted-class probability of 1.000.

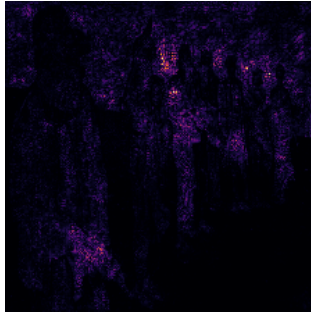
The heatmap and overlay provide a qualitative reference for comparing the spatial attribution pattern with the selected failure cases; agreement with the reference label does not validate either the model or the attribution method.



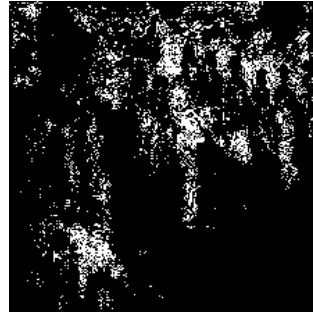
Input image
Original sample.



IG overlay
Attribution over input image.



IG heatmap
Normalized attribution map.



Top-attribution mask
Highest-attribution regions.

Case metadata: **scenario:** clean; **reference state:** weapon; **prediction:** non_weapon; **Max-P:** 0.692

Figure 6.12: False negative on clean input `xai_case_0006`. An image with reference operational state `weapon` is assigned to `non_weapon`

Clean false negative. Figure 6.12 reports a false negative on an unperturbed input. The reference operational assignment is `weapon`, whereas EfficientNet-B0 predicts `non_weapon` with a maximum predicted-class probability of 0.692.

The case is operationally relevant because the error occurs without adversarial or anti-forensic manipulation. A workflow that uses the automatic prediction to filter, rank, or suppress content could assign reduced priority to a potentially relevant image.

Because the map targets the incorrect predicted class, it supports case-level inspection but cannot determine whether the failure derives from appearance, scale, occlusion, context, source regularities, preprocessing, or their interaction.

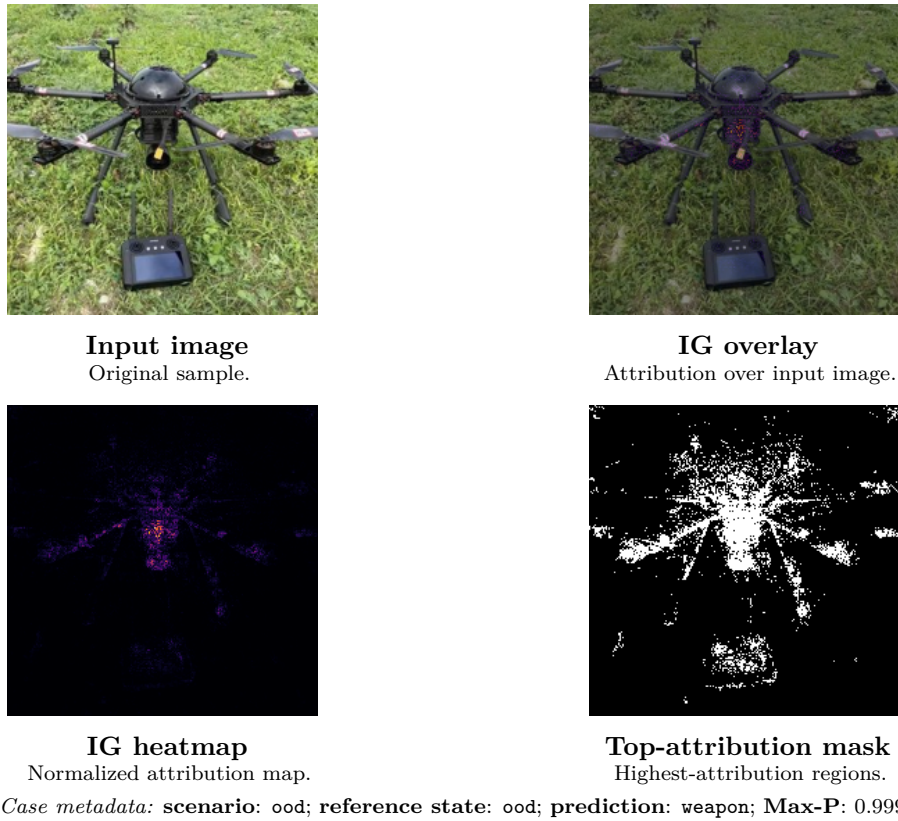
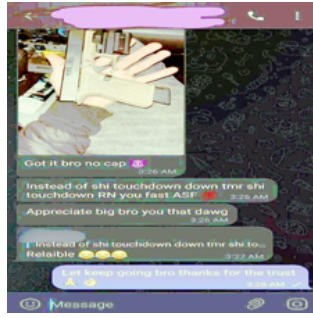


Figure 6.13: OOD case `xai_case_0009`. EfficientNet-B0 assigns an OOD input to `weapon` with a high maximum predicted-class probability

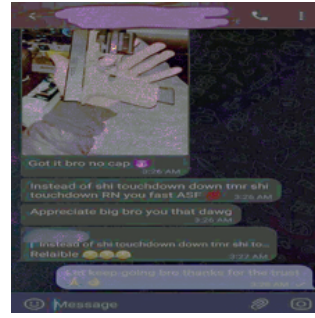
OOD input assigned to `weapon`. Figure 6.13 presents an input from the separate OOD branch. The sample has no supervised `weapon/non_weapon` reference assignment, but the closed-set proxy assigns it to `weapon` with a maximum predicted-class probability of 0.999.

The output is not a supervised false positive; it shows how an out-of-task input is absorbed into a positive closed-set class. Its 0.999 Max-P is a peaked checkpoint output, not proof of correctness, calibration, or forensic relevance, and the map does not establish a firearm-specific feature.

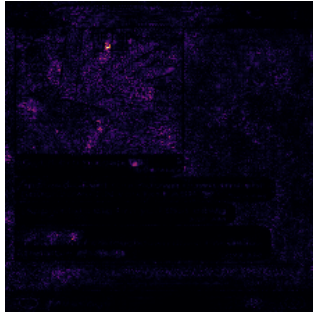
This case supports the methodological decision to preserve OOD as a separate evaluation branch. Merging the sample into the `non_weapon` class would incorrectly convert a closed-set assignment behavior into an ordinary supervised false positive and would obscure the specific uncertainty associated with out-of-task content.



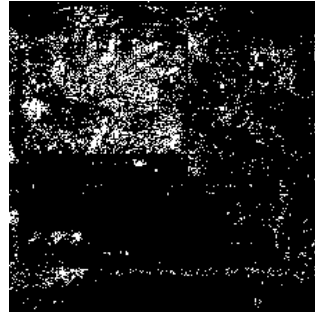
Input image
Original sample.



IG overlay
Attribution over input image.



IG heatmap
Normalized attribution map.



Top-attribution mask
Highest-attribution regions.

Case metadata: **scenario:** Histogram Modification; **reference state:** `weapon`; **prediction:** `non_weapon`;
Max-P: 0.852

Figure 6.14: Anti-forensic false negative `xai_case_0010`. An image with reference operational state `weapon`, transformed through Histogram Modification, is assigned to `non_weapon`

Anti-forensic false negative. Figure 6.14 reports a false negative observed after Histogram Modification. The reference operational assignment remains `weapon`, whereas EfficientNet-B0 predicts `non_weapon` with a maximum predicted-class probability of 0.852.

Unlike FGSM, Sigma-Zero, SuperDeepFool, and One Pixel, Histogram Modification is not optimized against the EfficientNet-B0 decision boundary. It is a deterministic, model-agnostic transformation affecting the tonal distribution of the input.

Operationally, the case illustrates that a comparatively ordinary image-level transformation can coincide with a missed target image. The model output must therefore remain subject to human verification even when the manipulation does not require model knowledge. The predicted-class map does not identify whether contrast redistribution, local or global intensity changes, preprocessing, or another internal effect caused the error.

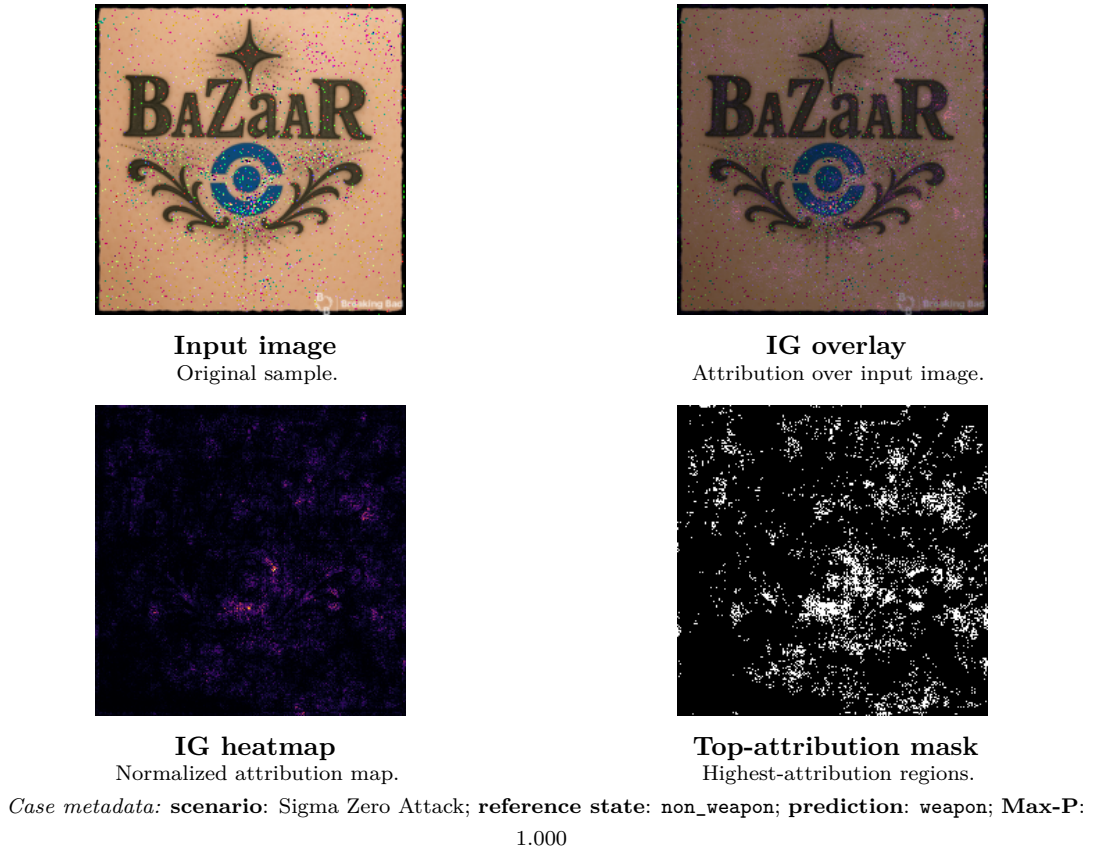


Figure 6.15: Maximum-probability adversarial false positive xai_case_0015. A non_weapon image perturbed using Sigma Zero Attack is assigned to weapon

Maximum-probability adversarial false positive. Figure 6.15 presents a Sigma-Zero false positive in the direct-target EfficientNet-B0 evaluation. The reference operational assignment is **non_weapon**, whereas the perturbed input is assigned to **weapon** with a maximum predicted-class probability of 1.000.

The reference assignment establishes that the strongly peaked output is wrong: Max-P 1.000 is not correctness, calibrated certainty, or forensic reliability. The predicted-class map neither explains the Sigma-Zero optimization nor identifies stable semantic features responsible for the error.

The historical attribution associated with this case also has a comparatively large legacy convergence delta. Its spatial pattern must consequently be interpreted with particular caution and only as a qualitative diagnostic asset. The classification failure itself is established by the matched prediction record; it does not depend on the attribution visualization.

The failure record, not the visualization, establishes the error and shows why output concentration cannot replace expert assessment.

Cross-case interpretation. Together, the cases contrast a clean reference, a nominal false negative, a peaked closed-set OOD assignment, a model-agnostic false negative, and a direct-target false positive at Max-P 1.000.

These cases must not be used to infer the frequency of the represented behaviors. They were deliberately selected to support qualitative discussion of distinct operating conditions and failure modes.

The analysis also does not establish that particular highlighted regions are semantically correct, causally decisive, or stable across alternative baselines, attribution targets, checkpoints,

integration-step counts, or explanation methods. A different attribution configuration could produce a different visualization for the same model output.

The principal contribution of the xAI layer is therefore diagnostic transparency at the proxy-model level. It preserves a traceable connection among the input, predicted class, maximum predicted-class probability, attribution target, and selected visualization.

This transparency does not convert the proxy prediction into evidence and does not extend to the selected black-box software systems. Within a DF workflow, the figures support critical review of the purposively selected cases while preserving the central role of the human analyst in the final interpretation.

6.9 Interpretive Boundaries of the Experimental Results

The results reported in this chapter characterize the observable behavior of the evaluated systems under the frozen experimental protocol. They must not be interpreted as universal estimates of robustness, as an absolute ranking of the evaluated systems, or as validation of their behavior outside the documented dataset, software versions, configurations, output mappings, and threat model.

Several distinctions are particularly important when interpreting the reported metrics. First, false negatives and false positives have different operational consequences for forensic triage, and aggregate accuracy may conceal changes in their direction or distribution. Second, a **weapon** assignment on an OOD input is not a supervised false positive, because the OOD population has no binary reference assignment. Third, the four model-dependent adversarial attacks were optimized against fold-specific EfficientNet-B0 checkpoints; their effects on ResNet18, the CLIP-based proxy, and the selected black-box configurations therefore represent empirical transfer rather than matched direct-target robustness.

The black-box results must likewise be interpreted as properties of the frozen observable configurations rather than intrinsic characteristics of the software families. Tags, bookmarks, semantic retrieval results, and exported classification labels were normalized through explicit operational mappings, and alternative defensible mappings or future software versions could produce different operating profiles. Maximum predicted-class probabilities available for the proxy models are also architecture-specific diagnostic outputs and are not treated as calibrated or directly comparable measures of correctness.

Finally, the manipulated populations are derived from the same 1 000 binary source images, and the proxy-model and black-box OOD rates use different prediction units. These properties constrain the statistical interpretation of the reported aggregate values and reinforce the protocol-specific nature of the comparison.

The broader methodological limitations of the study are discussed in Section 7.6, while the implications for forensic use, configuration management, traceability, and continued human review are developed in Section 7.5.

Chapter 7

Conclusions, Limitations, and Future Work

7.1 Summary of the Research

This thesis investigated the operational robustness of AI-based image classification and semantic-retrieval systems used in DF. Its starting point was that performance on clean, well-structured data does not fully describe how automated triage behaves when evidence is heterogeneous, degraded, outside the nominal task, or deliberately manipulated. In practice, the direction of an error matters: a missed **weapon** image may not receive appropriate priority, whereas an incorrect positive assignment may increase the analyst’s review workload.

The study therefore treated clean, adversarial, anti-forensic, and OOD inputs as separate evaluation conditions. It combined three transparent proxy models—EfficientNet-B0, ResNet18, and a CLIP-based proxy—with six frozen configurations of four selected black-box software systems: Magnet AXIOM/Magnet.AI, Magnet Griffeye/T3K CORE, Excire Foto 2025, and Cellebrite Inseyets. The proxy models provided inspectable predictions, maximum predicted-class probabilities, matched error transitions, and gradients. The black-box systems were assessed only through outputs observable to the analyst and explicitly documented normalization rules; no equivalence with the proxy architectures was assumed.

Both evaluation levels used the same frozen source population of 1 500 images: 500 **weapon**, 500 **non_weapon**, and 500 reserved for a separate OOD branch. The 1 000-image binary subset was used to generate 5 000 adversarial and 5 000 anti-forensic artifacts. Together with the clean binary and OOD inputs, these formed an 11 500-item forensic evaluation bundle. The perturbed files remained derived versions of the same 1 000 binary source images and were not treated as independent cases.

The complete procedure was formalized as the FAIRLab protocol. FAIRLab is neither a classifier nor a standalone software platform; it is a task-specific methodological procedure linking human-validated dataset construction, deterministic partitioning, proxy-model analysis, controlled perturbation, blind-bundle preparation, black-box normalization, quantitative and qualitative analysis, and traceability controls.

The central conclusion is that robustness is not an intrinsic property that can be inferred from clean accuracy. It is conditional on the task, input population, threat model, system version, configuration, observable output, and normalization rule. Automated image-analysis outputs can assist forensic triage, but they remain most defensible when they are traceable, reviewable, and subordinate to expert assessment [1, 2, 3].

7.2 Answers to the Research Questions

RQ1. To what extent do the transparent proxy models maintain their clean-base-line performance under the separate OOD, adversarial, and anti-forensic conditions? Only to a condition-dependent extent. Clean accuracies are 0.978 for EfficientNet-B0, 0.964 for ResNet18, and 0.927 for the CLIP-based proxy, but these values do not remain uniformly representative outside the clean branch.

The five anti-forensic transformations generally cause smaller aggregate variations than the strongest adversarial attacks. The largest anti-forensic accuracy reduction is 0.030 for ResNet18 under Histogram Modification. Aggregate stability is not equivalent to sample-level invariance, however, because new errors may be offset by previous errors that become correct. By contrast, direct-target attacks cause the most substantial degradation: EfficientNet-B0 accuracy decreases to 0.575 under FGSM and to 0.015 under Sigma Zero Attack.

The separate OOD branch reveals a further behavior that clean binary accuracy cannot predict. Fold-aggregated weapon assignment rates are 0.3696, 0.4376, and 0.8356 for EfficientNet-B0, ResNet18, and the CLIP-based proxy, respectively. These are forced closed-set assignments, not supervised correctness measures. Clean performance, anti-forensic sensitivity, direct adversarial vulnerability, and OOD behavior must therefore be evaluated as distinct dimensions (Sections 6.2–6.5).

RQ2. How do direct source-model attacks, cross-architecture transfer, and the model-agnostic Color Shift condition differ in their effects on the evaluated proxy models? Direct source-model attacks have substantially larger effects than the observed cross-architecture transfer. The four model-dependent attacks were generated against fold-specific EfficientNet-B0 checkpoints, so their evaluation on that architecture is direct-target, whereas results for ResNet18 and the CLIP-based proxy measure empirical transfer.

The clearest direct-target failure is Sigma Zero Attack: accuracy falls to 0.015 and 963 of 978 clean-correct samples become incorrect. FGSM reduces accuracy to 0.575 and introduces 406 errors, mainly in the `weapon-to-non_weapon` direction. On the non-target models, the largest transferred degradation occurs under FGSM, with accuracy decreasing from 0.964 to 0.918 for ResNet18 and from 0.927 to 0.914 for the CLIP-based proxy. Transferred Sigma Zero Attack artifacts do not reproduce the direct-target collapse.

Transferability must therefore be measured rather than assumed. Its limited effect here does not demonstrate direct robustness of the non-target models, because no matched attacks were optimized against them. Color Shift is also analytically separate: it is deterministic and model-agnostic and is interpreted as a condition-specific transformation rather than direct or transferred attack success (Section 6.5).

RQ3. How do the selected black-box software systems respond to the same frozen experimental corpus, and how do observable output semantics and configuration choices affect the resulting operational profiles? The systems produce heterogeneous operating profiles despite processing the same frozen bundle. Their decisions originate from tags, bookmarks, exported labels, or semantic-retrieval membership; the metrics are consequently conditional on the normalization adopted for each configuration.

Under the frozen protocol, Magnet Griffeye/T3K CORE has a selective profile, with bundle-weighted accuracy of 0.972 and FPR of 0.007, under the conservative use of only `CORE/Violence/Firearm` as the primary positive bookmark. Celebrite Inseyets is comparatively balanced, with bundle accuracy, weapon recall, and weapon precision close to 0.958. Magnet AXIOM/Magnet.AI has bundle accuracy of 0.933 and FNR of 0.099, with greater missed-target exposure under some conditions.

Configuration sensitivity is clearest for Excire Foto 2025. On clean inputs, moving from D20 to D80 reduces the FNR from 0.120 to 0.014 but increases the FPR from 0.026 to 0.156 and the

OOD weapon assignment rate from 0.238 to 0.522. Under transferred Sigma Zero Attack, the FNR falls from 0.180 to 0.024 while the FPR rises from 0.190 to 0.662.

No universal ranking follows from these results. Restrictive configurations may reduce review workload while increasing missed detections; permissive configurations may do the reverse. Software version, enabled functions, category or query settings, observable semantics, and normalization are all part of the reported operating profile (Section 6.6).

RQ4. Which error transitions, robustness degradations, OOD assignment behaviors, and unknown output states are most relevant to forensic triage and continued human review? The most consequential transition for triage is `weapon`→`non_weapon`, because it may prevent relevant content from receiving priority. The reverse transition remains important because it increases review workload. Aggregate change must therefore be interpreted together with error direction: the direct Sigma Zero Attack result combines near-total collapse with both types of error, while transferred conditions produce different false-negative and false-positive burdens across the commercial configurations.

The OOD branch creates a separate operational concern. A `weapon` assignment on an OOD input is not a supervised false positive, but it can still alter prioritization; every evaluated proxy and black-box configuration produces a non-zero rate. Maximum predicted-class probability must also remain separate from reliability: the representative Sigma Zero Attack failure reaches 1.000 despite being incorrect.

No `unknown` output occurs in the canonical 69 000 normalized black-box decisions, so it is not an observed failure mode in this experiment. Explicit unknown handling nevertheless remains necessary for future versions, alternative exports, and incomplete mappings. The operational response is to report missed-target risk, false-positive workload, and OOD assignments separately while preserving system versions, configurations, output mappings, source-to-decision traceability, and continued human review (Sections 6.7 and 6.9).

7.3 Main Contributions

The thesis makes four connected contributions.

1. **A forensic robustness protocol.** It formalizes FAIRLab as a methodological procedure for evaluating AI-based image analysis beyond clean accuracy. The protocol coordinates human-validated data construction, deterministic partitioning, transparent proxy analysis, perturbation generation, a separate OOD branch, black-box evaluation, output normalization, explainability, and traceability. Its contribution is the integration of these components into one reproducible and auditable process, not the proposal of a new classifier or defensive mechanism.
2. **A common frozen experimental basis.** The same 1 500-image source population supports both evaluation levels. Five adversarial or adversarial-style conditions and five anti-forensic conditions are generated from the 1 000-image binary subset and combined with the clean binary and OOD branches in the 11 500-item bundle. Stable source links ensure that the 10 000 perturbed artifacts remain identifiable as derived source-condition combinations rather than independent cases.
3. **A parallel transparent and black-box evaluation.** The proxy models provide inspectable checkpoints, predictions, matched error transitions, maximum predicted-class probabilities, and gradients. The commercial systems expose heterogeneous tags, bookmarks, classification labels, and semantic-query results. Explicit mappings normalize these outputs for operational comparison without implying common internal architectures or taxonomies. The design also distinguishes attacks evaluated directly against EfficientNet-B0 from empirical transfer to non-target models and black-box configurations.

4. **An auditable, error-oriented research artifact.** Identifiers, manifests, cryptographic digests, checkpoints, normalized predictions, metrics, and reporting assets preserve the path from each source image to its experimental outcomes. The analysis emphasizes missed-target risk, false-positive review workload, OOD assignment behavior, configuration sensitivity, and residual metadata limitations rather than reducing robustness to one score. Five purposively selected IG cases add qualitative proxy-model diagnostics without being treated as quantitative evidence or as explanations of the black-box systems.

Taken together, these contributions define and apply a traceable procedure for revealing how heterogeneous automated systems behave when the assumptions of clean, closed-set evaluation no longer hold.

7.4 Main Experimental Findings

The results reported in Chapter 6 show that nominal performance and operational robustness are related but non-equivalent properties. High clean accuracy provides an important baseline, but it does not predict how a system will behave under distributional mismatch, image-level transformations, adversarial manipulation, or alternative operational configurations.

Table 7.1 summarizes the principal findings and their forensic relevance.

Table 7.1: Summary of the main experimental findings

Area	Main finding	Operational relevance
Clean baseline	EfficientNet-B0 achieves the highest clean accuracy among the proxy models, at 0.978. Under the adopted conservative mapping, Magnet Griffeye/T3K CORE achieves the highest clean accuracy among the selected black-box configurations, also at 0.978. Cellebrite Inseyets produces slightly higher clean weapon recall than Magnet Griffeye/T3K CORE.	Clean performance establishes the nominal reference but does not determine behavior under OOD, adversarial, anti-forensic, or configuration-dependent conditions.
OOD	The CLIP-based proxy assigns 0.8356 of its fold-specific OOD responses to weapon . EfficientNet-B0 and ResNet18 produce lower weapon assignment rates but substantially higher proportions of maximum predicted-class probabilities above 0.90. The black-box weapon assignment rates range from 0.238 to 0.522 across the six configurations.	Inputs outside the supervised binary task may still be routed to the firearm-oriented review category. These assignments are closed-set operating behaviors rather than ordinary binary errors.
Anti-forensic	The five evaluated anti-forensic transformations generally produce smaller aggregate accuracy variations than the strongest direct-target adversarial conditions. Nevertheless, they introduce condition- and system-specific false negatives and false positives. Gaussian Blur is particularly relevant for EfficientNet-B0 and Magnet.AI, whereas Histogram Modification affects ResNet18, Griffeye/T3K CORE, Excire D50, and Cellebrite through different error profiles.	Operationally plausible compression, resizing, blurring, and tonal transformations may alter triage decisions even when the aggregate accuracy change appears limited. Error direction remains more informative than aggregate degradation alone.

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Area	Main finding	Operational relevance
Adversarial	EfficientNet-B0 undergoes severe direct-target degradation under Sigma Zero Attack, with accuracy decreasing to 0.015, and under FGSM, with accuracy decreasing to 0.575. The same artifacts produce more limited and heterogeneous transfer effects on ResNet18, the CLIP-based proxy, and the selected black-box configurations.	High clean accuracy does not guarantee direct adversarial robustness. Transfer must be measured rather than assumed and must remain distinct from attacks optimized directly against the evaluated system.
Black-box software	The selected configurations expose different operating profiles. Magnet Griffeye/T3K CORE is highly selective under the conservative primary bookmark mapping; Ccelebrite Inseyets presents a comparatively balanced profile; Magnet AXIOM/Magnet.AI shows greater missed-target exposure under some conditions; and Excire Foto 2025 varies materially across D20, D50, and D80.	No configuration minimizes false negatives, false positives, and OOD weapon assignments simultaneously. Software version, enabled functions, observable output semantics, configuration, and normalization mapping are part of the reported result.
Metadata sensitivity	Fifteen of the 11 500 blind-bundle files contain at least one predefined semantically or experimentally sensitive metadata-term match. Excluding the affected clean and OOD files produces only limited aggregate metric variation.	The leave-out results support the stability of the aggregate conclusions but do not establish that embedded metadata was unused or causally irrelevant to every affected decision.
xAI	The qualitative IG analysis examines one correct clean case, three supervised failure cases, and one OOD closed-set assignment. The cases include incorrect outputs associated with maximum predicted-class probabilities of up to 1.000.	Explainability supports qualitative diagnosis of selected proxy-model outputs but does not establish prediction correctness, calibrated certainty, causal feature relevance, or black-box system behavior.

The clean results already expose different error tradeoffs. Although EfficientNet-B0 makes the fewest clean errors among the proxy models, the CLIP-based proxy combines the lowest clean FNR (0.012) with the highest clean FPR (0.134). Reducing missed targets can therefore increase the quantity of material presented for review.

The separate OOD analysis adds a dimension that clean binary results do not capture. Both proxy models and black-box configurations assign some out-of-task inputs to the firearm-oriented state, but these assignments are not ordinary false positives. Assignment frequency and maximum predicted-class probability also describe different phenomena: the CLIP-based proxy has the highest OOD weapon assignment rate without any fold-specific response above 0.90, whereas ResNet18 produces fewer weapon assignments but the highest high-maximum-probability rate.

Anti-forensic transformations generally have smaller aggregate effects than the strongest direct-target attacks, yet their error direction remains operationally relevant. Gaussian Blur, for example, introduces 17 new EfficientNet-B0 errors, all in the **weapon-to-non_weapon** direction. A limited or negative accuracy drop is therefore not evidence of unchanged behavior, because new errors may be offset by previous errors that become correct.

The adversarial results provide the clearest direct failure: Sigma Zero Attack changes 963 of the 978 clean-correct EfficientNet-B0 inputs into errors, while FGSM concentrates more strongly on missed targets. The same frozen artifacts have smaller and heterogeneous effects on non-target

models and commercial configurations. Those observations quantify transfer only and cannot be generalized as direct resistance to attacks optimized for each system.

At the black-box level, the results remain inseparable from observable output semantics and frozen configuration. Excire’s D20–D80 settings demonstrate the clearest recall–selectivity tradeoff; Griffeye’s low FPR depends on the conservative primary bookmark; and the Cellebrite and Magnet profiles depend on their selected labels and tag. These are operating profiles, not version-independent properties of the software families.

Finally, excluding the 15 files with predefined sensitive metadata-term matches does not materially change aggregate results, but it cannot establish that embedded metadata was unused. The five xAI cases likewise support case-level inspection only. Together, the findings show that operational robustness requires joint consideration of error direction, OOD behavior, perturbation sensitivity, configuration, output mapping, traceability, and human review.

7.5 Operational and Forensic Implications

The evidence supports automated image analysis as a traceable triage aid, not as an autonomous mechanism for evidentiary decision-making. Its value lies in helping analysts organize and prioritize large collections while leaving each output open to reconstruction, source-image verification, and contextual interpretation. Four operational principles follow from the results.

1. **Evaluate consequences, not accuracy alone.** Systems with similar aggregate accuracy may distribute their errors very differently. A false negative may prevent firearm-oriented material from receiving priority; a false positive increases review workload and may dilute analyst attention. The appropriate balance is task-dependent: a context in which missed targets are especially consequential may favor a permissive configuration, whereas very large evidence volumes may require greater selectivity. Condition-level metrics and error transitions must therefore remain visible alongside aggregate performance.
2. **Keep OOD assignments and scores in their proper semantic scope.** An OOD image assigned to **weapon** has no supervised binary reference assignment and is not an ordinary false positive, but it can still trigger a flag or increase review workload. Anomalous, ambiguous, degraded, synthetic, and semantically adjacent content should consequently be reported separately and, where supported, exposed through rejection or unknown states. Likewise, proxy-model maximum predicted-class probabilities are architecture-specific diagnostic values, not cross-model measures of correctness. Tags, bookmarks, exported percentages, labels, and retrieval membership from commercial systems have different semantics and must not be collapsed into a common probability scale without independent validation.
3. **Treat configuration and version as part of the system.** Excire’s **Distance limit**, Griffeye’s positive bookmark, Cellebrite’s selected export labels, and Magnet.AI’s **Possible weapons** tag directly shape the observed profiles. A result is therefore inseparable from the software or model version, enabled functions, checkpoint or configuration, category or query, output format, and normalization rule. Updates may alter preprocessing, weights, thresholds, taxonomies, retrieval behavior, or export structures; validation from one frozen version must not be transferred to a later version without verification.
4. **Preserve source-to-decision traceability.** Each output should remain linked to the source item and its integrity record, the processing condition, the observable system output, the applied mapping, and the subsequent analyst review. This chain helps distinguish genuine system behavior from incomplete exports, missing records, data-processing errors, or incorrect normalization. A result whose provenance cannot be reconstructed has limited analytical value in a forensic workflow.

The metadata analysis illustrates the need to define traceability and blindness precisely. Opaque names and directories conceal experimental assignments at the file-system level but do not remove information embedded in the image bytes. The leave-out analysis supports the stability of the aggregate conclusions; it does not prove that metadata was never accessed. Claims of blind processing should therefore identify whether they concern paths, embedded metadata, pixels, or all available channels.

The qualitative xAI layer provides limited diagnostic transparency for selected EfficientNet-B0 cases by linking input, prediction, maximum predicted-class probability, attribution target, and visualization. The maps do not identify causal features, certify correctness, or explain the black-box systems. Explainability remains supplementary to quantitative robustness tests and expert assessment.

These principles define a *human-in-the-loop* use model. The analyst retains responsibility for interpreting flagged and, where appropriate, unflagged material, corroborating automated outputs with the remaining evidence, and determining their final analytical significance. The findings do not argue against AI-assisted forensic triage; they specify the conditions that make it more defensible: task-specific validation, an explicit threat model, controlled configurations, version management, traceability, and continued visibility of limitations.

7.6 Limitations

The findings apply within the scope of the frozen protocol and the defined case study. They must not be interpreted as universal validation of the evaluated models or software systems. The principal limitations are summarized below.

- **Dataset and semantic validation.** The 1 500 manually selected source images support a controlled case study, not the full diversity or prevalence of operational evidence. The binary subset is artificially balanced, and the `non_weapon` branch is predominantly derived from the `05_deepweb` source group. Moreover, the documented *human-in-the-loop* assignments were produced by one reviewer without independent annotation or an inter-rater agreement measure. The labels are traceable but remain dependent on the adopted criteria and reviewer judgment.
- **Definition and measurement of OOD behavior.** The OOD branch intentionally combines borderline, degraded, synthetic, non-firearm weapon, and semantically external content. It is neither a homogeneous supervised class nor a universal representation of distributional shift. Proxy rates aggregate 2 500 fold-specific responses—five checkpoints for each of 500 images—whereas each black-box configuration produces one decision per image. The reported proportions are parallel operating tendencies, not identical prediction units.
- **Derived-sample dependence and finite conditions.** The 5 000 adversarial and 5 000 anti-forensic files derive from the same 1 000 clean binary images. Bundle-level results therefore concern source-condition combinations rather than an independent 11 000-case binary population. Each perturbation also uses one frozen parameterization; different budgets, severities, encodings, implementations, or transformation chains may produce different profiles.
- **Direct-target scope and artifact representation.** The four model-dependent attacks were optimized only against fold-specific EfficientNet-B0 checkpoints. Results for ResNet18, the CLIP-based proxy, and the black-box systems measure transfer and do not establish direct robustness. In addition, FGSM, Sigma Zero Attack, and SuperDeepFool were exported as 224×224 PNG files, while clean sources retained heterogeneous formats and dimensions; One Pixel Attack also used PNG serialization. Black-box responses may therefore combine perturbation, resizing, and serialization effects.
- **Proxy and black-box comparability.** The proxy models are transparent experimental references, not validated surrogates for proprietary internal models. Their gradients,

probabilities, and attributions cannot explain black-box behavior. Conversely, black-box tags, bookmarks, labels, percentages, and retrieval membership have heterogeneous semantics. Their operational recoding does not establish a shared internal binary task, and alternative defensible mappings could produce different profiles. All commercial findings remain specific to the frozen software versions, enabled functions, configurations, exports, and normalization rules.

- **Qualitative explainability and residual metadata.** The xAI analysis contains five purposively selected EfficientNet-B0 cases and is not statistically representative. The historical IG assets use 32 integration steps and a zero baseline without established numerical convergence, so they remain qualitative diagnostics. The blind bundle also retains embedded image metadata: 15 files contain predefined sensitive-term matches. Their exclusion has little aggregate effect but does not determine whether a system accessed the embedded fields.
- **Controlled-data and public-artifact boundary.** Legal, ethical, institutional, licensing, and proprietary restrictions prevent complete public redistribution of all source images, generated trees, commercial exports, and prediction-level evidence. The repository preserves scripts, schemas, manifests, configurations, digests, normalized summaries, metrics, selected assets, and validation reports. This supports procedural reproducibility and auditability, but full re-execution requires authorized access to controlled data and proprietary outputs.
- **Legal and evidentiary scope.** The study evaluates technical and operational triage behavior. It does not determine admissibility, probative value, or the evidentiary weight of an automated classification. Those questions require the applicable legal framework, chain-of-custody analysis, independent validation, disclosure obligations, and case-specific interpretation.

These limitations delimit rather than invalidate the experiment. The findings describe the observable behavior of the frozen systems, configurations, mappings, population, and threat model. They do not certify a product, establish a universal ranking, validate a production firearm detector, or support autonomous evidentiary deployment.

7.7 Future Work

Future research directions derive directly from the experimental findings and from the limitations of the frozen protocol. Six directions are especially useful.

Broader data and stronger semantic validation. Future studies should use larger multi-source populations, realistic class prevalence, additional firearm-related subcategories, and more varied negative samples from different devices, platforms, acquisition procedures, and media-processing pipelines. Multiple independent reviewers, formal agreement measures, and an adjudication procedure would reduce reliance on a single judgment and quantify ambiguity. The OOD population should also be stratified into categories such as borderline firearm content, non-firearm weapons, synthetic imagery, degraded media, video-game material, military equipment, and semantically unrelated inputs. This would show whether closed-set assignments concentrate in particular visual or semantic groups.

Explicit OOD handling and analyst-centered evaluation. Calibrated model-specific estimates, abstention thresholds, dedicated OOD detectors, open-set recognition, and selective classification should be compared as mechanisms for exposing unknown or ambiguous inputs. Their value should be measured not only through classification metrics but also through missed detections, false-positive workload, deferred cases, review time, and final decision quality. Controlled user studies could compare workflows with and without automated triage, measure analyst overrides, and test whether uncertainty indicators, explicit unknown states, or explanation aids improve the quality of human review.

Matched adversarial evaluation and representation controls. Direct-target attacks should be generated independently against ResNet18 and the CLIP-based proxy, using harmonized access assumptions and perturbation budgets where technically meaningful. This would permit a clearer comparison of direct vulnerability and transferability. Black-box transfer studies should also include clean controls matched to each artifact’s resolution and serialization format, so that perturbation effects can be separated from resizing and encoding. Broader parameter sweeps, alternative objectives, targeted attacks, patches, and physical or hybrid delivery conditions would test whether the observed effects persist under different threat models.

Anti-forensic chains and operationally evaluated defenses. The anti-forensic branch should cover multiple severities and sequential transformations, including repeated JPEG encoding, platform-specific recompression, varied resize ratios, cropping, partial occlusion, screenshots, color-space conversion, and combined tonal and geometric changes. Defensive strategies—such as adversarial training, robust preprocessing, input randomization, ensembles, and anomaly detection—should be evaluated against the same operational criteria. Recovery of aggregate accuracy is insufficient if it increases missed targets, OOD assignments, computational cost, or analyst workload.

Longitudinal black-box validation and metadata controls. Commercial systems should be re-evaluated across software versions, processing modules, categories, and documented configurations. Longitudinal studies would distinguish persistent behavior from release-specific effects and support change-management rules for updates that modify preprocessing, thresholds, taxonomies, retrieval logic, or export formats. Alternative defensible output mappings should be compared, including broader bookmark or label combinations and different Excire semantic queries and distance limits. A paired experiment using metadata-preserving and metadata-neutralized copies of the same visual content should separately measure the effect of embedded non-pixel information.

Explainability, replication, and extension of FAIRLab. The five historical IG cases should be regenerated with the adaptive convergence-validation procedure described in Section 5.7.2, explicit convergence criteria, multiple baselines, and a larger systematic sample. Comparing attribution methods and their stability across checkpoints and perturbations could improve diagnostic reliability, while remaining supplementary to quantitative testing rather than evidence of correctness or causality.

The protocol could then be applied to other forensic visual tasks, including documents, screenshots, illicit substances, explicit or extremist content, vehicles, faces, and logos. Each application would require a task-specific definition, semantic criteria, threat model, output mapping, and operational interpretation. Independent replication across laboratories, datasets, software versions, and tasks is ultimately necessary to assess generalizability.

Where complete redistribution is restricted, replication should still be supported through schemas, scripts, configuration and version records, manifests, stable identifiers, cryptographic digests, sanitized outputs, condition-level metrics, and validation reports. This preserves the distinction between procedural reproducibility and unrestricted access to controlled or proprietary material.

7.8 Closing Remarks

This thesis shows that clean accuracy is a necessary baseline but an insufficient account of operational robustness in forensic image analysis. Systems that perform well under nominal conditions may still miss relevant content, increase false-positive workload, assign OOD inputs to a closed operational category, respond differently after manipulation, or change materially with configuration.

The parallel evaluation of transparent proxy models and selected black-box configurations on the same frozen source population makes these differences observable without conflating inspectable model behavior with proprietary outputs. The FAIRLab protocol provides the traceable structure needed to connect data selection, perturbations, configurations, outputs, normalization, metrics, and reviewable analytical records.

No evaluated system minimizes every operational risk. High recall may increase review workload; greater selectivity may increase missed detections; strong clean performance may coexist with direct adversarial failure. These tradeoffs rule out a universal ranking and make validation conditional on the task, threat model, version, configuration, and output mapping.

Automated classifications, tags, bookmarks, and semantic retrieval results should therefore be used as triage indicators rather than autonomous evidentiary conclusions. Their usefulness depends on source-to-output traceability, separate visibility of error directions and OOD behavior, and corroboration within the wider investigative context.

Operational robustness must be demonstrated under the conditions in which a system is expected to operate; it cannot be presumed from nominal performance, proprietary status, or an apparently confident output. In forensic image analysis, automation is most defensible as a controlled, auditable, and reviewable component of a *human-in-the-loop* workflow.

Appendix A

Repository and Reproducibility Artifacts

This appendix documents the technical artifacts that support the reproducibility, verifiability, and traceability of the experimental work described in the thesis. It does not replace the methodology presented in Chapter 4, nor does it anticipate the analysis of the results. Rather, it provides an operational reference for connecting the academic description of the pipeline to the scripts, manifests, hashes, and outputs maintained in the repository associated with the project. This approach is consistent with the general principles of DF, in which the preservation of provenance, artifact integrity, and documentation of operations are essential requirements for making a digital analysis process verifiable [4, 5, 8, 9].

The complete source code is not reported in full within the document, since duplicating it in the thesis would make the text redundant and less maintainable. The repository instead represents the appropriate location for consulting the scripts, manifests, and operational documentation. The appendix therefore describes the logical structure of the project, the official artifacts of the pipeline, the main execution sequence, the traceability mechanisms, and the boundaries within which reproducibility must be interpreted. In particular, reproducibility concerns the controlled reconstruction of the experimental artifacts starting from the official manifests and scripts, whereas it does not necessarily imply the public redistribution of all raw images, which may be limited by legal, ethical, or source-dependent constraints.

A.1 Repository Structure

The repository is organized according to a functional separation among raw data, prepared data, the final dataset, experimental splits, perturbations, proxy models, evaluation, explainability, results, and thesis documentation. This organization makes it possible to distinguish the phases of acquisition, preparation, manual selection, split generation, attack production, automated evaluation, and preparation of the bundle intended for selected black-box software systems.

```

1 msc-thesis-ai-robustness-in-digital-forensics/
2   datasets/
3     raw/
4     prepared/
5     final/
6     splits/
7     forensic_evaluation_bundle/
8     scripts/
9       acquisition/
10      prepared/
11      final/
12      splits/
13      attacks/
14      bundle/
15      utils/
16   attacks/
17     adversarial/
18     anti_forensic/
19     manifests/
20   models/
21     scripts/
22     checkpoints/
23     reports/
24   evaluation/
25     scripts/
26     proxy_models/
27     forensic_tools/
28   forensic_tools/
29     magnet_axiom/
30     griffeye/
31     excire_foto_2025/
32     cellebrite_insejets/
33     scripts/
34   explainability/
35     scripts/
36     manifests/
37     logs/
38   results/
39     scripts/
40     metrics/
41     figures/
42     latex_image_audit/
43   docs/
44     LatexThesis/
45     LatexSlides/
46     artifact/
47     assets/
48     maintenance/
49     slides/
50   tools/
51     latex/
52   .github/

```

Listing A.1: Simplified structure of the experimental repository.

The `datasets/` directory contains the data pipeline, from the initial collection phase to the forensic evaluation bundle. The `attacks/` directory collects the outputs of adversarial perturbations and anti-forensic transformations, whereas `models/` contains the scripts and checkpoints of the proxy models. The `evaluation/` directory collects the scripts and outputs of automated evaluation and normalization of the exports produced by selected black-box software systems. The `forensic_tools/` directory contains the normalized black-box evaluation perimeter and related exported artifacts, whereas the `results/scripts/` directory contains the reporting and audit scripts used to generate thesis-ready figures, tables, and sensitivity checks. The `explainability/` directory is reserved for the generation of xAI-based case studies; complete generated visualization trees remain local when they cannot be redistributed, while public manifests, scripts, logs, and selected reporting assets remain auditable. The `docs/LatexThesis/` directory contains the official LaTeX version of the thesis, and `tools/latex/` contains supporting layout-audit utilities.

A.2 Official Experimental Artifacts

Table A.1 summarizes the main artifacts of the pipeline. These files are not merely intermediate outputs, but constitute the anchoring points between the methodology described in the body of the thesis and its experimental implementation. In particular, the manifests preserve the connection among image identifier, source, final label, hash, split, possible perturbation, and path of the generated file.

Table A.1: Main official artifacts of the experimental pipeline

Artifact	Role in the pipeline
<code>datasets/prepared/final_pool/metadata.csv</code>	Technical metadata of the prepared pool, generated after image validation, hash computation, and removal of exact duplicates.
<code>datasets/prepared/manifests/review_manifest_full.csv</code>	Complete manifest used as the basis for the <i>human-in-the-loop</i> manual review protocol.
<code>datasets/final/manifests/manual_selection_protocol_db.csv</code>	Operational database of the manual selection process, including review status and semantic assignments.
<code>datasets/final/manifests/manual_selection_final_1500.csv</code>	Official manifest of the frozen source population, containing 1 500 images organized into three equally sized branches: 500 weapon images, 500 non_weapon images, and 500 images reserved for separate OOD evaluation.
<code>datasets/final/manifests/manual_selection_adversarial_subset.csv</code>	Official manifest of the binary subset, composed of the weapon and non_weapon classes, used for clean splits, adversarial attacks, and anti-forensic transformations.
<code>datasets/final/manifests/manual_selection_removed.csv</code>	Manifest of the samples excluded during the final selection, useful for auditing and checking review decisions.
<code>datasets/splits/manifests/clean_folds_manifest.csv</code>	Manifest of the five clean folds, with traceability by class, source, path, hash, and sample type.
<code>datasets/splits/manifests/ood_eval_manifest.csv</code>	Manifest of the OOD set, kept separate from the binary subset and used for out-of-distribution evaluation.
<code>attacks/manifests/</code>	Directory of the manifests produced to connect each perturbed image to the clean sample, fold, true class, and type of attack or transformation.

Continued on the next page

Artifact	Role in the pipeline
<code>datasets/forensic_evaluation_bundle/meta_data/bundle_manifest.csv</code>	Global manifest of the bundle intended for evaluation with selected black-box software systems.
<code>datasets/forensic_evaluation_bundle/meta_data/bundle_hashes_sha256.csv</code>	Control file containing the SHA-256 hashes of the artifacts included in the forensic bundle.
<code>datasets/forensic_evaluation_bundle/meta_data/bundle_summary.json</code>	Structured summary of the bundle, including counts, consistency checks, and operational instructions for import into selected black-box software systems.
<code>datasets/forensic_evaluation_bundle/meta_data/embedded_metadata_audit.csv</code>	Audit file generated during bundle construction to document residual embedded metadata in the blind input view.
<code>results/figures/chapter_5/chapter5_figures_manifest.csv</code>	Manifest of the thesis-ready figures generated from the consolidated metric files by the reporting layer.
<code>results/figures/chapter_5/chapter5_figures_summary.json</code>	Structured summary of the experimental reporting assets generated for Chapter 6.
<code>results/figures/chapter_5/embedded_metadata_sensitivity_summary.json</code>	Structured summary of the embedded-metadata sensitivity check used to support the audit discussion in Chapter 6.

The directory names containing `chapter5` or `chapter_5` are retained as historical repository identifiers. They refer to artifacts used by the current experimental-results chapter and do not indicate the present chapter number.

A.3 Dataset Traceability and Human-in-the-Loop Selection

The data pipeline is based on a combination of automatic checks and manual review. Images acquired from the original sources are technically validated, deduplicated through hashes, and inserted into a prepared pool. Only afterward are they included in the review manifest, in which the final classification is assigned through a *human-in-the-loop* protocol. This methodological choice is consistent with the forensic purpose of the work: the dataset is not treated as a purely automatic product, but as a verifiable, frozen, and documented experimental artifact.

The final source population is represented by the `manual_selection_final_1500.csv` manifest and contains 1 500 images organized into three equally sized branches.

The supervised binary component comprises 500 **weapon** images and 500 **non_weapon** images. The **weapon** class contains realistic individual firearms and clearly recognizable firearm-related visual content, whereas the **non_weapon** class contains realistic negative samples consistent with the binary task.

A separate 500-image OOD branch contains borderline, anomalous, degraded, synthetic, semantically adjacent, or otherwise out-of-task content. Examples include knives, swords, replicas, airsoft items, synthetic renderings, video game images, heavy military vehicles, explosions, and war scenes. This branch is not a third supervised class and is excluded from proxy-model training and perturbation generation.

The official adversarial subset, stored in `manual_selection_adversarial_subset.csv`, therefore excludes the OOD branch and retains only the two supervised operational classes, **weapon** and **non_weapon**, for a total of 1 000 images.

Stable identifiers, SHA-256 hashes, relative paths, and intermediate manifests make it possible to reconstruct the derivation of each sample from the prepared pool to the splits, perturbed

versions, and evaluation outputs. This connection is particularly relevant in the black-box evaluation phase, since selected software systems may rename files, modify the structure of exports, or produce reports in heterogeneous formats. The use of hashes therefore makes it possible to maintain a criterion that is independent of the file name and export path.

A.4 Official Script Sequence

The operational pipeline uses numbered scripts as official entry points. This numbering documents the logical order of the operations and reduces ambiguity in the reconstruction of the workflow. Listing A.2 reports the main sequence aligned with the current state of the repository.

```

1 # Raw acquisition and source preparation
2 python datasets/scripts/acquisition/00_download_raw_datasets_bundle.py
3 python datasets/scripts/acquisition/01_download_kaggle.py
4 python datasets/scripts/acquisition/02_download_github.py
5 python datasets/scripts/acquisition/03_build_subset_deepfirearm.py
6 python datasets/scripts/acquisition/04_scrape_google.py
7 python datasets/scripts/acquisition/05_scrape_telegram.py
8 python datasets/scripts/acquisition/06_scrape_youtube.py
9 python datasets/scripts/acquisition/07_scrape_deepweb.py
10
11 # Prepared dataset and human-in-the-loop selection
12 python datasets/scripts/prepared/08_build_prepared_dataset.py
13 python datasets/scripts/prepared/09_generate_review_manifest_full.py
14 python datasets/scripts/final/10_manual_selection_protocol_reviewer.py
15
16 # Clean/OOD split generation
17 python datasets/scripts/splits/11_generate_clean_and_ood_splits.py
18
19 # Proxy model training and perturbation generation
20 python models/scripts/12_train_proxy_models.py
21 python datasets/scripts/attacks/13_generate_anti_forensic_attacks.py
22 python datasets/scripts/attacks/14_generate_adversarial_attacks.py
23
24 # Proxy model evaluation and forensic bundle construction
25 python evaluation/scripts/15_evaluate_proxy_models.py
26 python datasets/scripts/bundle/16_build_forensic_evaluation_bundle.py
27
28 # Explainability case-study preparation
29 python explainability/scripts/17_generate_integrated_gradients_case_studies.py
30 python explainability/scripts/18_xai_interactive_launcher.py
31
32 # Black-box software export normalization
33 python evaluation/scripts/19_normalize_forensic_ai_tool_predictions.py
34
35 # Experimental reporting assets and metadata sensitivity
36 python results/scripts/20_generate_experimental_reporting_assets.py
37 python results/scripts/21_generate_embedded_metadata_sensitivity_check.py

```

Listing A.2: Official sequence of the main pipeline scripts.

The script `evaluation/scripts/19_normalize_forensic_ai_tool_predictions.py` constitutes the connection point between the exports produced by selected black-box software systems and the final comparative metrics. In the final protocol, it exclusively supports outputs attributable to `magnet_axiom`, `griffeye`, `excire_foto_2025`, and `celebrite_inseyets`. Its role is to normalize the exports, map predictions back to the records of the forensic evaluation bundle, and produce results that are comparable with the proxy level through mapping based, in preferential order, on `bundle_id`, SHA-256 hash, MD5 hash, file name, and manual audit.

The final two scripts belong to the reporting and audit layer of the pipeline. The script `results/scripts/20_generate_experimental_reporting_assets.py` reads the consolidated metric files and produces thesis-ready figures, tables, manifest files, and summaries for Chapter 6. The script `results/scripts/21_generate_embedded_metadata_sensitivity`

`ity_check.py` connects the embedded-metadata audit generated during bundle construction with the normalized black-box predictions, producing the sensitivity tables and summary used to document the residual metadata limitation.

A.5 Models, Attacks, and Evaluation Outputs

The consolidated experimental protocol uses three proxy models: `efficientnet_b0`, `resnet18`, and CLIP. These models are used as transparent and inspectable baselines to observe, in a controlled manner, the behavior of classifiers with respect to clean, perturbed, and out-of-distribution inputs. Any preliminary components or exploratory baselines not included in the final evaluation are not considered part of the consolidated experimental protocol. The use of proxy models places the analysis within the broader context of robustness in ML and AI systems [15, 39, 45].

The pipeline generates four model-dependent adversarial attacks—`fgsm`, `superdeepfool`, `sigma_zero`, and `one_pixel`—and the model-agnostic `color_shift` comparison condition. The anti-forensic transformations are `jpeg_recompression`, `resample_resize`, `gaussian_blur`, `histogram_modification`, and `contrast_stretching`. This distinction is methodologically important: model-dependent adversarial attacks evaluate decision vulnerability under the specified access assumptions, Color Shift provides a shared model-agnostic perturbation condition, and anti-forensic transformations simulate operationally plausible manipulations of the image content or file that may alter the visibility, quality, or technical traces of the artifact [16, 59, 86].

The outputs of the proxy model evaluation are collected in the `evaluation/proxy_models/` and `results/metrics/` directories. These outputs include predictions, aggregate metrics, confusion matrices, and comparisons among clean, adversarial, anti-forensic, and OOD conditions. Their function is not only quantitative, but also methodological: they make it possible to compare the controlled behavior of the proxy models with that of the AI-based software evaluated in *black-box* mode. The downstream reporting assets generated under `results/figures/chapter_5/` provide the thesis-ready figures, figure manifests, and summary files derived from these consolidated metrics.

A.6 Forensic Evaluation Bundle

The *forensic evaluation bundle* constitutes the operational artifact used to submit a common set of images to the AI-based software evaluated in *black-box* mode, without revealing the corresponding ground truth in advance. Its methodological function is to prevent file names, directory structure, or experimental metadata from influencing the analysis procedure or making the true class and applied transformation immediately recognizable. The bundle is generated by the script `datasets/scripts/bundle/16_build_forensic_evaluation_bundle.py` and is saved in the `datasets/forensic_evaluation_bundle/` directory; its structure strictly separates the blind inputs intended for the software systems from the metadata required for the subsequent evaluation through hash-based mapping.

```
1 datasets/forensic_evaluation_bundle/  
2   blind_tool_input/  
3     files/  
4   metadata/  
5     bundle_manifest.csv  
6     bundle_hashes_sha256.csv  
7     bundle_summary.json  
8   structured_audit_view/
```

Listing A.3: Logical structure of the forensic evaluation bundle.

In the *black-box* evaluation, only the files contained in `blind_tool_input/files/` must be imported into the considered software systems. The `metadata/` and `structured_audit_view/`

directories must not be imported, since they contain information that would reveal true classes, sources, attacks, transformations, and experimental mapping. This separation reduces the risk of bias induced by file names, directory structure, or prior knowledge of the ground truth.

The current bundle includes clean images, OOD samples, adversarial perturbations, and anti-forensic transformations. The operational summary of the bundle documents 11 500 files in total: 1 000 clean images, 500 OOD samples, 5 000 adversarial samples, and 5 000 anti-forensic samples. The validation checks include uniqueness of the `bundle_id`, uniqueness of the actual hashes, correspondence of hashes when present in the manifest, semantic cleanliness of the blind paths, and separation of metadata from the inputs intended for the software systems.

A.7 Software Environment and Dependencies

The pipeline is implemented primarily in Python and uses libraries for tabular manifest management, image processing, model execution, and production of experimental outputs. Table A.2 summarizes the main software components. The exact versions of the dependencies should be maintained in the repository environment files or in the experimental logs, since computational reproducibility may be affected by variations in libraries, numerical backends, Graphics Processing Unit (GPU) drivers, and hardware availability.

Table A.2: Main software components used in the consolidated pipeline

Component	Main function
Python	Main language used to orchestrate the experimental pipeline.
pandas	Management of manifests, metadata, tabular reports, and experimental results.
NumPy	Numerical support for arrays, transformations, and matrix operations.
SciPy	Support for numerical operations and transformations used in the experimental pipeline.
Pillow	Image opening, validation, conversion, and saving.
PyTorch	Training, inference, and perturbation generation on neural models.
torchvision	CNN architectures, preprocessing, and support for proxy model evaluation.
open_clip_torch	Implementation of the CLIP model used as a proxy model.
Captum	Generation of attribution maps through IG.
matplotlib	Production of plots, curves, experimental figures, and diagnostic visualizations.

The operational-forensic evaluation also includes four selected software systems observed in *black-box* mode: Magnet AXIOM, version 10.1.0.48673, with Magnet.AI functionality; Magnet Griffeye x64, version 26.2.108, with the T3K CORE v1.18.0 module; Excire Foto 2025, version 4.1.5, used as a stand-alone general-purpose application with AI-based functionality; and Cellebrite Inseyets, version 10.9, with exports produced in the *Cellebrite Physical Analyzer* 10.9.0.3029 environment. Unlike Python scripts and proxy models, these systems introduce a component that is not fully automatable, due to licenses, graphical interfaces, software versions, operational configurations, and export formats. For this reason, their evaluation is documented through the blind bundle, operational instructions, normalized exports, and hash-based mapping.

A.8 Logs, Reports, and Audit Outputs

In addition to the main manifests, the repository preserves logs, summaries, and intermediate reports that are useful for verifying the operations performed. These artifacts do not all have the same role: some describe the status of the manual selection process, others summarize split generation, and others document the composition of the bundle or the evaluation of proxy

models. Their overall function is to make the experimental path directly observable and to support any subsequent checks.

```

1 [INFO] Scanning raw dataset sources...
2 [INFO] Valid images detected: ...
3 [INFO] Invalid images discarded: ...
4 [INFO] SHA256 hashes computed: ...
5 [INFO] Global duplicates removed: ...
6 [INFO] Prepared image pool written to datasets/prepared/final_pool/images/
7 [INFO] Metadata written to datasets/prepared/final_pool/metadata.csv

```

Listing A.4: Schematic example of output from the dataset preparation phase.

The most relevant audit artifacts include `manual_selection_log.csv`, `manual_selection_state.json`, `manual_selection_summary.json`, `split_generation_summary.json`, the attack manifests, `bundle_summary.json`, `results/figures/chapter_5/chapter5_figures_manifest.csv`, `results/figures/chapter_5/chapter5_figures_summary.json`, and `results/figures/chapter_5/embedded_metadata_sensitivity_summary.json`. These files make it possible to reconstruct not only the final output, but also the status and operational assumptions of the main phases of the pipeline.

A.9 Reproducibility Boundaries

The reproducibility of the pipeline must be interpreted consistently with the nature of the case study. The deterministically re-executable phases include the technical preparation of the dataset, manifest generation, split construction, production of perturbed outputs, and construction of the bundle, within the limits imposed by software dependencies and input availability. The final selection of the dataset, instead, includes a deliberate, documented, and motivated manual component, which must not be concealed as if it were a fully automatic procedure. The scientific value of this component lies in its traceability, in the explicit definition of the criteria, and in the freezing of the final manifest.

Similarly, the evaluation with selected black-box software systems is not entirely reproducible through command-line execution, since it depends on proprietary software, specific versions, interface settings, internal AI-based modules, and export procedures. To mitigate this limitation, the pipeline uses a common blind input, preserves file hashes, separates the ground truth from the inputs submitted to the software systems, and includes a normalization phase for the exports. In this way, reproducibility is not understood as a bit-for-bit replica of every graphical interaction, but as verifiability of the protocol, inputs, mappings, and final metrics.

A.10 Repository Availability

The repository associated with the thesis is available at the following address:

<https://github.com/lmolinario/msc-thesis-ai-robustness-in-digital-forensics>

The repository contains the main scripts, official manifests, operational documentation, and artifacts required to reconstruct the experimental workflow. Before the final public release, the availability of raw images, generated outputs, and sensitive materials must be assessed according to the legal, ethical, and licensing constraints of the original sources. The source code of the repository is released under the MIT license, as indicated in the `LICENSE` file. This license applies to the code and experimental scripts, but it does not automatically imply the free redistribution of the thesis text, figures, datasets, raw images, selected software exports, model checkpoints, or generated artifacts. These materials must be considered separately, according to their respective legal, ethical, institutional, and source-dependent licensing constraints.

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